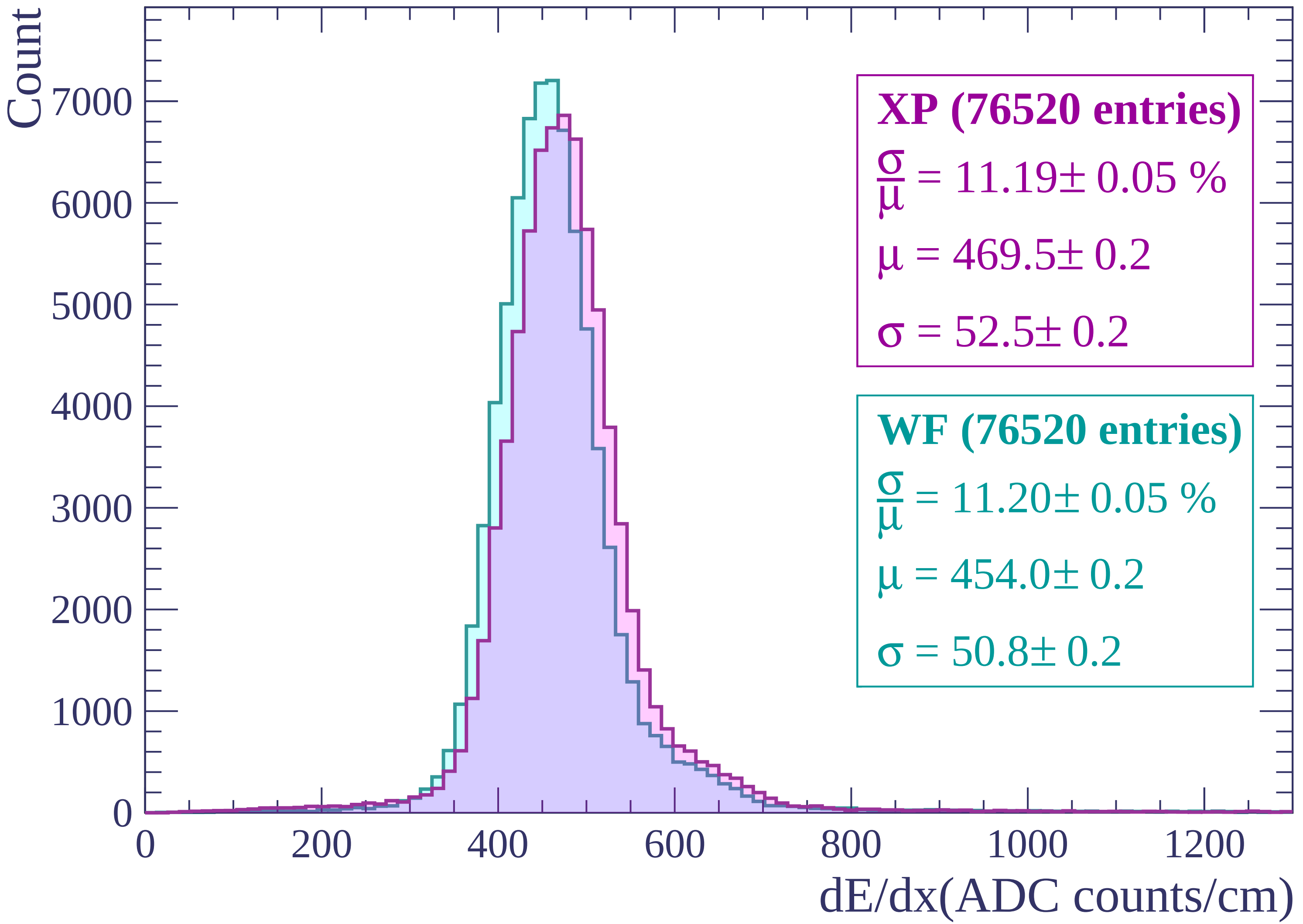
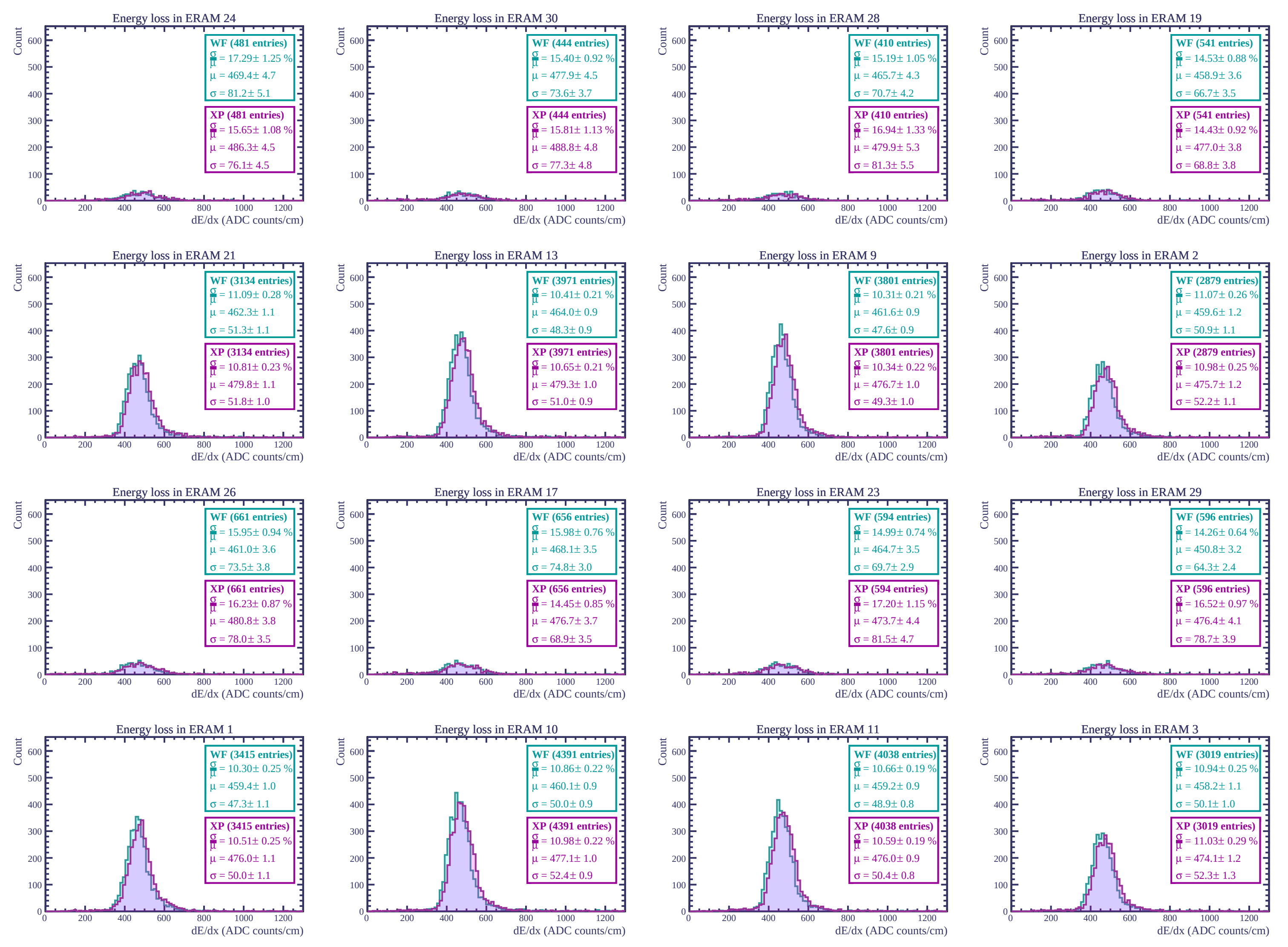
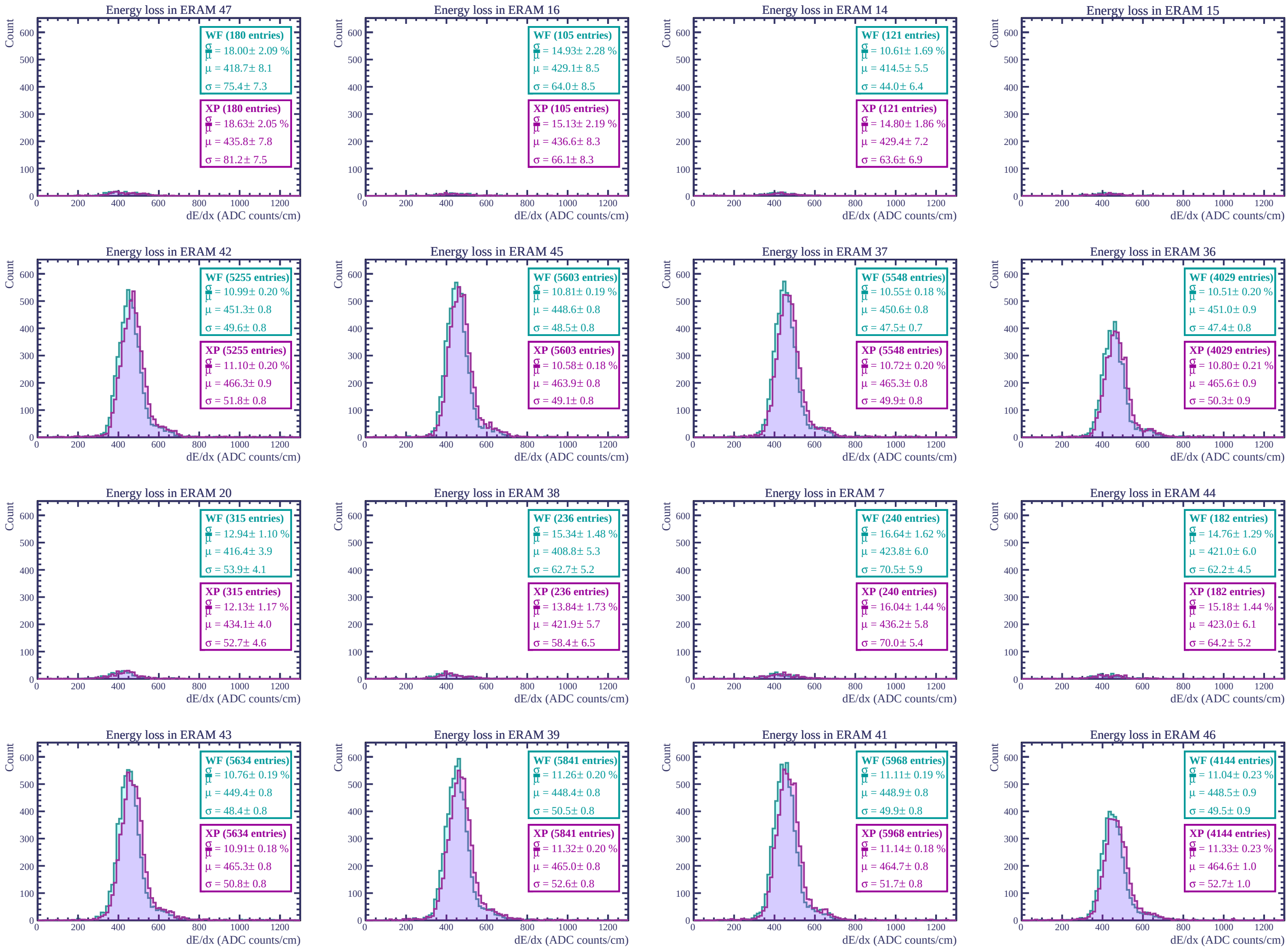
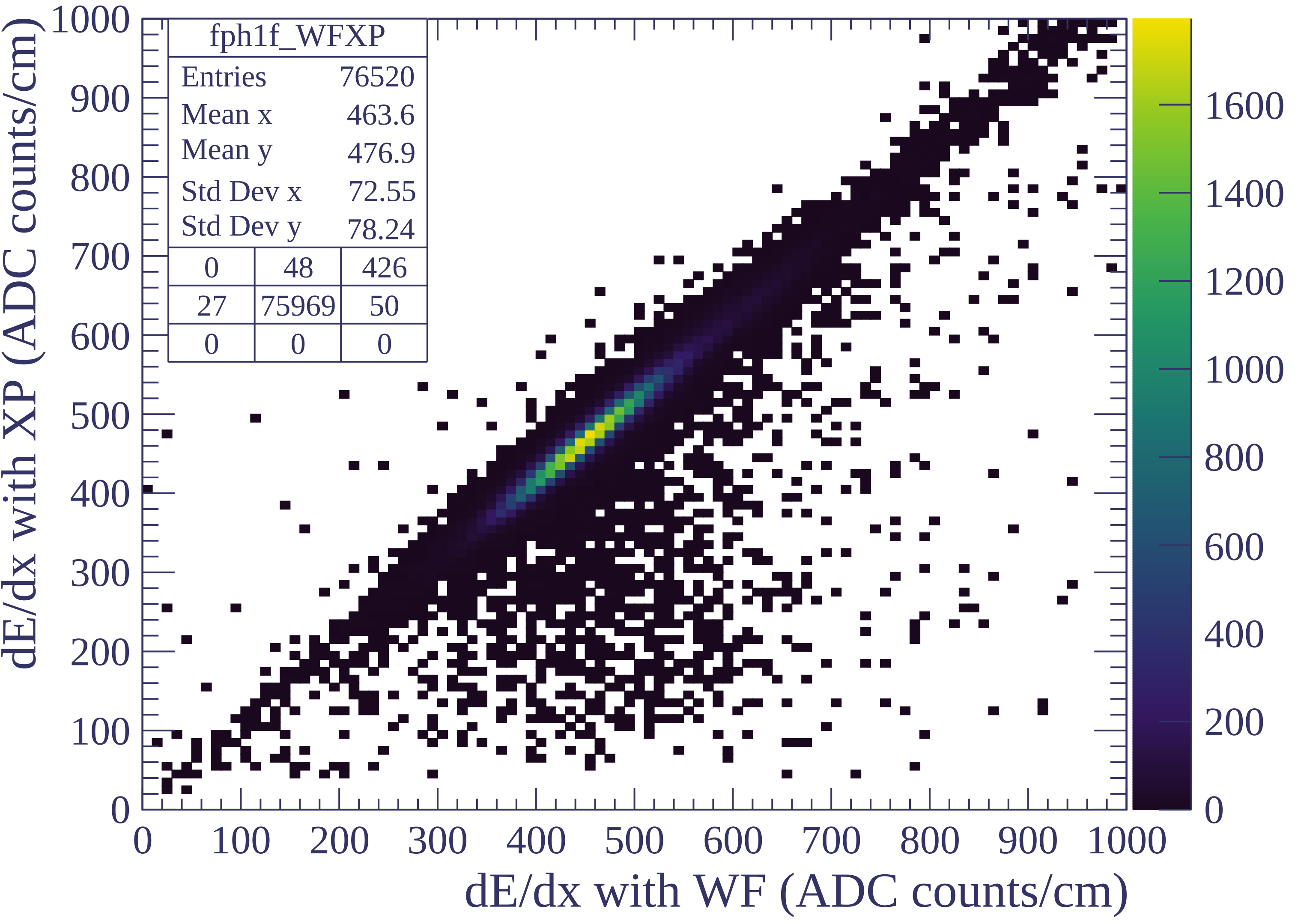


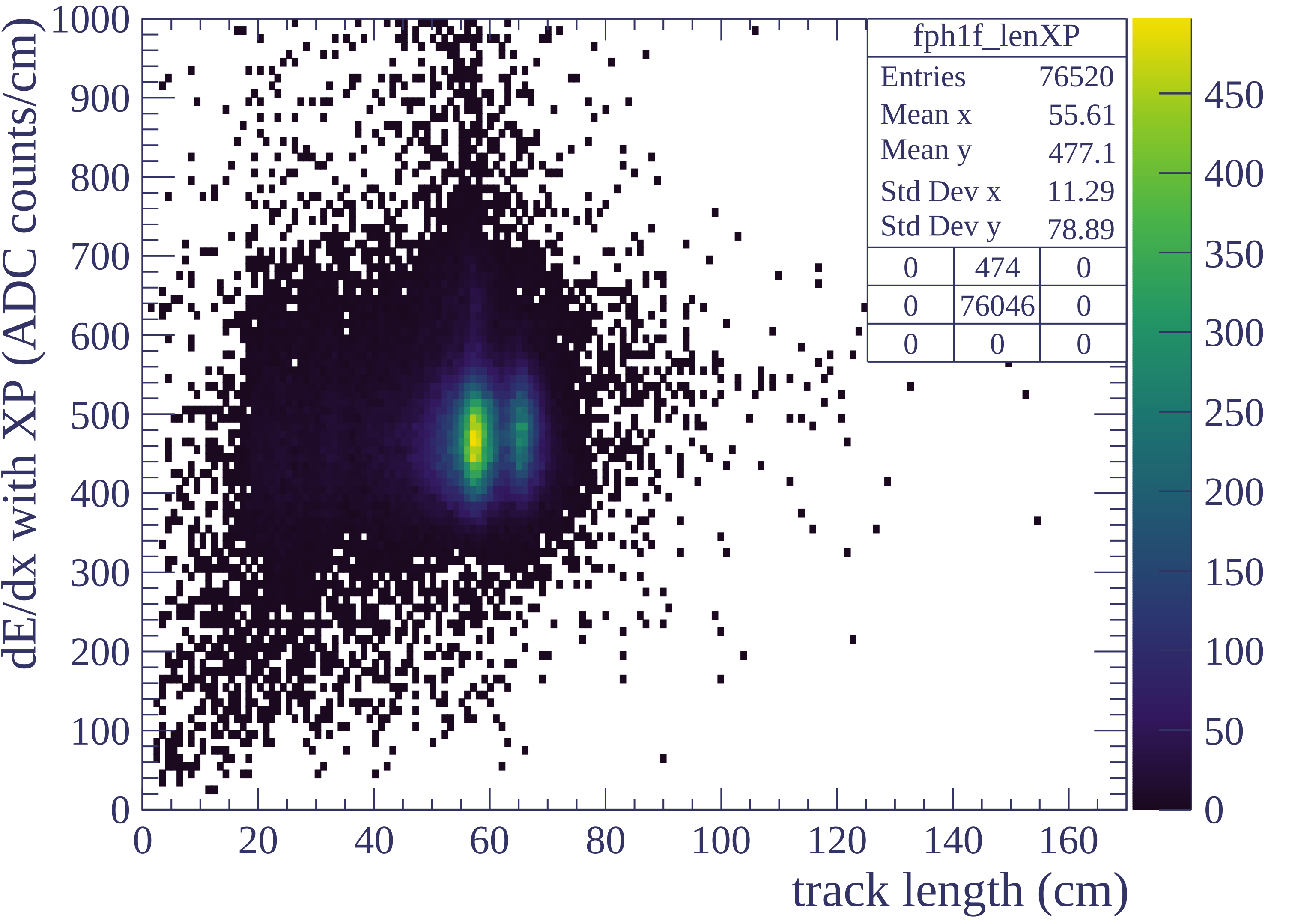
Count
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (76520 entries)
σ/μ = 11.19± 0.05 %
μ = 469.5± 0.2
σ = 52.5± 0.2
WF (76520 entries)
σ/μ = 11.20± 0.05 %
μ = 454.0± 0.2
σ = 50.8± 0.2



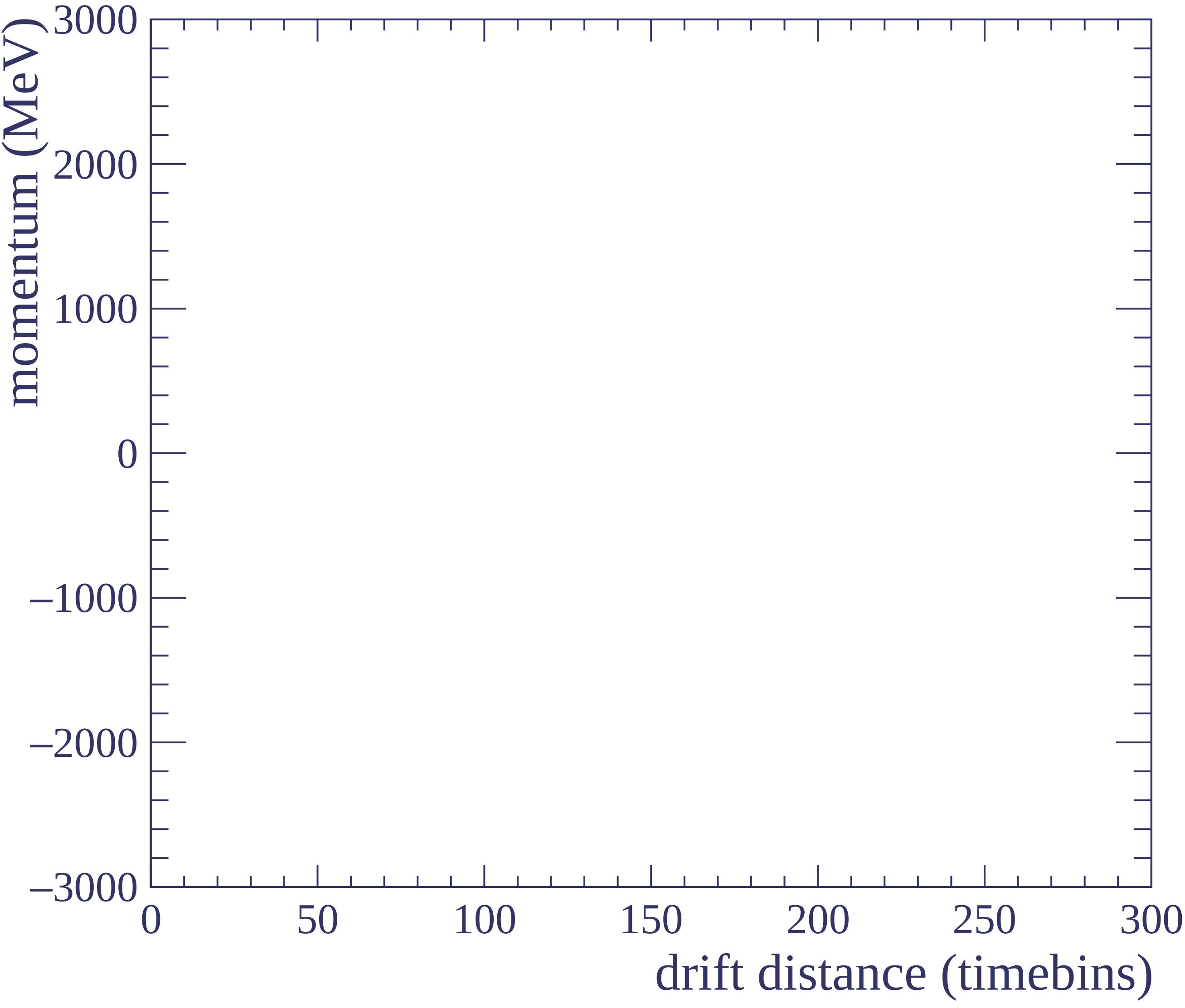
Energy loss in ERAM 24
WF (481 entries)
σ/μ = 17.29± 1.25 %
μ = 469.4± 4.7
σ = 81.2± 5.1
XP (481 entries)
σ/μ = 15.65± 1.08 %
μ = 486.3± 4.5
σ = 76.1± 4.5
Energy loss in ERAM 30
WF (444 entries)
σ/μ = 15.40± 0.92 %
μ = 477.9± 4.5
σ = 73.6± 3.7
XP (444 entries)
σ/μ = 15.81± 1.13 %
μ = 488.8± 4.8
σ = 77.3± 4.8
Energy loss in ERAM 28
WF (410 entries)
σ/μ = 15.19± 1.05 %
μ = 465.7± 4.3
σ = 70.7± 4.2
XP (410 entries)
σ/μ = 16.94± 1.33 %
μ = 479.9± 5.3
σ = 81.3± 5.5
Energy loss in ERAM 19
WF (541 entries)
σ/μ = 14.53± 0.88 %
μ = 458.9± 3.6
σ = 66.7± 3.5
XP (541 entries)
σ/μ = 14.43± 0.92 %
μ = 477.0± 3.8
σ = 68.8± 3.8
Energy loss in ERAM 21
WF (3134 entries)
σ/μ = 11.09± 0.28 %
μ = 462.3± 1.1
σ = 51.3± 1.1
XP (3134 entries)
σ/μ = 10.81± 0.23 %
μ = 479.8± 1.1
σ = 51.8± 1.0
Energy loss in ERAM 13
WF (3971 entries)
σ/μ = 10.41± 0.21 %
μ = 464.0± 0.9
σ = 48.3± 0.9
XP (3971 entries)
σ/μ = 10.65± 0.21 %
μ = 479.3± 1.0
σ = 51.0± 0.9
Energy loss in ERAM 9
WF (3801 entries)
σ/μ = 10.31± 0.21 %
μ = 461.6± 0.9
σ = 47.6± 0.9
XP (3801 entries)
σ/μ = 10.34± 0.22 %
μ = 476.7± 1.0
σ = 49.3± 1.0
Energy loss in ERAM 2
WF (2879 entries)
σ/μ = 11.07± 0.26 %
μ = 459.6± 1.2
σ = 50.9± 1.1
XP (2879 entries)
σ/μ = 10.98± 0.25 %
μ = 475.7± 1.2
σ = 52.2± 1.1
Energy loss in ERAM 26
WF (661 entries)
σ/μ = 15.95± 0.94 %
μ = 461.0± 3.6
σ = 73.5± 3.8
XP (661 entries)
σ/μ = 16.23± 0.87 %
μ = 480.8± 3.8
σ = 78.0± 3.5
Energy loss in ERAM 17
WF (656 entries)
σ/μ = 15.98± 0.76 %
μ = 468.1± 3.5
σ = 74.8± 3.0
XP (656 entries)
σ/μ = 14.45± 0.85 %
μ = 476.7± 3.7
σ = 68.9± 3.5
Energy loss in ERAM 23
WF (594 entries)
σ/μ = 14.99± 0.74 %
μ = 464.7± 3.5
σ = 69.7± 2.9
XP (594 entries)
σ/μ = 17.20± 1.15 %
μ = 473.7± 4.4
σ = 81.5± 4.7
Energy loss in ERAM 29
WF (596 entries)
σ/μ = 14.26± 0.64 %
μ = 450.8± 3.2
σ = 64.3± 2.4
XP (596 entries)
σ/μ = 16.52± 0.97 %
μ = 476.4± 4.1
σ = 78.7± 3.9
Energy loss in ERAM 1
WF (3415 entries)
σ/μ = 10.30± 0.25 %
μ = 459.4± 1.0
σ = 47.3± 1.1
XP (3415 entries)
σ/μ = 10.51± 0.25 %
μ = 476.0± 1.1
σ = 50.0± 1.1
Energy loss in ERAM 10
WF (4391 entries)
σ/μ = 10.86± 0.22 %
μ = 460.1± 0.9
σ = 50.0± 0.9
XP (4391 entries)
σ/μ = 10.98± 0.22 %
μ = 477.1± 1.0
σ = 52.4± 0.9
Energy loss in ERAM 11
WF (4038 entries)
σ/μ = 10.66± 0.19 %
μ = 459.2± 0.9
σ = 48.9± 0.8
XP (4038 entries)
σ/μ = 10.59± 0.19 %
μ = 476.0± 0.9
σ = 50.4± 0.8
Energy loss in ERAM 3
WF (3019 entries)
σ/μ = 10.94± 0.25 %
μ = 458.2± 1.1
σ = 50.1± 1.0
XP (3019 entries)
σ/μ = 11.03± 0.29 %
μ = 474.1± 1.2
σ = 52.3± 1.3
Count
dE/dx (ADC counts/cm)

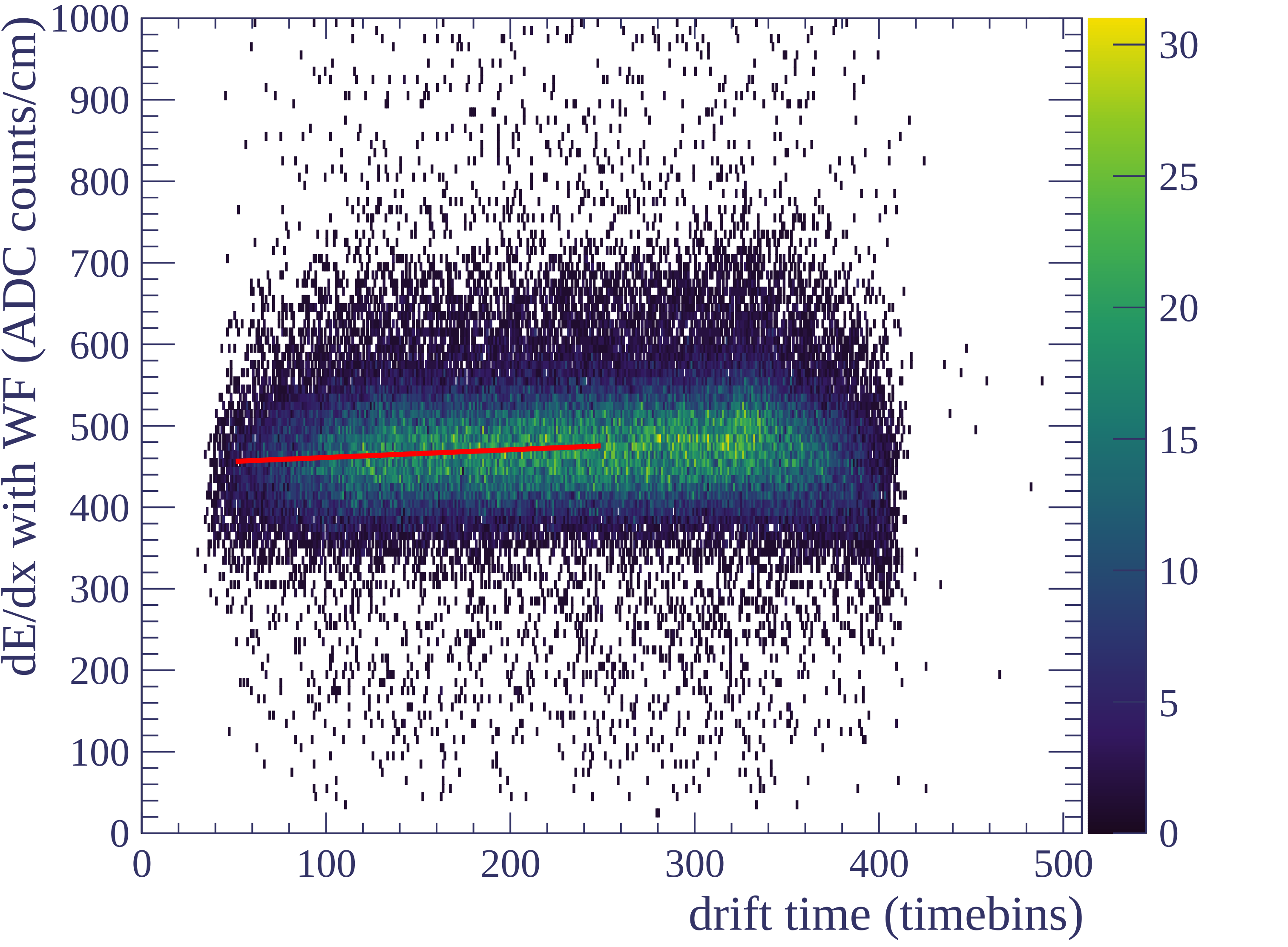

Energy loss in ERAM 47
WF (180 entries)
σ/μ = 18.00± 2.09 %
μ = 418.7± 8.1
σ = 75.4± 7.3
XP (180 entries)
σ/μ = 18.63± 2.05 %
μ = 435.8± 7.8
σ = 81.2± 7.5
Energy loss in ERAM 16
WF (105 entries)
σ/μ = 14.93± 2.28 %
μ = 429.1± 8.5
σ = 64.0± 8.5
XP (105 entries)
σ/μ = 15.13± 2.19 %
μ = 436.6± 8.3
σ = 66.1± 8.3
Energy loss in ERAM 14
WF (121 entries)
σ/μ = 10.61± 1.69 %
μ = 414.5± 5.5
σ = 44.0± 6.4
XP (121 entries)
σ/μ = 14.80± 1.86 %
μ = 429.4± 7.2
σ = 63.6± 6.9
Energy loss in ERAM 15
Energy loss in ERAM 42
WF (5255 entries)
σ/μ = 10.99± 0.20 %
μ = 451.3± 0.8
σ = 49.6± 0.8
XP (5255 entries)
σ/μ = 11.10± 0.20 %
μ = 466.3± 0.9
σ = 51.8± 0.8
Energy loss in ERAM 45
WF (5603 entries)
σ/μ = 10.81± 0.19 %
μ = 448.6± 0.8
σ = 48.5± 0.8
XP (5603 entries)
σ/μ = 10.58± 0.18 %
μ = 463.9± 0.8
σ = 49.1± 0.8
Energy loss in ERAM 37
WF (5548 entries)
σ/μ = 10.55± 0.18 %
μ = 450.6± 0.8
σ = 47.5± 0.7
XP (5548 entries)
σ/μ = 10.72± 0.20 %
μ = 465.3± 0.8
σ = 49.9± 0.8
Energy loss in ERAM 36
WF (4029 entries)
σ/μ = 10.51± 0.20 %
μ = 451.0± 0.9
σ = 47.4± 0.8
XP (4029 entries)
σ/μ = 10.80± 0.21 %
μ = 465.6± 0.9
σ = 50.3± 0.9
Energy loss in ERAM 20
WF (315 entries)
σ/μ = 12.94± 1.10 %
μ = 416.4± 3.9
σ = 53.9± 4.1
XP (315 entries)
σ/μ = 12.13± 1.17 %
μ = 434.1± 4.0
σ = 52.7± 4.6
Energy loss in ERAM 38
WF (236 entries)
σ/μ = 15.34± 1.48 %
μ = 408.8± 5.3
σ = 62.7± 5.2
XP (236 entries)
σ/μ = 13.84± 1.73 %
μ = 421.9± 5.7
σ = 58.4± 6.5
Energy loss in ERAM 7
WF (240 entries)
σ/μ = 16.64± 1.62 %
μ = 423.8± 6.0
σ = 70.5± 5.9
XP (240 entries)
σ/μ = 16.04± 1.44 %
μ = 436.2± 5.8
σ = 70.0± 5.4
Energy loss in ERAM 44
WF (182 entries)
σ/μ = 14.76± 1.29 %
μ = 421.0± 6.0
σ = 62.2± 4.5
XP (182 entries)
σ/μ = 15.18± 1.44 %
μ = 423.0± 6.1
σ = 64.2± 5.2
Energy loss in ERAM 43
WF (5634 entries)
σ/μ = 10.76± 0.19 %
μ = 449.4± 0.8
σ = 48.4± 0.8
XP (5634 entries)
σ/μ = 10.91± 0.18 %
μ = 465.3± 0.8
σ = 50.8± 0.8
Energy loss in ERAM 39
WF (5841 entries)
σ/μ = 11.26± 0.20 %
μ = 448.4± 0.8
σ = 50.5± 0.8
XP (5841 entries)
σ/μ = 11.32± 0.20 %
μ = 465.0± 0.8
σ = 52.6± 0.8
Energy loss in ERAM 41
WF (5968 entries)
σ/μ = 11.11± 0.19 %
μ = 448.9± 0.8
σ = 49.9± 0.8
XP (5968 entries)
σ/μ = 11.14± 0.18 %
μ = 464.7± 0.8
σ = 51.7± 0.8
Energy loss in ERAM 46
WF (4144 entries)
σ/μ = 11.04± 0.23 %
μ = 448.5± 0.9
σ = 49.5± 0.9
XP (4144 entries)
σ/μ = 11.33± 0.23 %
μ = 464.6± 1.0
σ = 52.7± 1.0
Count
dE/dx (ADC counts/cm)


fph1f_WFXP
Entries 76520
Mean x 463.6
Mean y 476.9
Std Dev x 72.55
Std Dev y 78.24
0 48 426
27 75969 50
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
200
400
600
800
1000
1200
1400
1600

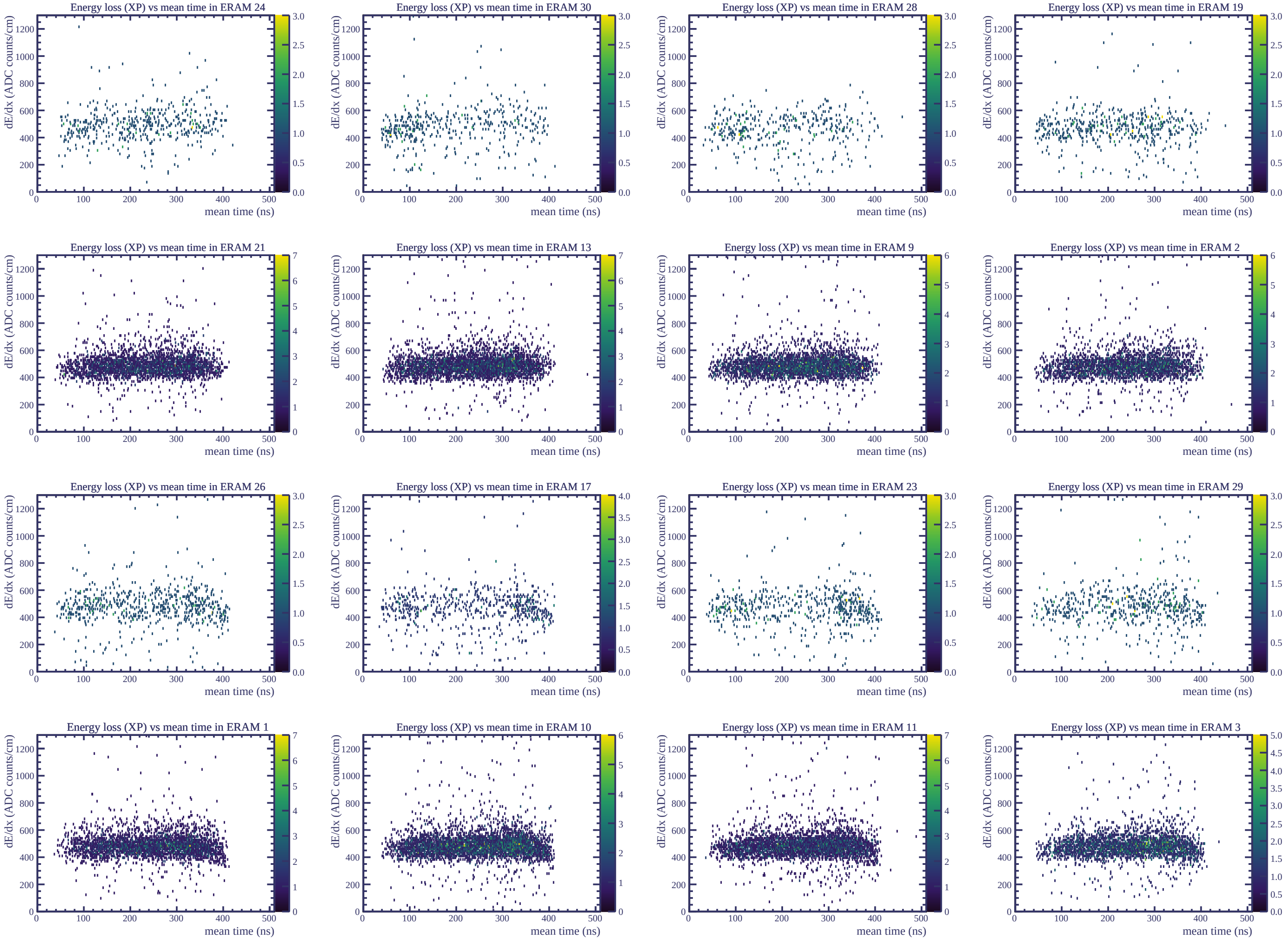


fph1f_lenXP
Entries 76520
Mean x 55.61
Mean y 477.1
Std Dev x 11.29
Std Dev y 78.89
0 474 0
0 76046 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250
300
350
400
450

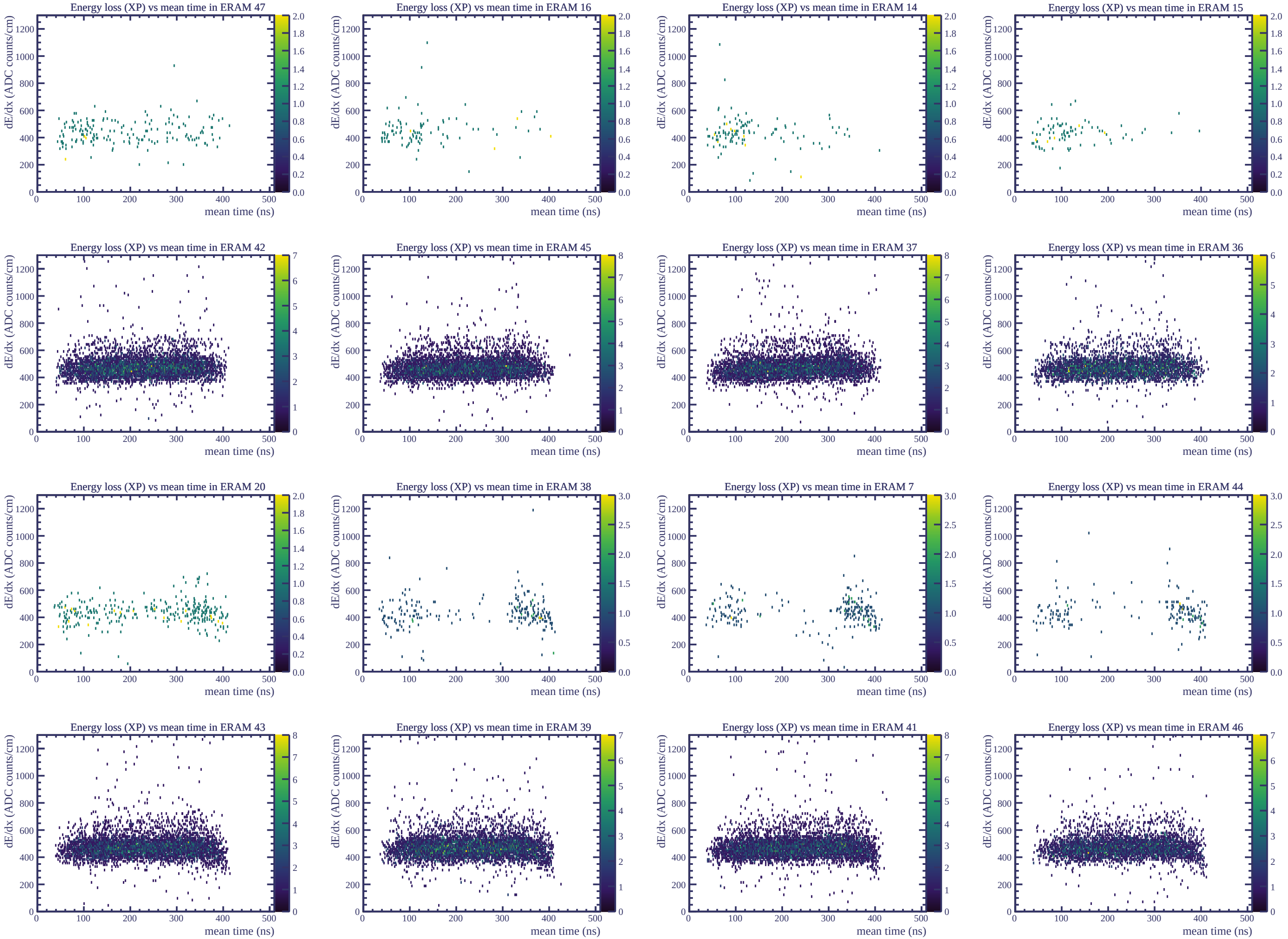

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
0
50
100
150
200
250
300
drift distance (timebins)

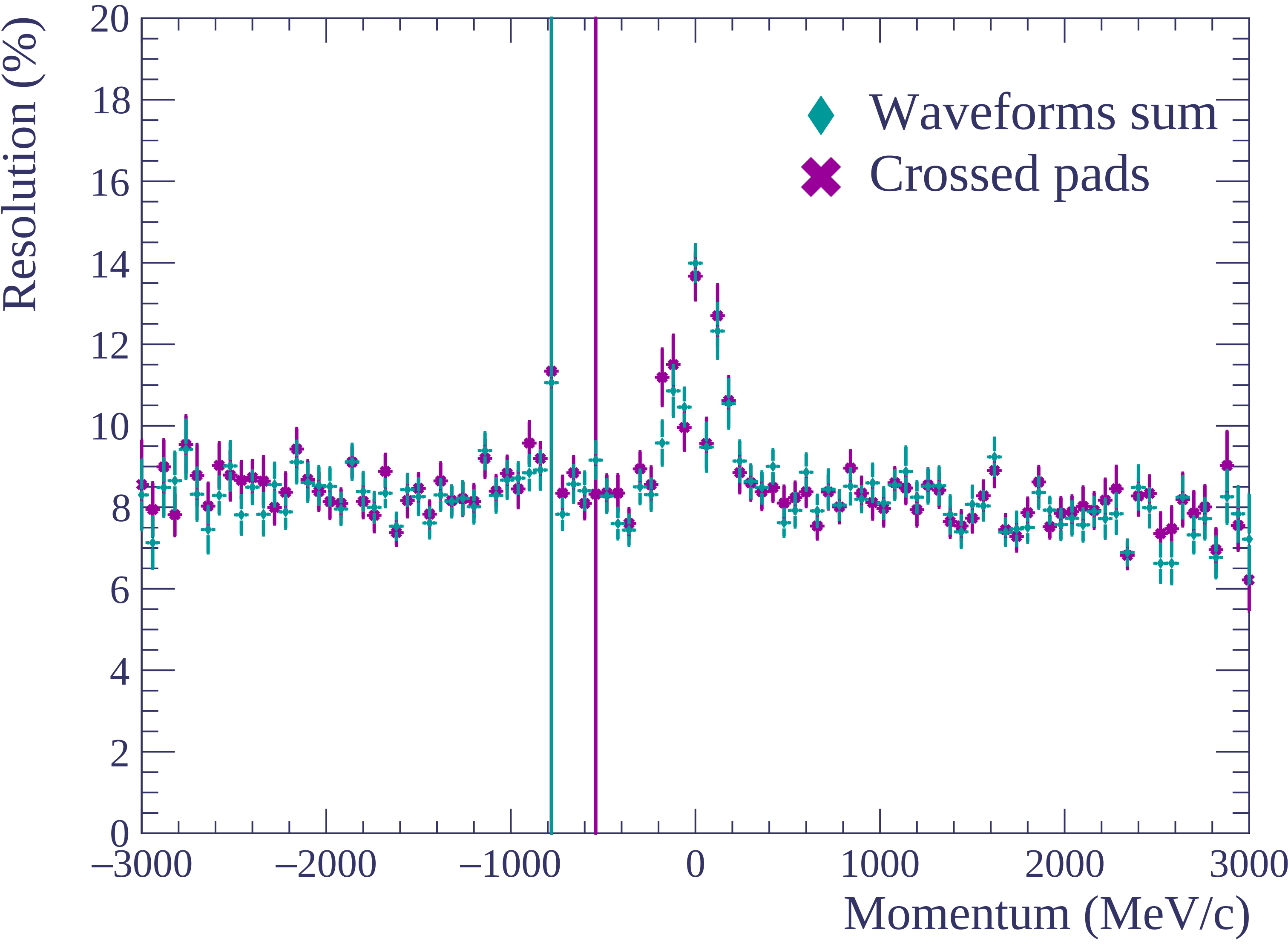


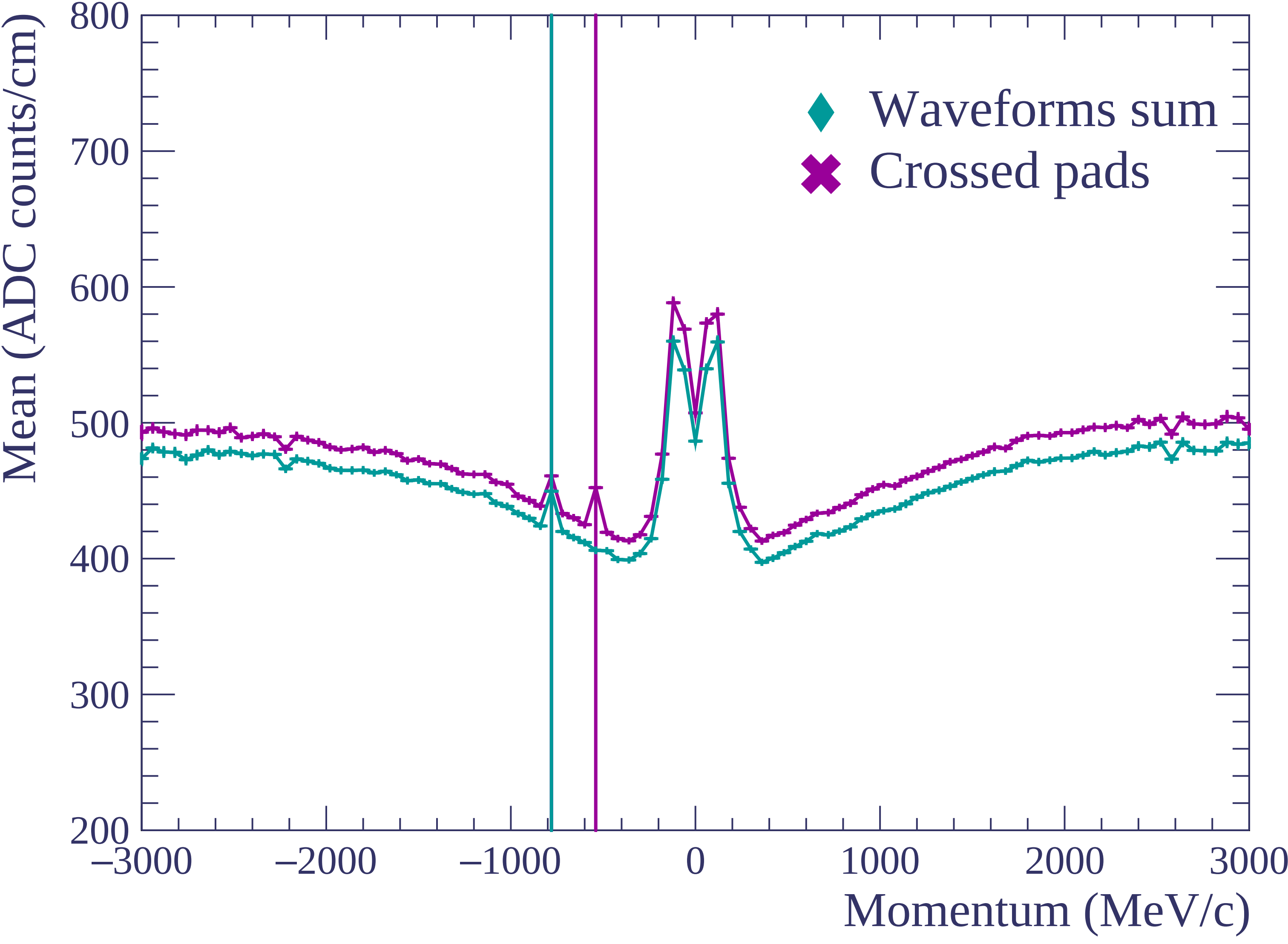
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30



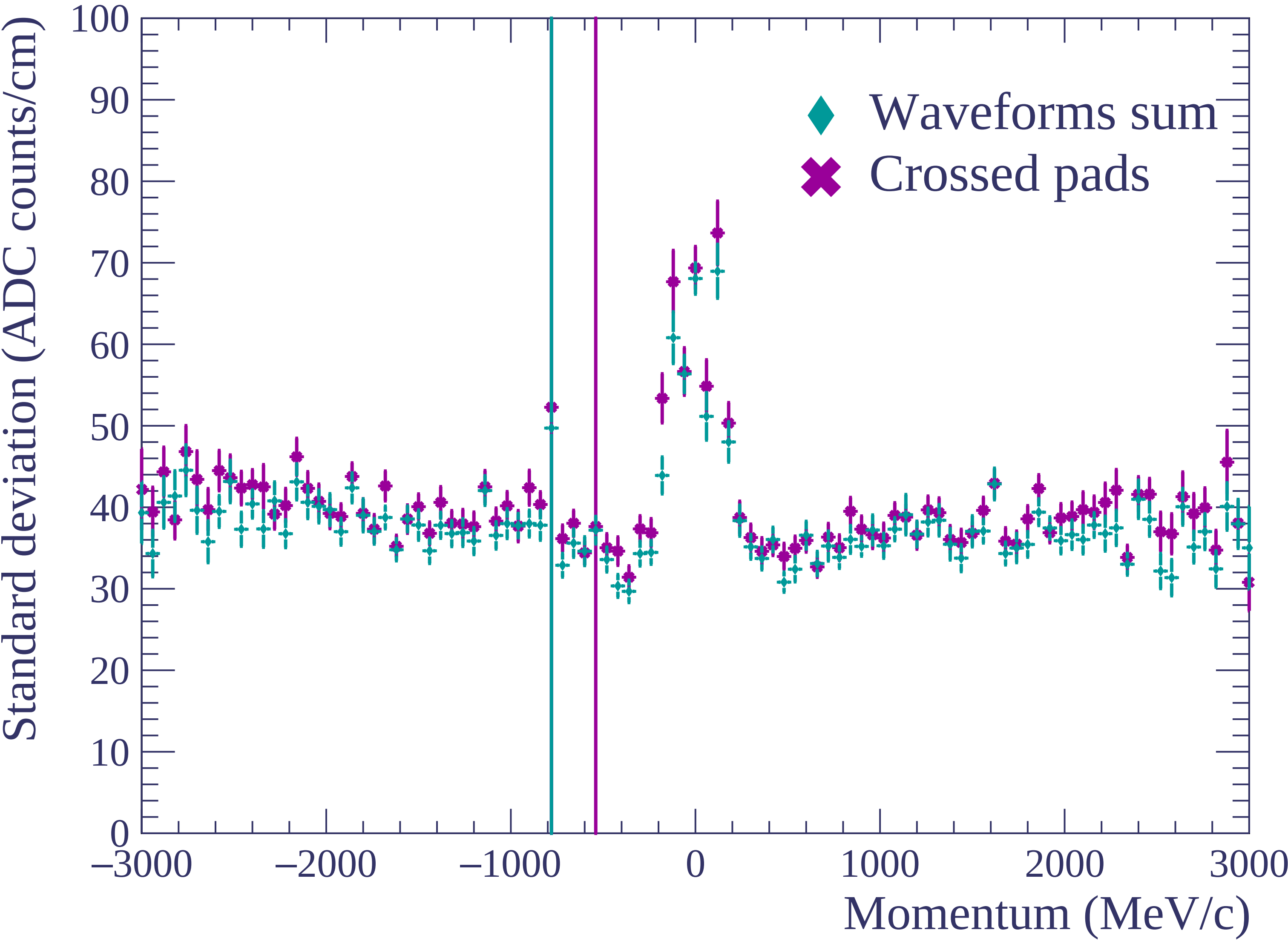
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

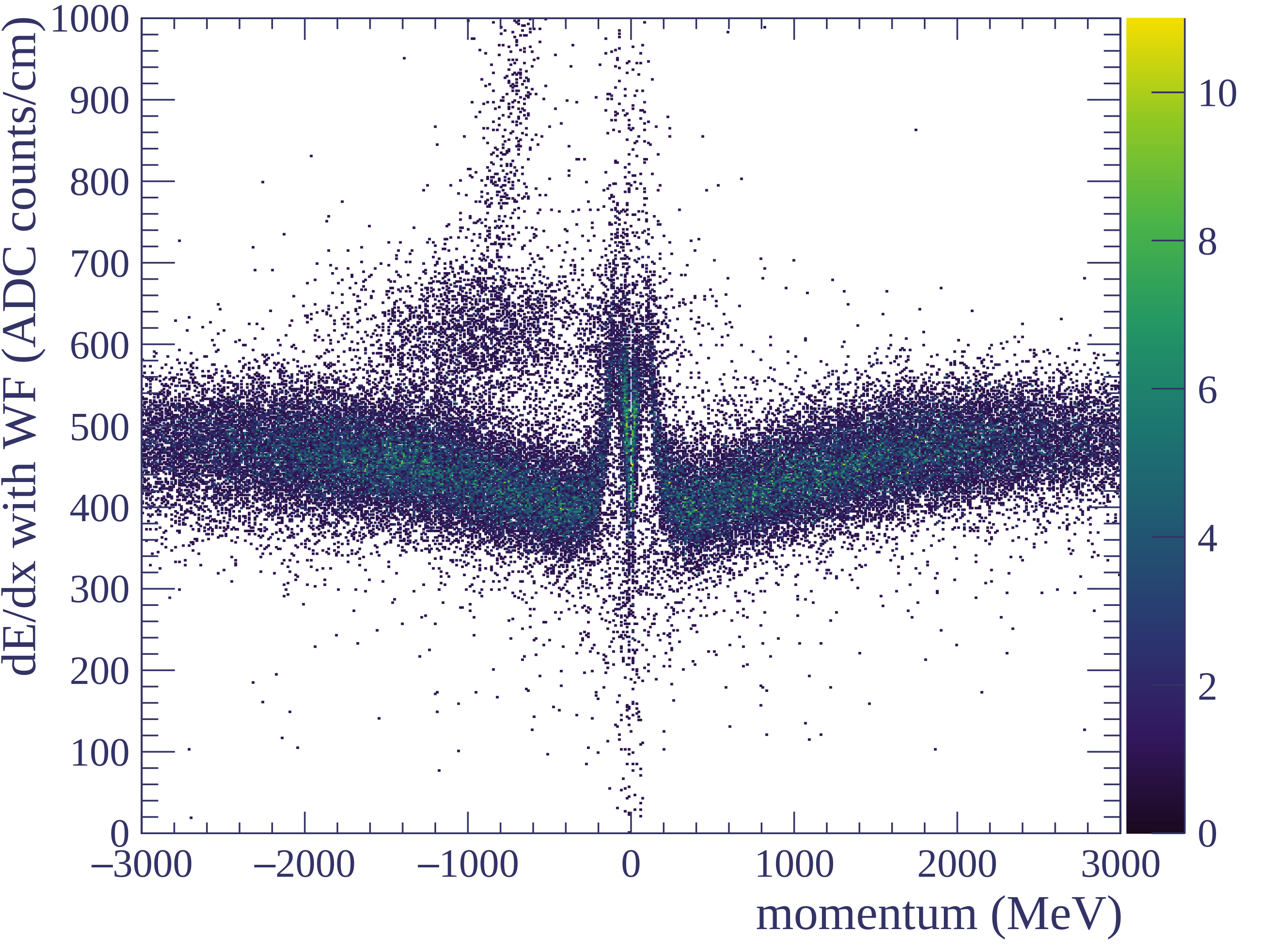

Resolution (%)
20
18
16
14
12
10
8
6
4
2
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



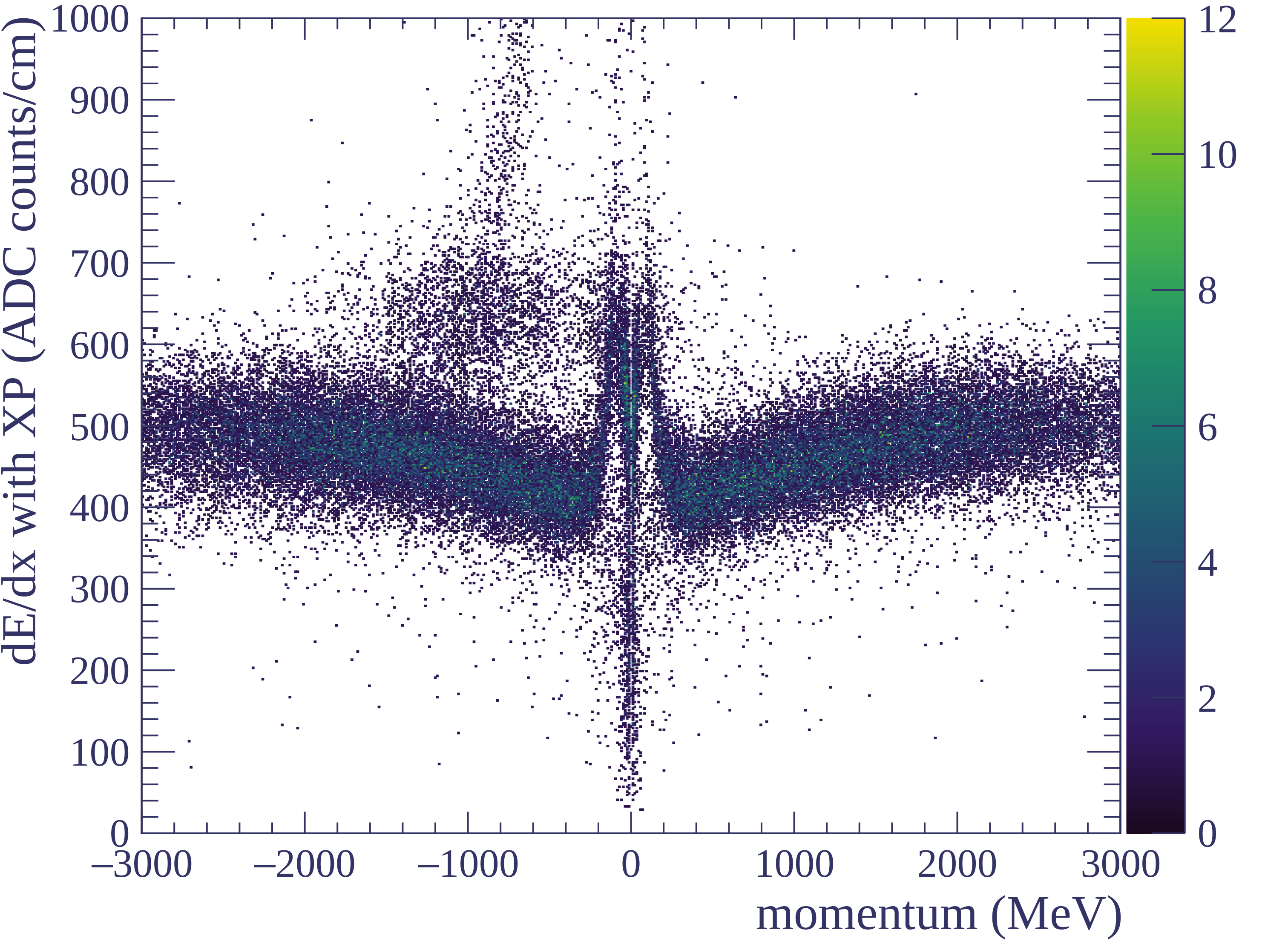
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



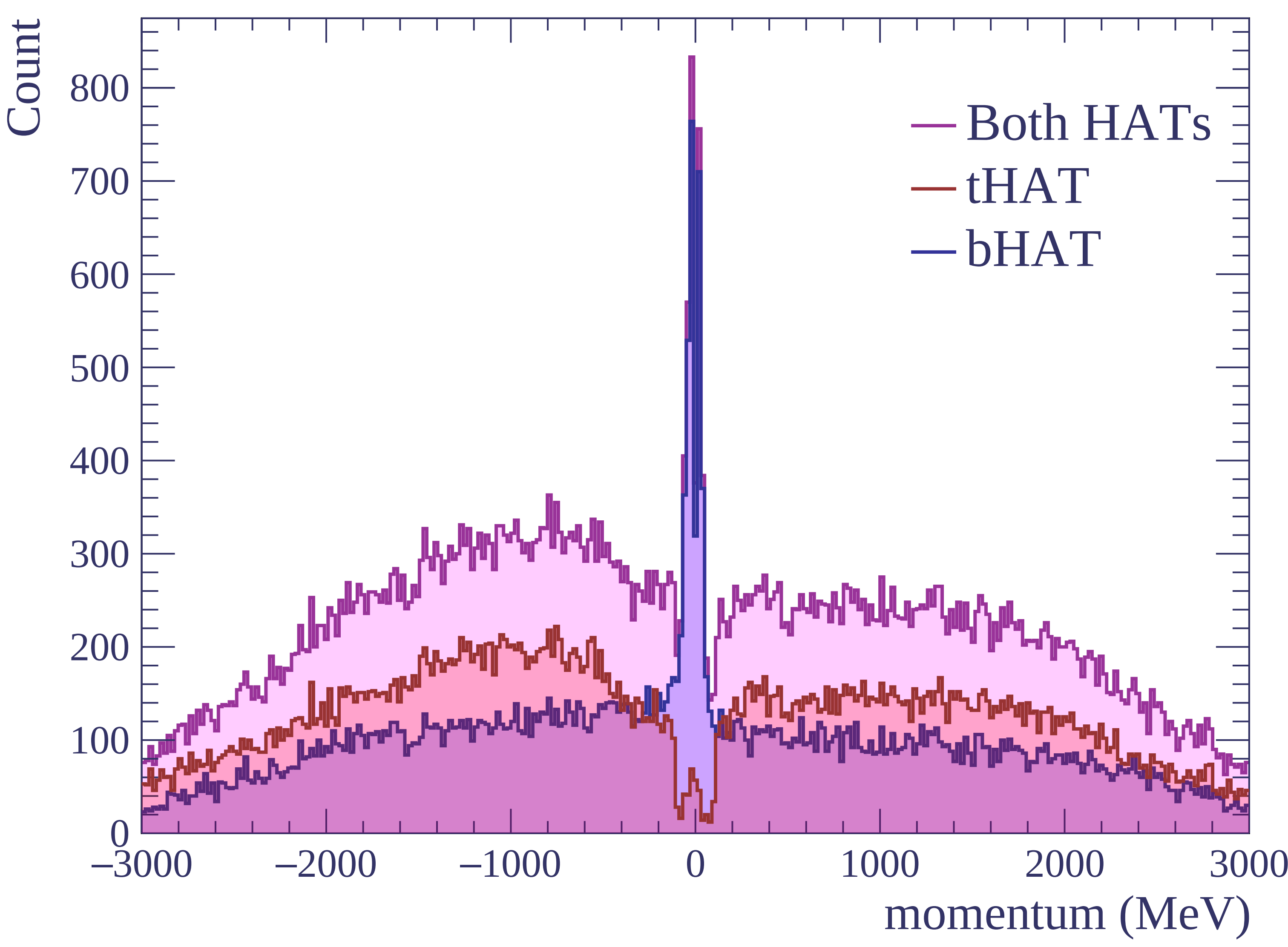
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000

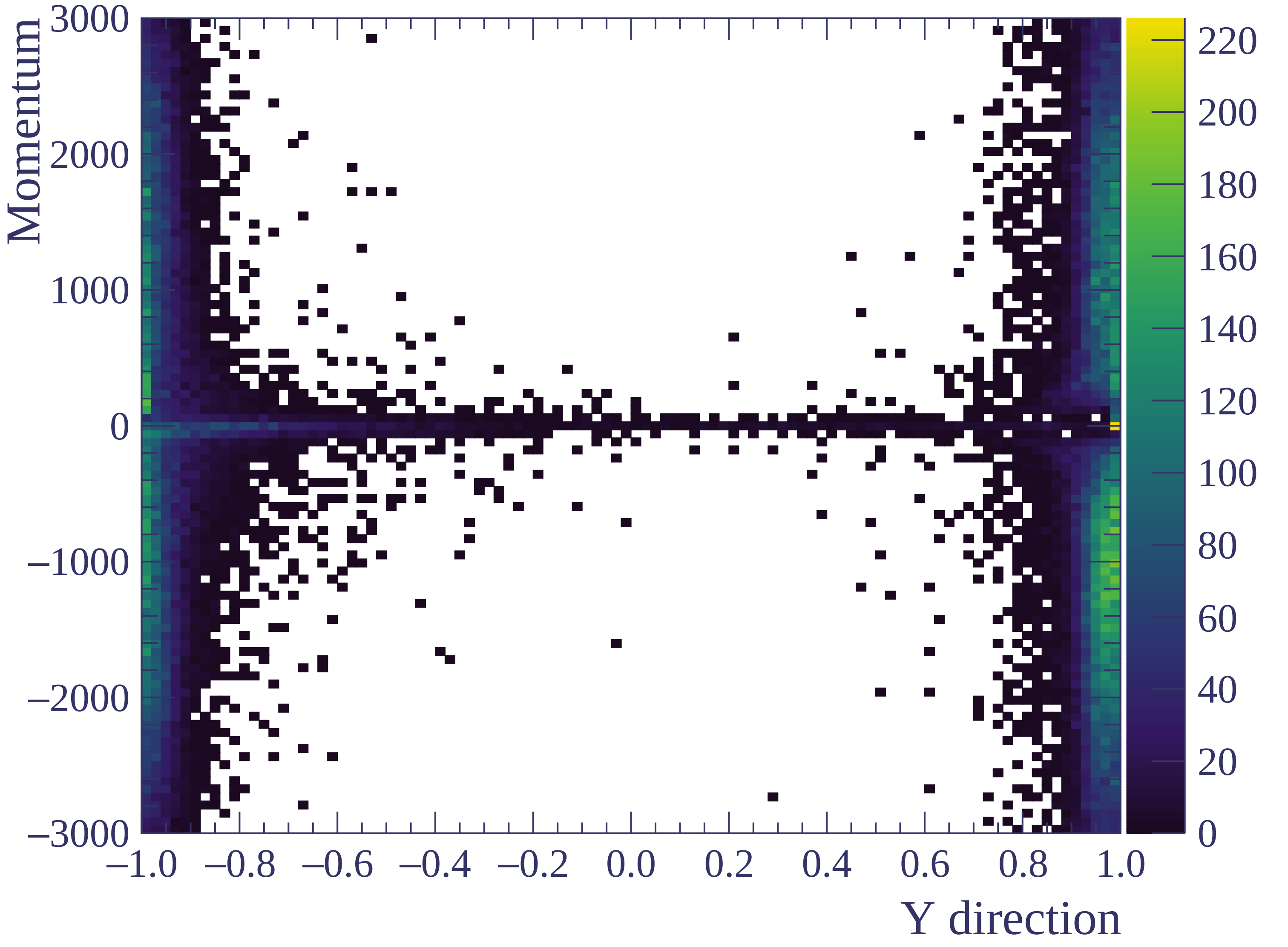


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
2
4
6
8
10

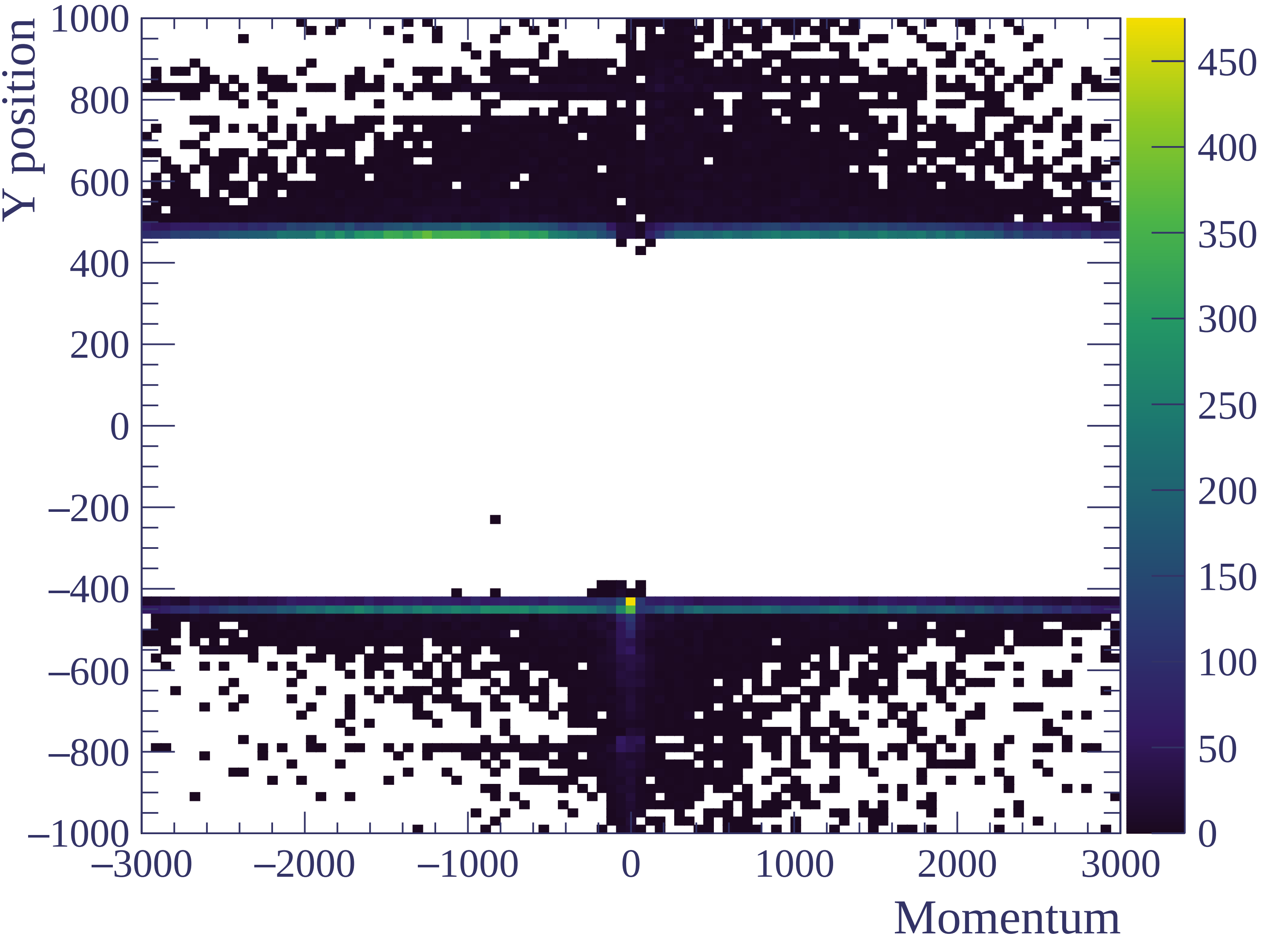


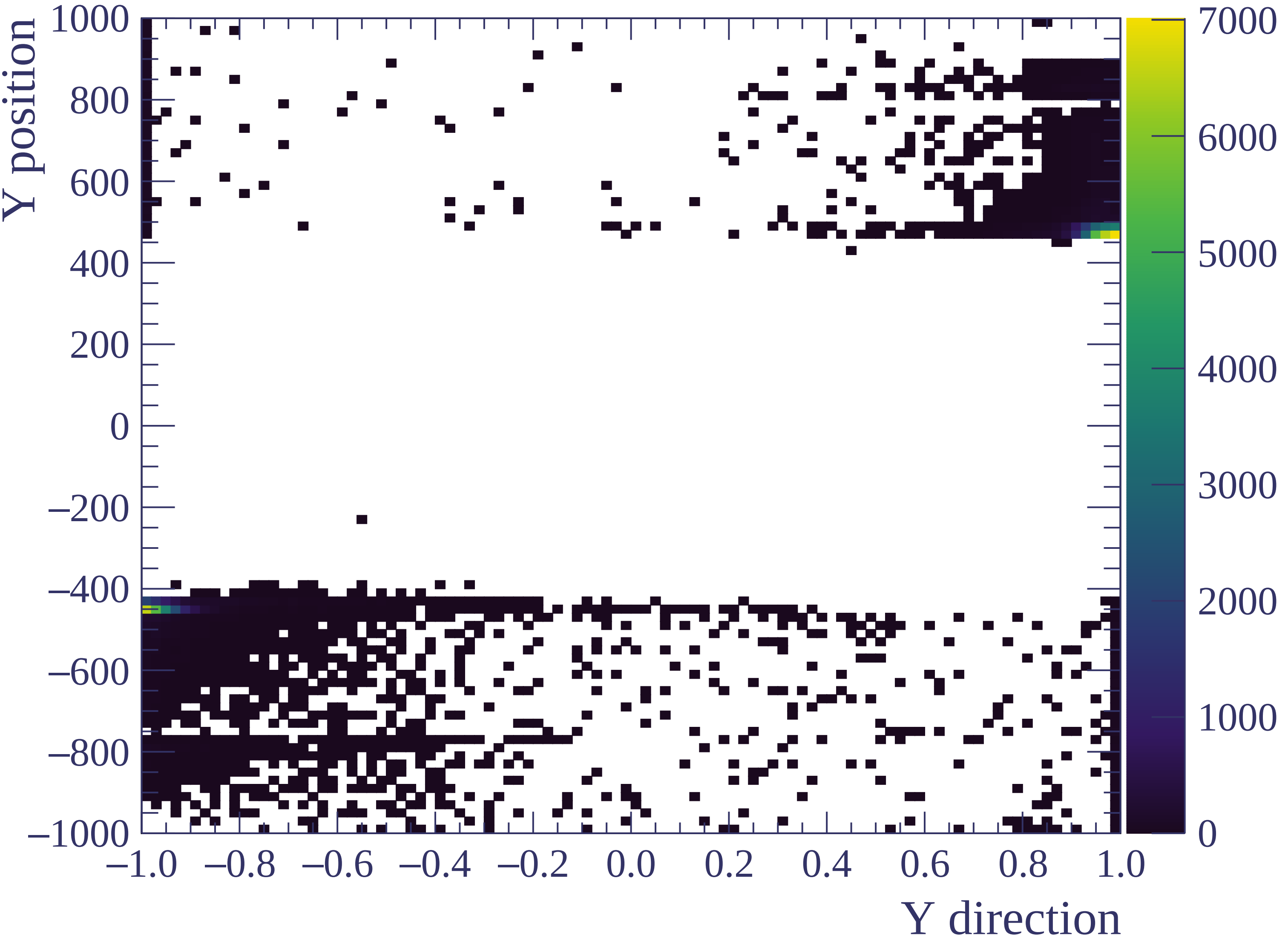
dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
2
4
6
8
10
12


Count
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

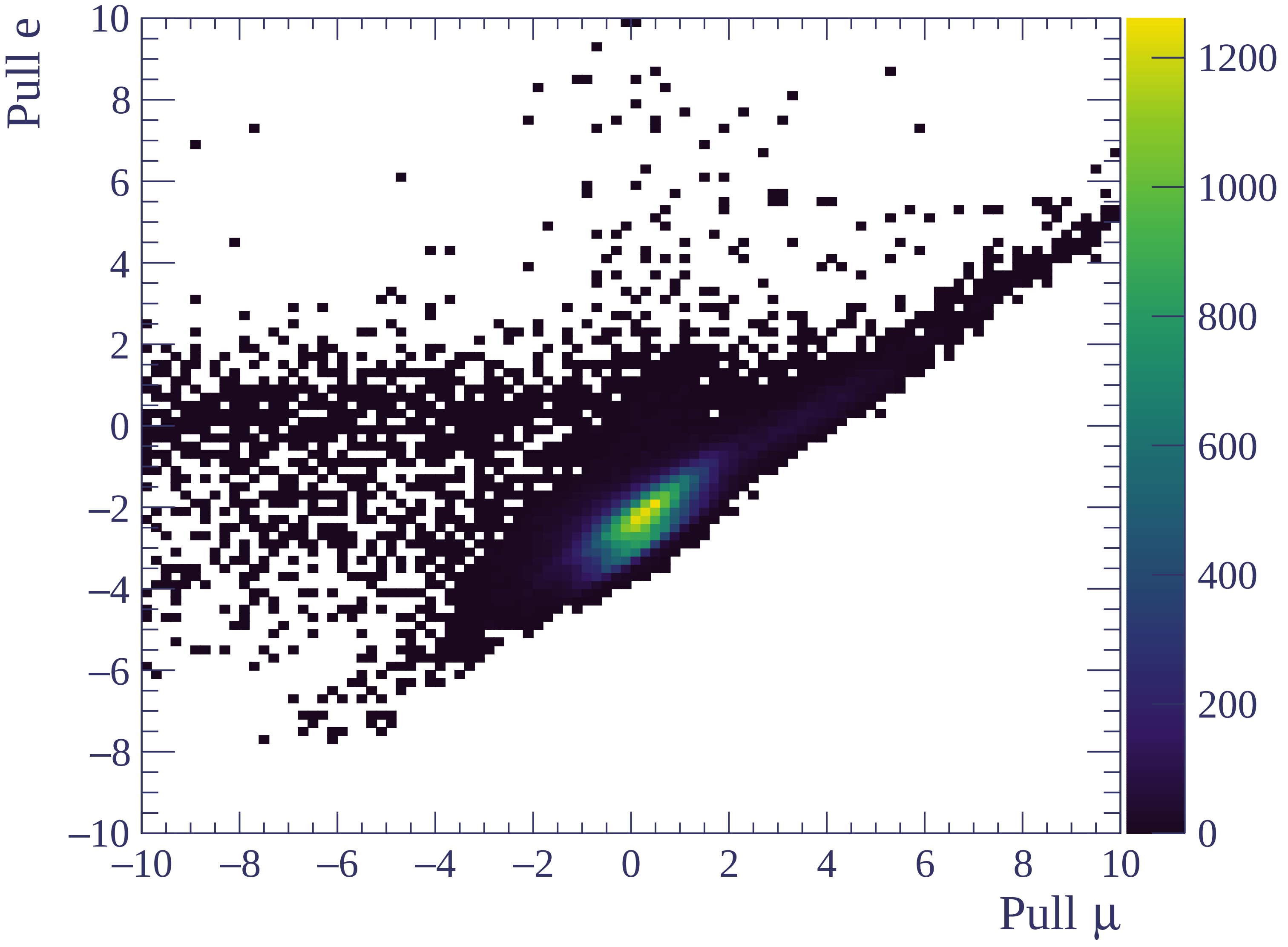


Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
220
200
180
160
140
120
100
80
60
40
20
0

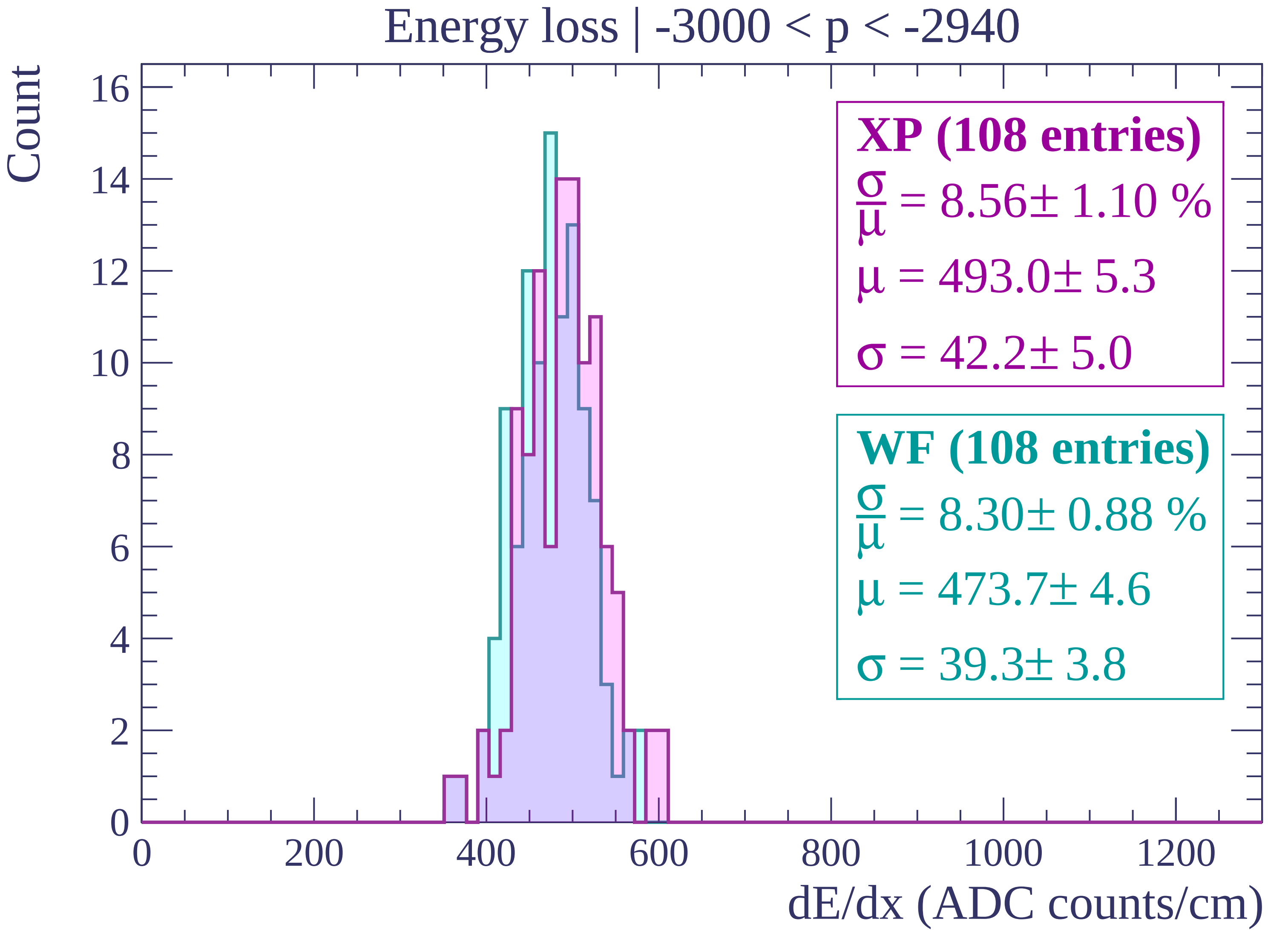

Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
Momentum
450
400
350
300
250
200
150
100
50
0



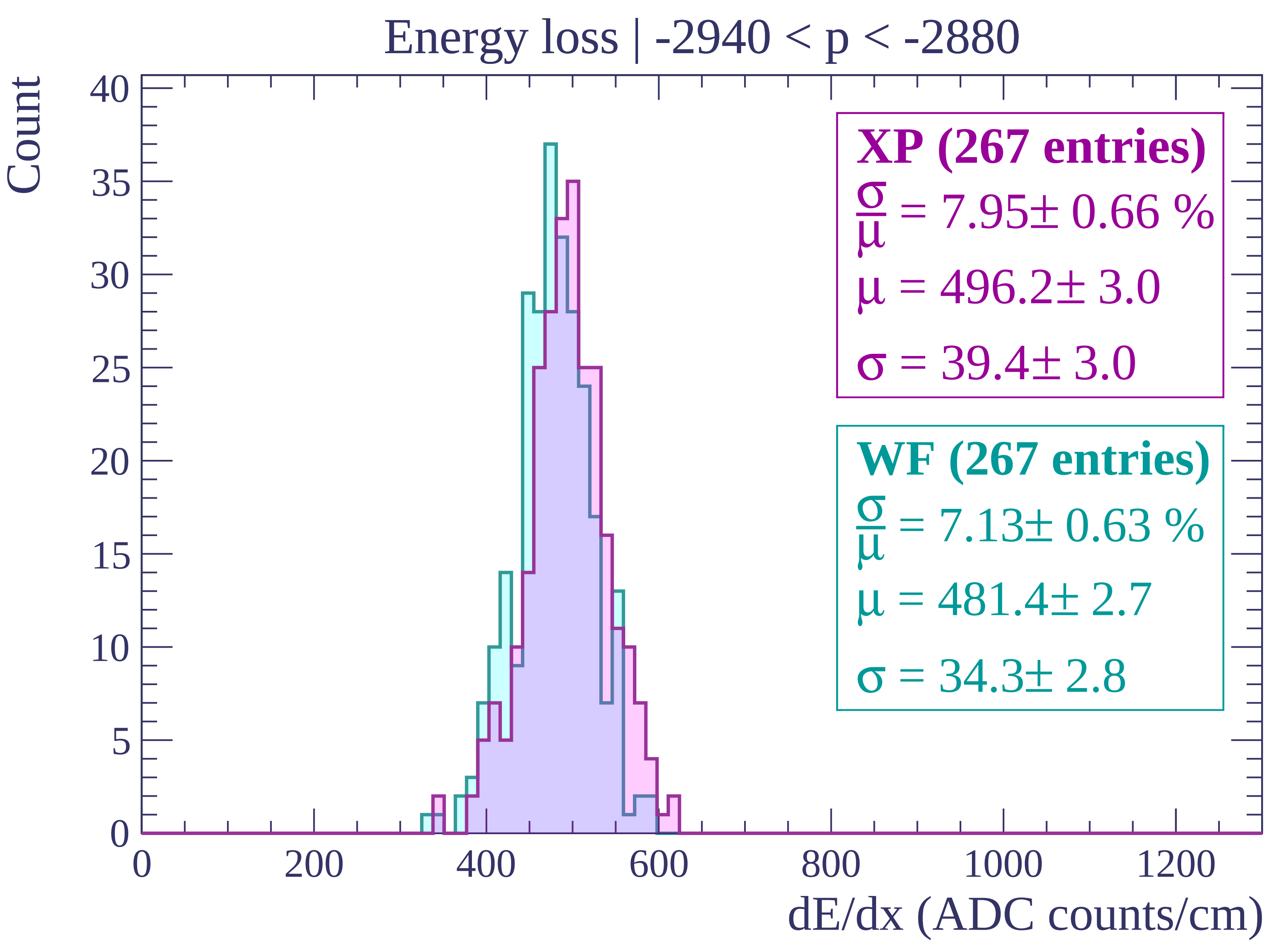
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
7000
6000
5000
4000
3000
2000
1000
0



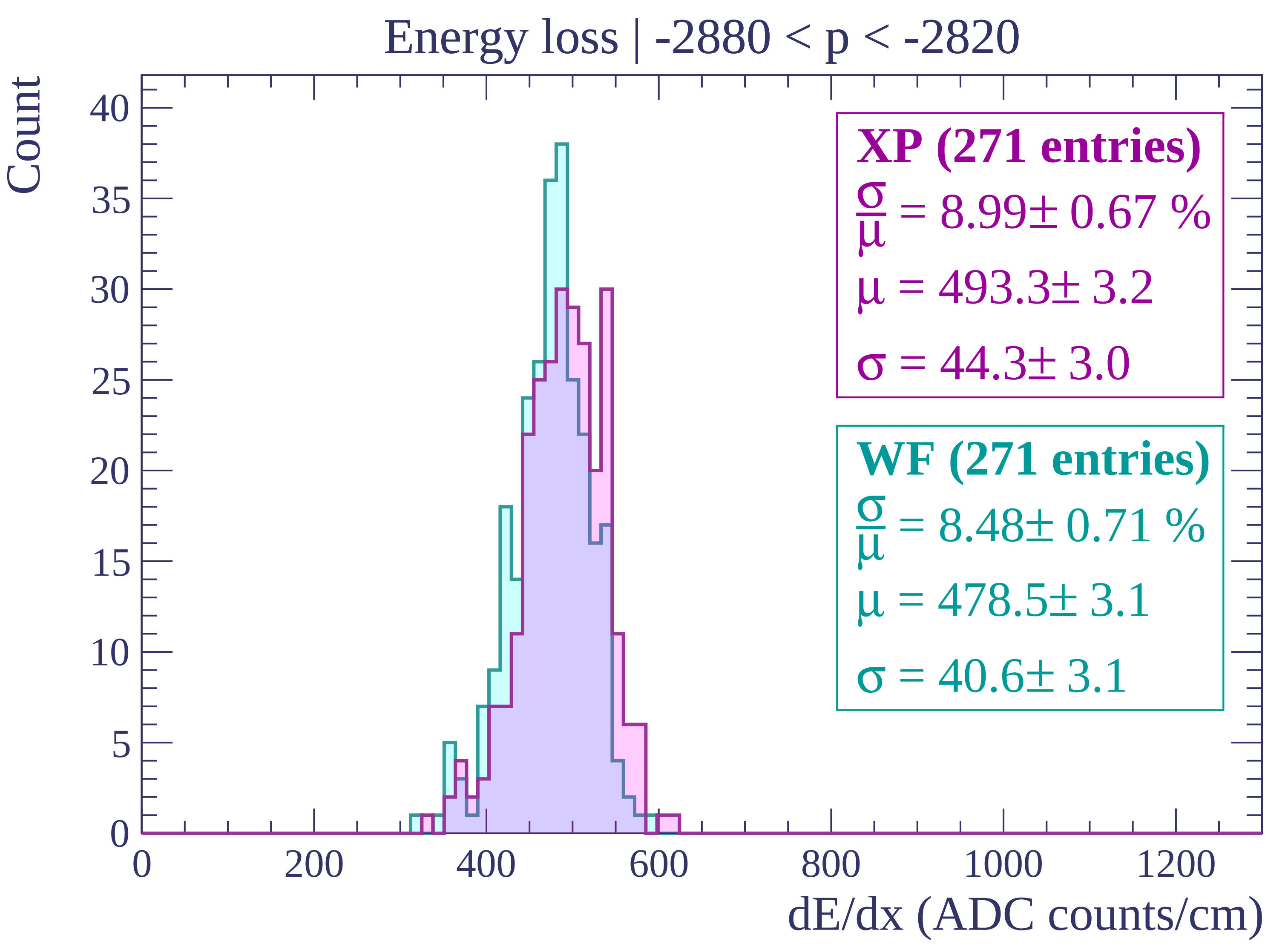
Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
1200
1000
800
600
400
200

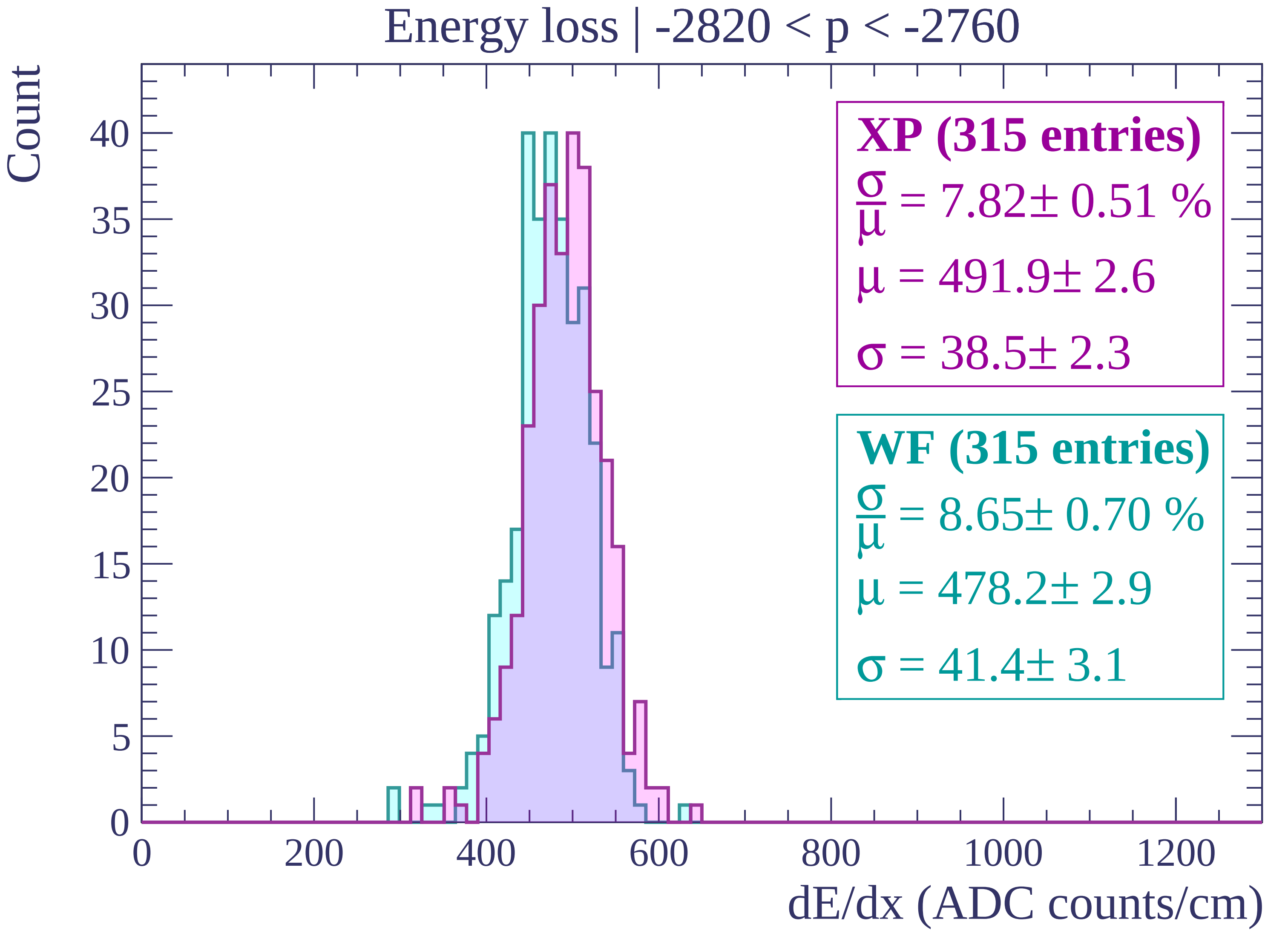


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP (108 entries)
σ/μ = 8.56± 1.10 %
μ = 493.0± 5.3
σ = 42.2± 5.0
WF (108 entries)
σ/μ = 8.30± 0.88 %
μ = 473.7± 4.6
σ = 39.3± 3.8

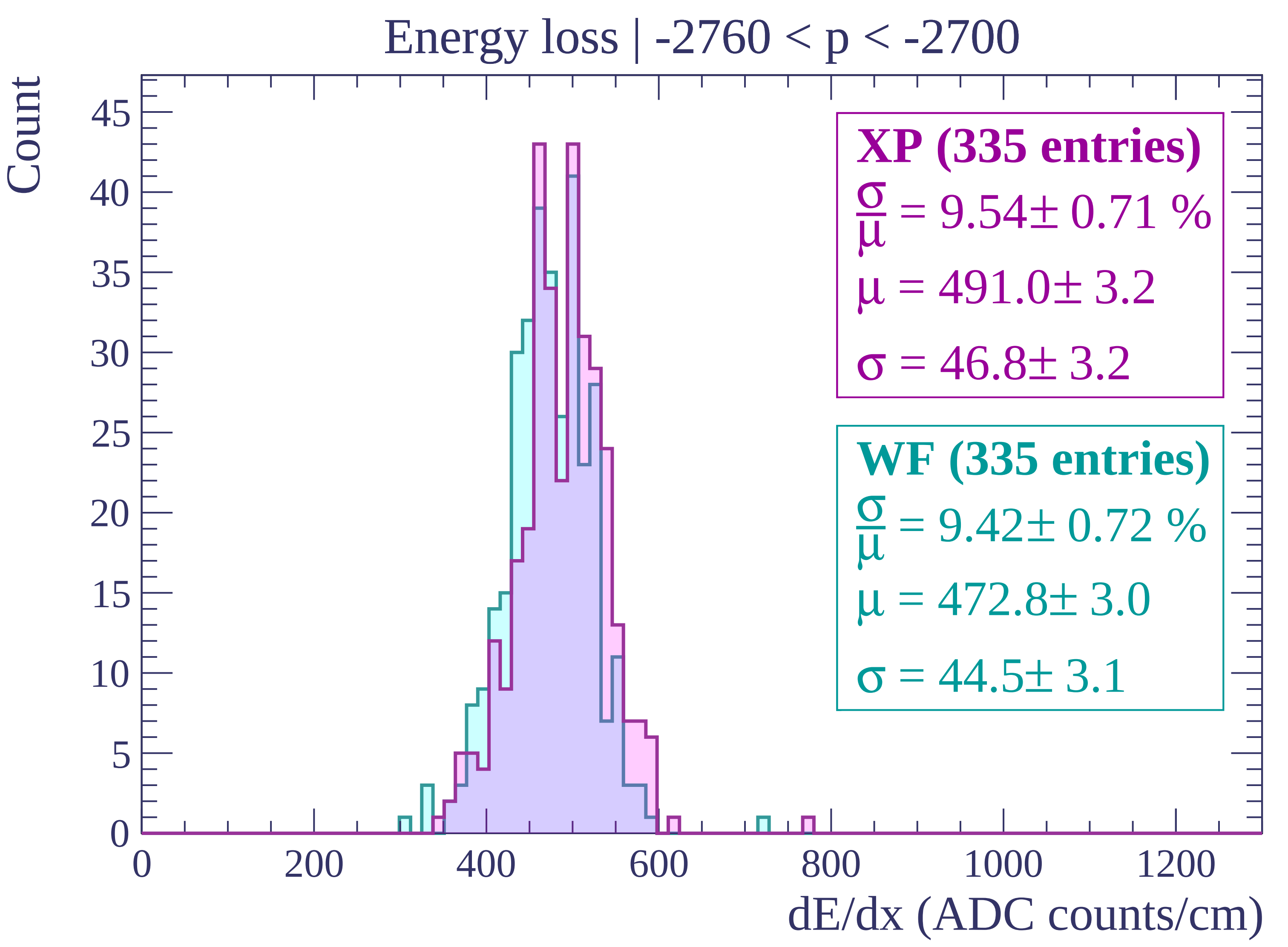


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (267 entries)
σ/μ = 7.95± 0.66 %
μ = 496.2± 3.0
σ = 39.4± 3.0
WF (267 entries)
σ/μ = 7.13± 0.63 %
μ = 481.4± 2.7
σ = 34.3± 2.8

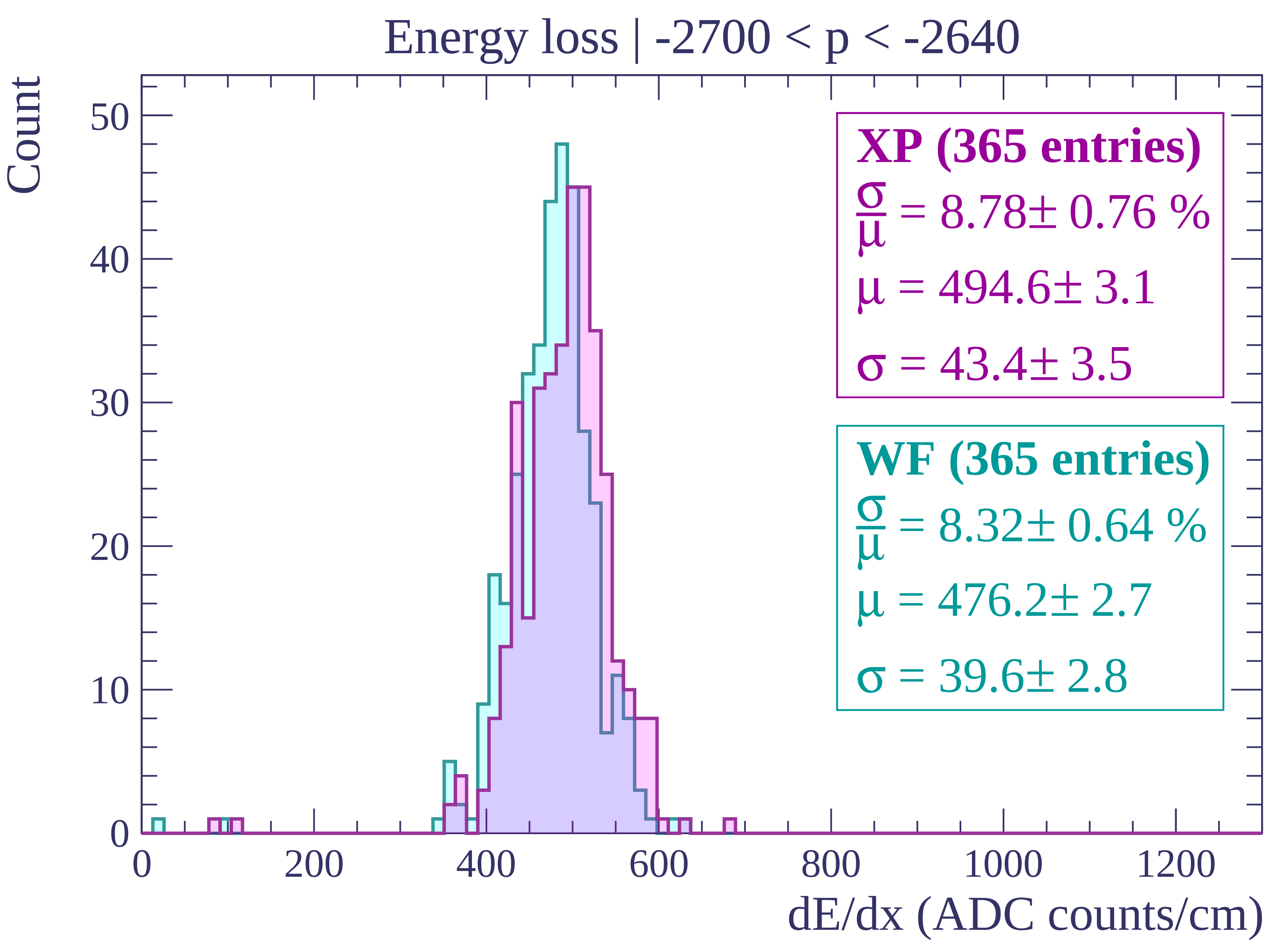

Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP (271 entries)
σ/μ = 8.99± 0.67 %
μ = 493.3± 3.2
σ = 44.3± 3.0
WF (271 entries)
σ/μ = 8.48± 0.71 %
μ = 478.5± 3.1
σ = 40.6± 3.1



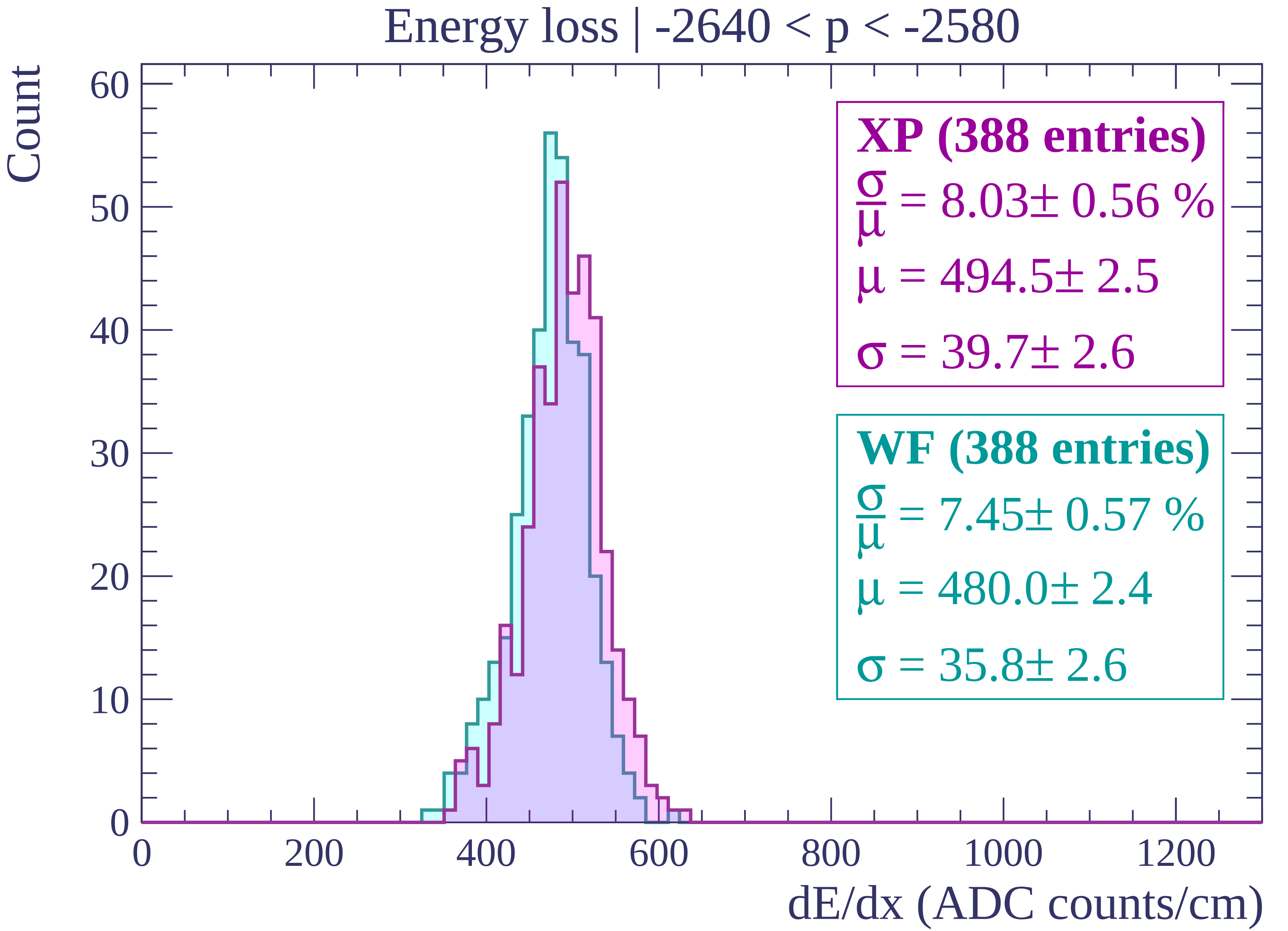
Energy loss | -2820 < p < -2760
Count
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (315 entries)
σ/μ = 7.82± 0.51 %
μ = 491.9± 2.6
σ = 38.5± 2.3
WF (315 entries)
σ/μ = 8.65± 0.70 %
μ = 478.2± 2.9
σ = 41.4± 3.1

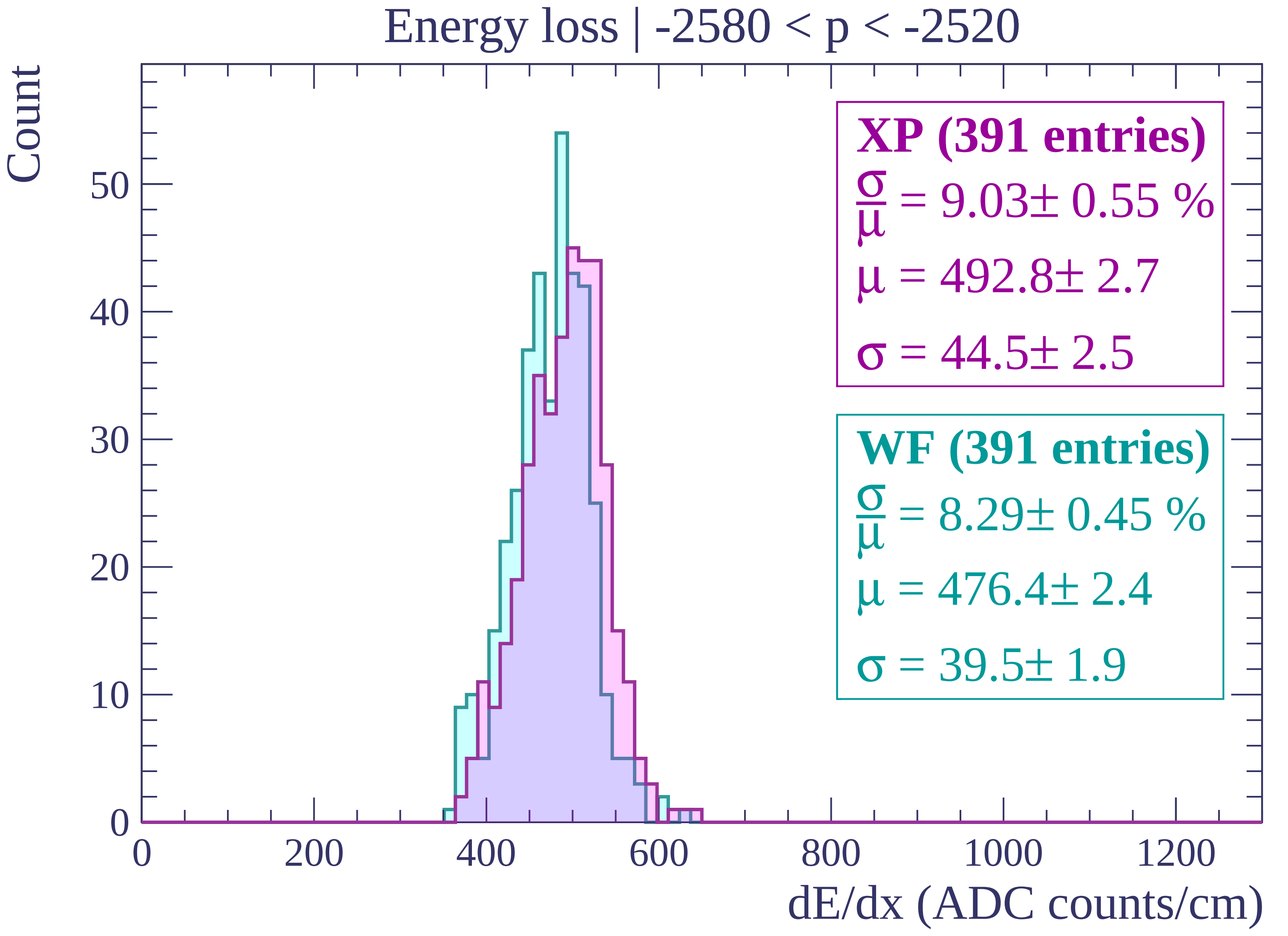


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP (335 entries)
σ/μ = 9.54± 0.71 %
μ = 491.0± 3.2
σ = 46.8± 3.2
WF (335 entries)
σ/μ = 9.42± 0.72 %
μ = 472.8± 3.0
σ = 44.5± 3.1

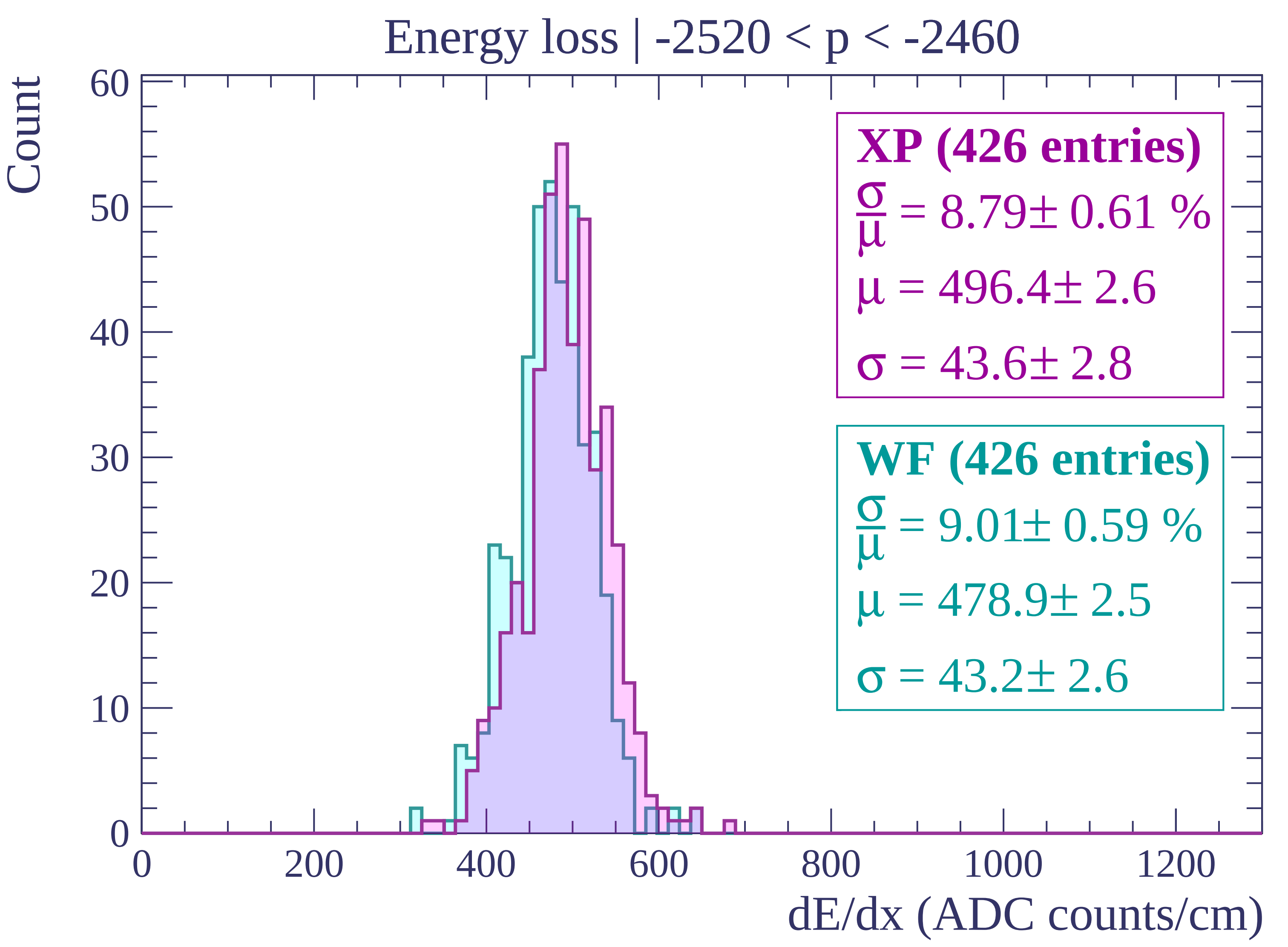


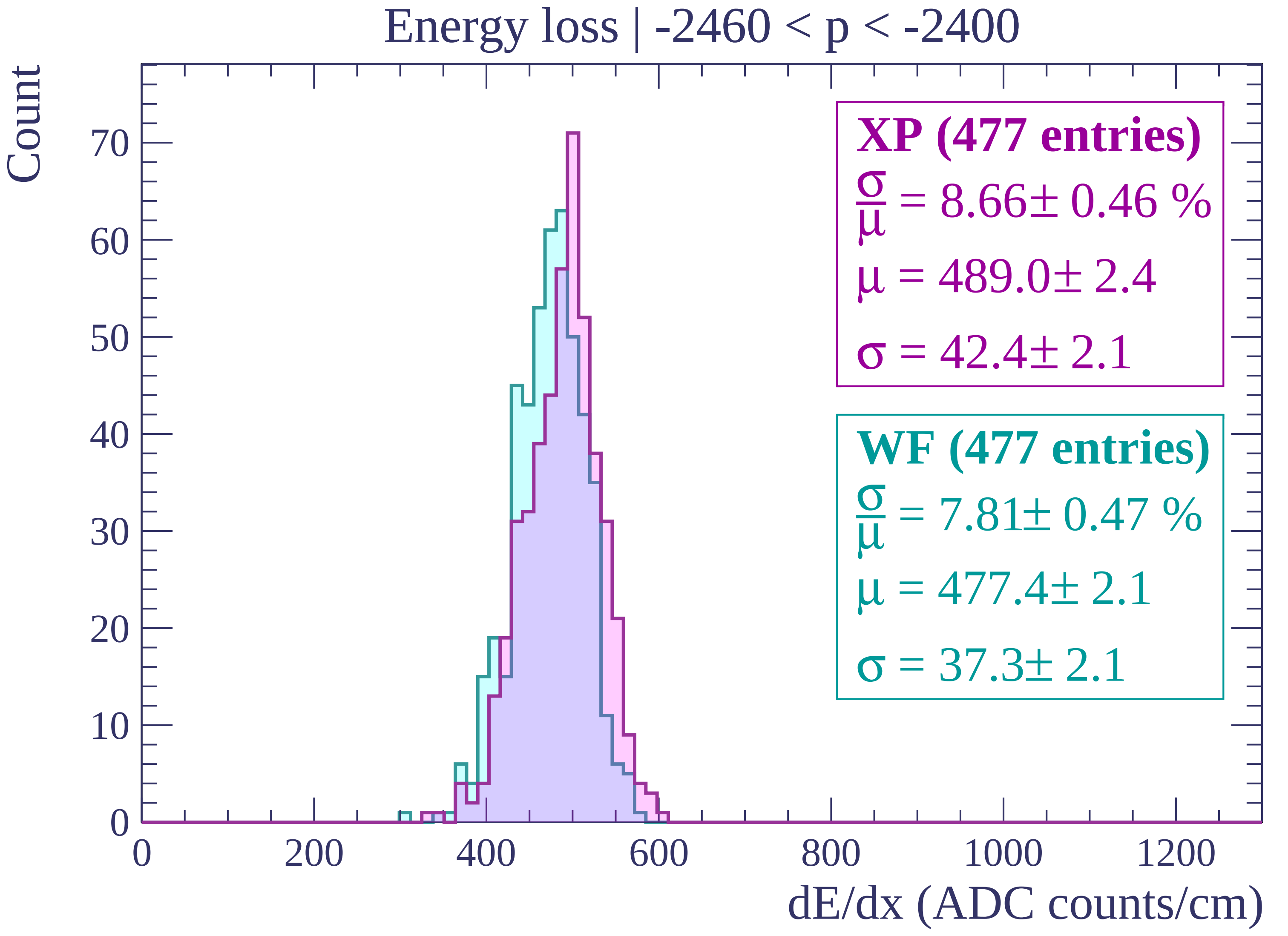
Energy loss | -2700 < p < -2640
Count
XP (365 entries)
σ/μ = 8.78± 0.76 %
μ = 494.6± 3.1
σ = 43.4± 3.5
WF (365 entries)
σ/μ = 8.32± 0.64 %
μ = 476.2± 2.7
σ = 39.6± 2.8
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (388 entries)
σ/μ = 8.03± 0.56 %
μ = 494.5± 2.5
σ = 39.7± 2.6
WF (388 entries)
σ/μ = 7.45± 0.57 %
μ = 480.0± 2.4
σ = 35.8± 2.6

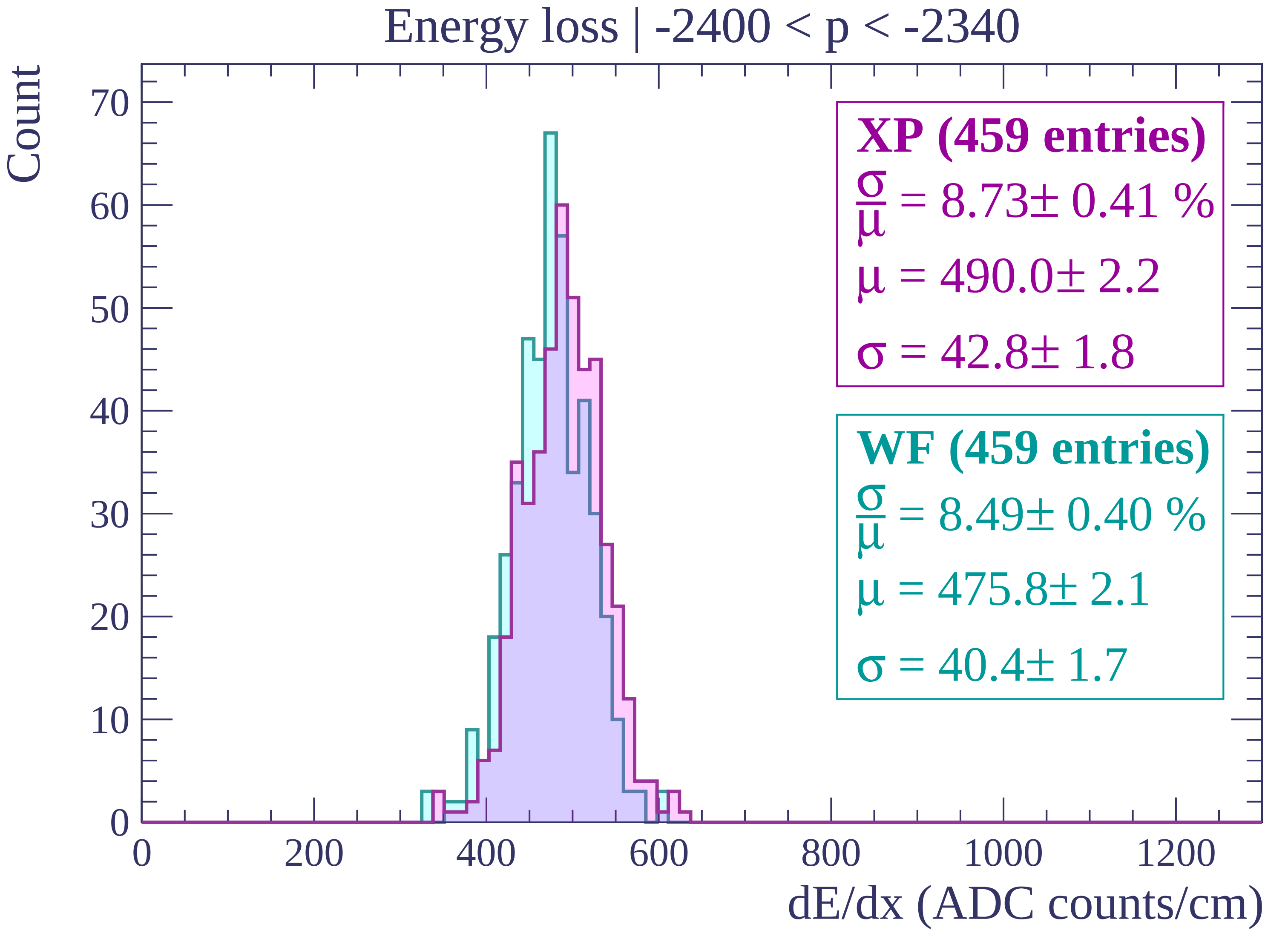


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
XP (391 entries)
$\frac{\sigma}{\mu}$ = 9.03± 0.55 %
μ = 492.8± 2.7
σ = 44.5± 2.5
WF (391 entries)
$\frac{\sigma}{\mu}$ = 8.29± 0.45 %
μ = 476.4± 2.4
σ = 39.5± 1.9
0
10
20
30
40
50
0
200
400
600
800
1000
1200

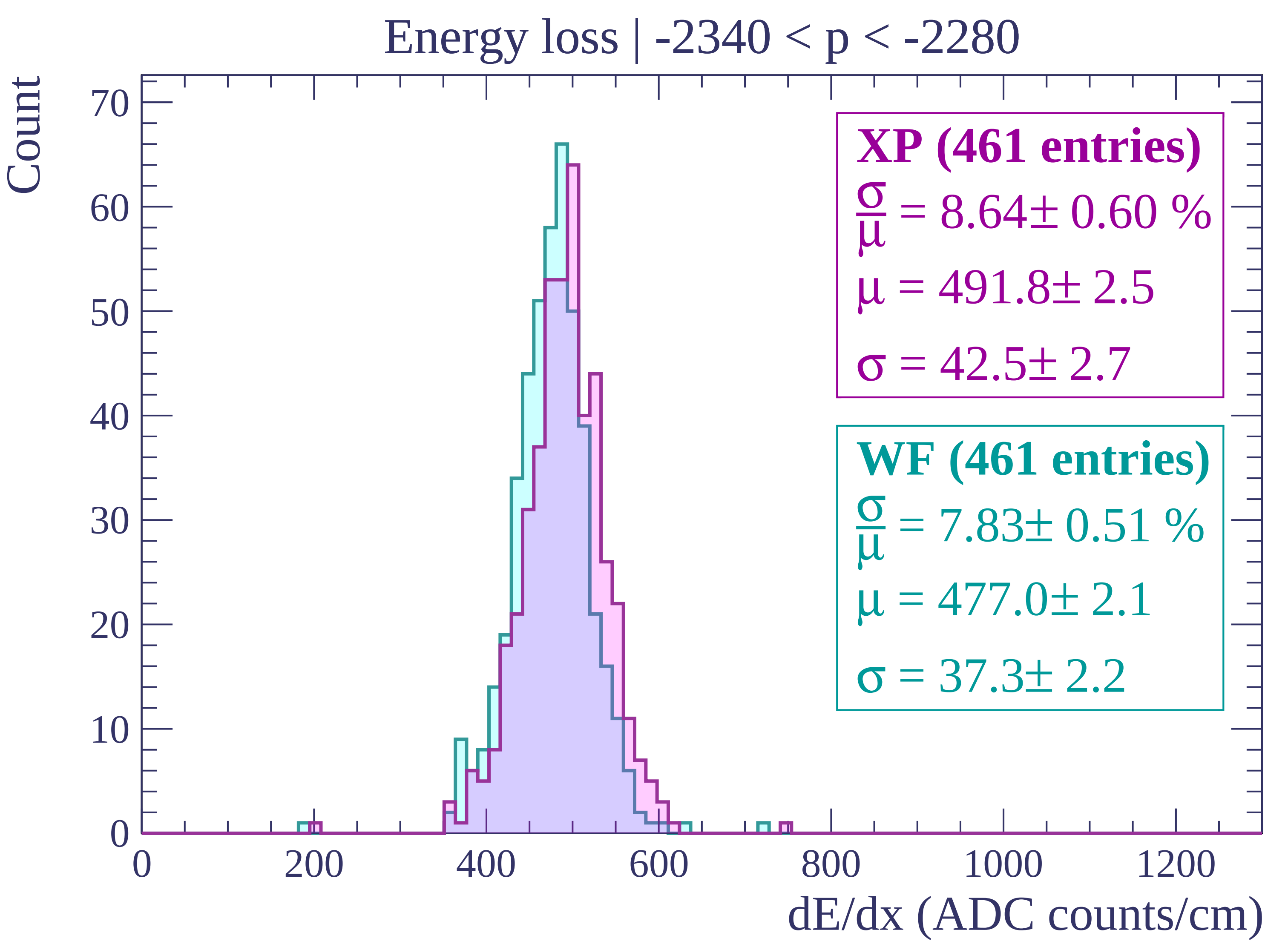

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (426 entries)
$\frac{\sigma}{\mu}$ = 8.79± 0.61 %
μ = 496.4± 2.6
σ = 43.6± 2.8
WF (426 entries)
$\frac{\sigma}{\mu}$ = 9.01± 0.59 %
μ = 478.9± 2.5
σ = 43.2± 2.6

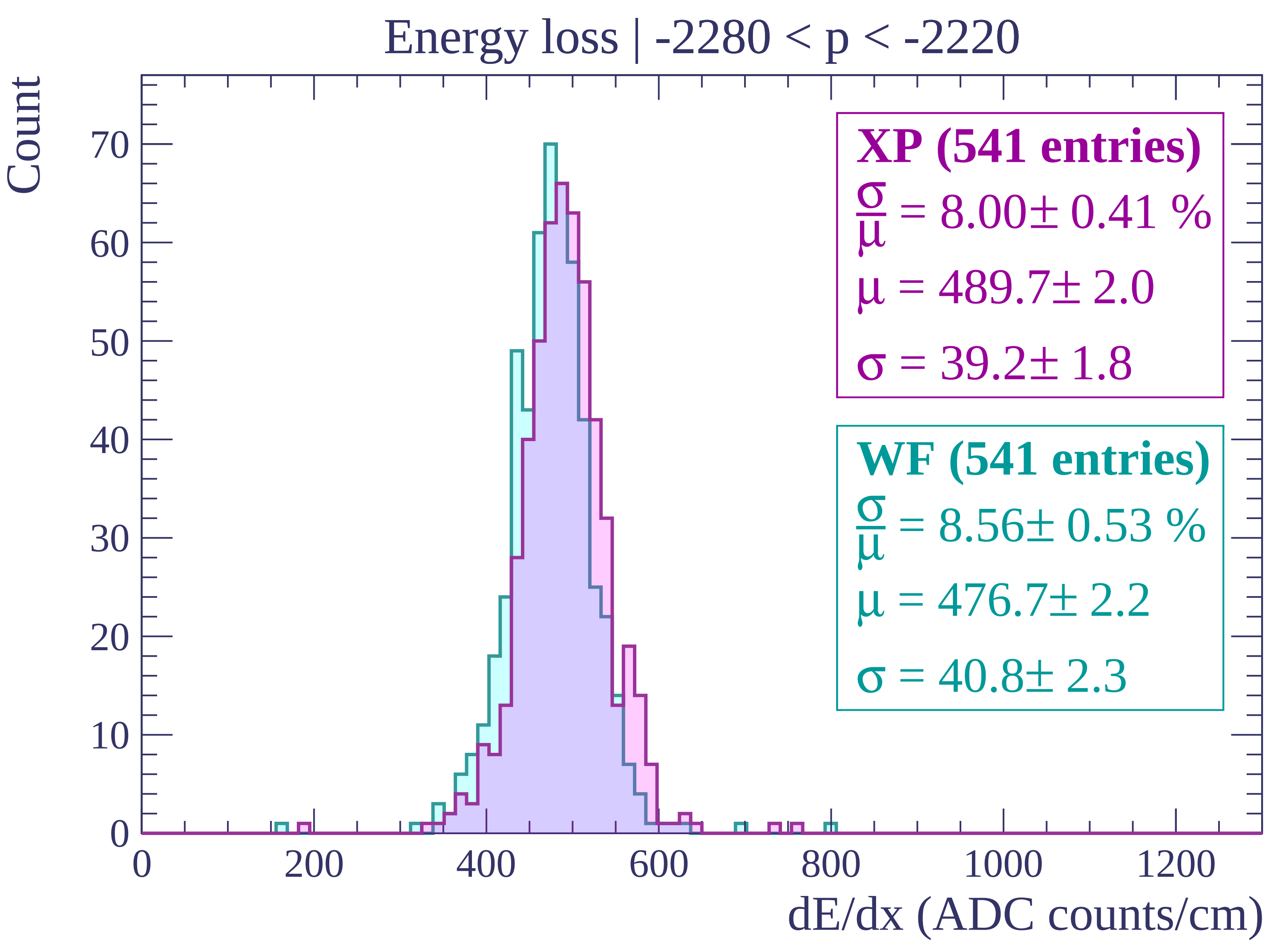


Energy loss | -2460 < p < -2400
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (477 entries)
σ/μ = 8.66± 0.46 %
μ = 489.0± 2.4
σ = 42.4± 2.1
WF (477 entries)
σ/μ = 7.81± 0.47 %
μ = 477.4± 2.1
σ = 37.3± 2.1

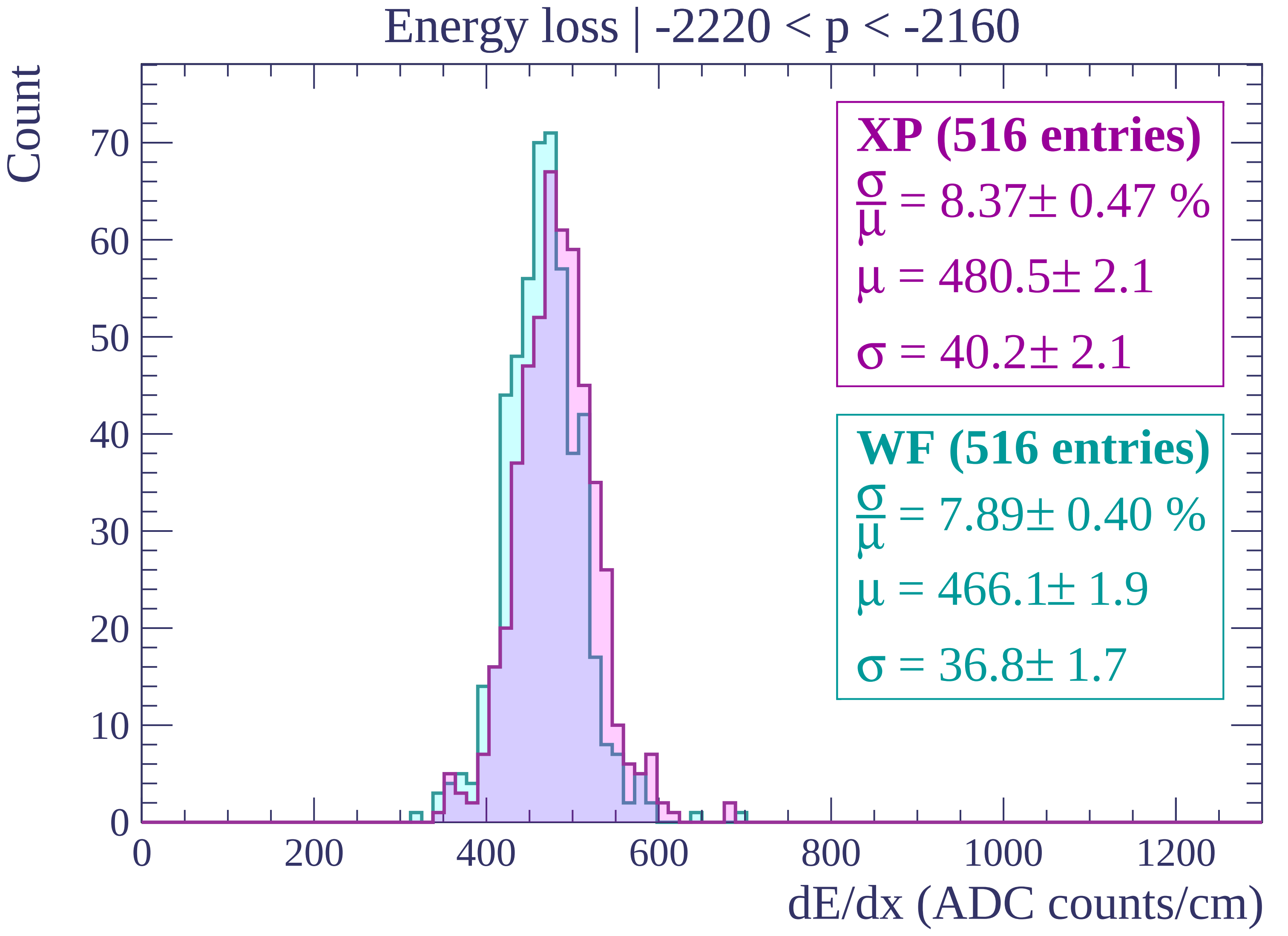


Energy loss | -2400 < p < -2340
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (459 entries)
σ/μ = 8.73± 0.41 %
μ = 490.0± 2.2
σ = 42.8± 1.8
WF (459 entries)
σ/μ = 8.49± 0.40 %
μ = 475.8± 2.1
σ = 40.4± 1.7

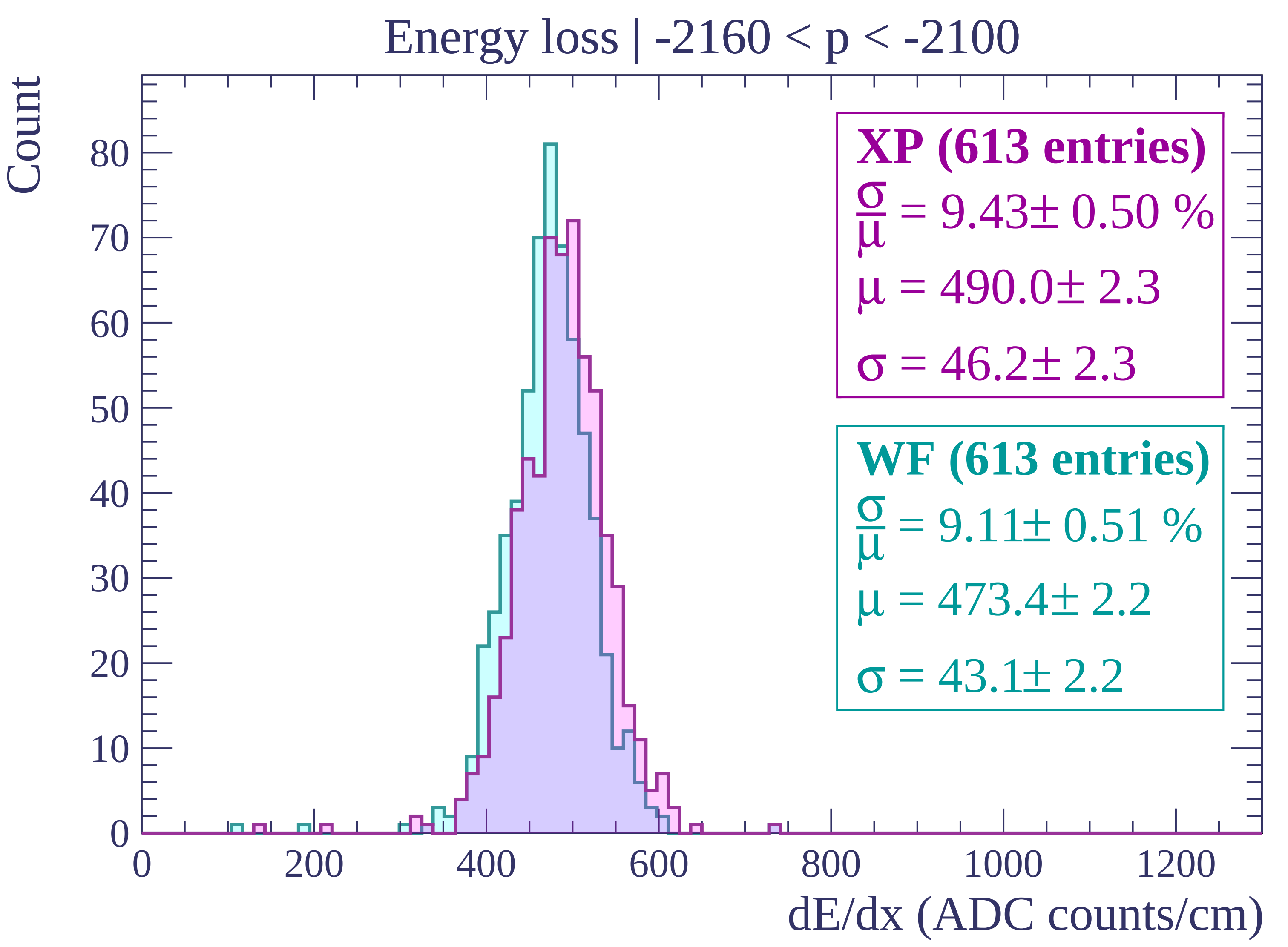

Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (461 entries)
σ/μ = 8.64± 0.60 %
μ = 491.8± 2.5
σ = 42.5± 2.7
WF (461 entries)
σ/μ = 7.83± 0.51 %
μ = 477.0± 2.1
σ = 37.3± 2.2



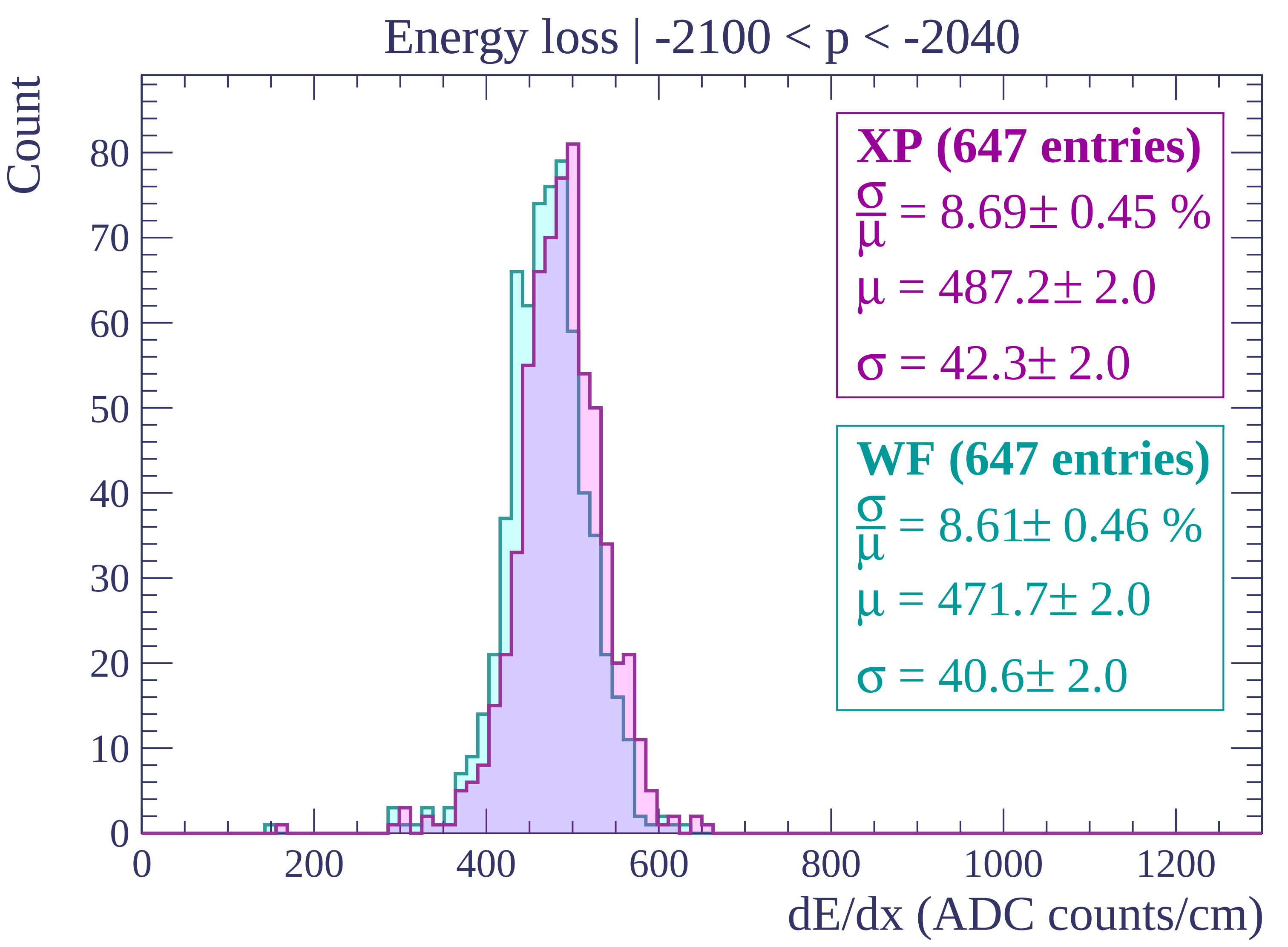
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
XP (541 entries)
σ/μ = 8.00± 0.41 %
μ = 489.7± 2.0
σ = 39.2± 1.8
WF (541 entries)
σ/μ = 8.56± 0.53 %
μ = 476.7± 2.2
σ = 40.8± 2.3
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200



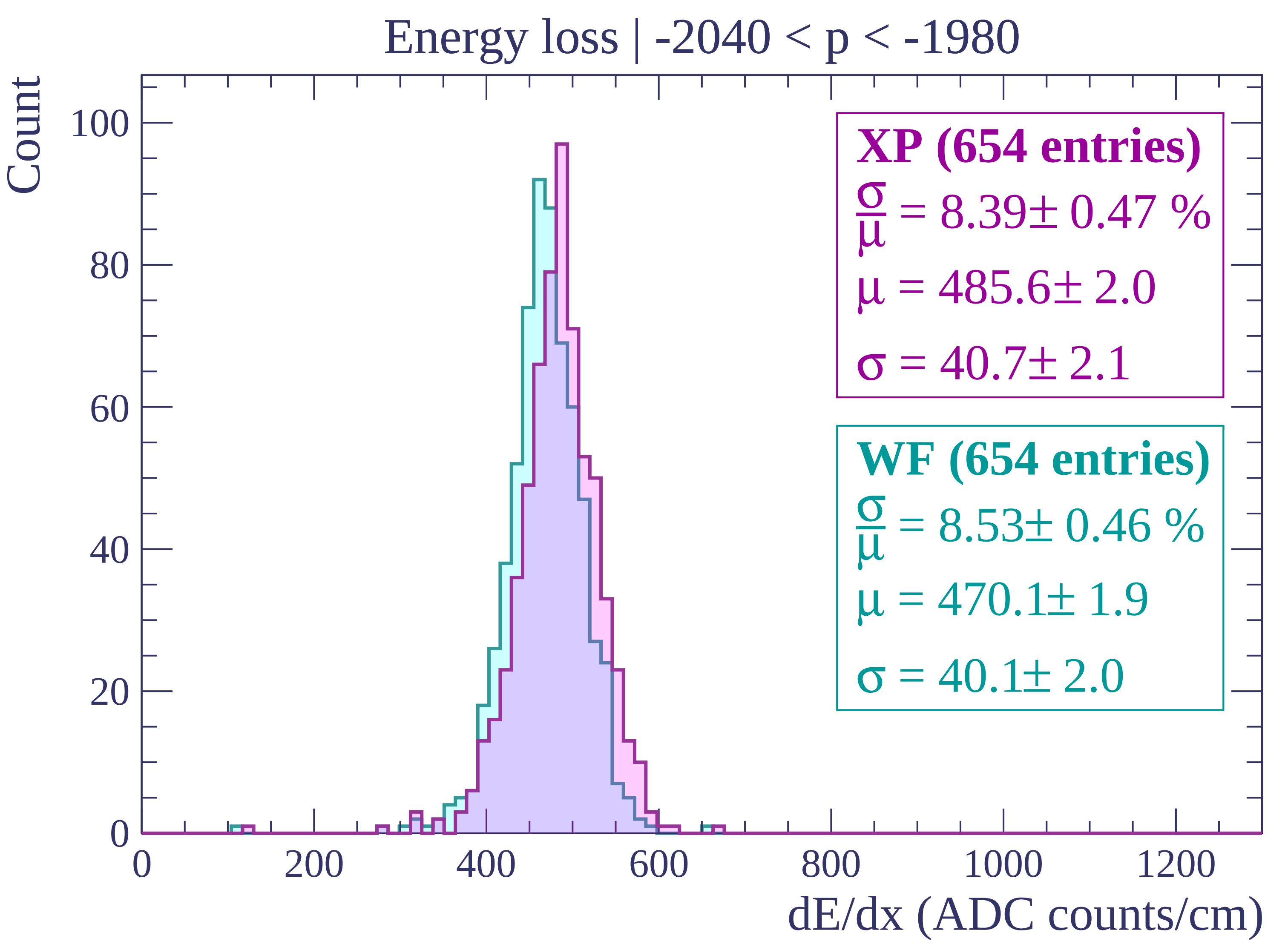
Energy loss | -2220 < p < -2160
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (516 entries)
$\frac{\sigma}{\mu}$ = 8.37± 0.47 %
μ = 480.5± 2.1
σ = 40.2± 2.1
WF (516 entries)
$\frac{\sigma}{\mu}$ = 7.89± 0.40 %
μ = 466.1± 1.9
σ = 36.8± 1.7



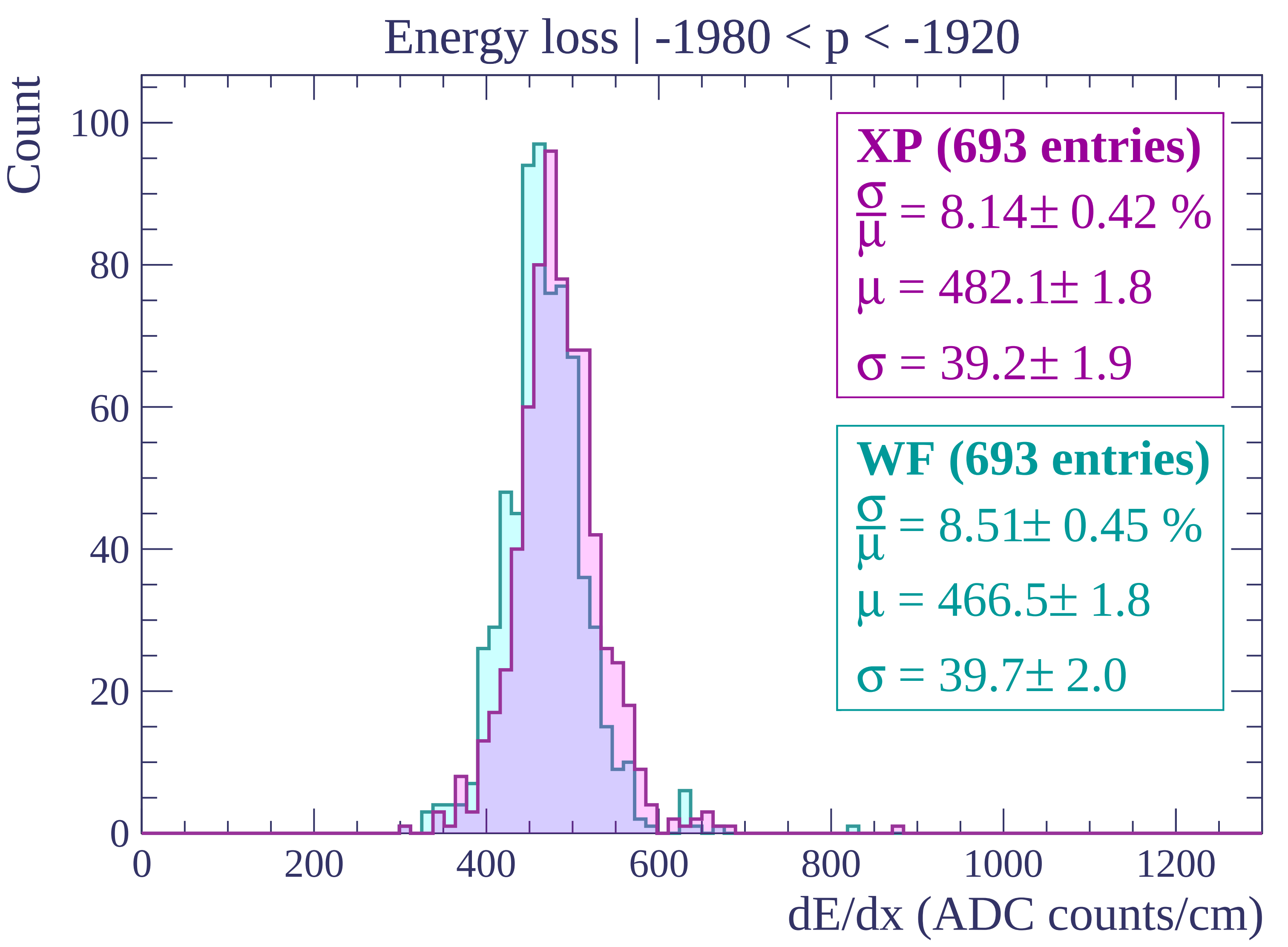
Energy loss | -2160 < p < -2100
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (613 entries)
σ/μ = 9.43± 0.50 %
μ = 490.0± 2.3
σ = 46.2± 2.3
WF (613 entries)
σ/μ = 9.11± 0.51 %
μ = 473.4± 2.2
σ = 43.1± 2.2



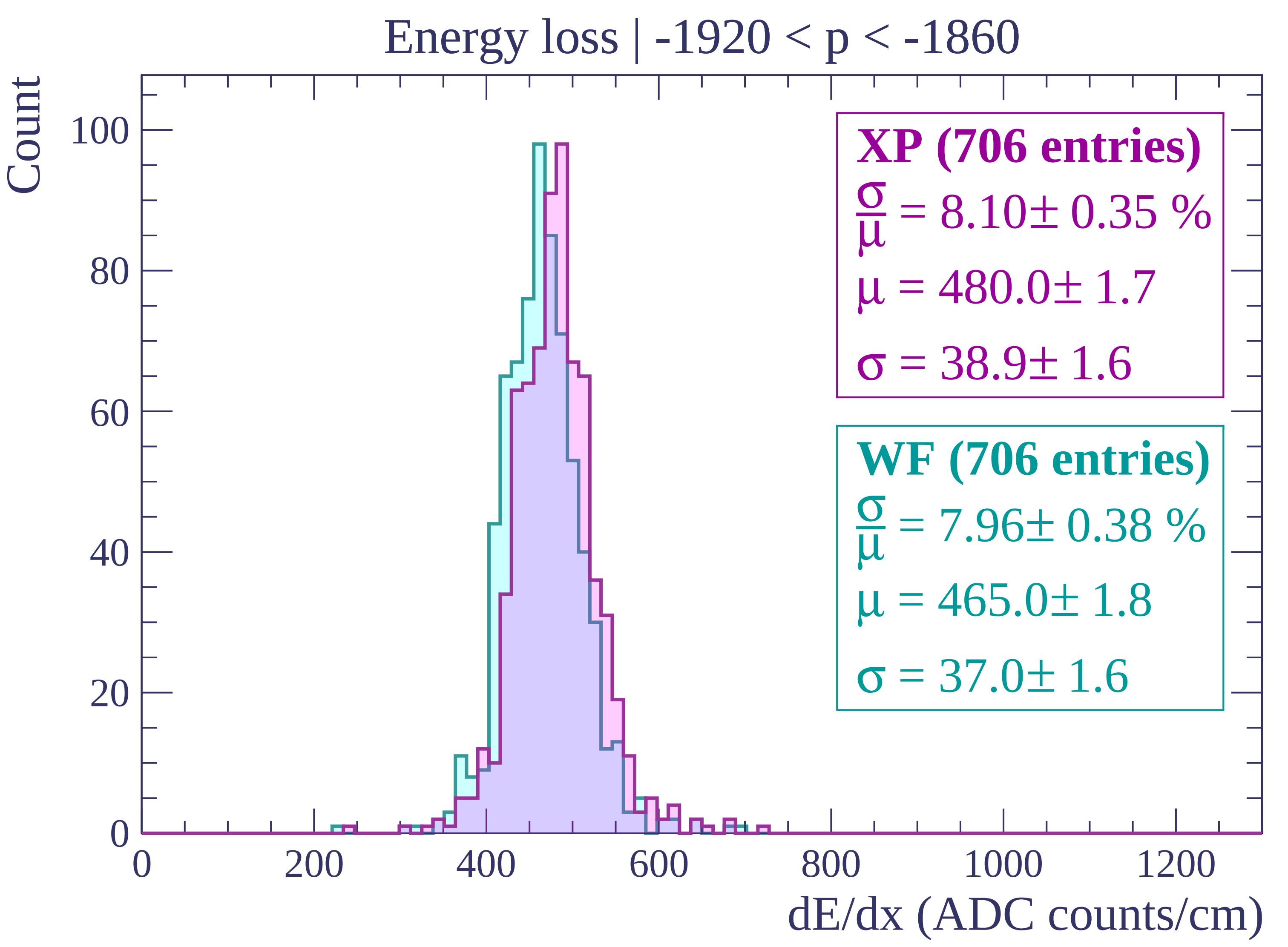
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
XP (647 entries)
$\frac{\sigma}{\mu}$ = 8.69± 0.45 %
μ = 487.2± 2.0
σ = 42.3± 2.0
WF (647 entries)
$\frac{\sigma}{\mu}$ = 8.61± 0.46 %
μ = 471.7± 2.0
σ = 40.6± 2.0



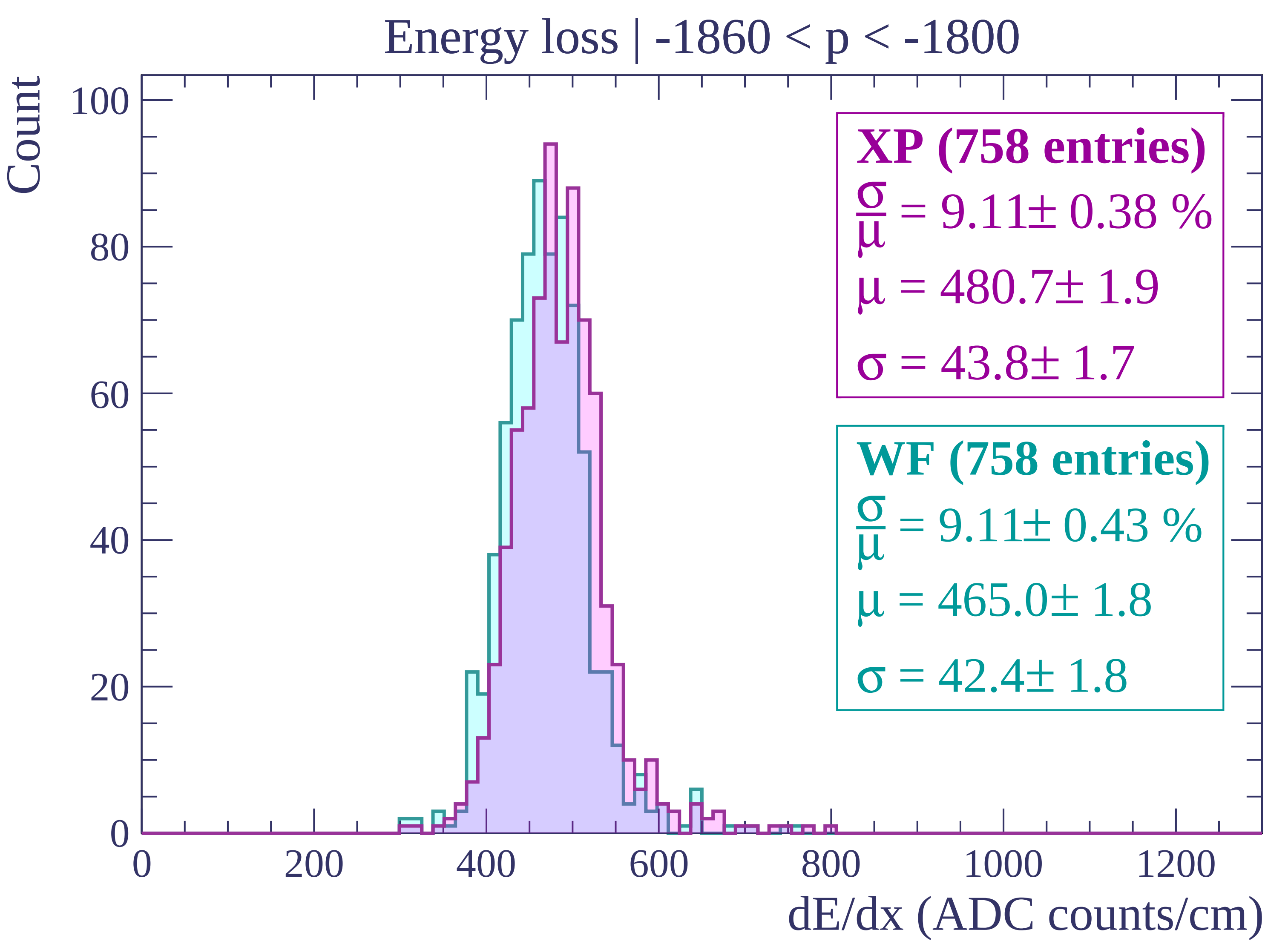
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
XP (654 entries)
σ/μ = 8.39± 0.47 %
μ = 485.6± 2.0
σ = 40.7± 2.1
WF (654 entries)
σ/μ = 8.53± 0.46 %
μ = 470.1± 1.9
σ = 40.1± 2.0
0
20
40
60
80
100
0
200
400
600
800
1000
1200



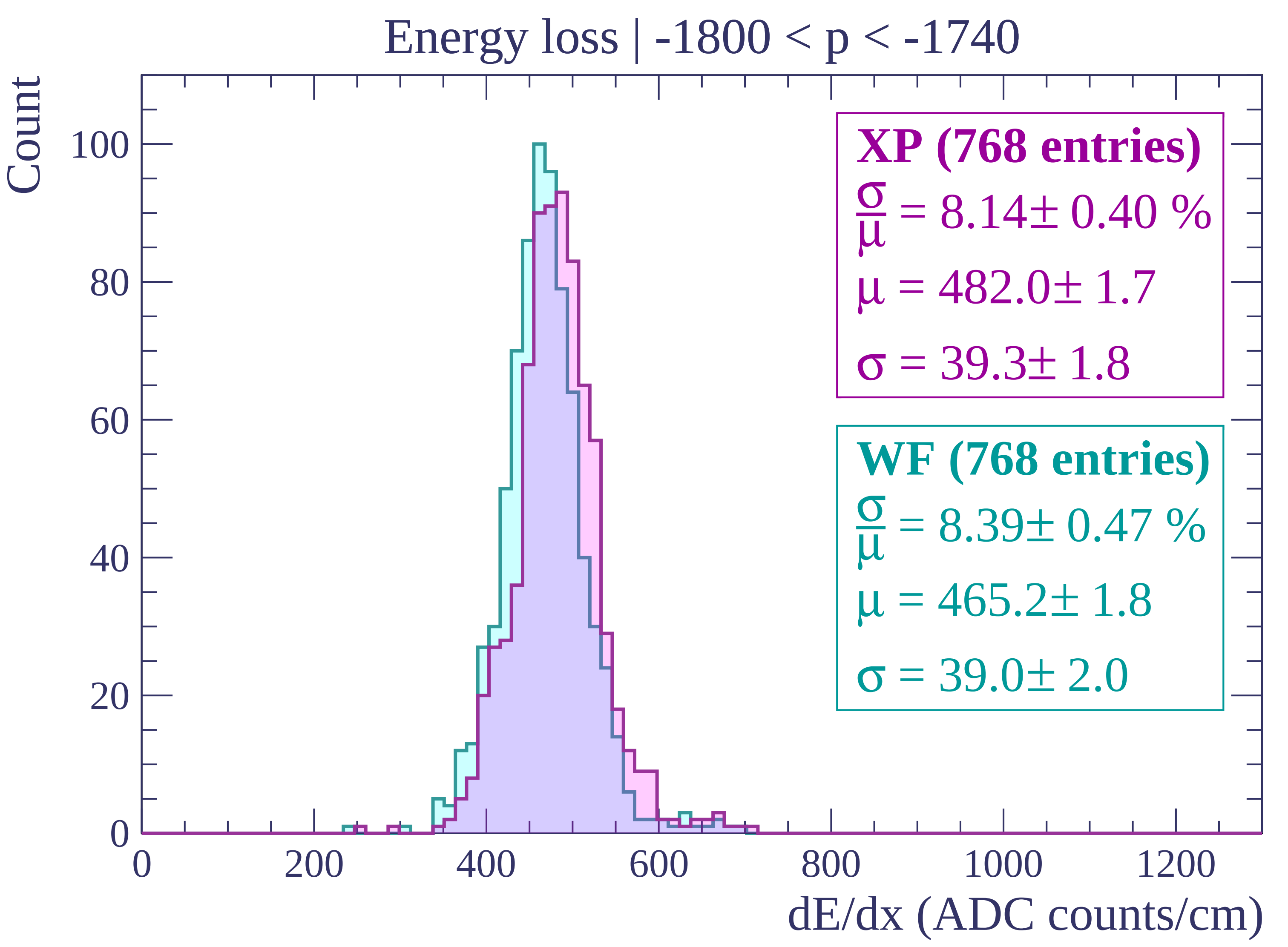
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (693 entries)
σ/μ = 8.14± 0.42 %
μ = 482.1± 1.8
σ = 39.2± 1.9
WF (693 entries)
σ/μ = 8.51± 0.45 %
μ = 466.5± 1.8
σ = 39.7± 2.0

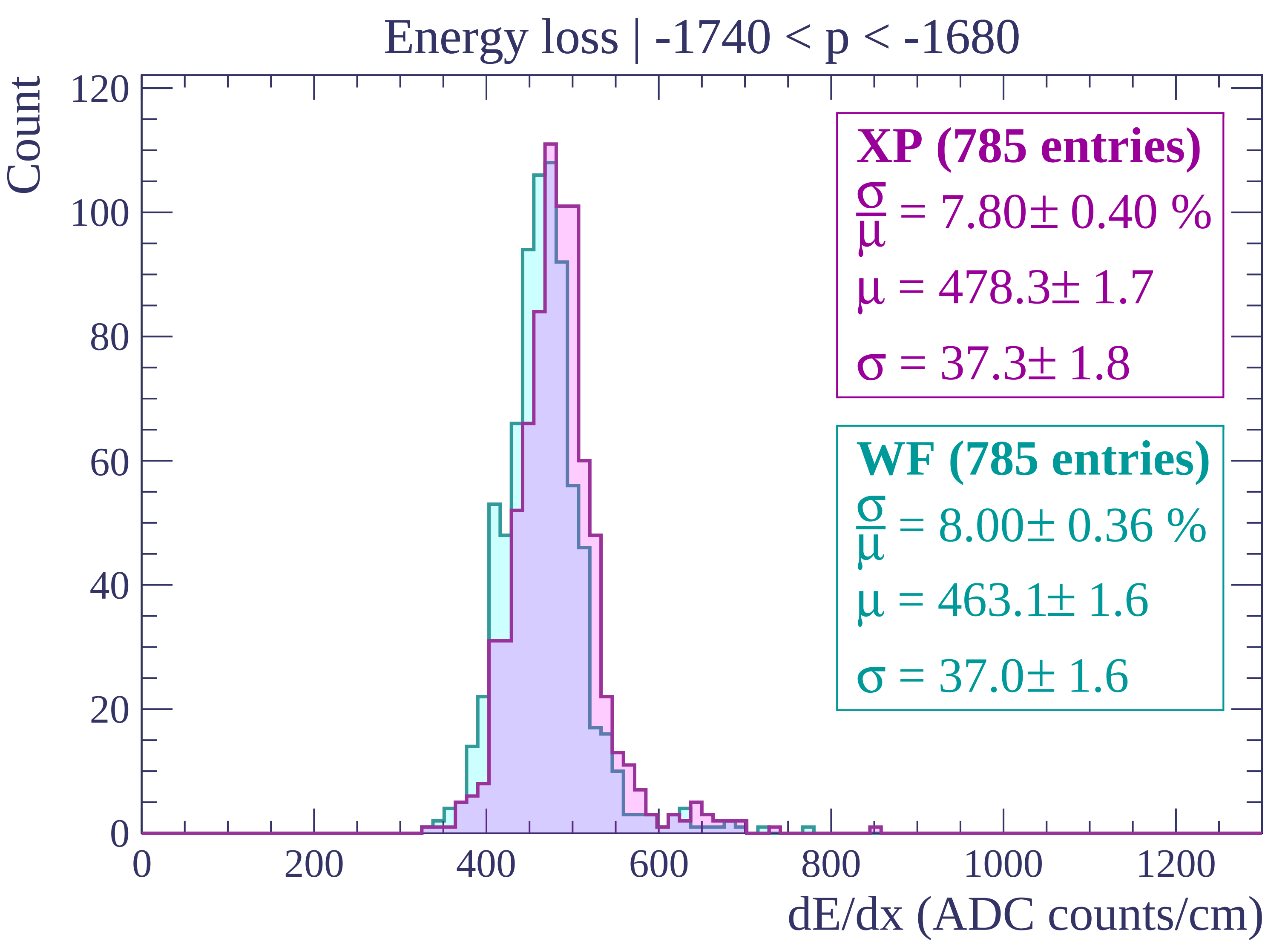


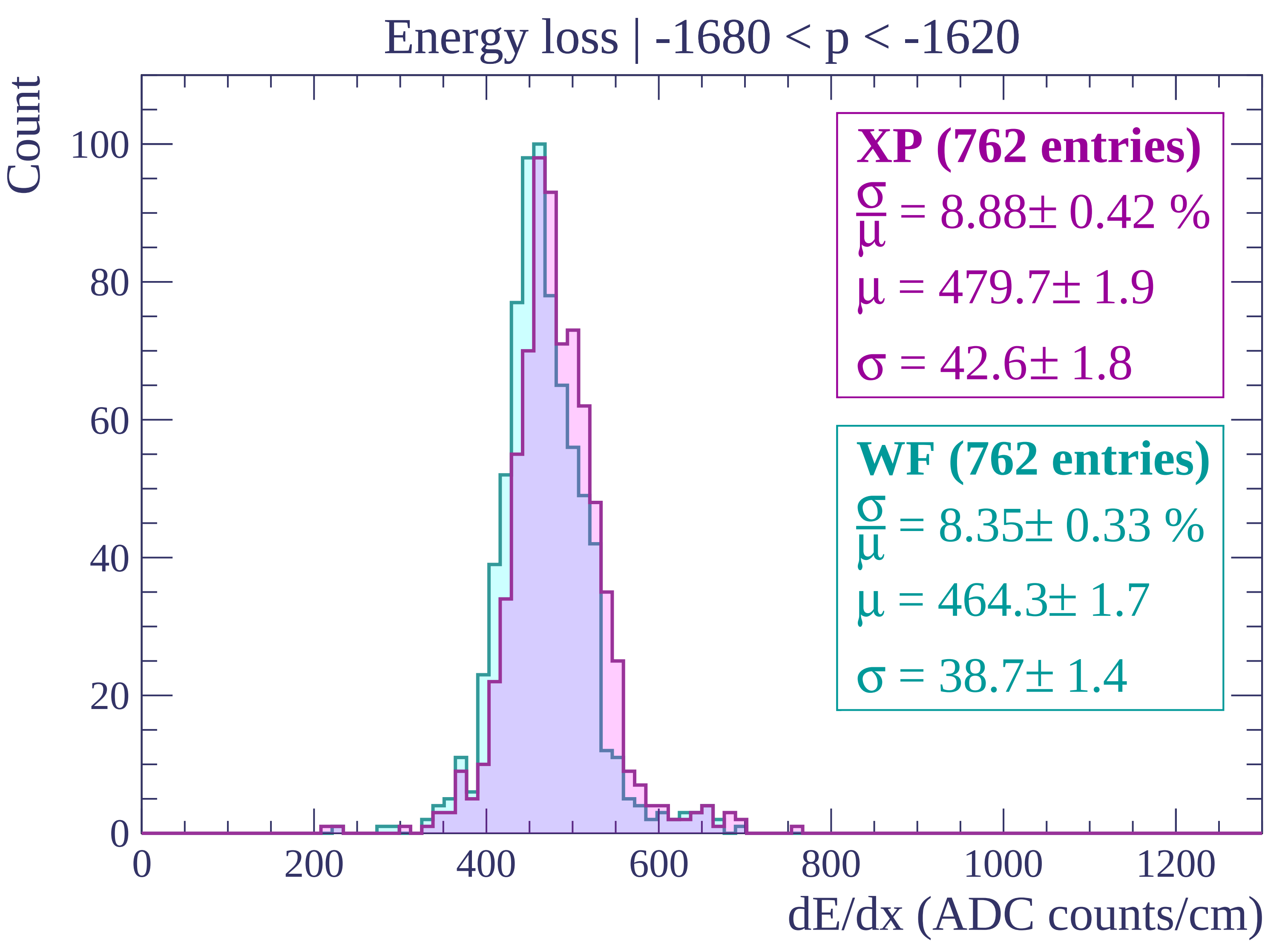
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (706 entries)
$\frac{\sigma}{\mu}$ = 8.10± 0.35 %
μ = 480.0± 1.7
σ = 38.9± 1.6
WF (706 entries)
$\frac{\sigma}{\mu}$ = 7.96± 0.38 %
μ = 465.0± 1.8
σ = 37.0± 1.6

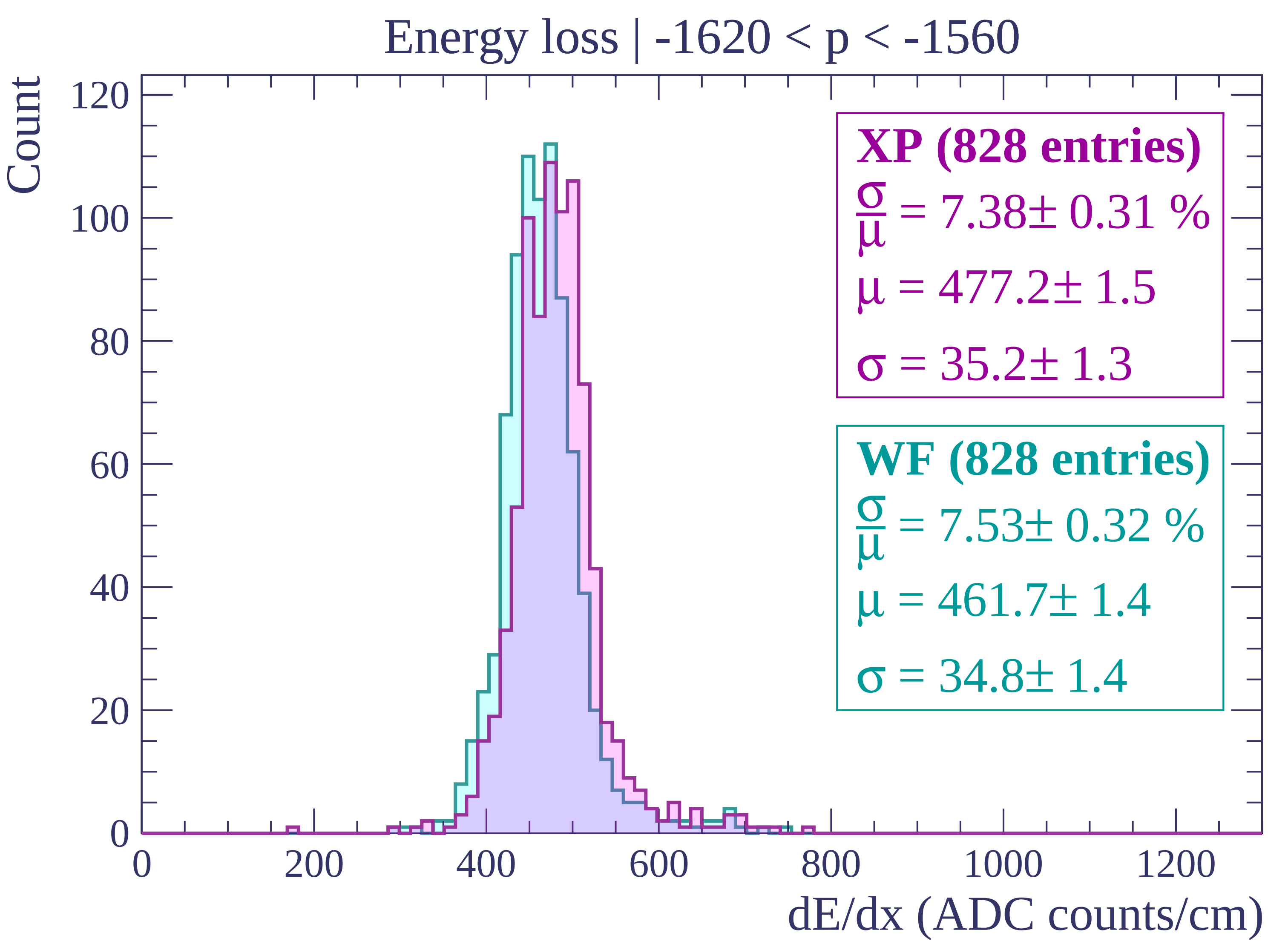


Energy loss | -1860 < p < -1800
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (758 entries)
σ/μ = 9.11± 0.38 %
μ = 480.7± 1.9
σ = 43.8± 1.7
WF (758 entries)
σ/μ = 9.11± 0.43 %
μ = 465.0± 1.8
σ = 42.4± 1.8

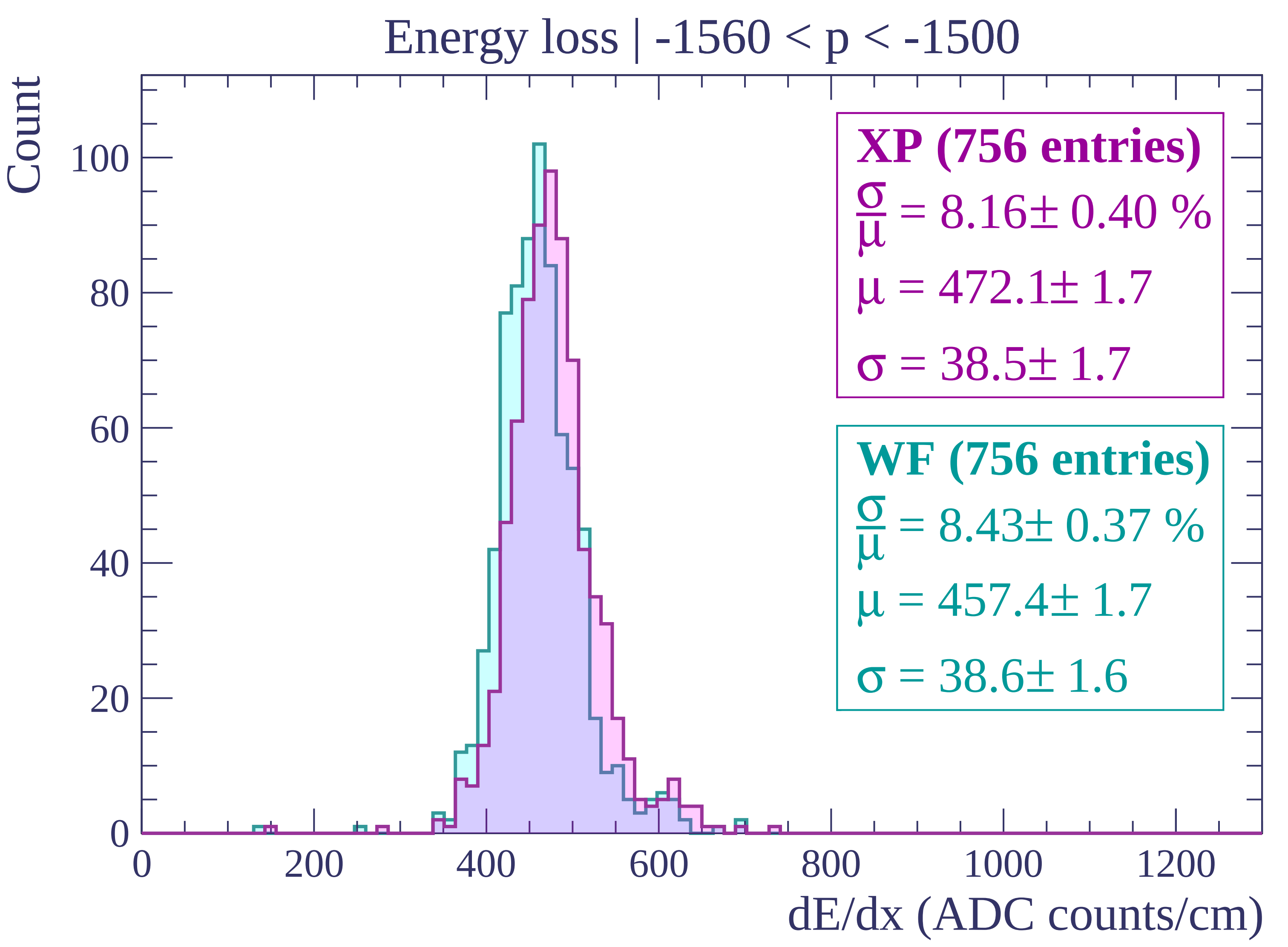

Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (768 entries)
σ/μ = 8.14± 0.40 %
μ = 482.0± 1.7
σ = 39.3± 1.8
WF (768 entries)
σ/μ = 8.39± 0.47 %
μ = 465.2± 1.8
σ = 39.0± 2.0


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (785 entries)
σ/μ = 7.80± 0.40 %
μ = 478.3± 1.7
σ = 37.3± 1.8
WF (785 entries)
σ/μ = 8.00± 0.36 %
μ = 463.1± 1.6
σ = 37.0± 1.6

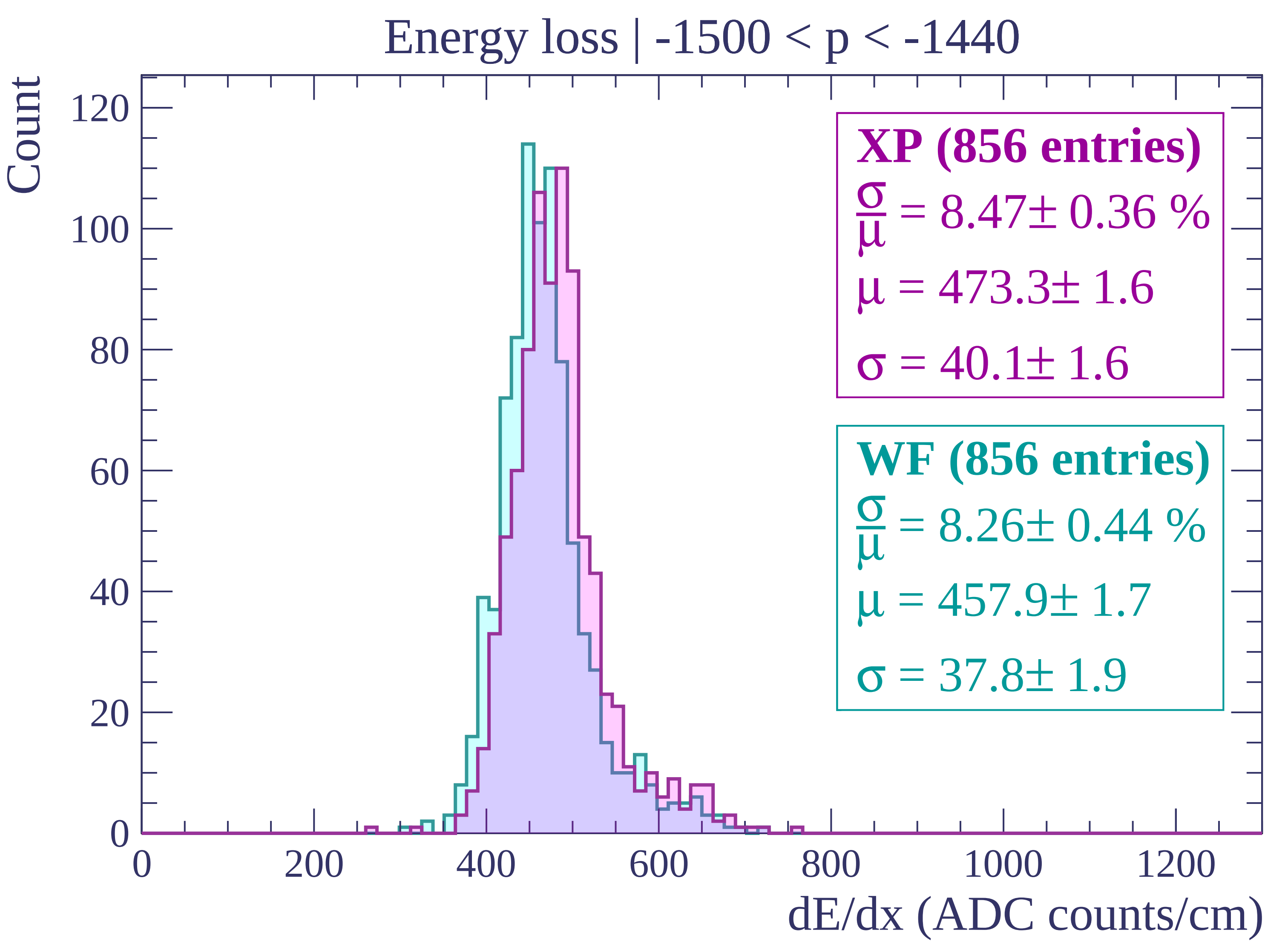

Energy loss | -1680 < p < -1620
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (762 entries)
σ/μ = 8.88± 0.42 %
μ = 479.7± 1.9
σ = 42.6± 1.8
WF (762 entries)
σ/μ = 8.35± 0.33 %
μ = 464.3± 1.7
σ = 38.7± 1.4



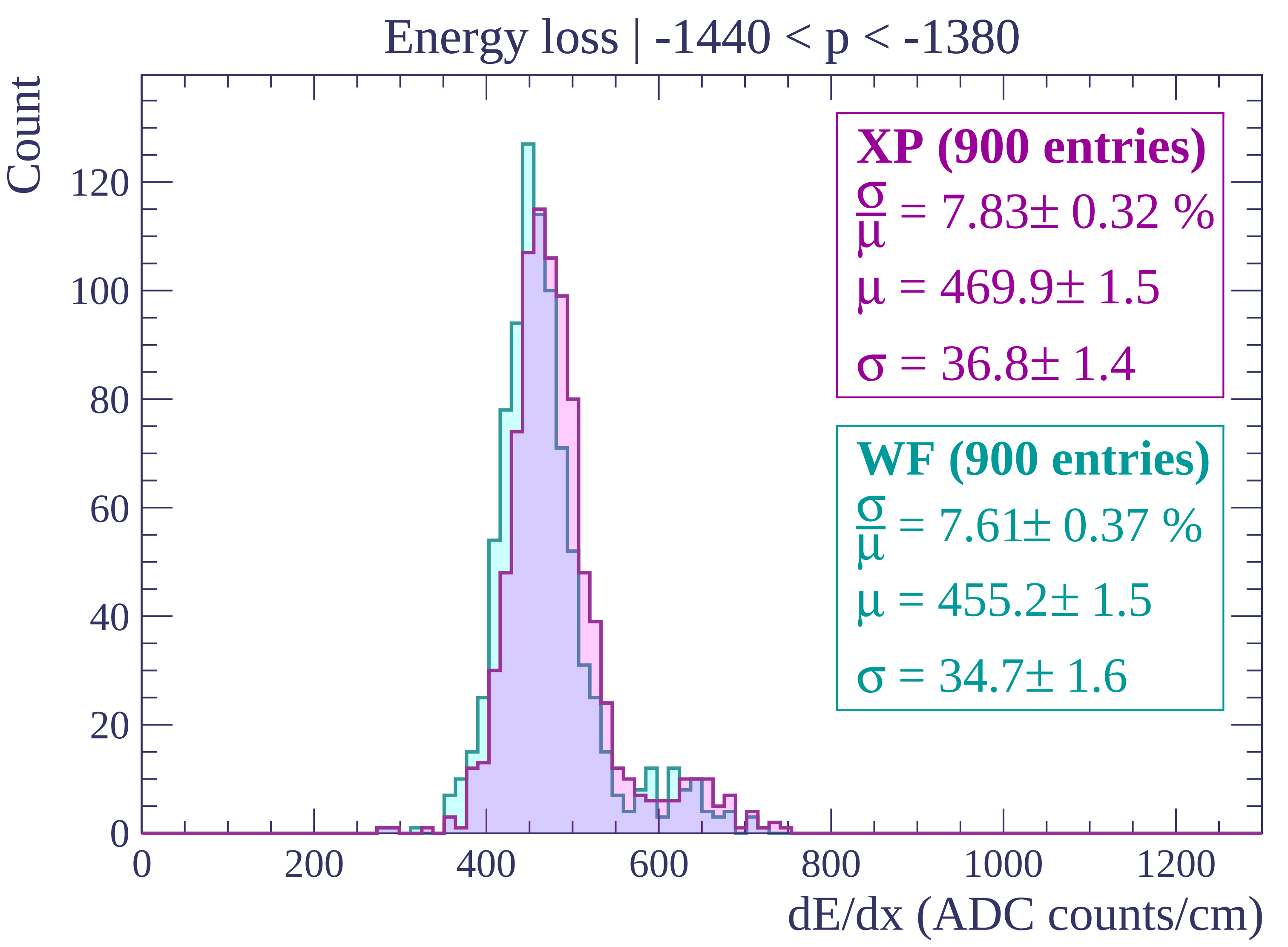
Energy loss | -1620 < p < -1560
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (828 entries)
σ/μ = 7.38± 0.31 %
μ = 477.2± 1.5
σ = 35.2± 1.3
WF (828 entries)
σ/μ = 7.53± 0.32 %
μ = 461.7± 1.4
σ = 34.8± 1.4



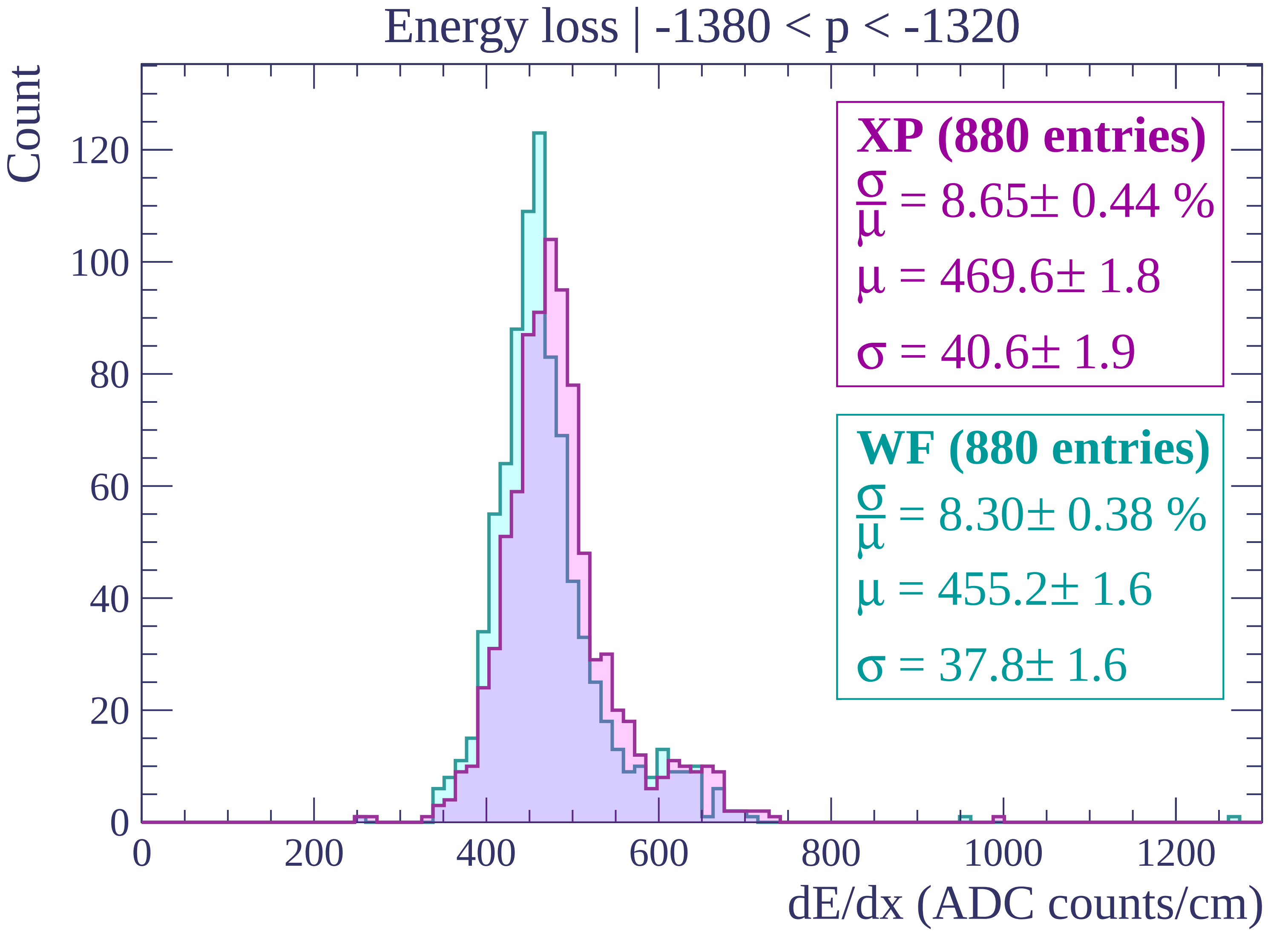
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (756 entries)
σ/μ = 8.16± 0.40 %
μ = 472.1± 1.7
σ = 38.5± 1.7
WF (756 entries)
σ/μ = 8.43± 0.37 %
μ = 457.4± 1.7
σ = 38.6± 1.6



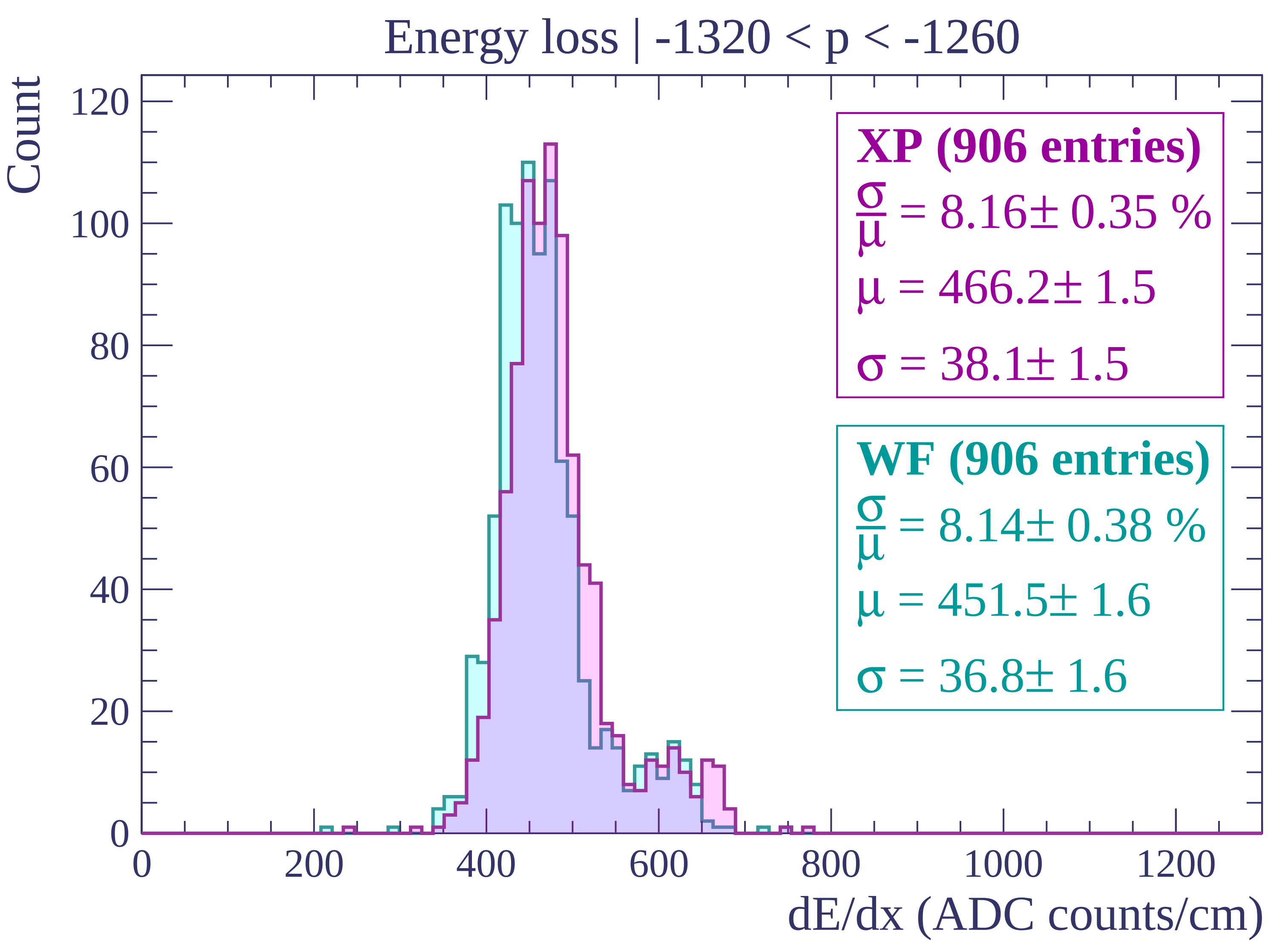
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (856 entries)
σ/μ = 8.47± 0.36 %
μ = 473.3± 1.6
σ = 40.1± 1.6
WF (856 entries)
σ/μ = 8.26± 0.44 %
μ = 457.9± 1.7
σ = 37.8± 1.9

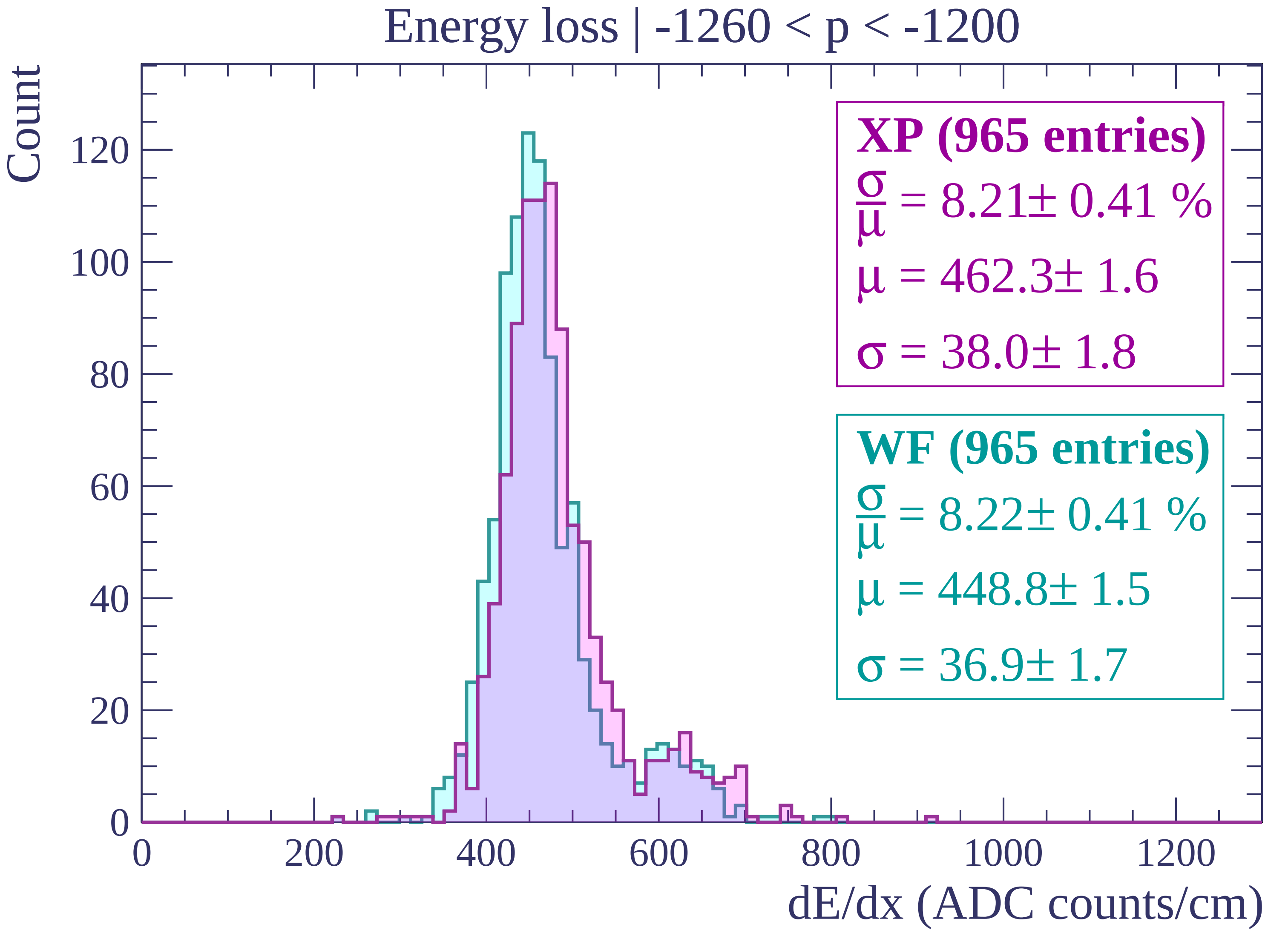


Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (900 entries)
σ/μ = 7.83± 0.32 %
μ = 469.9± 1.5
σ = 36.8± 1.4
WF (900 entries)
σ/μ = 7.61± 0.37 %
μ = 455.2± 1.5
σ = 34.7± 1.6

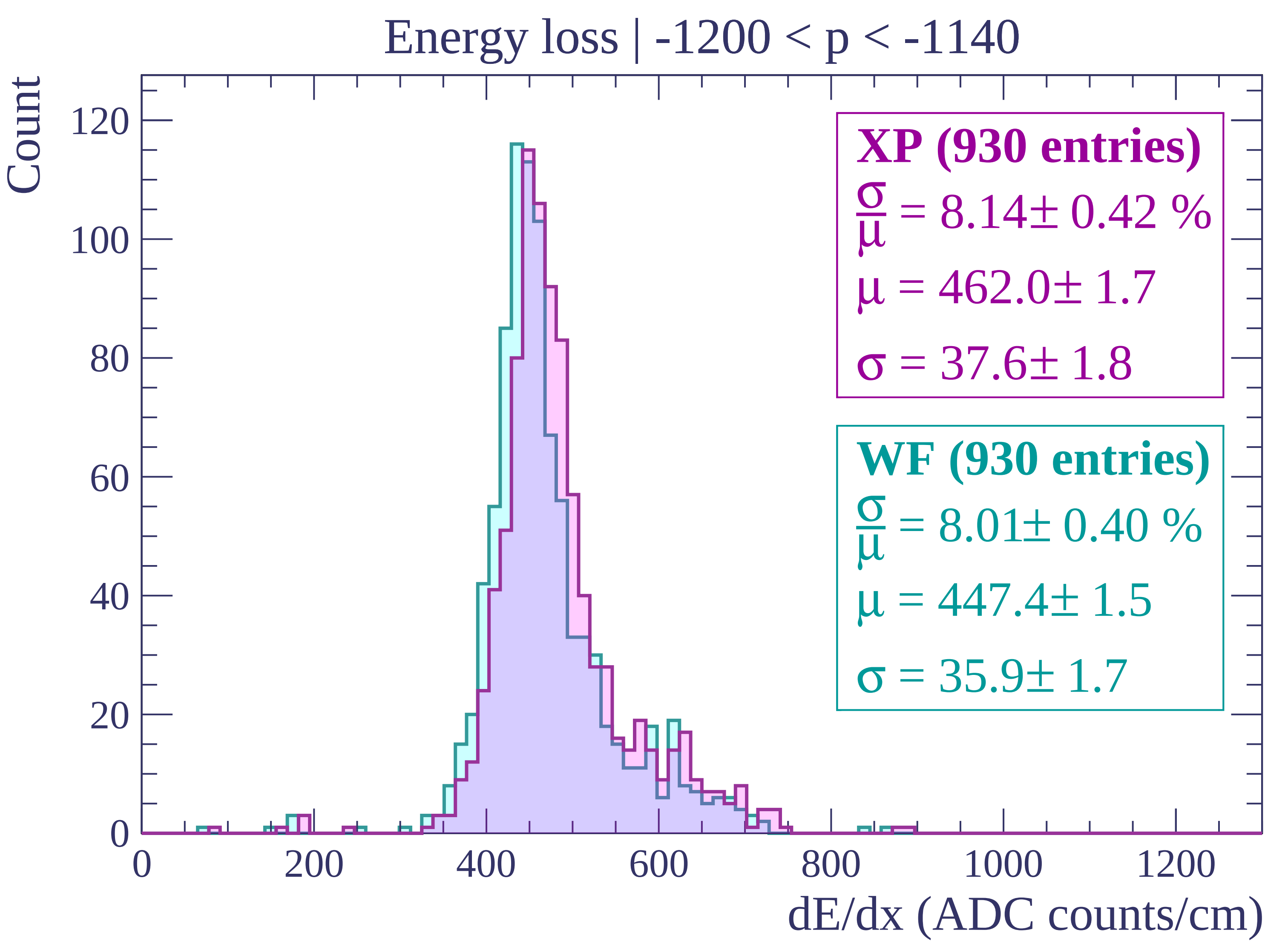


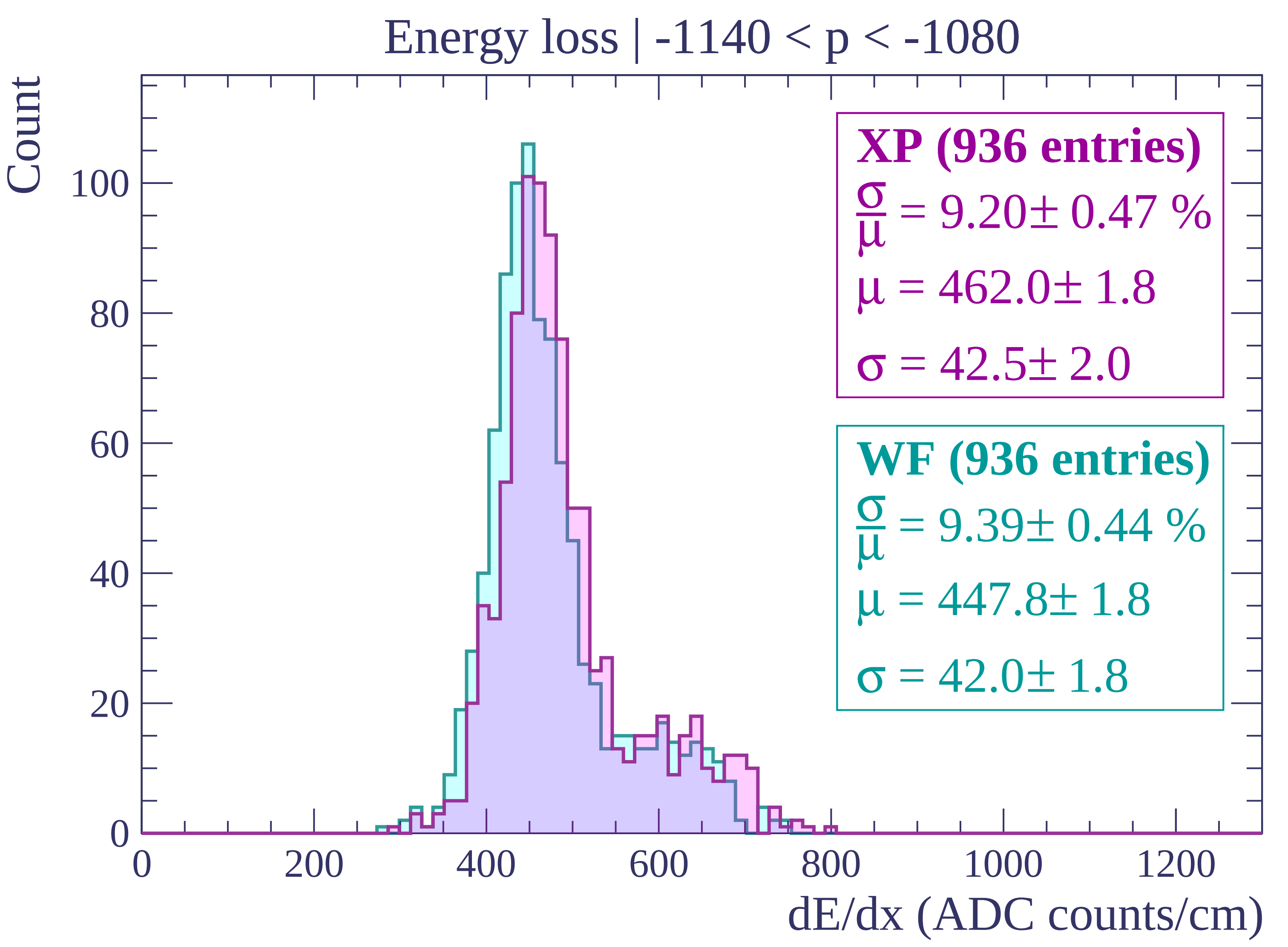
Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (880 entries)
σ/μ = 8.65± 0.44 %
μ = 469.6± 1.8
σ = 40.6± 1.9
WF (880 entries)
σ/μ = 8.30± 0.38 %
μ = 455.2± 1.6
σ = 37.8± 1.6


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
XP (906 entries)
σ/μ = 8.16± 0.35 %
μ = 466.2± 1.5
σ = 38.1± 1.5
WF (906 entries)
σ/μ = 8.14± 0.38 %
μ = 451.5± 1.6
σ = 36.8± 1.6
0
20
40
60
80
100
120
200
400
600
800
1000
1200

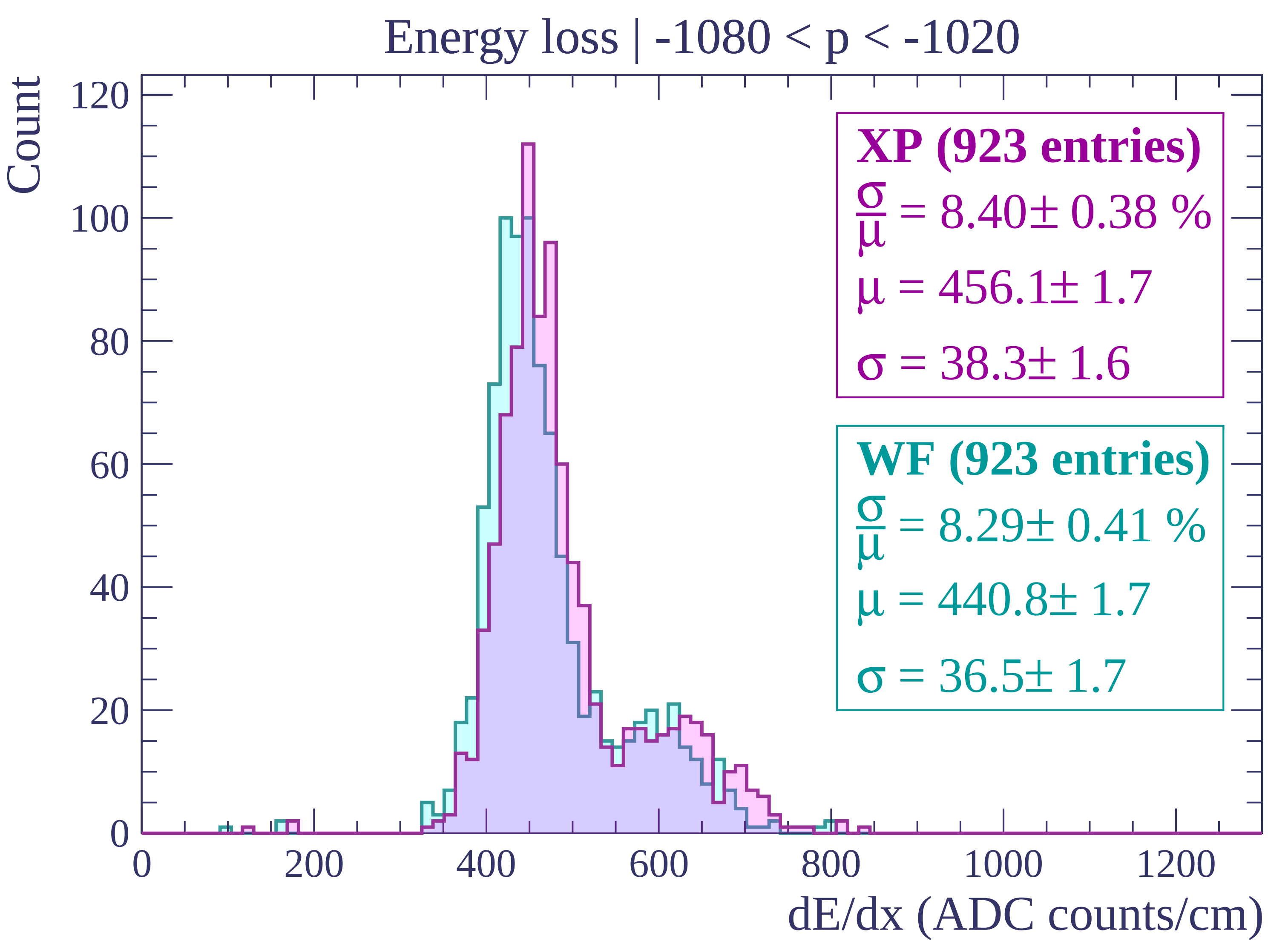


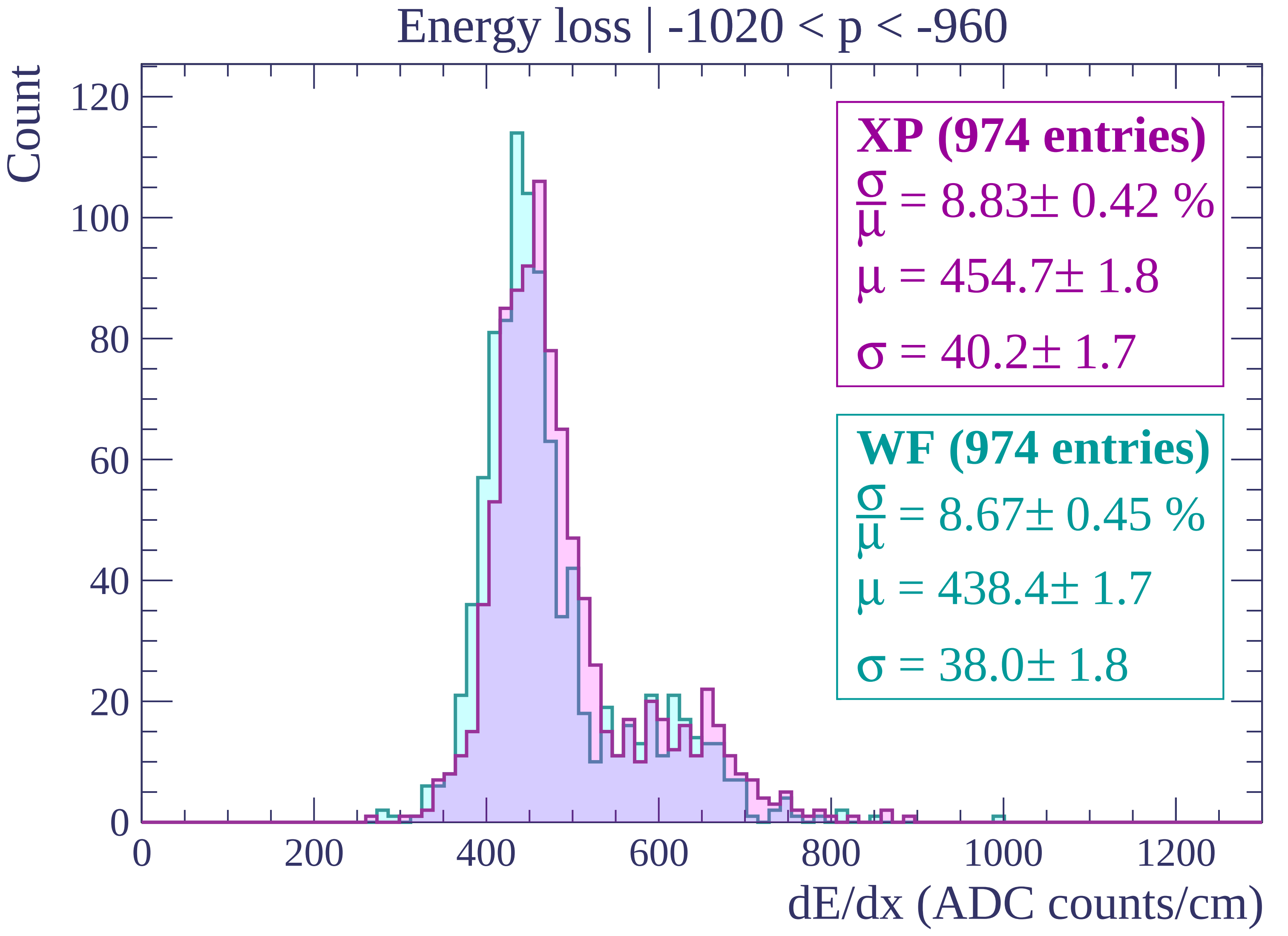
Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
XP (965 entries)
$\frac{\sigma}{\mu}$ = 8.21± 0.41 %
μ = 462.3± 1.6
σ = 38.0± 1.8
WF (965 entries)
$\frac{\sigma}{\mu}$ = 8.22± 0.41 %
μ = 448.8± 1.5
σ = 36.9± 1.7
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

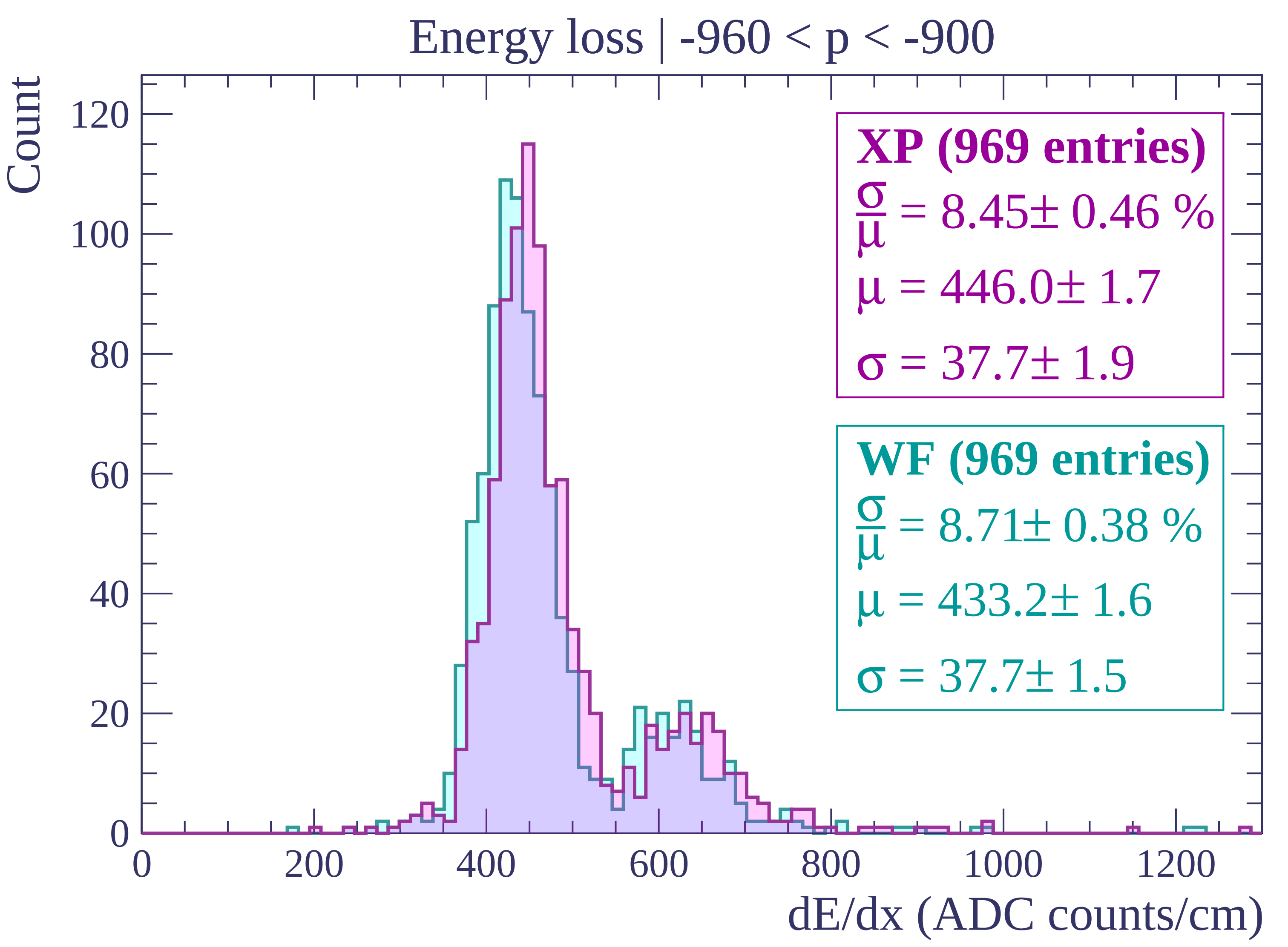

Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
200
400
600
800
1000
1200
XP (930 entries)
σ/μ = 8.14± 0.42 %
μ = 462.0± 1.7
σ = 37.6± 1.8
WF (930 entries)
σ/μ = 8.01± 0.40 %
μ = 447.4± 1.5
σ = 35.9± 1.7



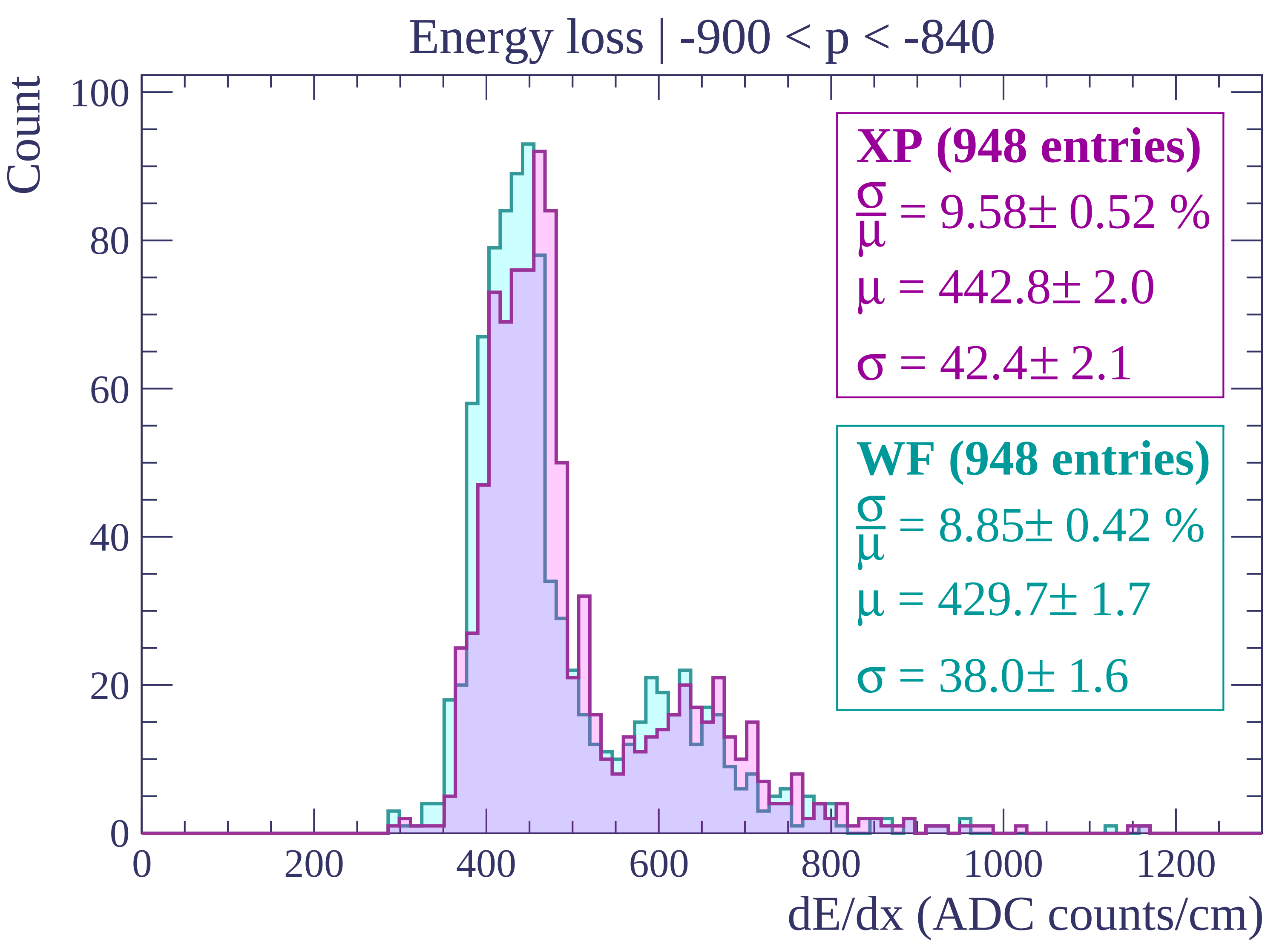
Energy loss | -1140 < p < -1080
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (936 entries)
σ/μ = 9.20± 0.47 %
μ = 462.0± 1.8
σ = 42.5± 2.0
WF (936 entries)
σ/μ = 9.39± 0.44 %
μ = 447.8± 1.8
σ = 42.0± 1.8

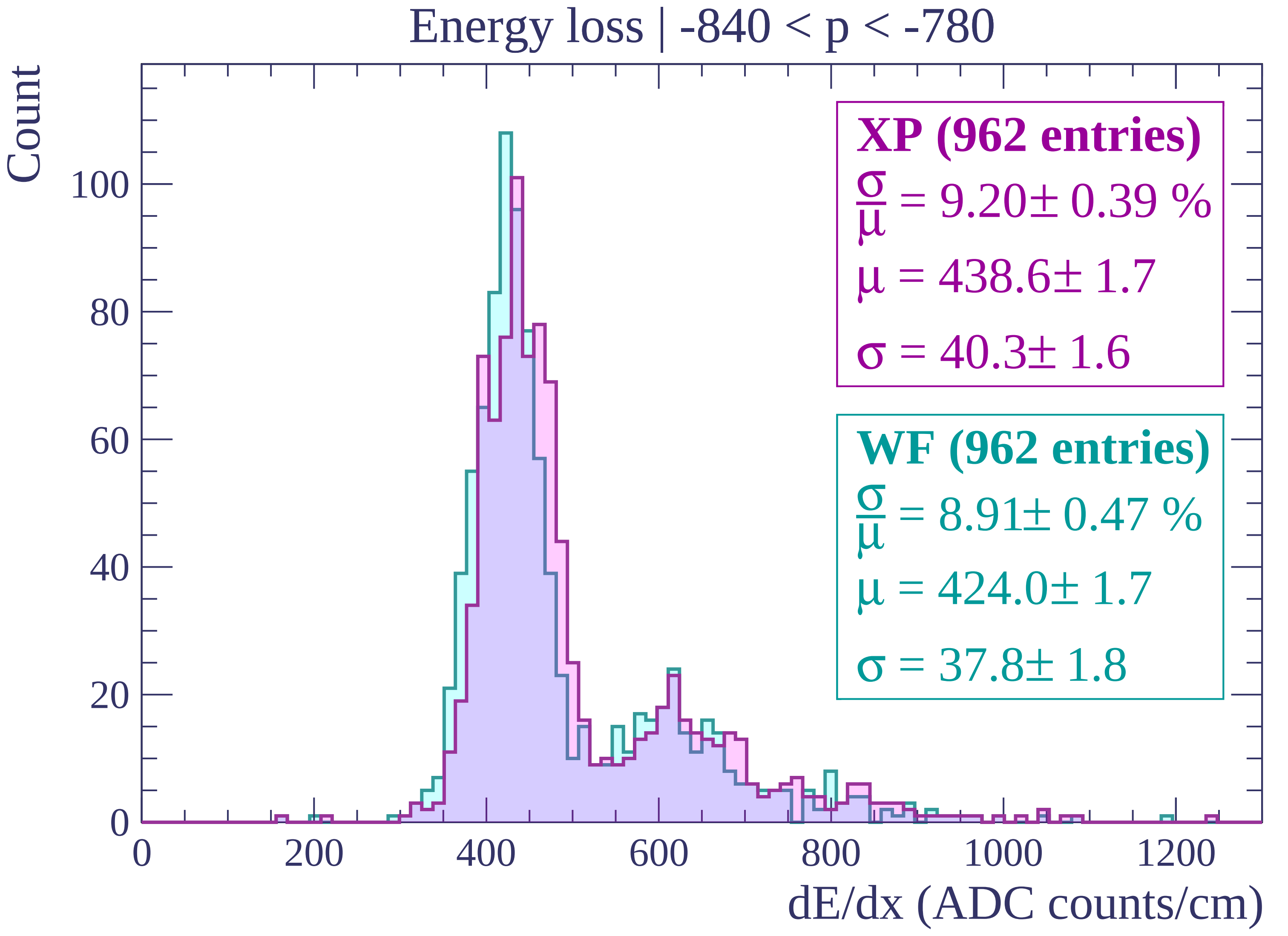

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (923 entries)
σ/μ = 8.40± 0.38 %
μ = 456.1± 1.7
σ = 38.3± 1.6
WF (923 entries)
σ/μ = 8.29± 0.41 %
μ = 440.8± 1.7
σ = 36.5± 1.7
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

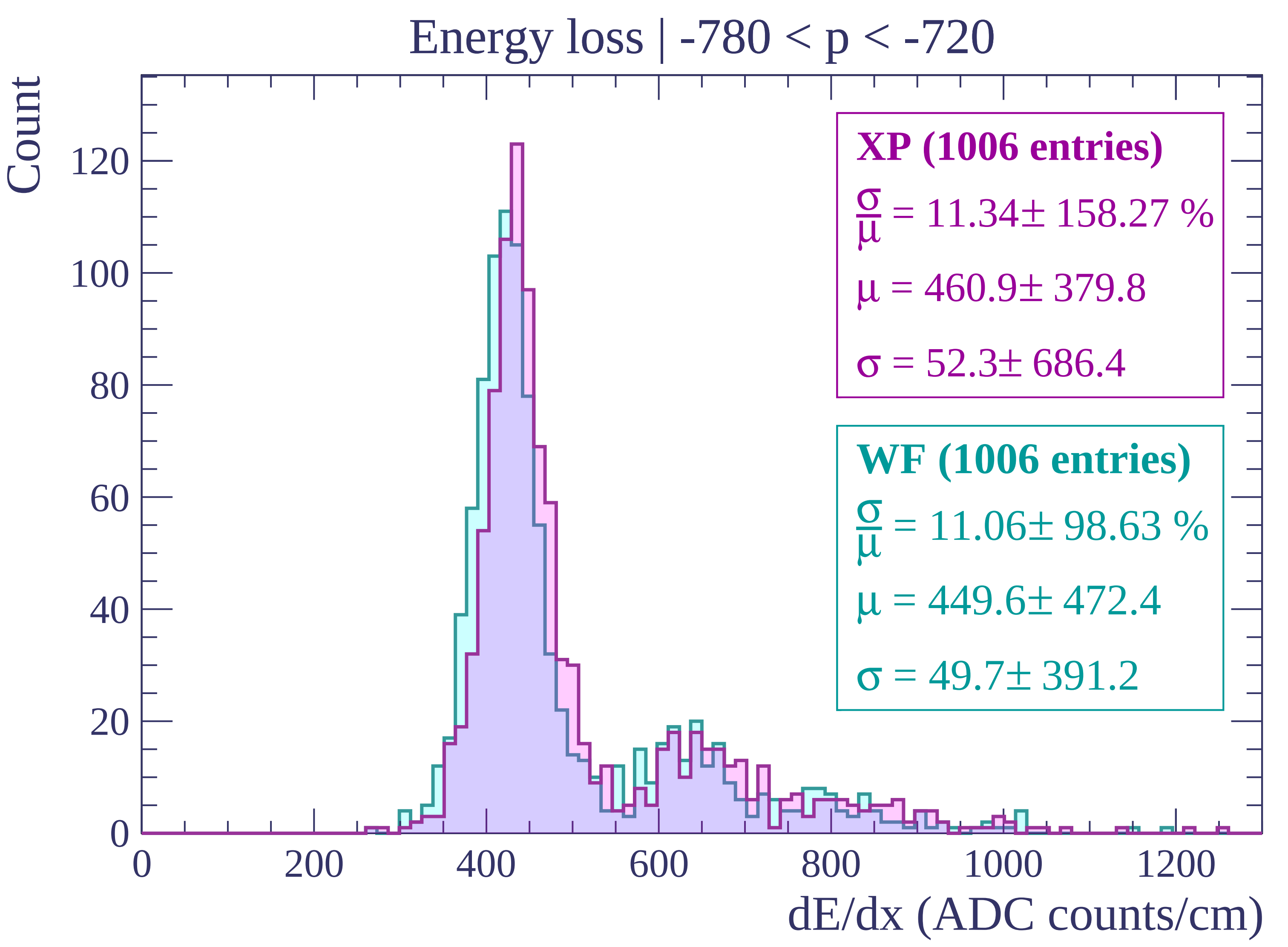

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (974 entries)
σ/μ = 8.83± 0.42 %
μ = 454.7± 1.8
σ = 40.2± 1.7
WF (974 entries)
σ/μ = 8.67± 0.45 %
μ = 438.4± 1.7
σ = 38.0± 1.8



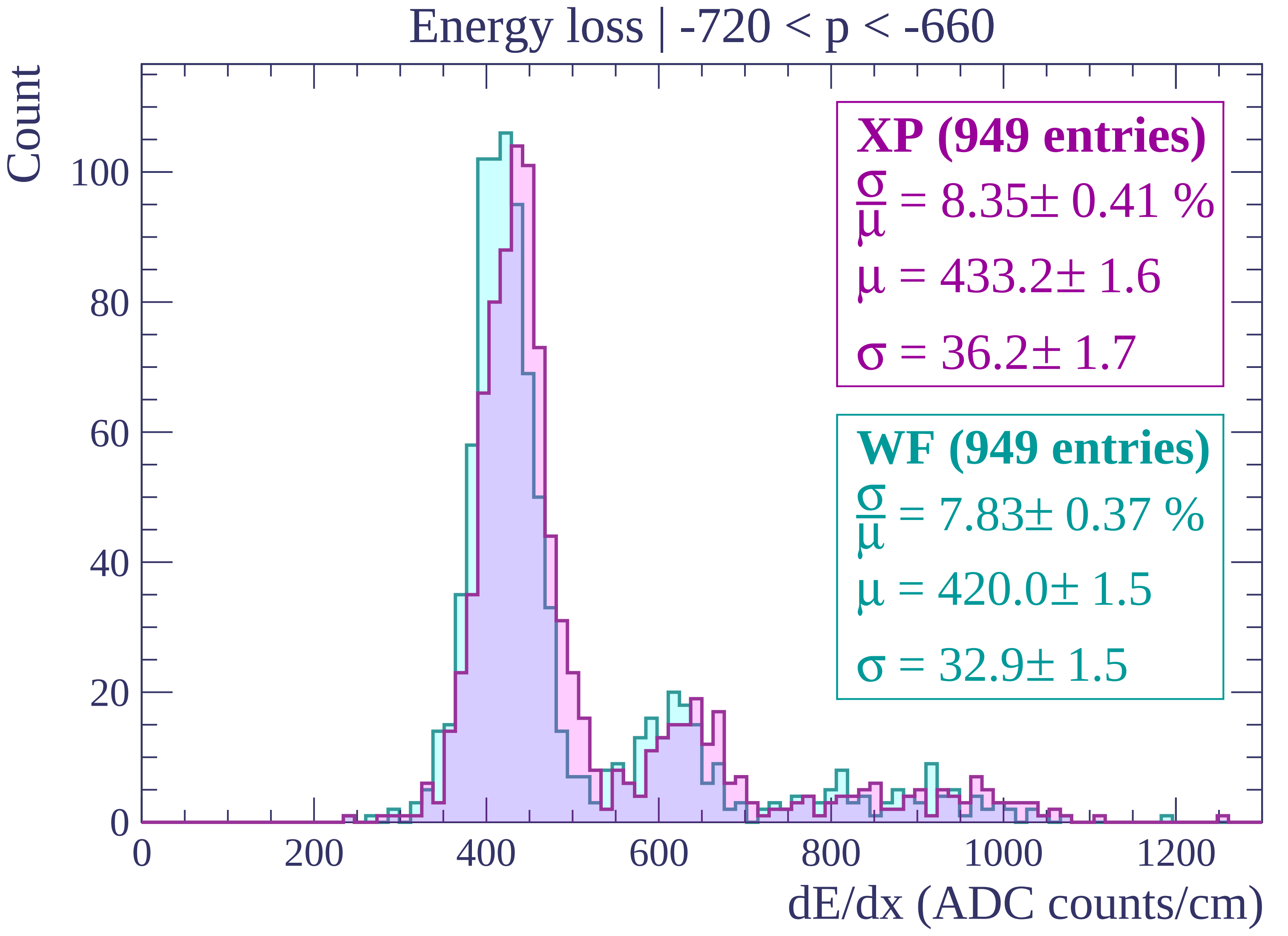
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
XP (969 entries)
σ/μ = 8.45± 0.46 %
μ = 446.0± 1.7
σ = 37.7± 1.9
WF (969 entries)
σ/μ = 8.71± 0.38 %
μ = 433.2± 1.6
σ = 37.7± 1.5


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (948 entries)
σ/μ = 9.58± 0.52 %
μ = 442.8± 2.0
σ = 42.4± 2.1
WF (948 entries)
σ/μ = 8.85± 0.42 %
μ = 429.7± 1.7
σ = 38.0± 1.6
0
20
40
60
80
100
0
200
400
600
800
1000
1200

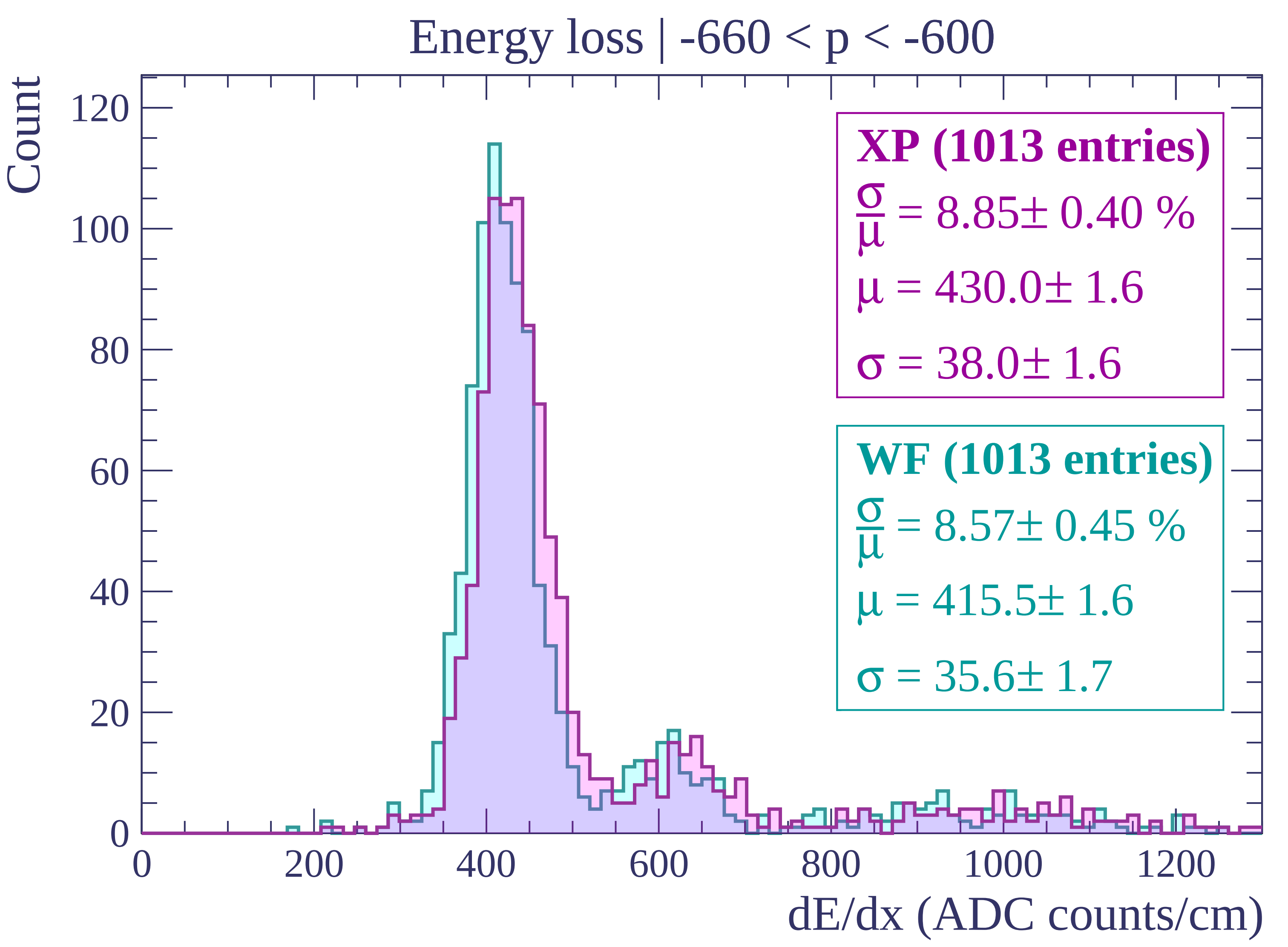

Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (962 entries)
σ/μ = 9.20± 0.39 %
μ = 438.6± 1.7
σ = 40.3± 1.6
WF (962 entries)
σ/μ = 8.91± 0.47 %
μ = 424.0± 1.7
σ = 37.8± 1.8
0
20
40
60
80
100
0
200
400
600
800
1000
1200

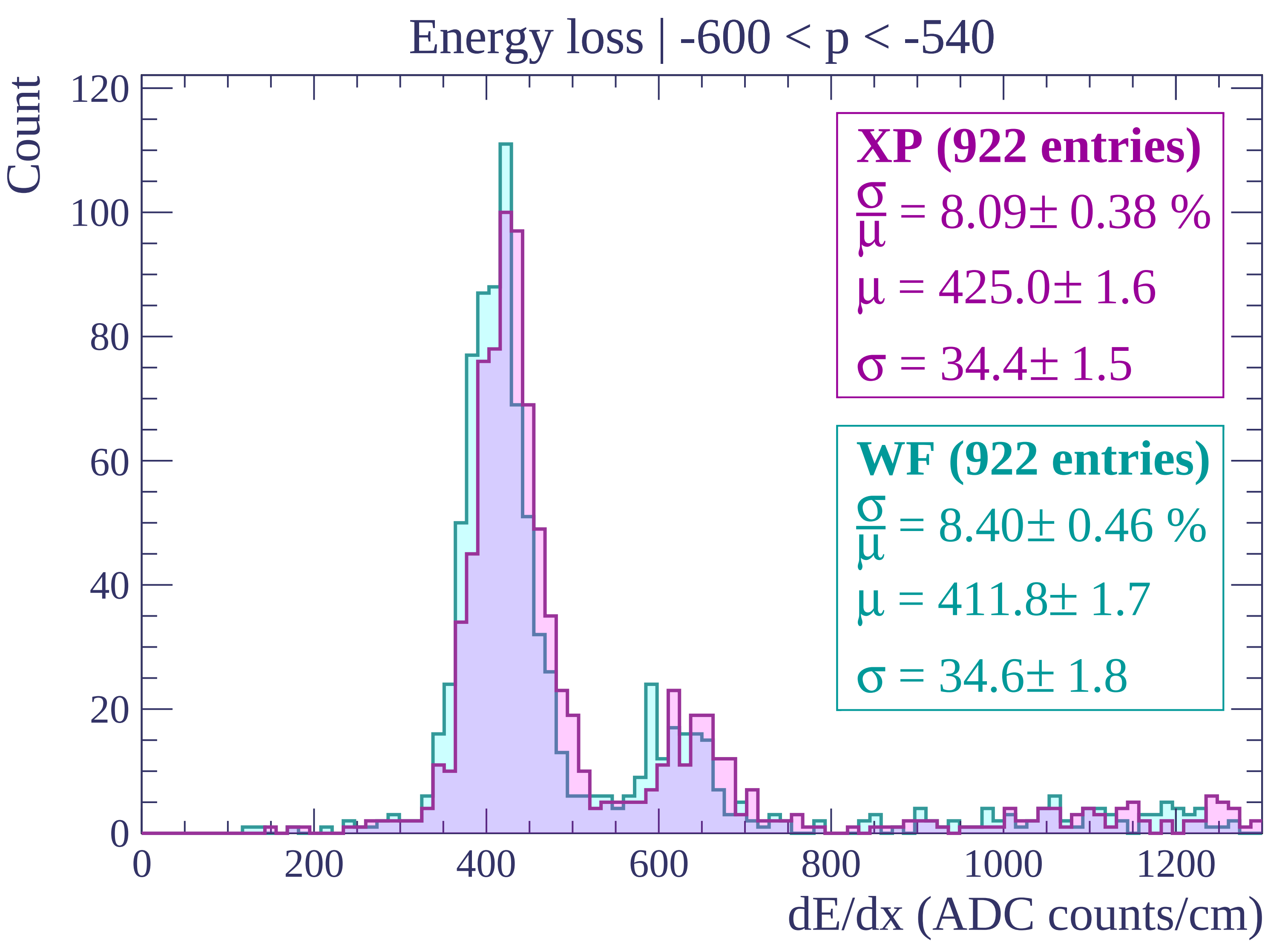


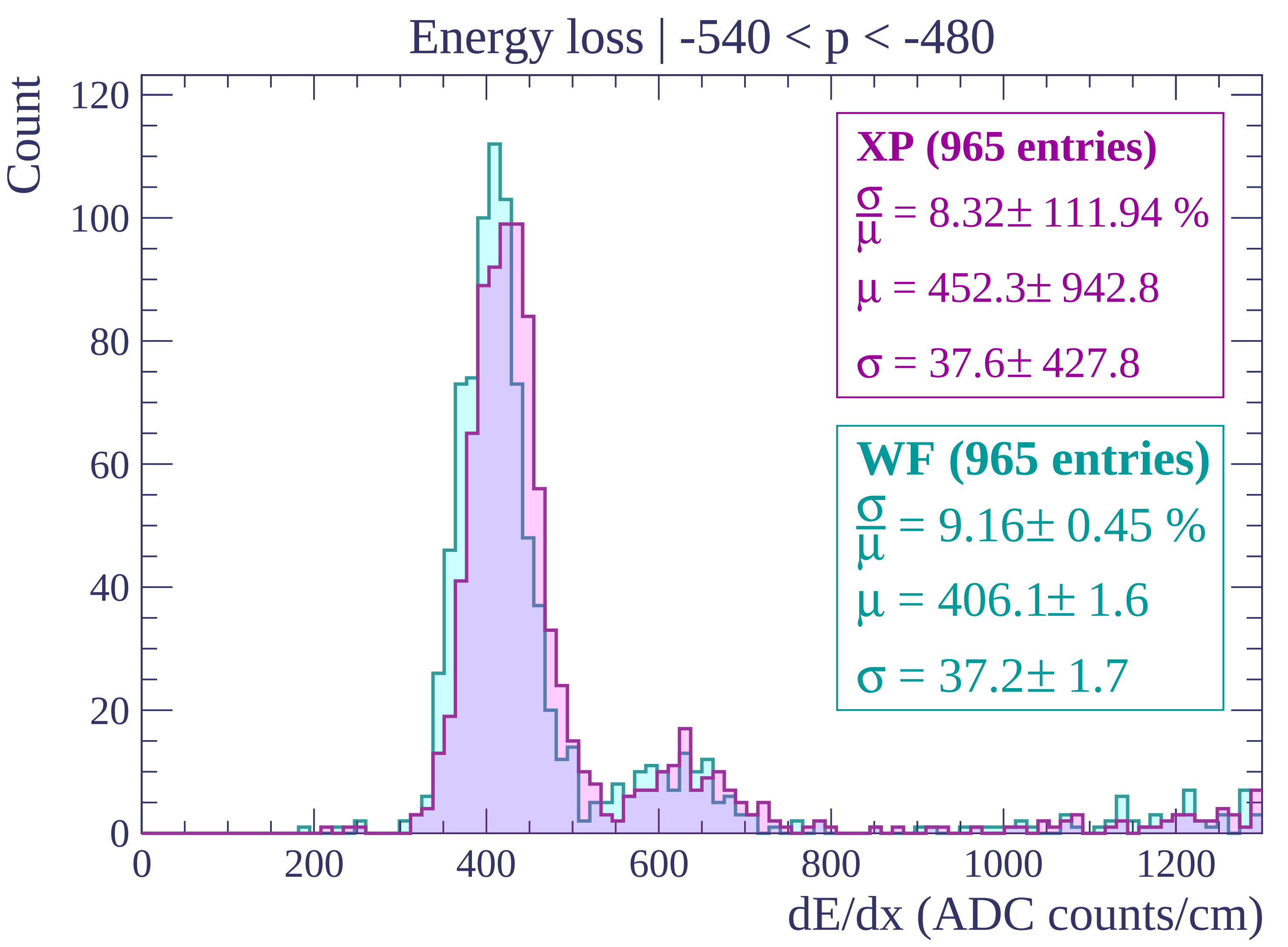
Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (1006 entries)
σ/μ = 11.34± 158.27 %
μ = 460.9± 379.8
σ = 52.3± 686.4
WF (1006 entries)
σ/μ = 11.06± 98.63 %
μ = 449.6± 472.4
σ = 49.7± 391.2
0
200
400
600
800
1000
1200
20
40
60
80
100
120



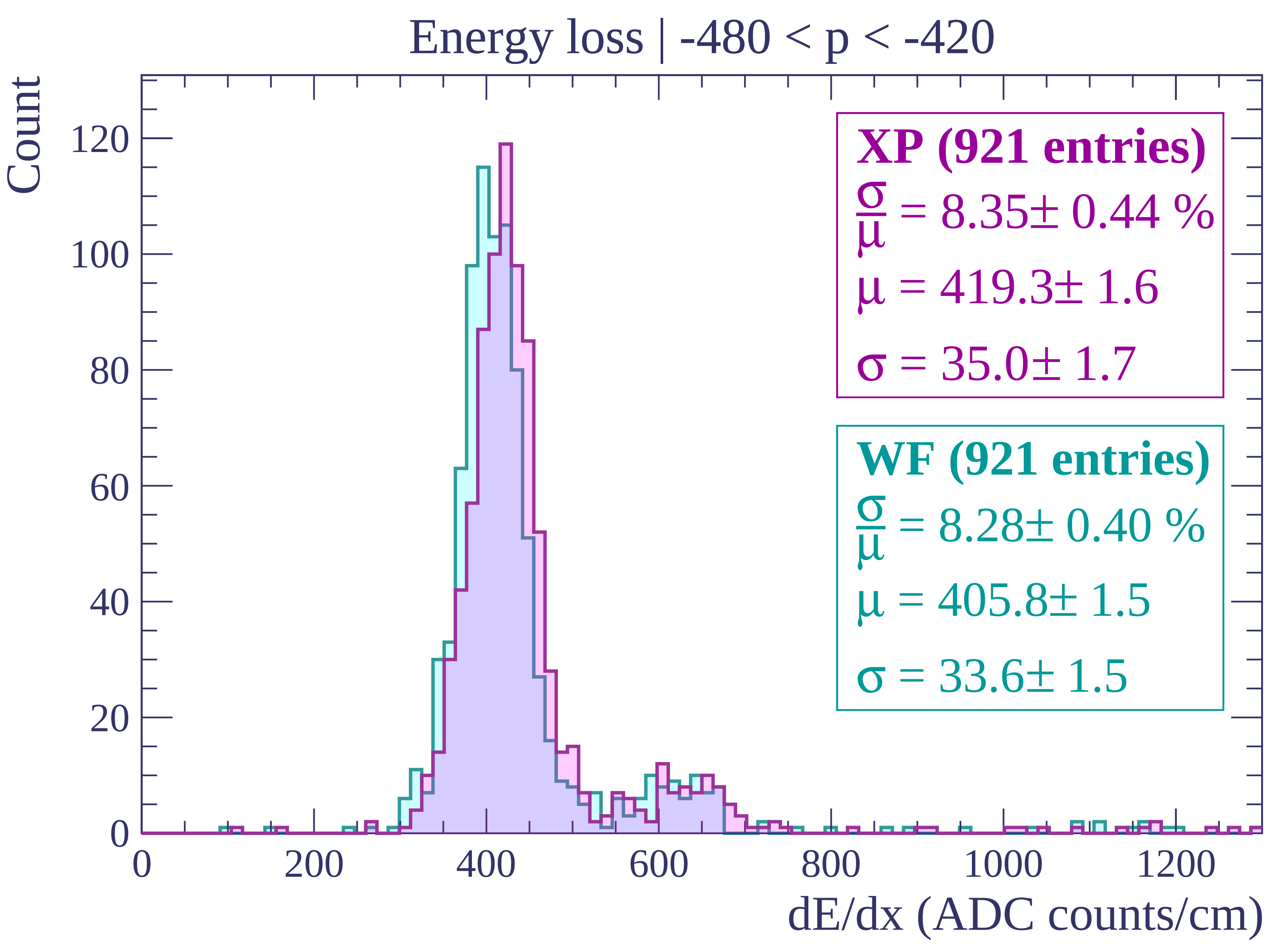
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (949 entries)
σ/μ = 8.35± 0.41 %
μ = 433.2± 1.6
σ = 36.2± 1.7
WF (949 entries)
σ/μ = 7.83± 0.37 %
μ = 420.0± 1.5
σ = 32.9± 1.5


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (1013 entries)
σ/μ = 8.85± 0.40 %
μ = 430.0± 1.6
σ = 38.0± 1.6
WF (1013 entries)
σ/μ = 8.57± 0.45 %
μ = 415.5± 1.6
σ = 35.6± 1.7
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

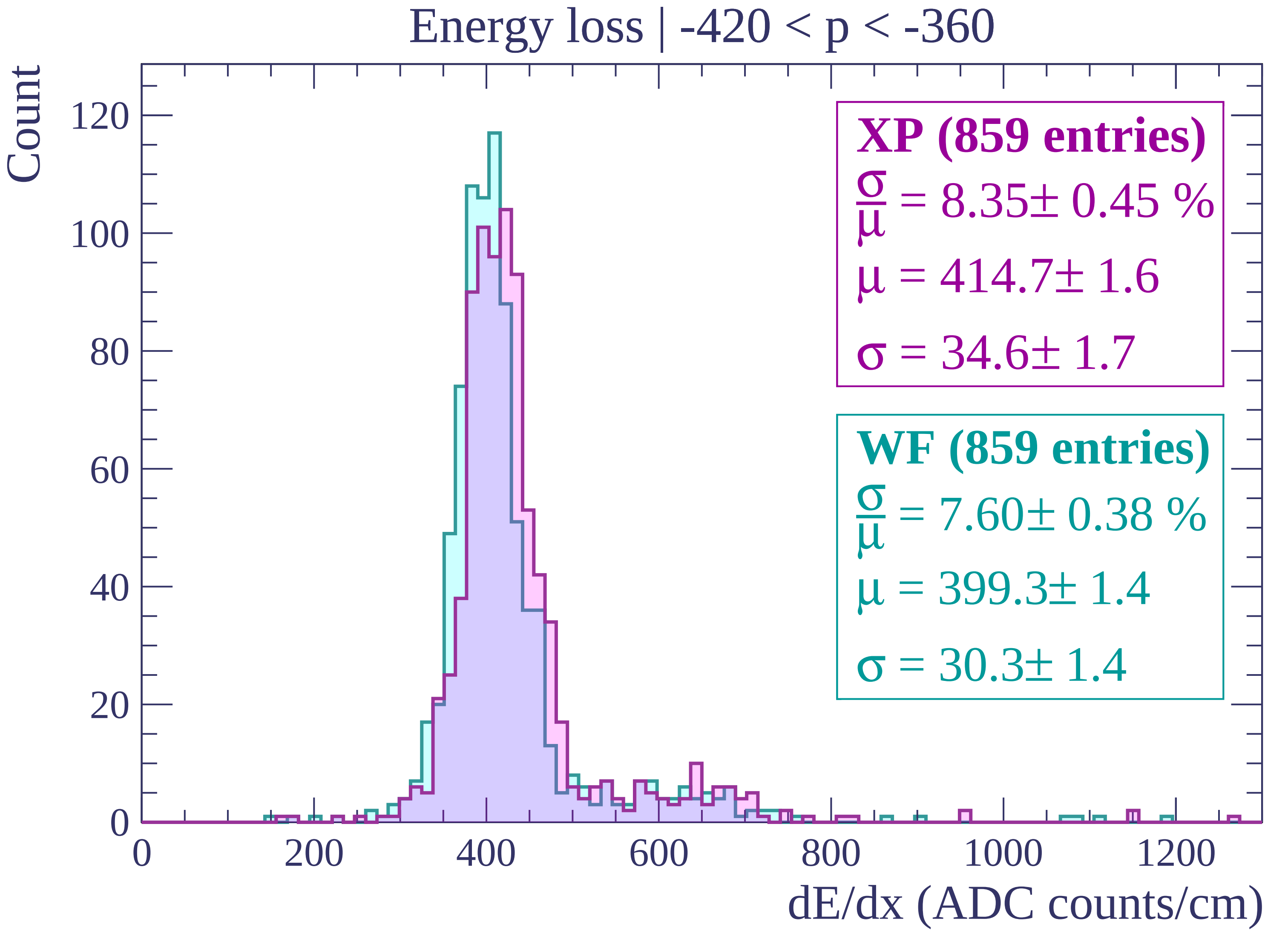

Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
XP (922 entries)
σ/μ = 8.09± 0.38 %
μ = 425.0± 1.6
σ = 34.4± 1.5
WF (922 entries)
σ/μ = 8.40± 0.46 %
μ = 411.8± 1.7
σ = 34.6± 1.8
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

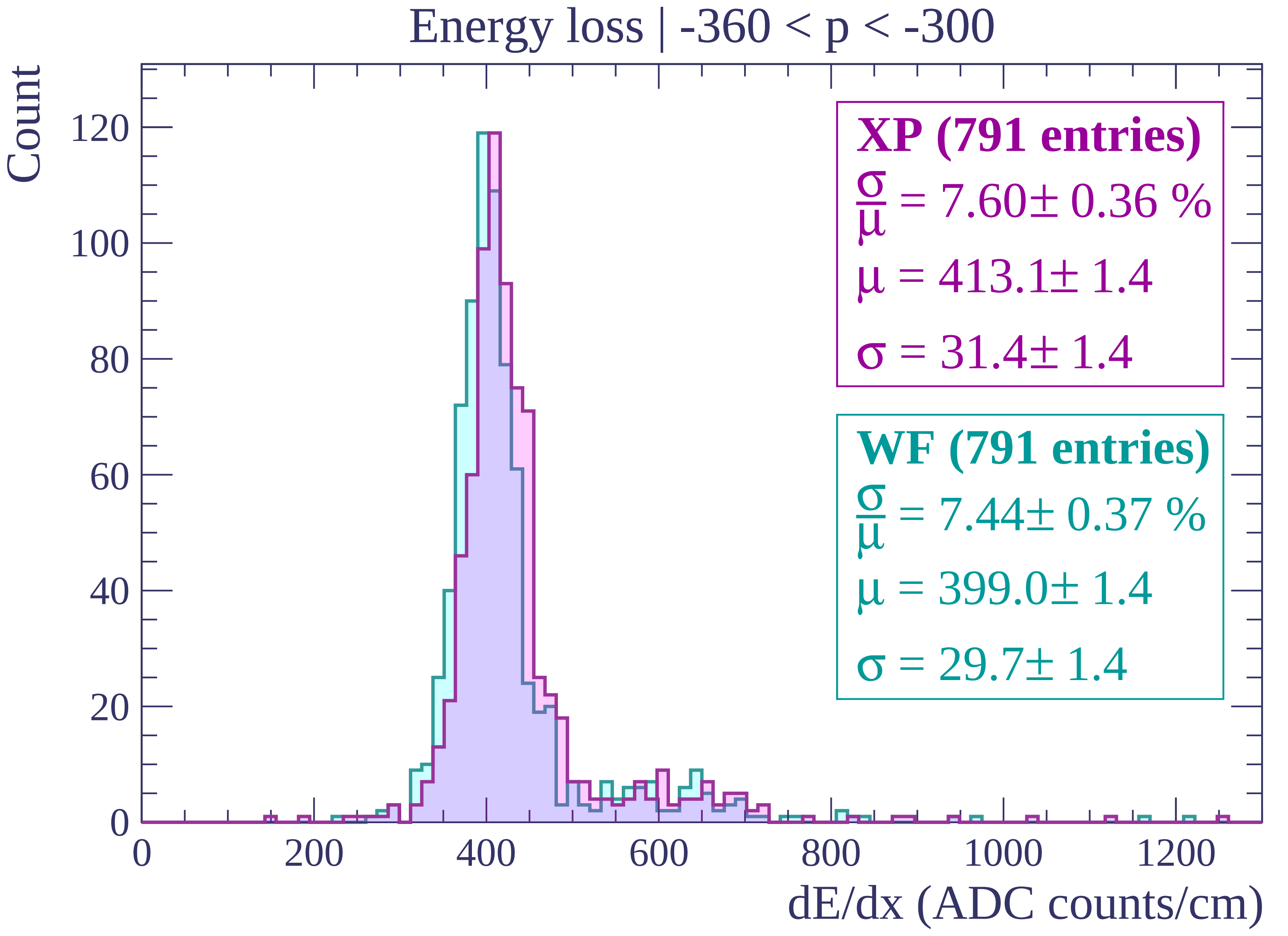


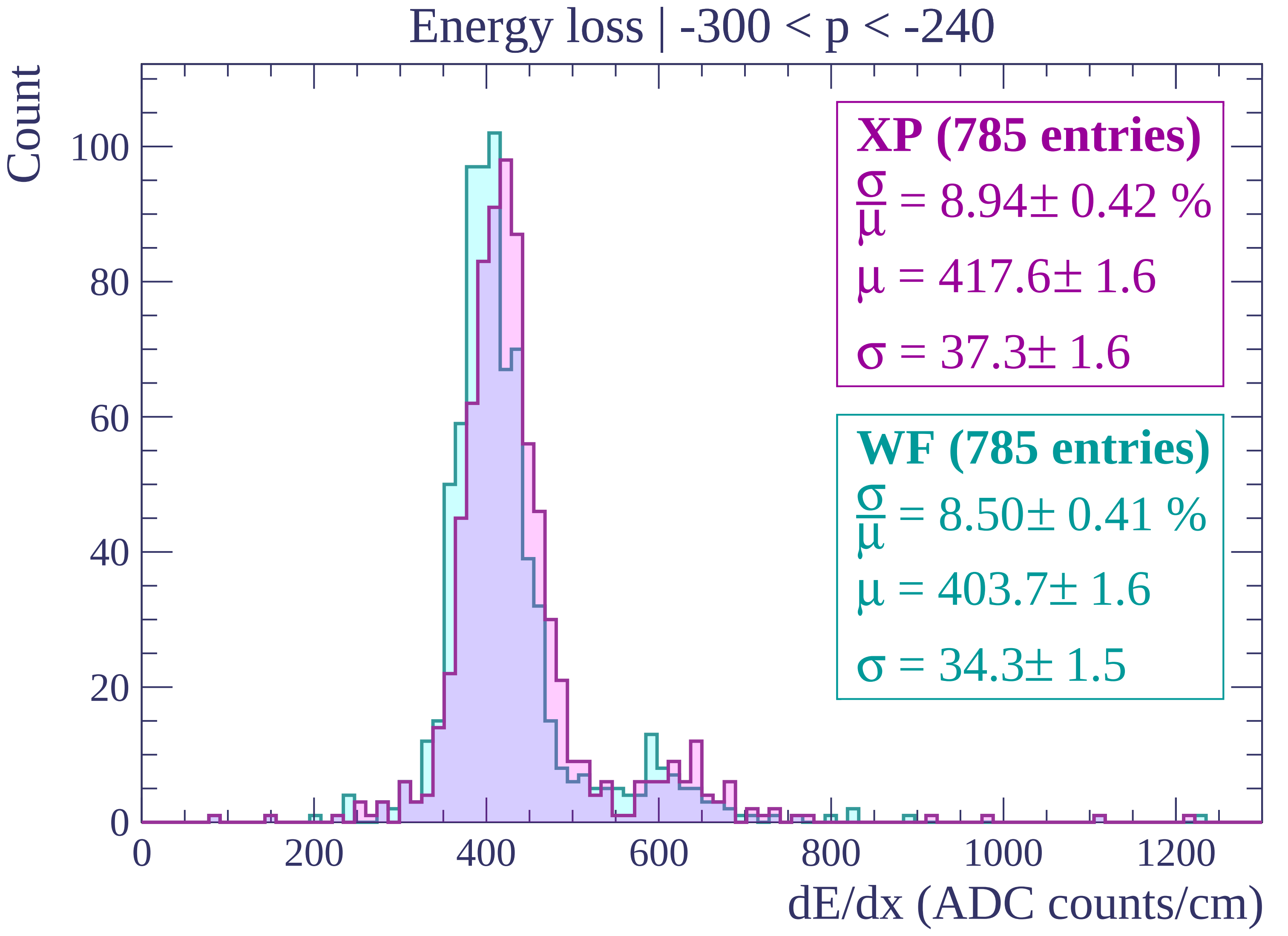
Energy loss | -540 < p < -480
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (965 entries)
σ/μ = 8.32± 111.94 %
μ = 452.3± 942.8
σ = 37.6± 427.8
WF (965 entries)
σ/μ = 9.16± 0.45 %
μ = 406.1± 1.6
σ = 37.2± 1.7



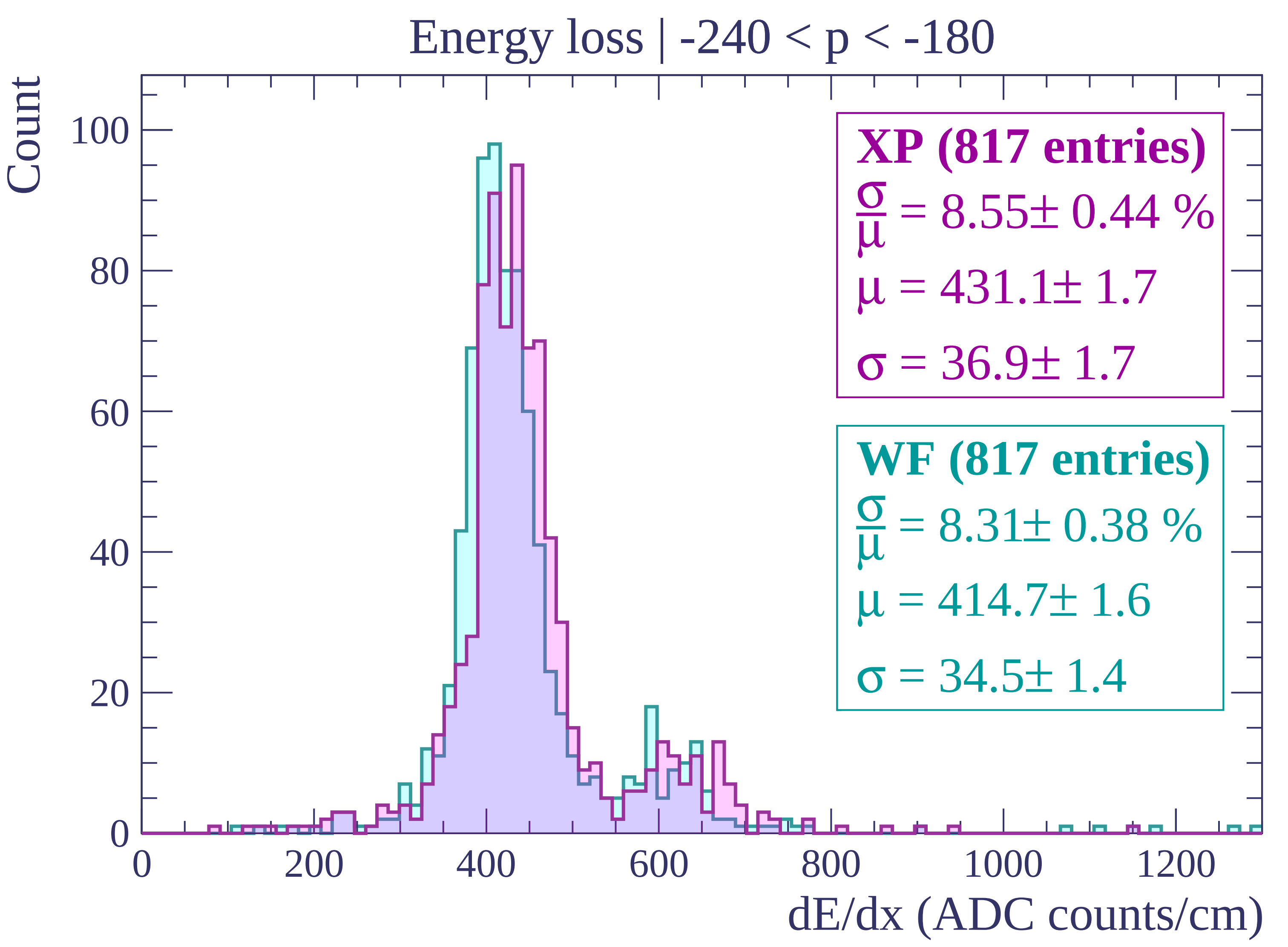
Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (921 entries)
$\frac{\sigma}{\mu}$ = 8.35± 0.44 %
μ = 419.3± 1.6
σ = 35.0± 1.7
WF (921 entries)
$\frac{\sigma}{\mu}$ = 8.28± 0.40 %
μ = 405.8± 1.5
σ = 33.6± 1.5

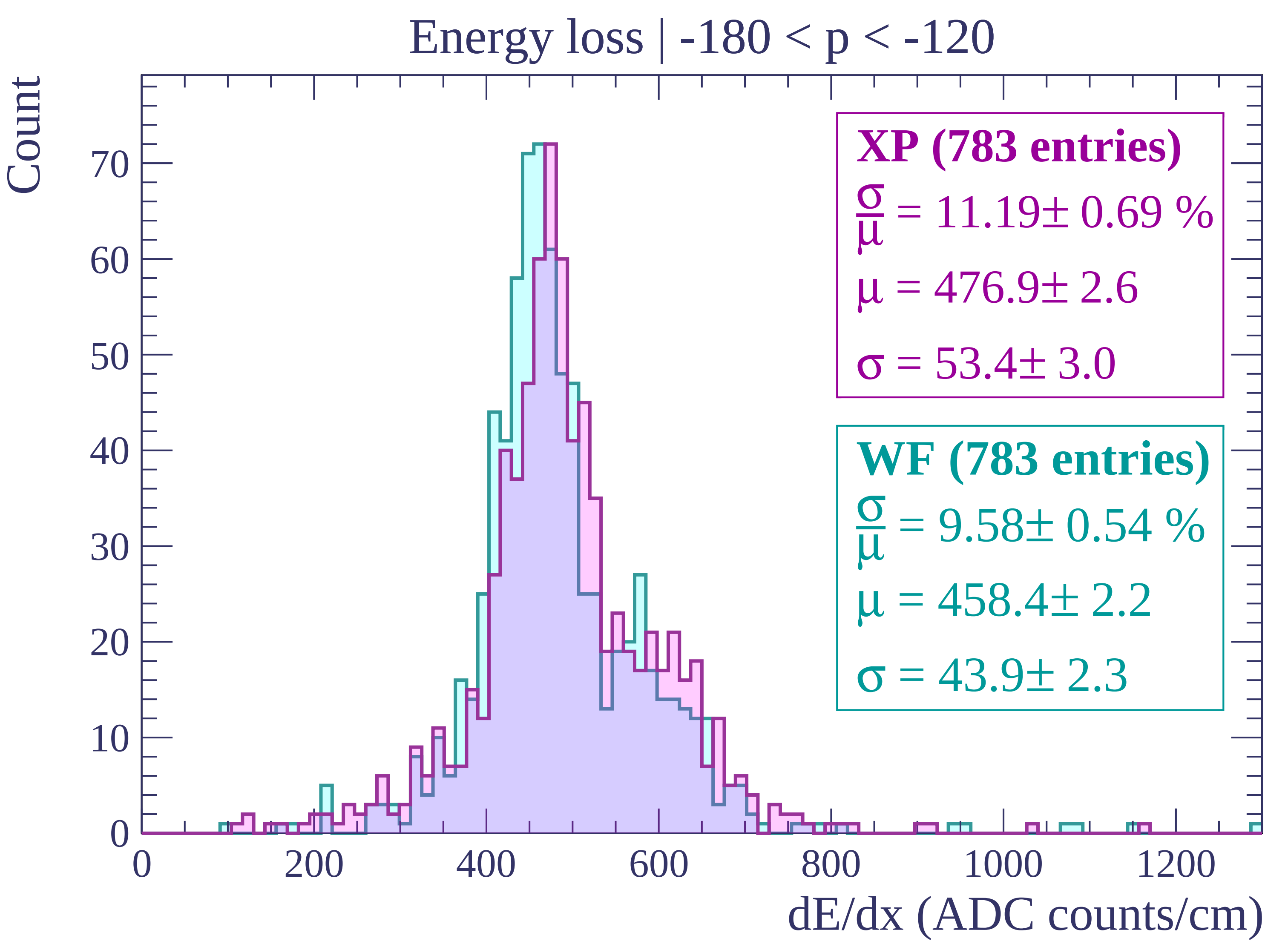

Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (859 entries)
σ/μ = 8.35± 0.45 %
μ = 414.7± 1.6
σ = 34.6± 1.7
WF (859 entries)
σ/μ = 7.60± 0.38 %
μ = 399.3± 1.4
σ = 30.3± 1.4


Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (791 entries)
σ/μ = 7.60± 0.36 %
μ = 413.1± 1.4
σ = 31.4± 1.4
WF (791 entries)
σ/μ = 7.44± 0.37 %
μ = 399.0± 1.4
σ = 29.7± 1.4

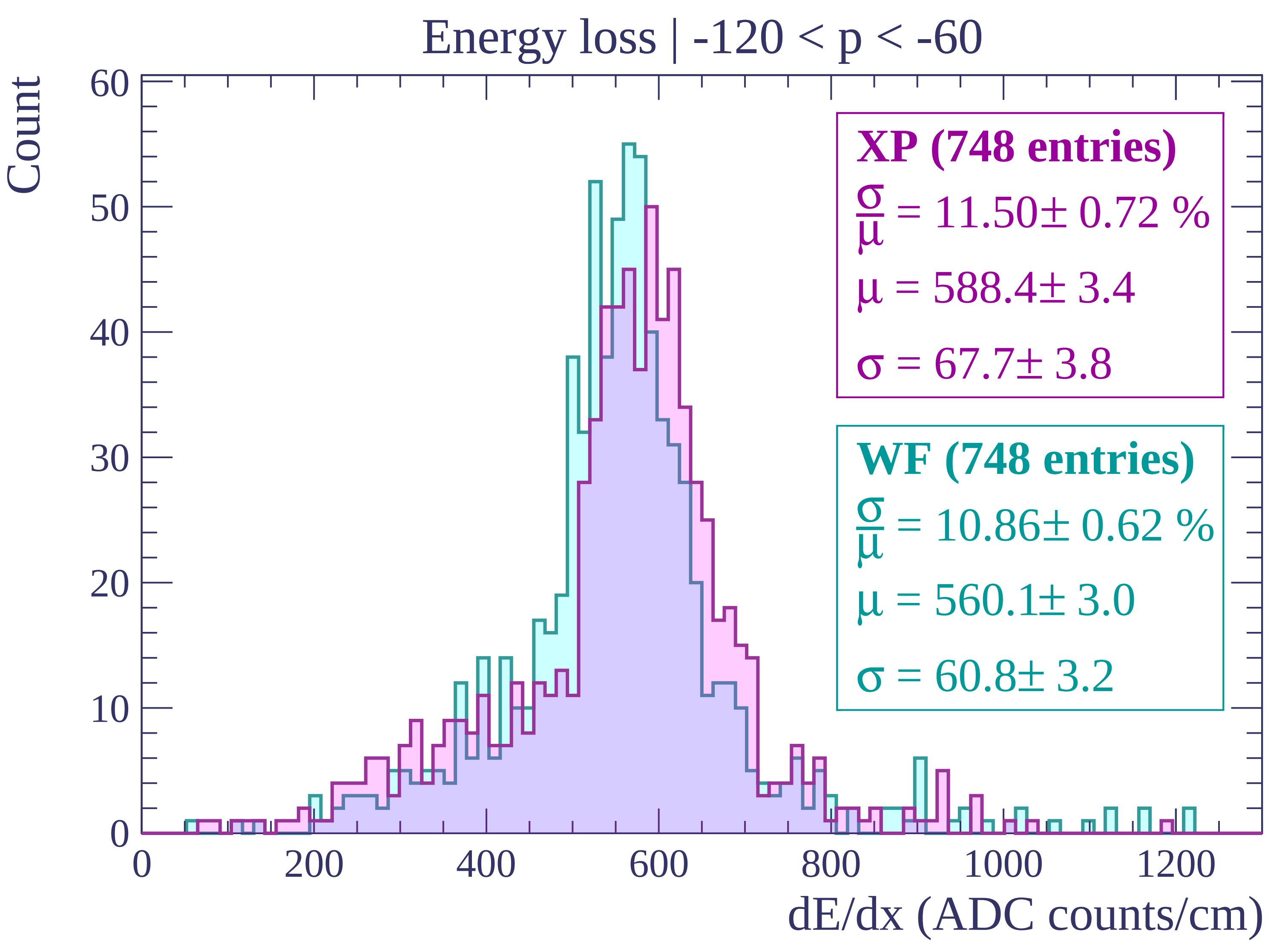


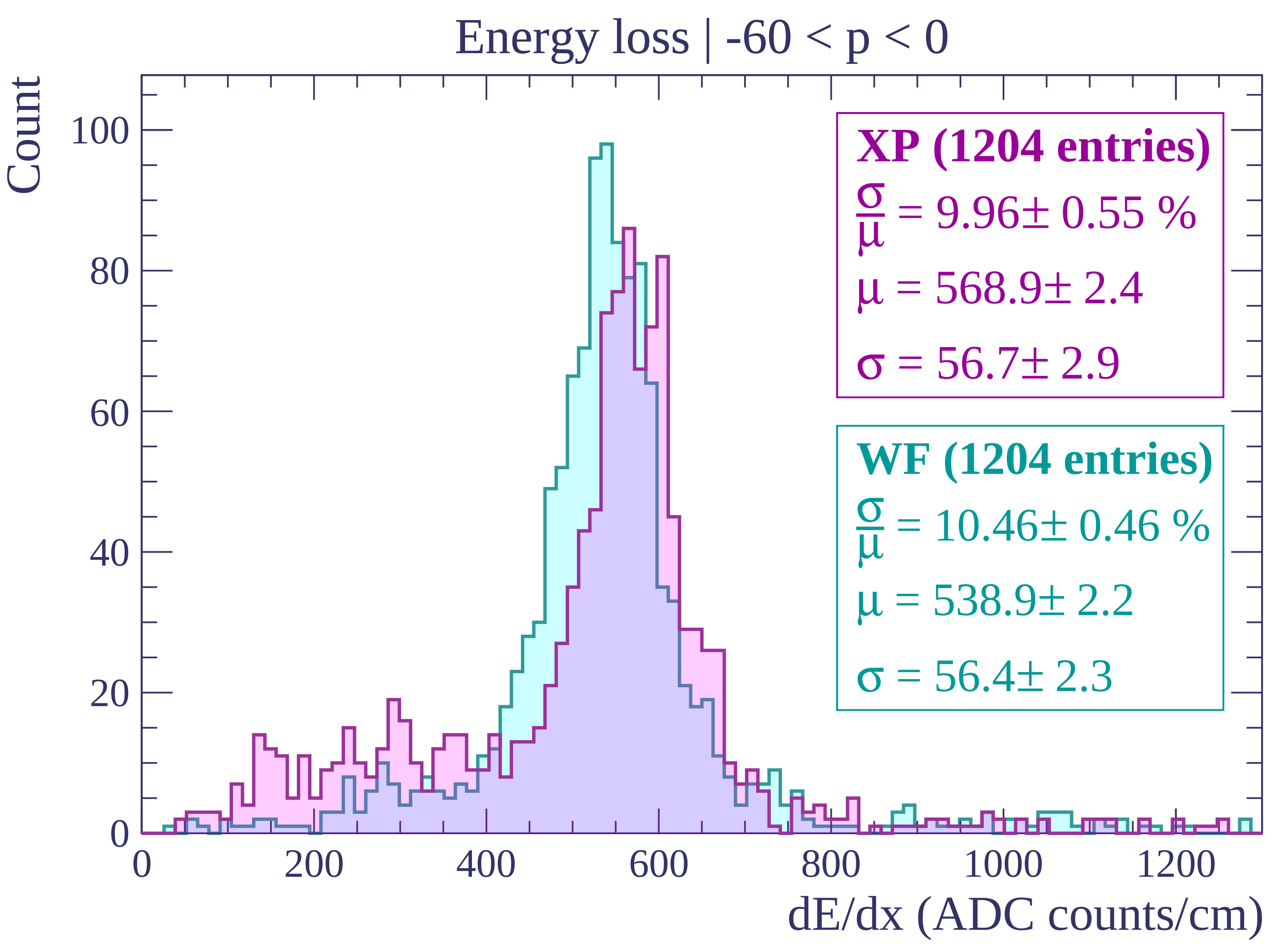
Energy loss | -300 < p < -240
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (785 entries)
σ/μ = 8.94± 0.42 %
μ = 417.6± 1.6
σ = 37.3± 1.6
WF (785 entries)
σ/μ = 8.50± 0.41 %
μ = 403.7± 1.6
σ = 34.3± 1.5


Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (817 entries)
σ/μ = 8.55± 0.44 %
μ = 431.1± 1.7
σ = 36.9± 1.7
WF (817 entries)
σ/μ = 8.31± 0.38 %
μ = 414.7± 1.6
σ = 34.5± 1.4

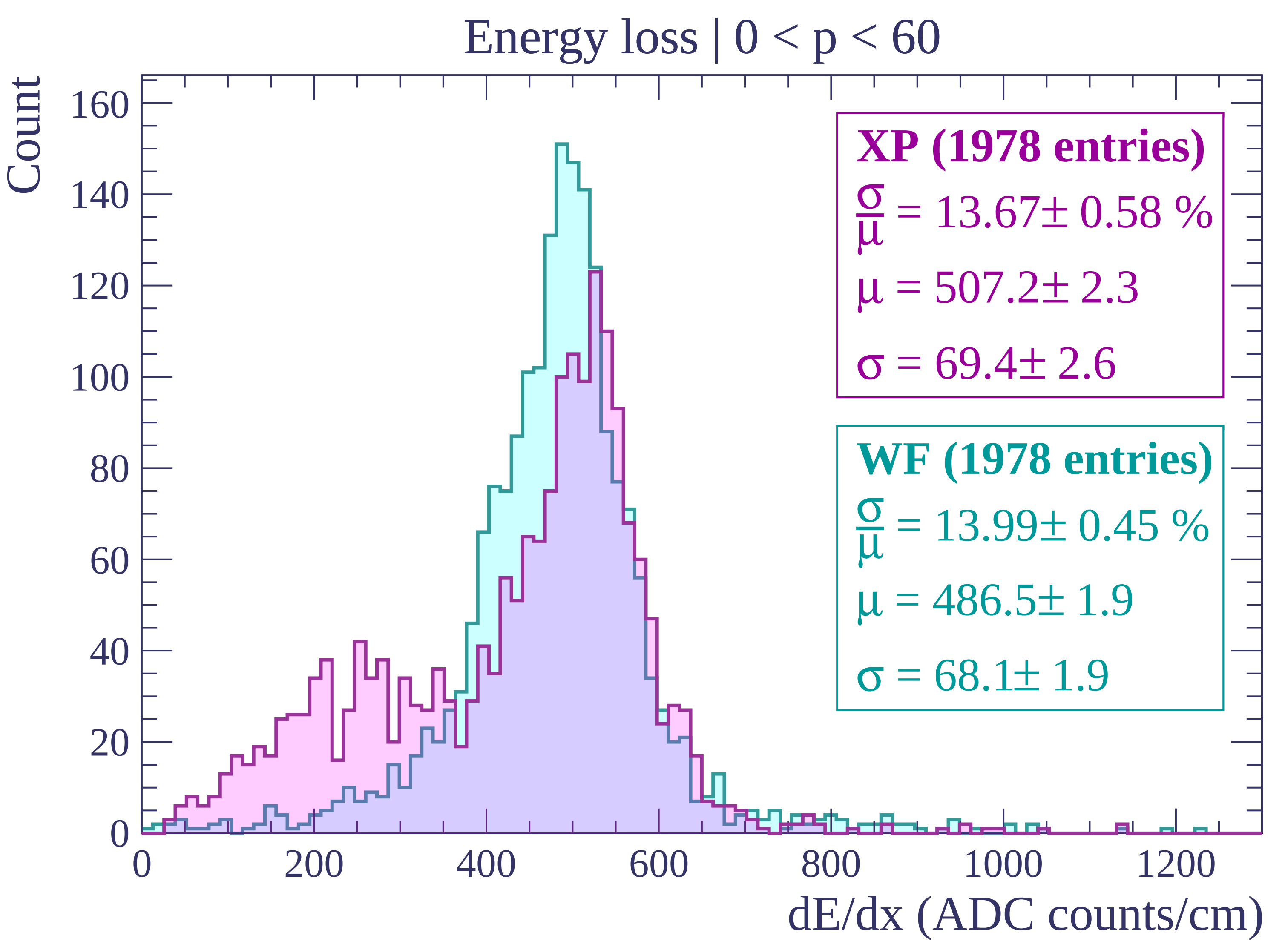


Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (783 entries)
σ/μ = 11.19± 0.69 %
μ = 476.9± 2.6
σ = 53.4± 3.0
WF (783 entries)
σ/μ = 9.58± 0.54 %
μ = 458.4± 2.2
σ = 43.9± 2.3

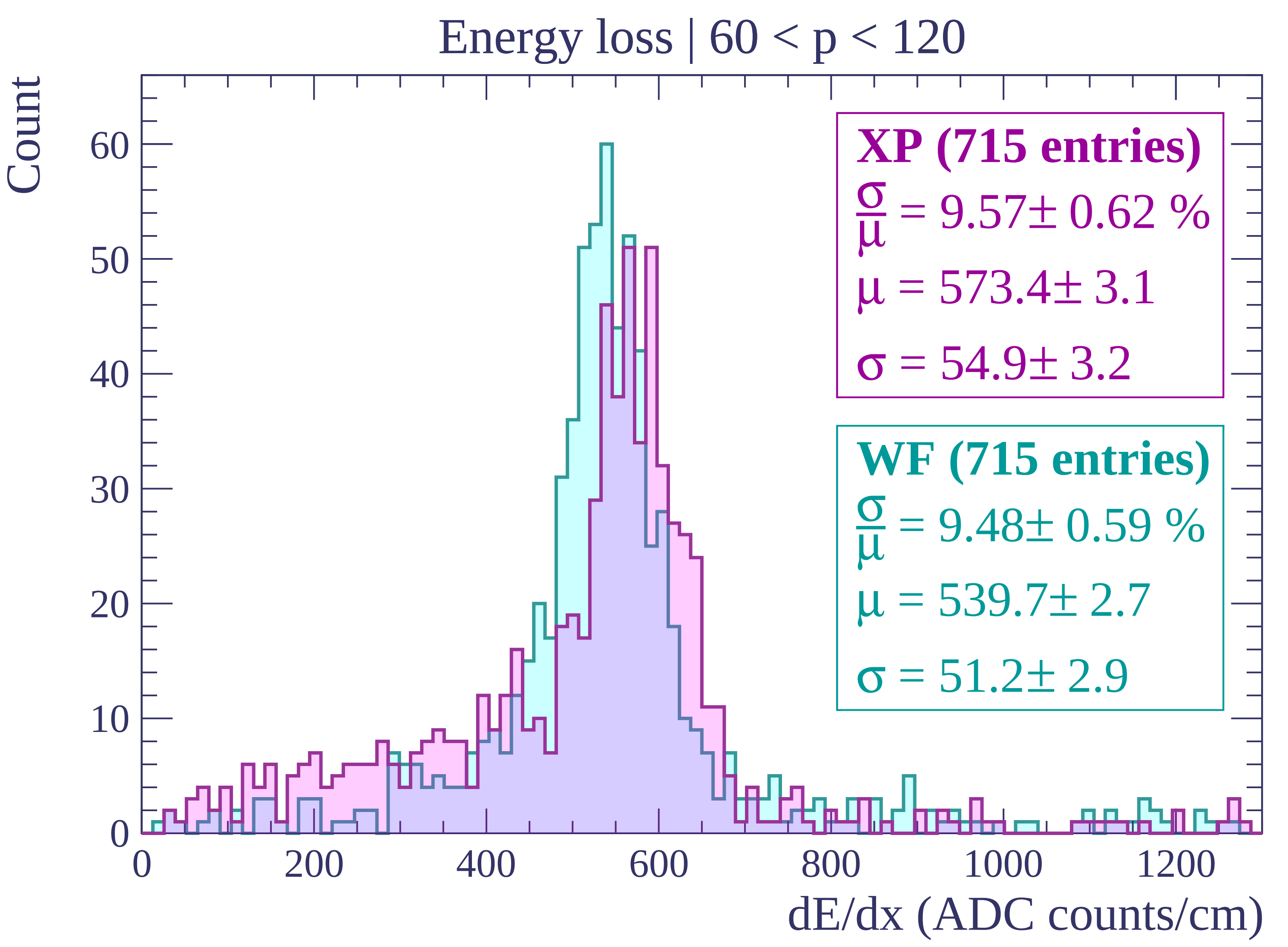

Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
XP (748 entries)
$\frac{\sigma}{\mu}$ = 11.50± 0.72 %
μ = 588.4± 3.4
σ = 67.7± 3.8
WF (748 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.62 %
μ = 560.1± 3.0
σ = 60.8± 3.2
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200



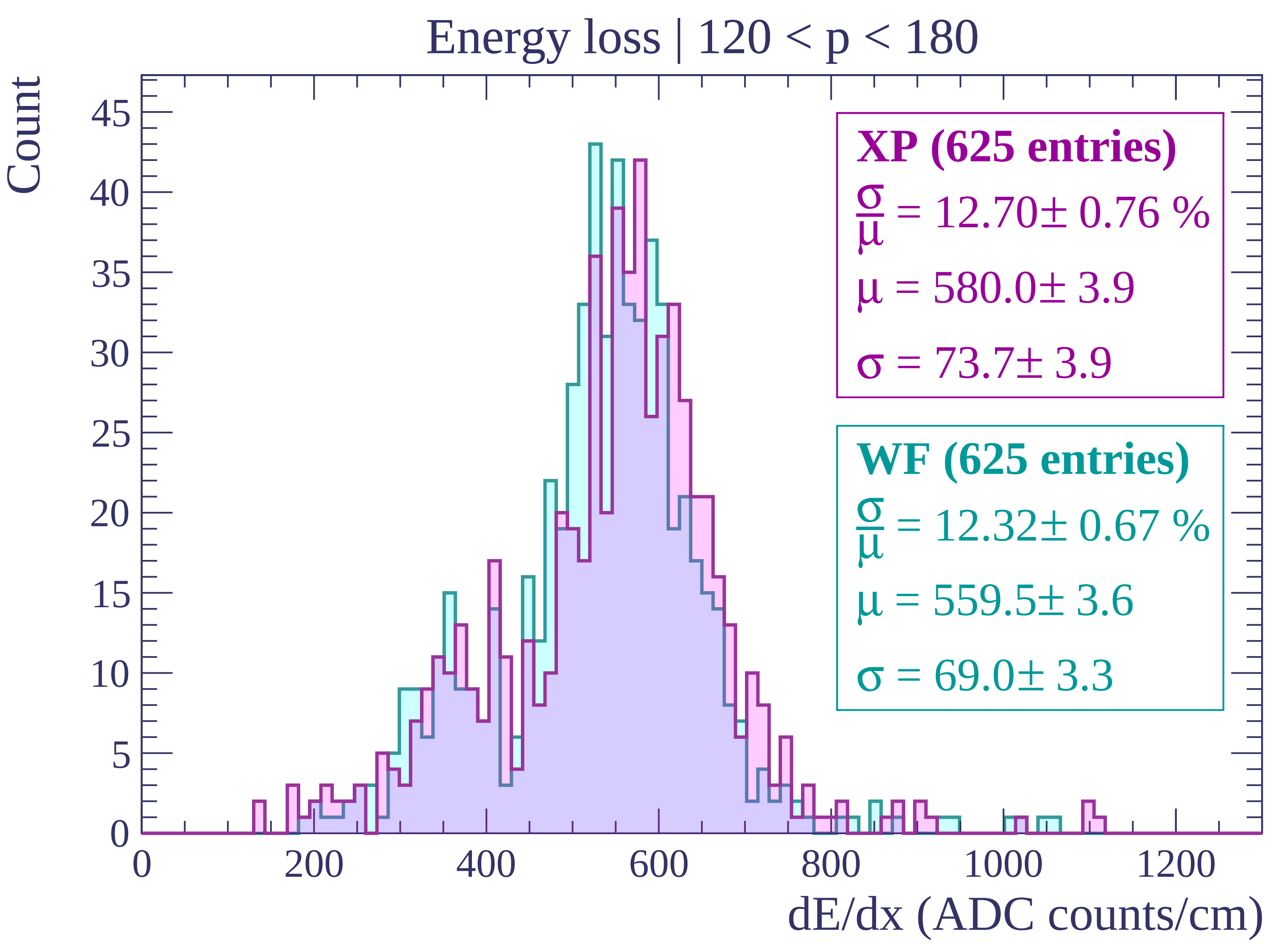
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1204 entries)
σ/μ = 9.96± 0.55 %
μ = 568.9± 2.4
σ = 56.7± 2.9
WF (1204 entries)
σ/μ = 10.46± 0.46 %
μ = 538.9± 2.2
σ = 56.4± 2.3



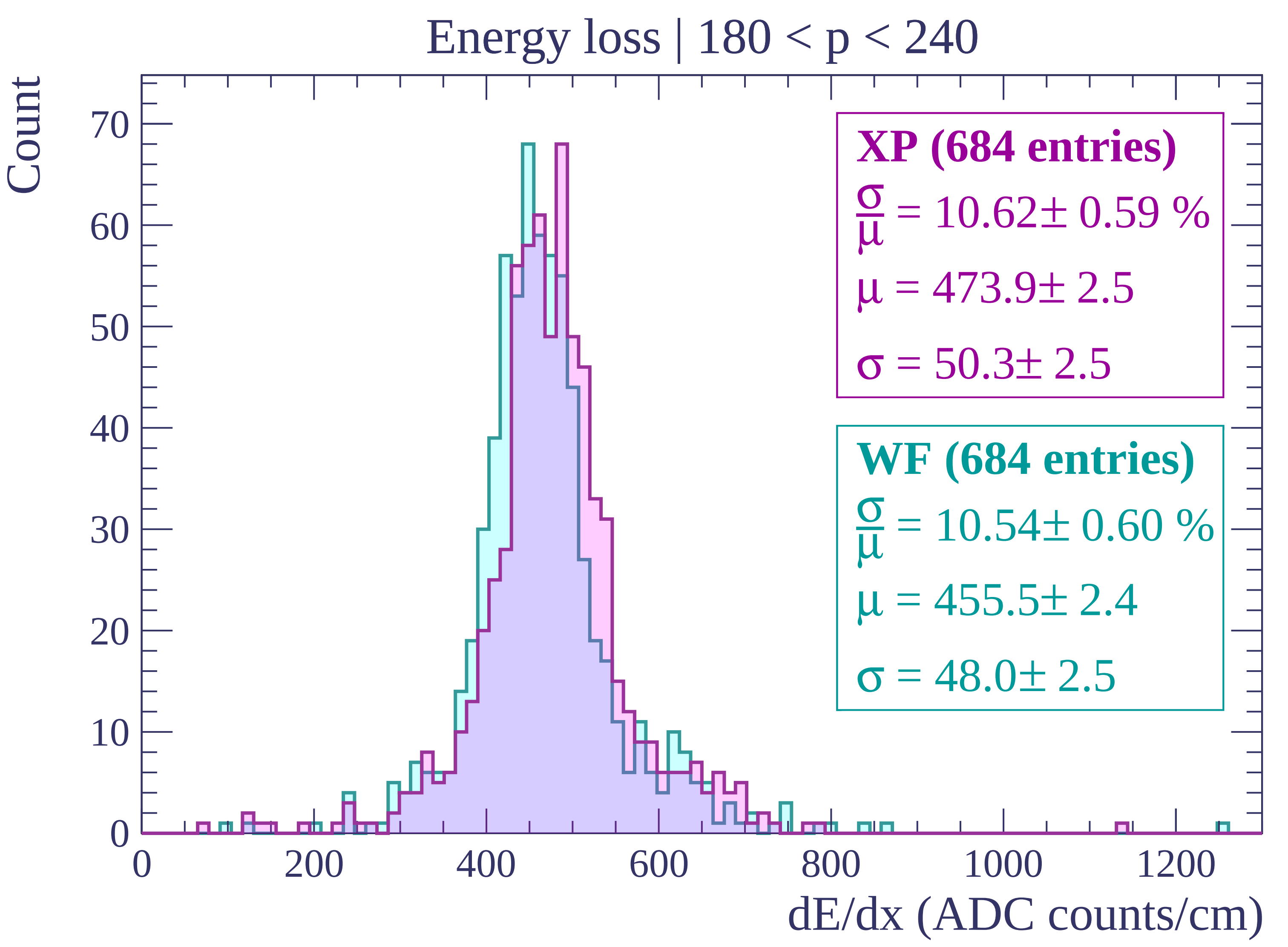
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1978 entries)
σ/μ = 13.67± 0.58 %
μ = 507.2± 2.3
σ = 69.4± 2.6
WF (1978 entries)
σ/μ = 13.99± 0.45 %
μ = 486.5± 1.9
σ = 68.1± 1.9

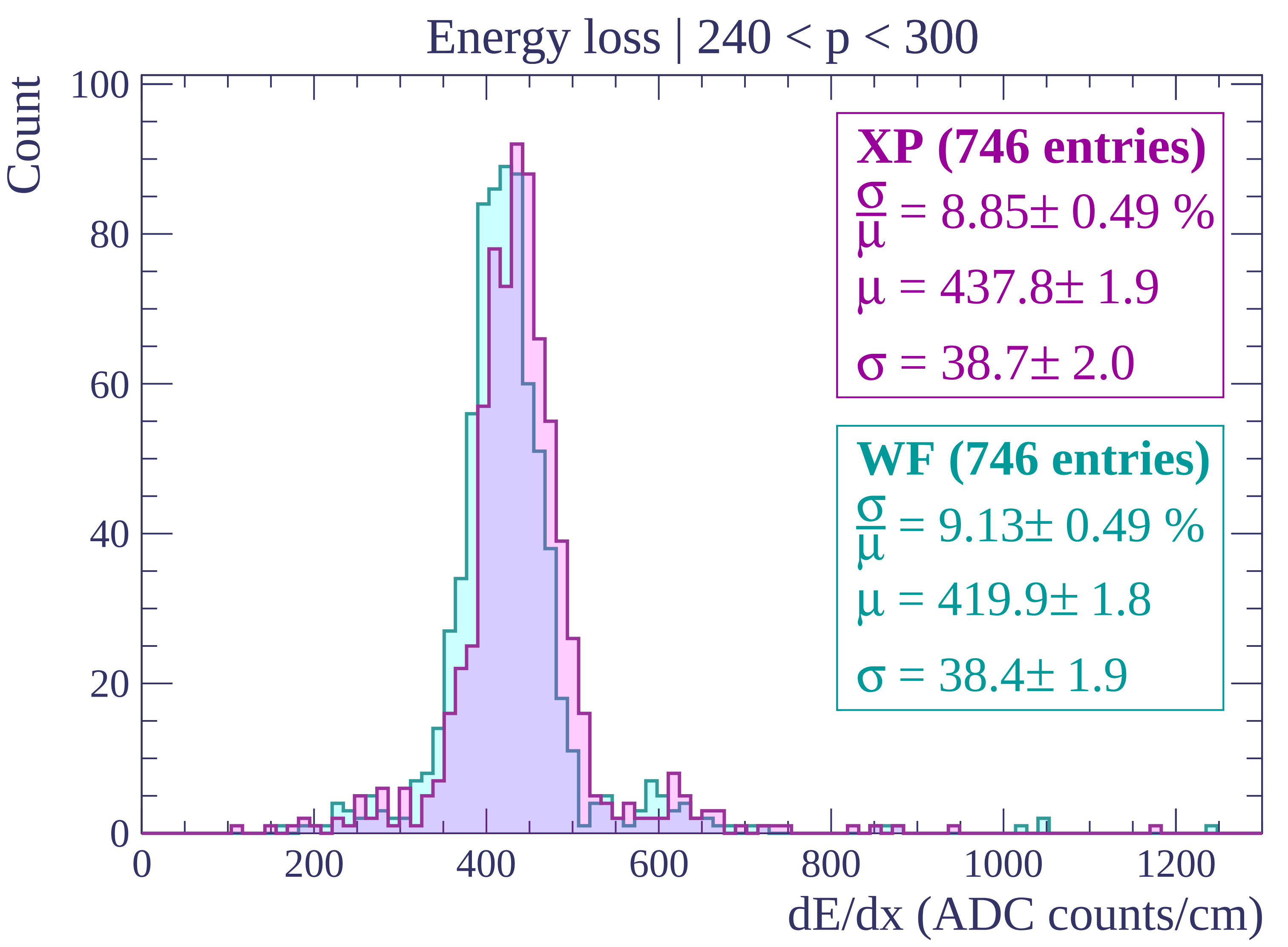


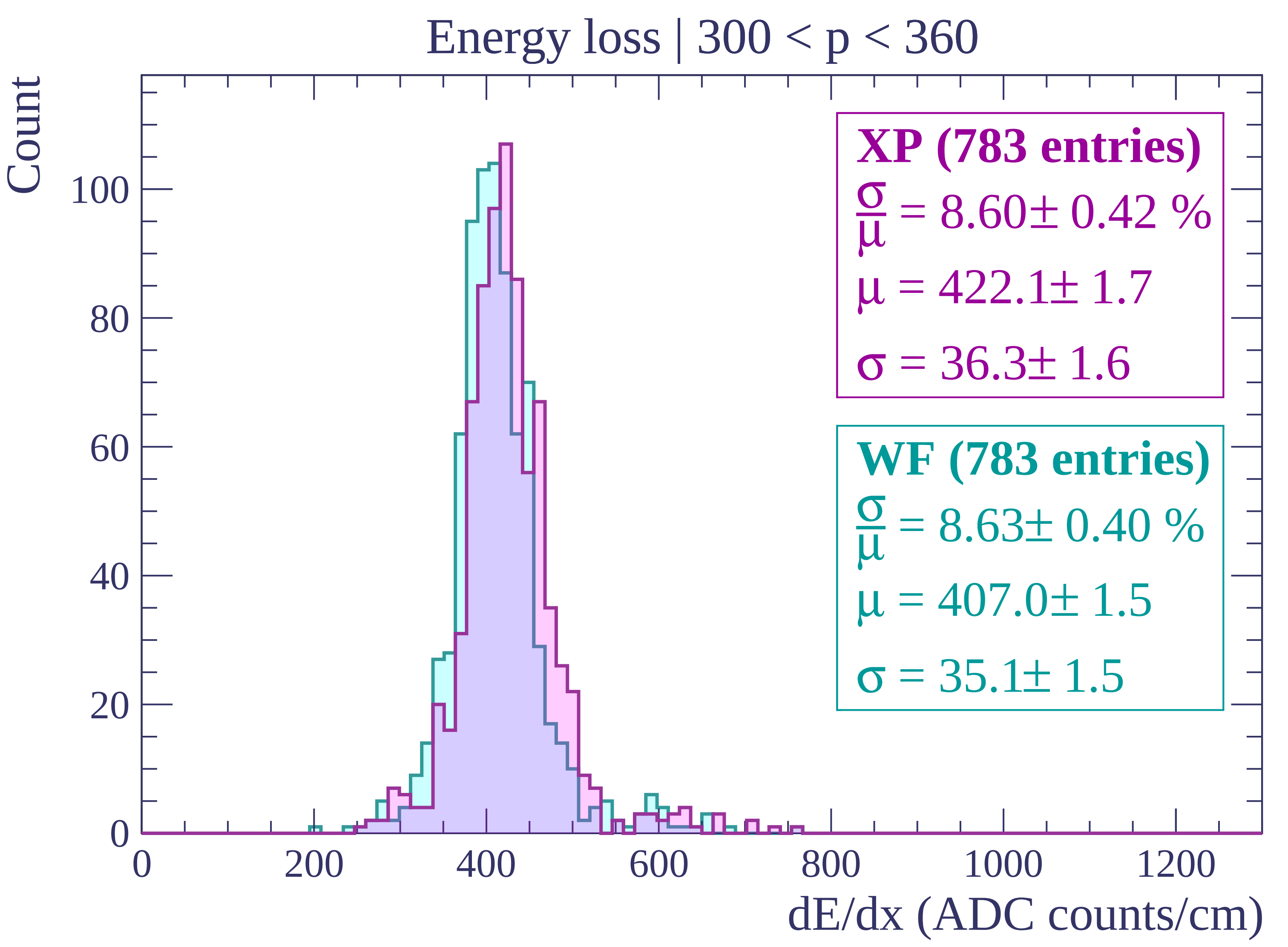
Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (715 entries)
$\frac{\sigma}{\mu}$ = 9.57± 0.62 %
μ = 573.4± 3.1
σ = 54.9± 3.2
WF (715 entries)
$\frac{\sigma}{\mu}$ = 9.48± 0.59 %
μ = 539.7± 2.7
σ = 51.2± 2.9



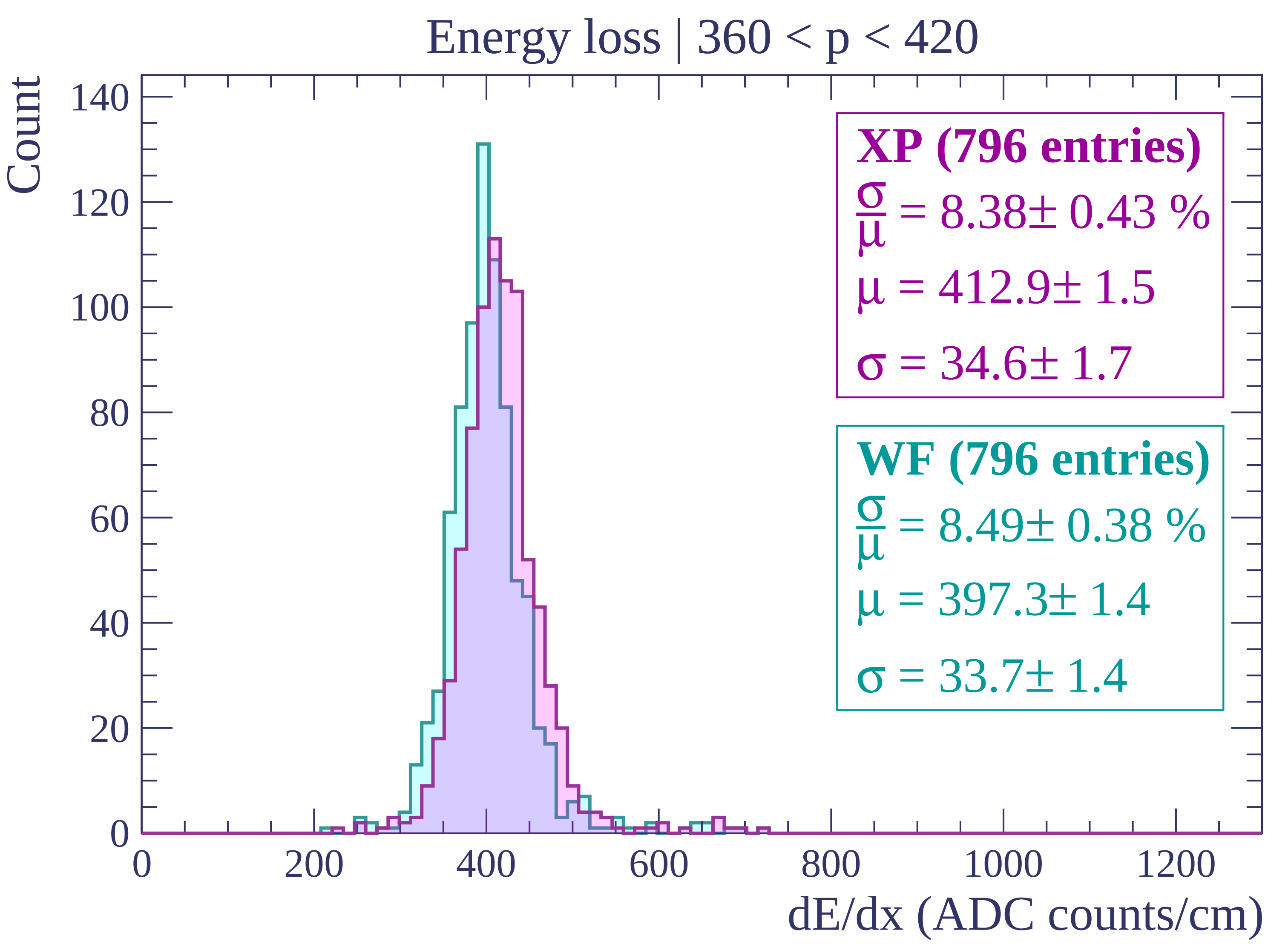
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (625 entries)
σ/μ = 12.70± 0.76 %
μ = 580.0± 3.9
σ = 73.7± 3.9
WF (625 entries)
σ/μ = 12.32± 0.67 %
μ = 559.5± 3.6
σ = 69.0± 3.3


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
XP (684 entries)
$\frac{\sigma}{\mu}$ = 10.62± 0.59 %
μ = 473.9± 2.5
σ = 50.3± 2.5
WF (684 entries)
$\frac{\sigma}{\mu}$ = 10.54± 0.60 %
μ = 455.5± 2.4
σ = 48.0± 2.5

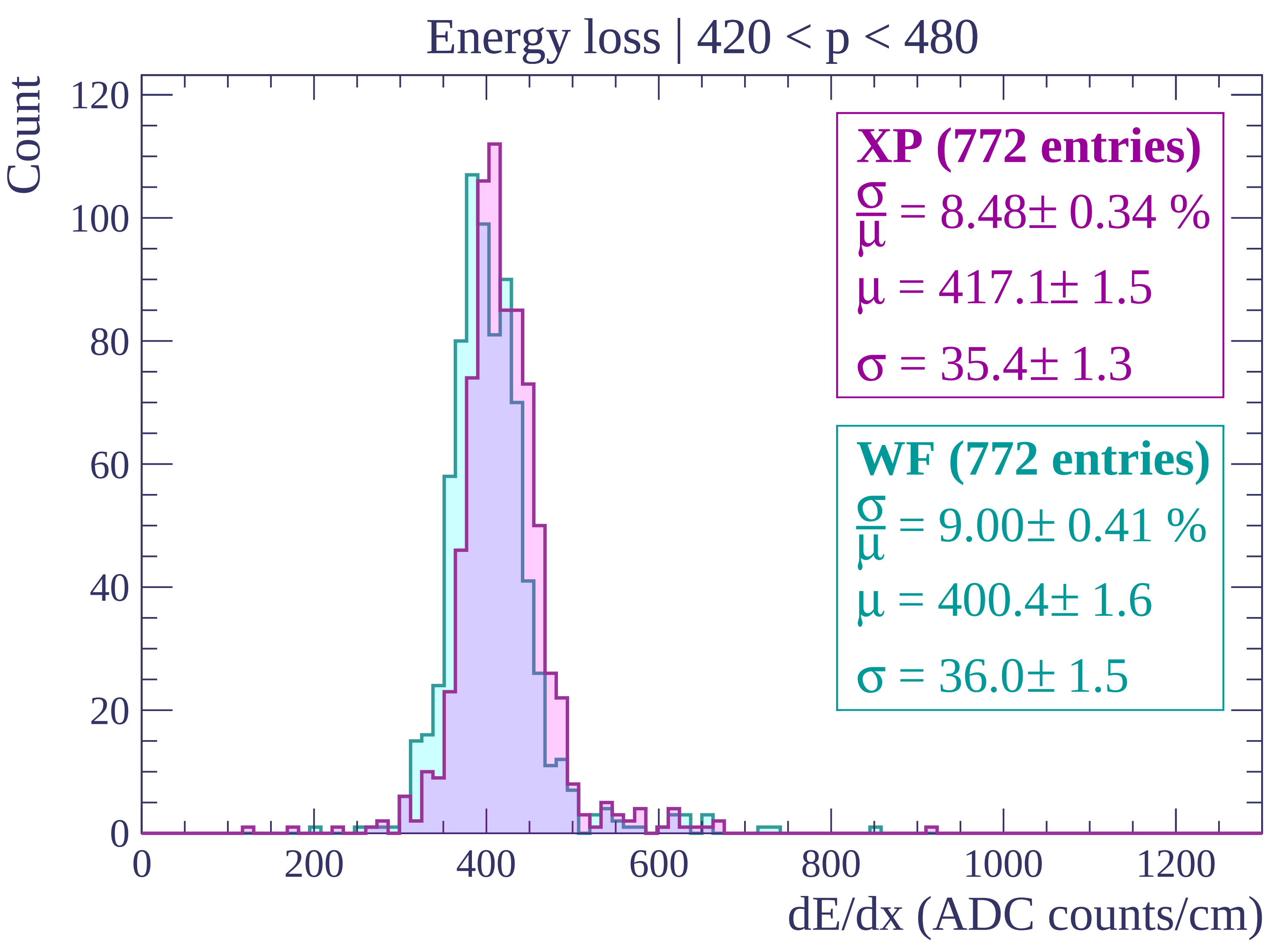

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
XP (746 entries)
σ/μ = 8.85± 0.49 %
μ = 437.8± 1.9
σ = 38.7± 2.0
WF (746 entries)
σ/μ = 9.13± 0.49 %
μ = 419.9± 1.8
σ = 38.4± 1.9
0
20
40
60
80
100
0
200
400
600
800
1000
1200

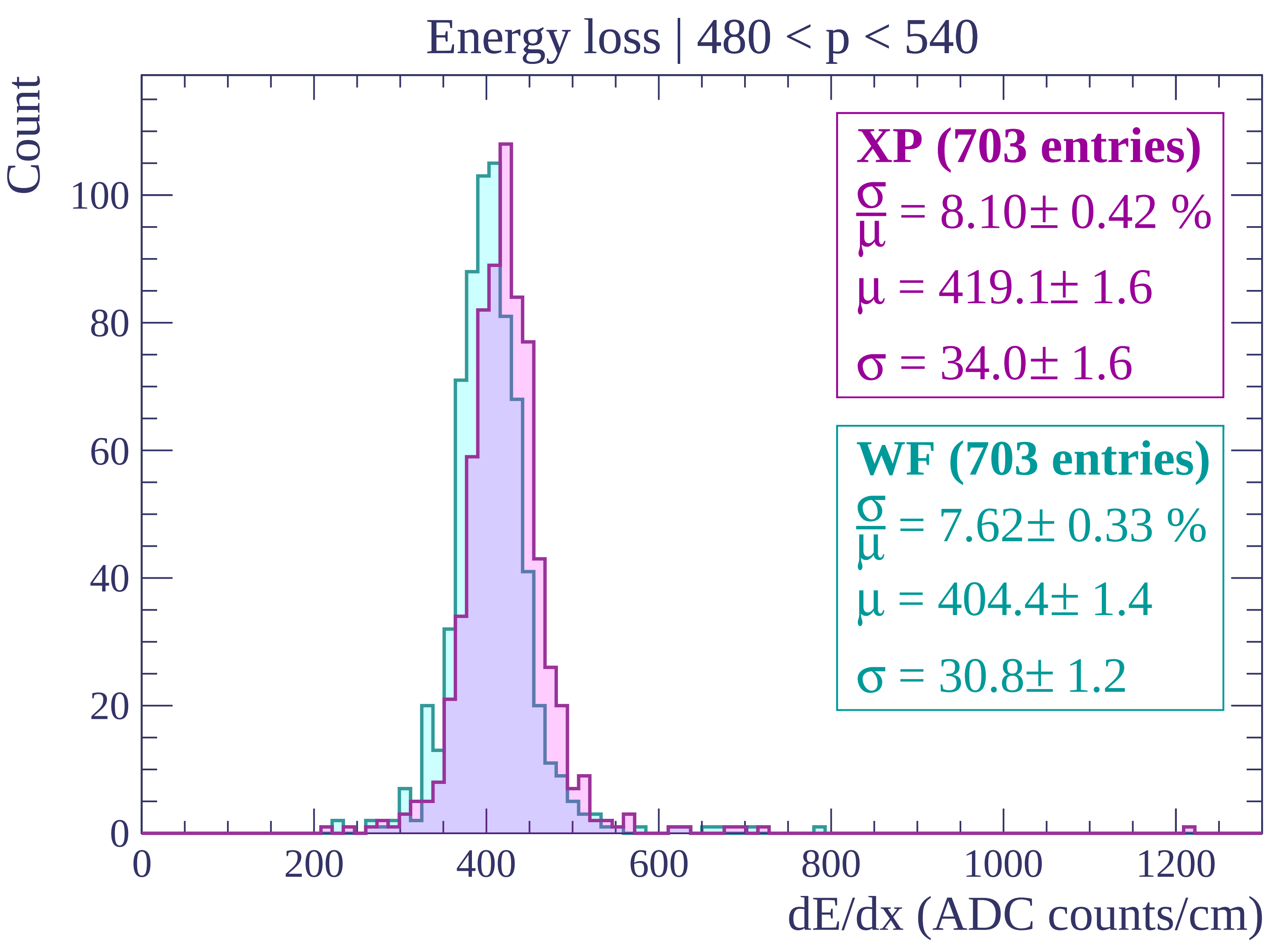


Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (783 entries)
σ/μ = 8.60± 0.42 %
μ = 422.1± 1.7
σ = 36.3± 1.6
WF (783 entries)
σ/μ = 8.63± 0.40 %
μ = 407.0± 1.5
σ = 35.1± 1.5

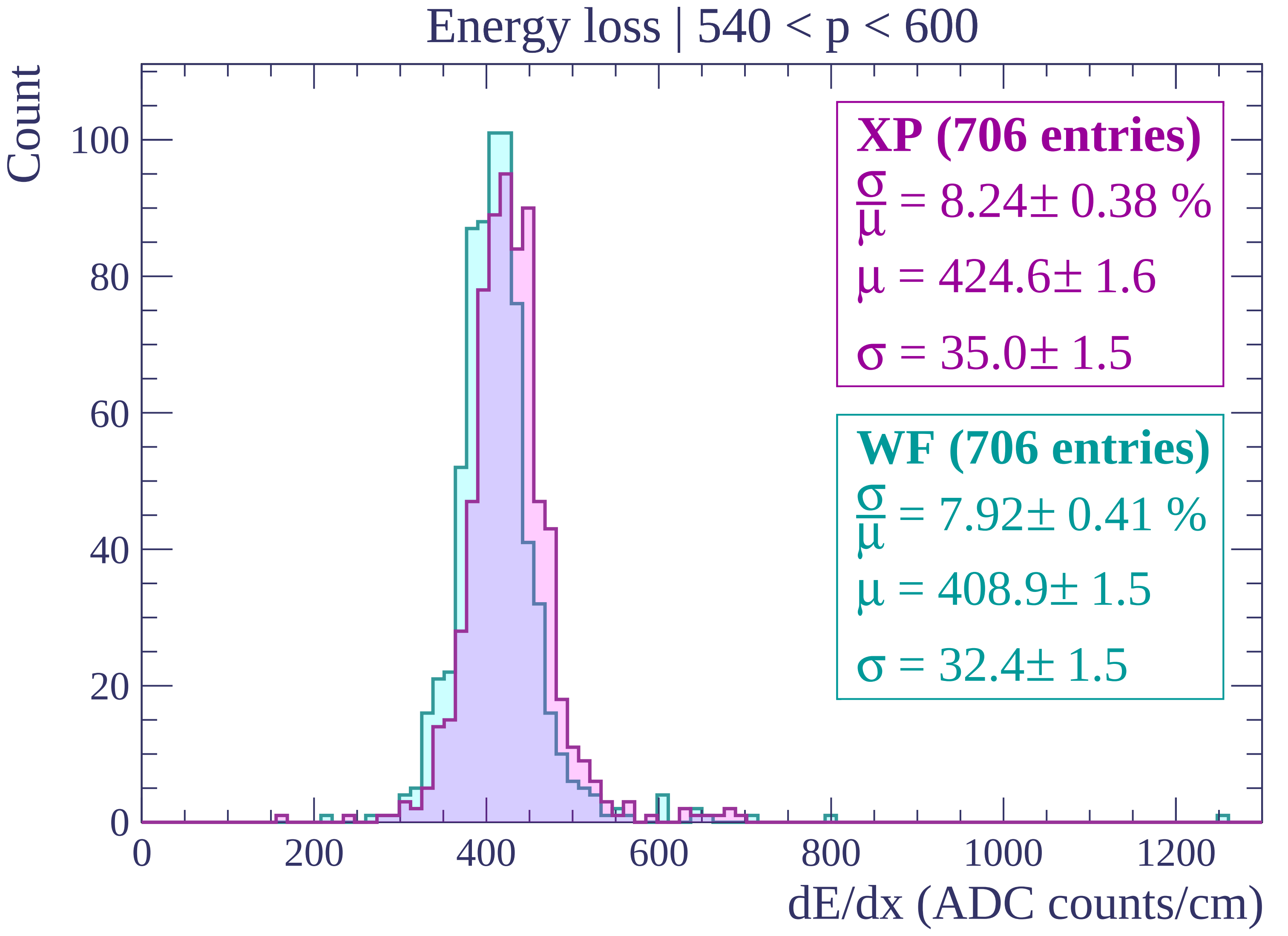


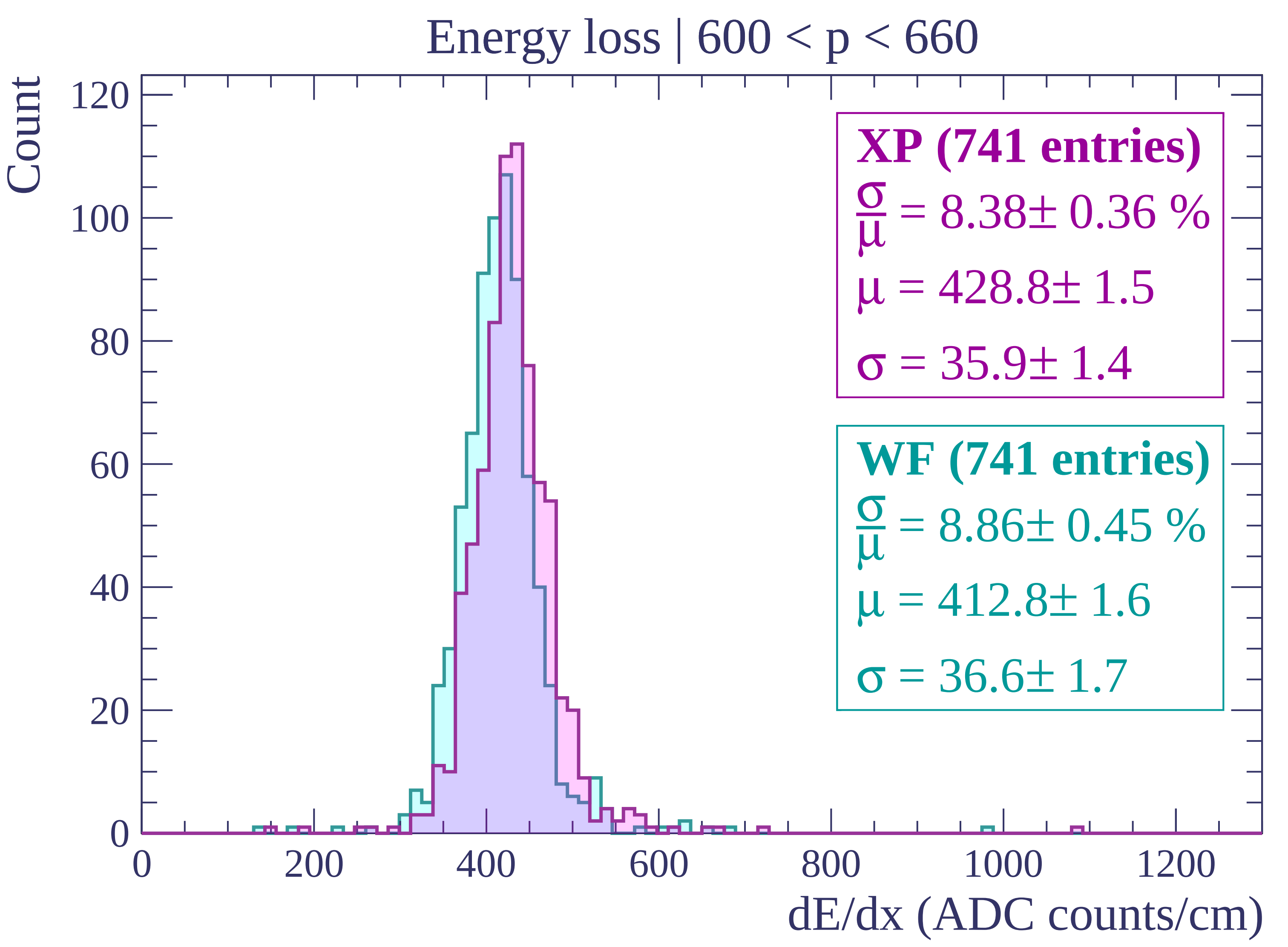
Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
XP (796 entries)
σ/μ = 8.38± 0.43 %
μ = 412.9± 1.5
σ = 34.6± 1.7
WF (796 entries)
σ/μ = 8.49± 0.38 %
μ = 397.3± 1.4
σ = 33.7± 1.4


Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (772 entries)
$\frac{\sigma}{\mu}$ = 8.48± 0.34 %
μ = 417.1± 1.5
σ = 35.4± 1.3
WF (772 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.41 %
μ = 400.4± 1.6
σ = 36.0± 1.5

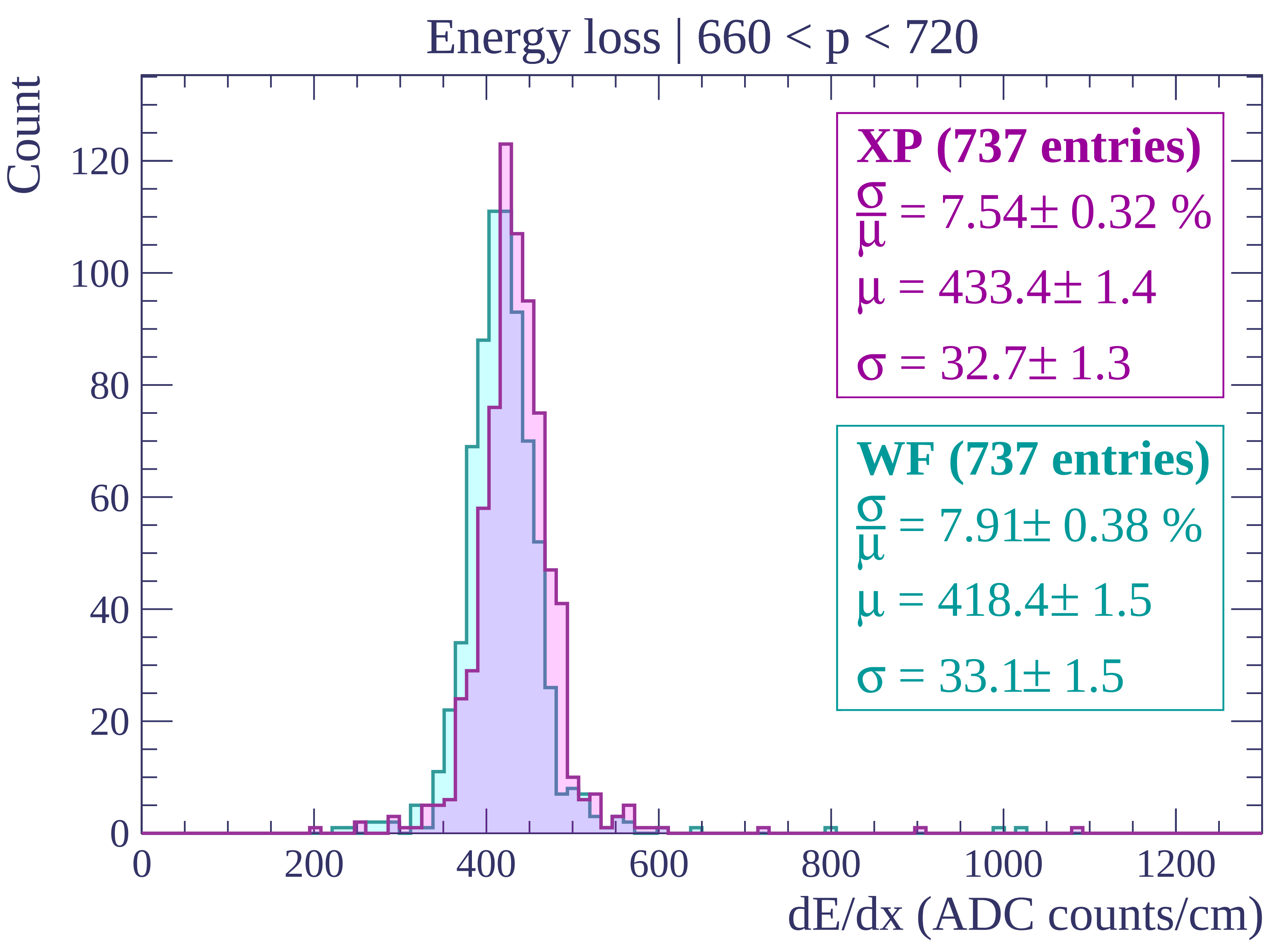


Energy loss | 480 < p < 540
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (703 entries)
$\frac{\sigma}{\mu}$ = 8.10± 0.42 %
μ = 419.1± 1.6
σ = 34.0± 1.6
WF (703 entries)
$\frac{\sigma}{\mu}$ = 7.62± 0.33 %
μ = 404.4± 1.4
σ = 30.8± 1.2

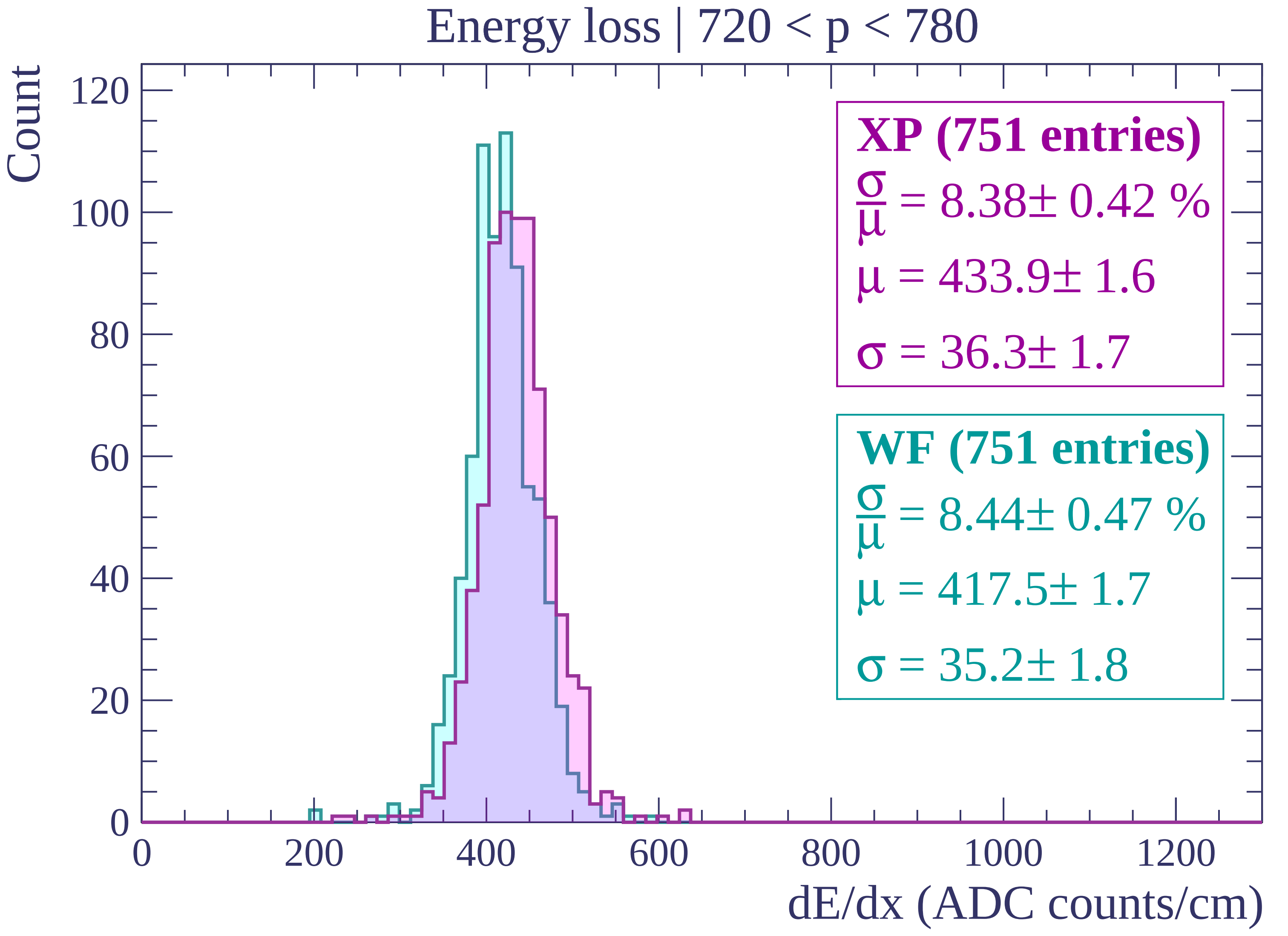

Energy loss | 540 < p < 600
Count
XP (706 entries)
σ/μ = 8.24± 0.38 %
μ = 424.6± 1.6
σ = 35.0± 1.5
WF (706 entries)
σ/μ = 7.92± 0.41 %
μ = 408.9± 1.5
σ = 32.4± 1.5
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

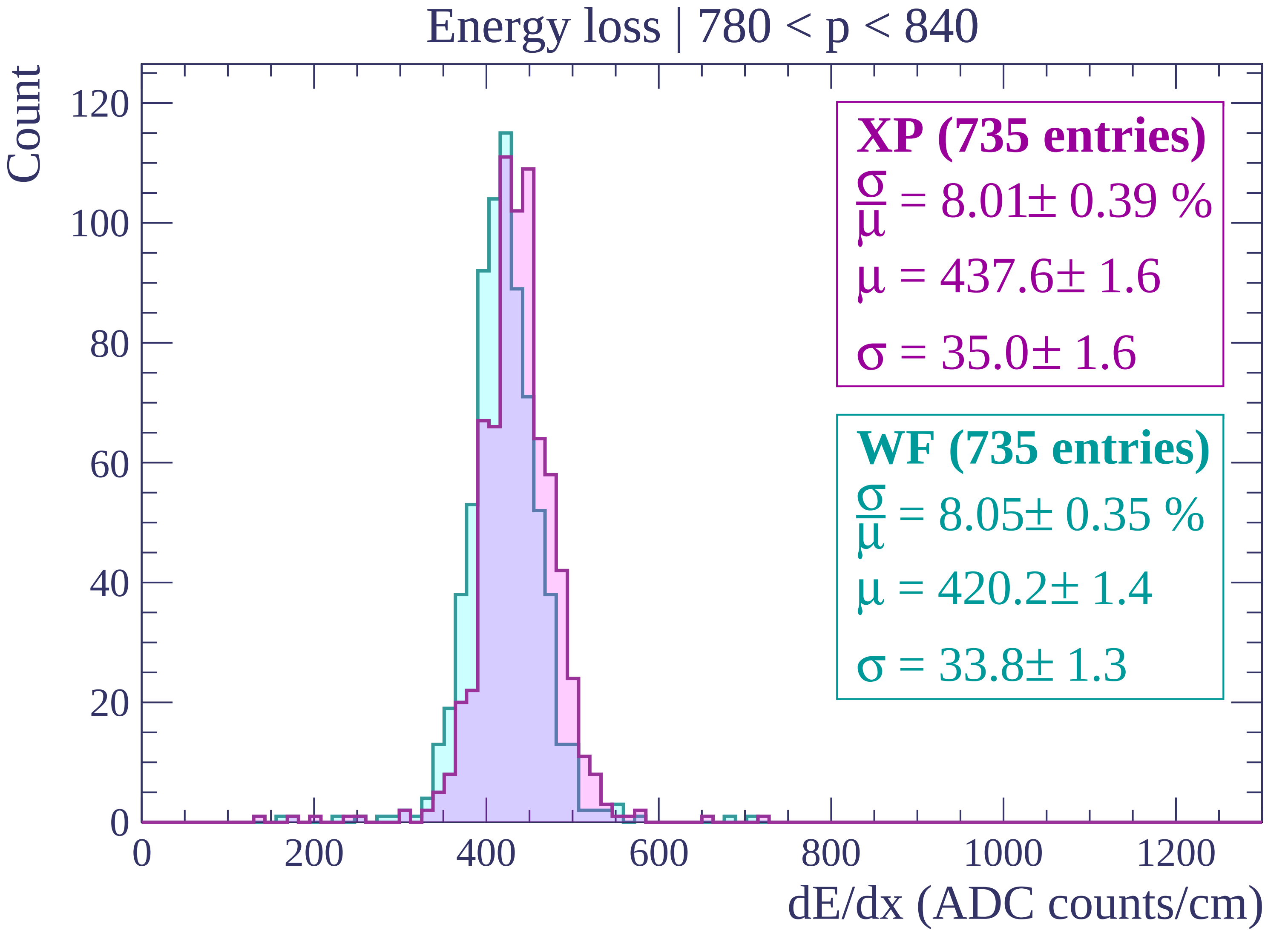


Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (741 entries)
$\frac{\sigma}{\mu}$ = 8.38± 0.36 %
μ = 428.8± 1.5
σ = 35.9± 1.4
WF (741 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.45 %
μ = 412.8± 1.6
σ = 36.6± 1.7

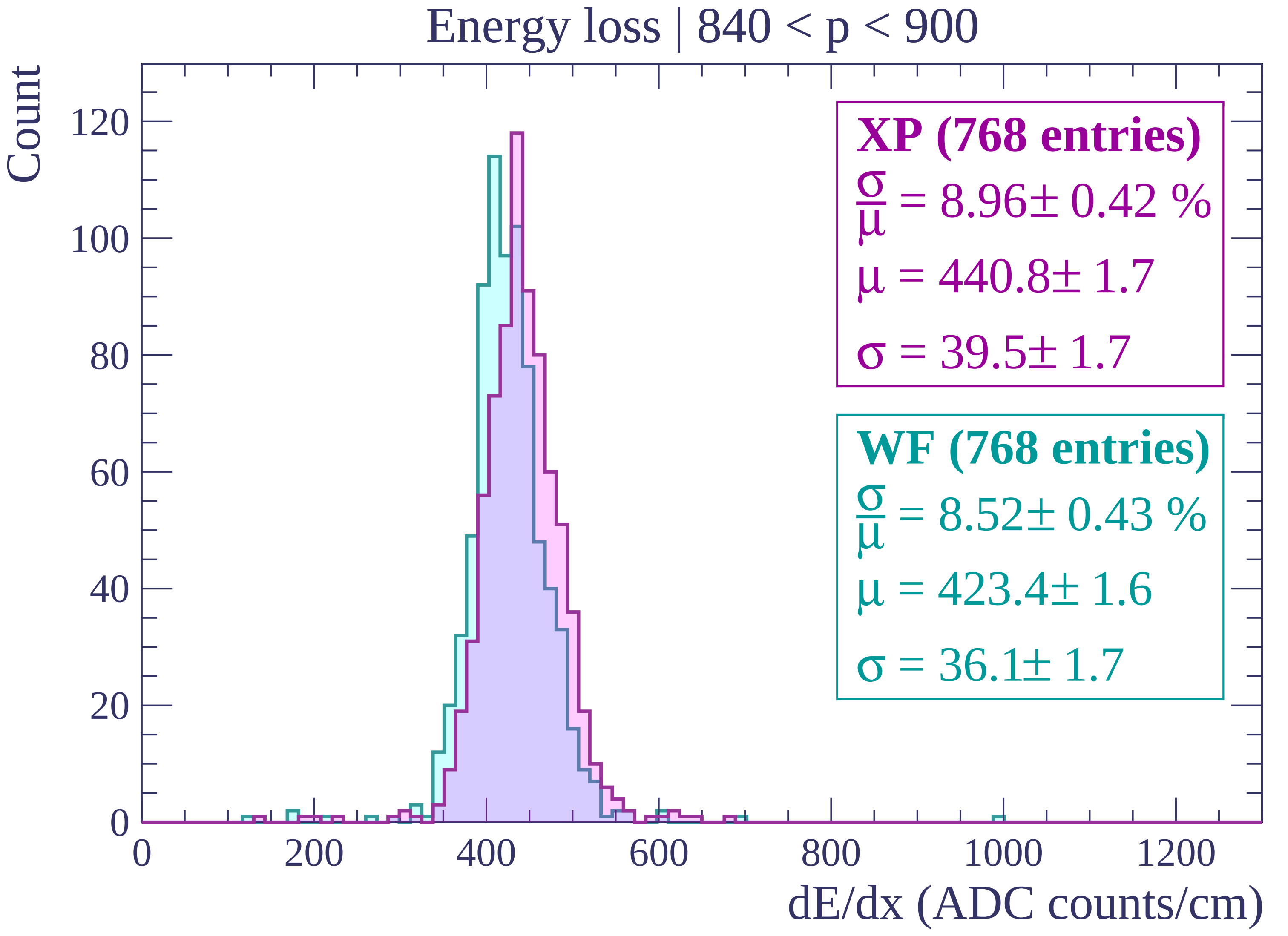


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (737 entries)
$\frac{\sigma}{\mu}$ = 7.54± 0.32 %
μ = 433.4± 1.4
σ = 32.7± 1.3
WF (737 entries)
$\frac{\sigma}{\mu}$ = 7.91± 0.38 %
μ = 418.4± 1.5
σ = 33.1± 1.5

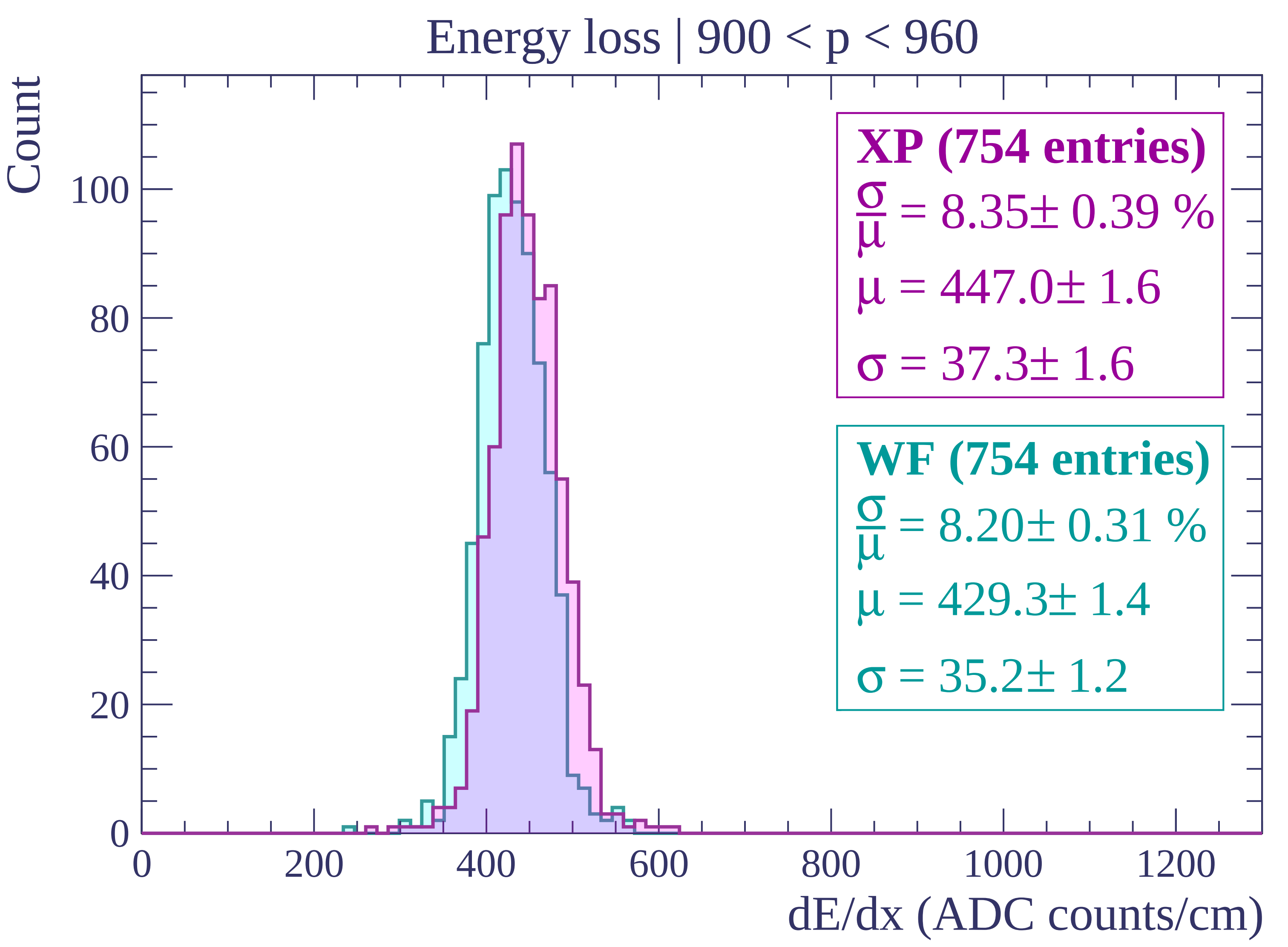

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (751 entries)
σ/μ = 8.38± 0.42 %
μ = 433.9± 1.6
σ = 36.3± 1.7
WF (751 entries)
σ/μ = 8.44± 0.47 %
μ = 417.5± 1.7
σ = 35.2± 1.8



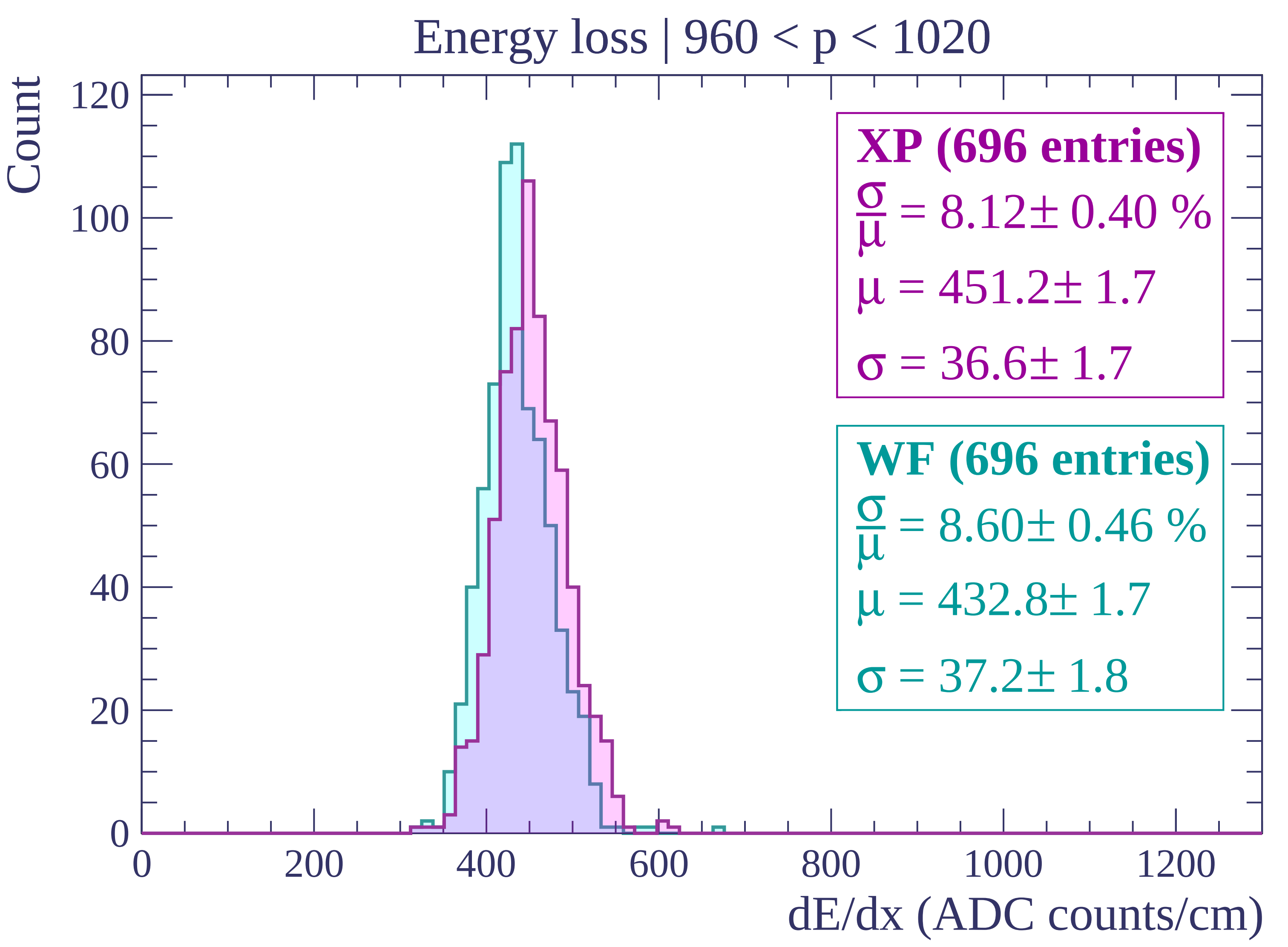
Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (735 entries)
σ/μ = 8.01± 0.39 %
μ = 437.6± 1.6
σ = 35.0± 1.6
WF (735 entries)
σ/μ = 8.05± 0.35 %
μ = 420.2± 1.4
σ = 33.8± 1.3



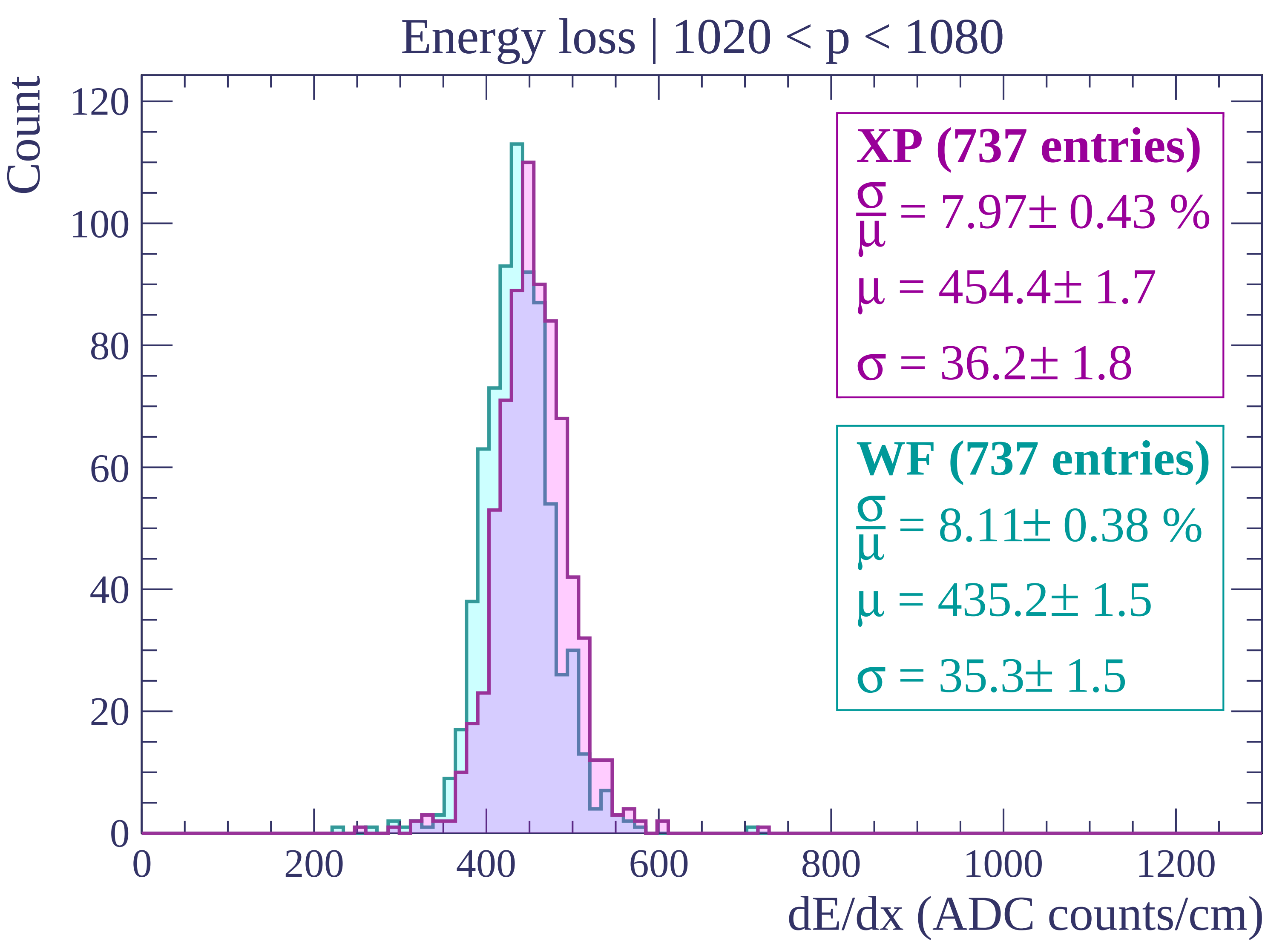
Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (768 entries)
$\frac{\sigma}{\mu}$ = 8.96± 0.42 %
μ = 440.8± 1.7
σ = 39.5± 1.7
WF (768 entries)
$\frac{\sigma}{\mu}$ = 8.52± 0.43 %
μ = 423.4± 1.6
σ = 36.1± 1.7

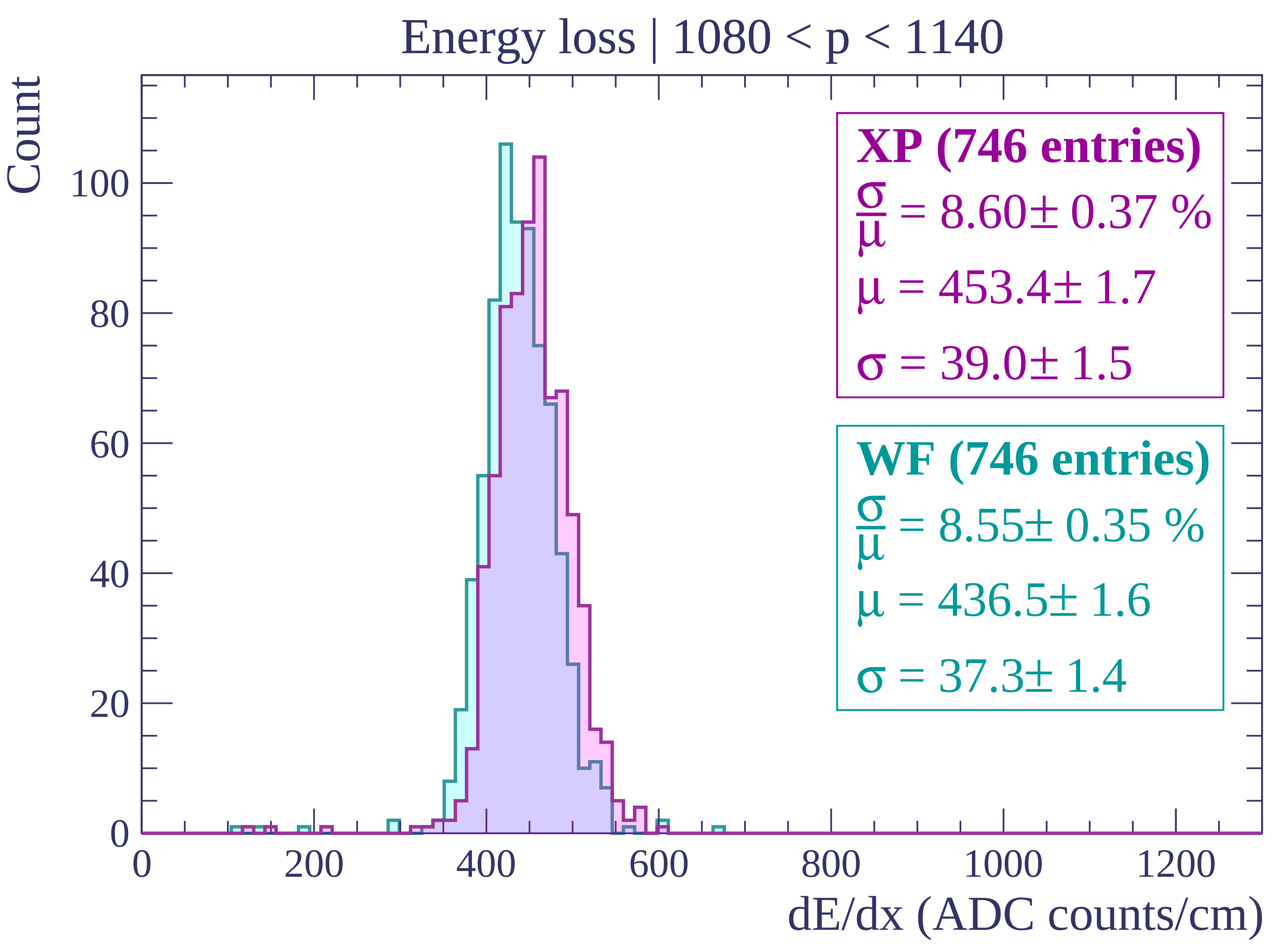


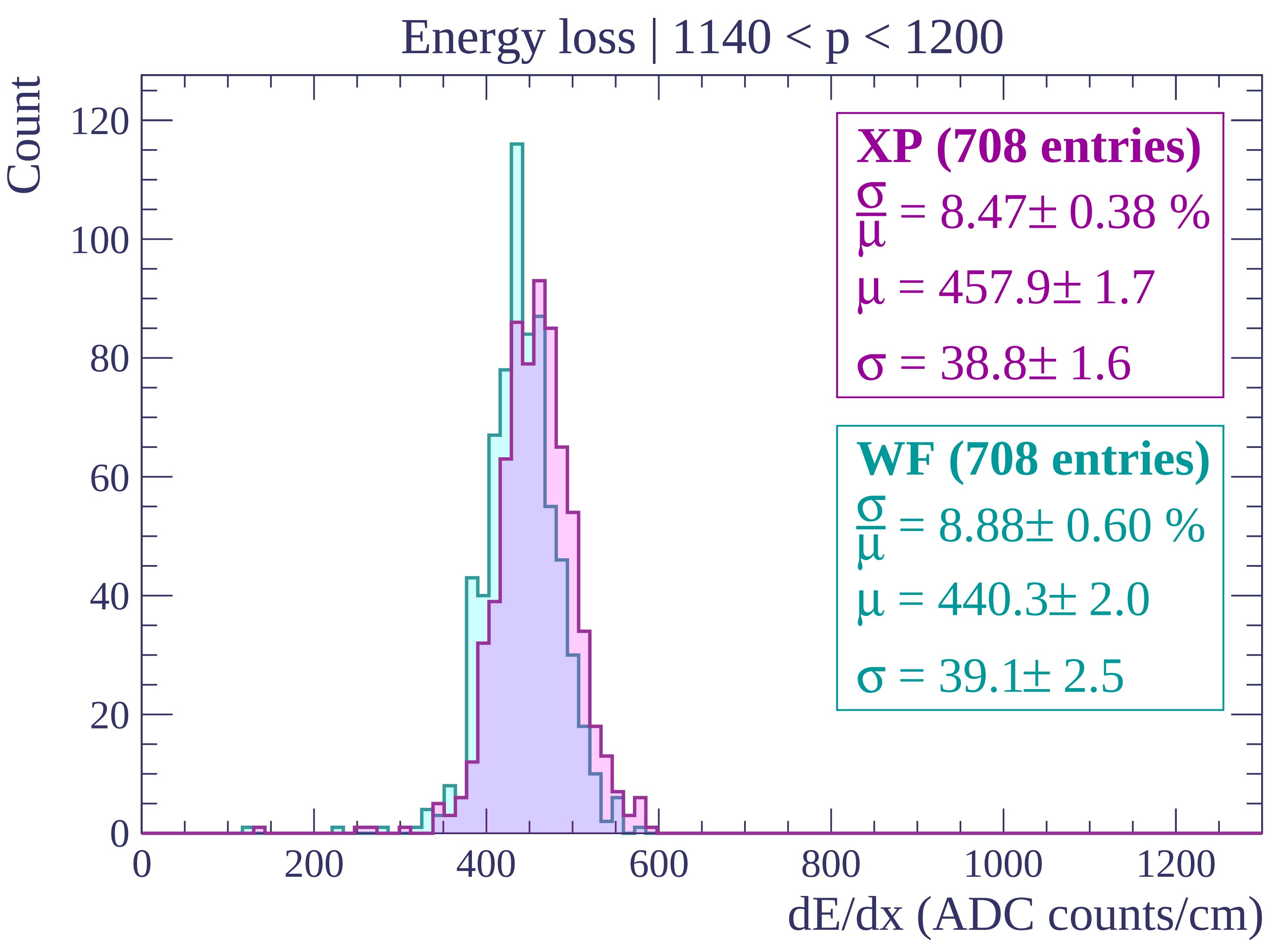
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (754 entries)
σ/μ = 8.35± 0.39 %
μ = 447.0± 1.6
σ = 37.3± 1.6
WF (754 entries)
σ/μ = 8.20± 0.31 %
μ = 429.3± 1.4
σ = 35.2± 1.2

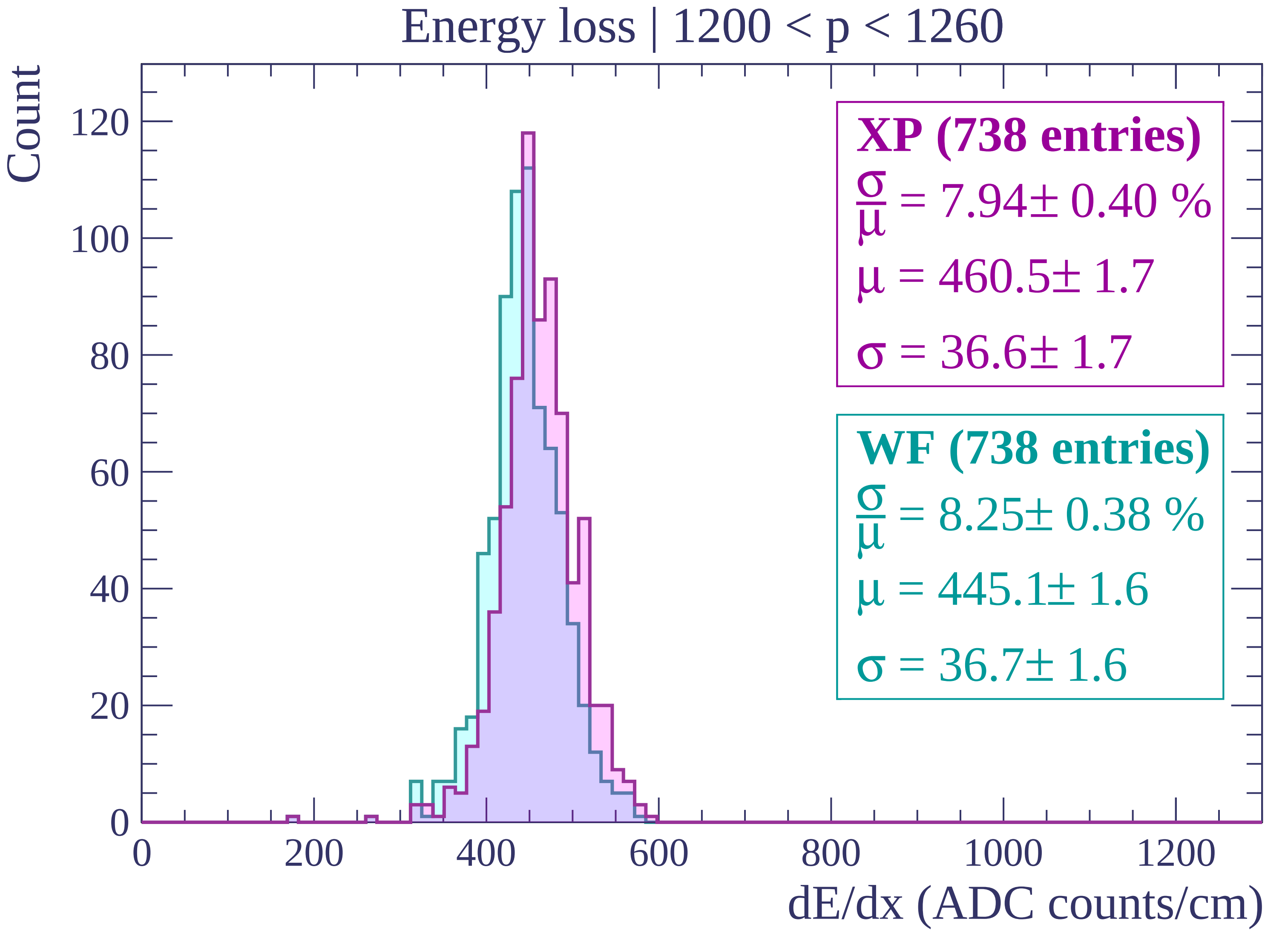


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (696 entries)
σ/μ = 8.12± 0.40 %
μ = 451.2± 1.7
σ = 36.6± 1.7
WF (696 entries)
σ/μ = 8.60± 0.46 %
μ = 432.8± 1.7
σ = 37.2± 1.8

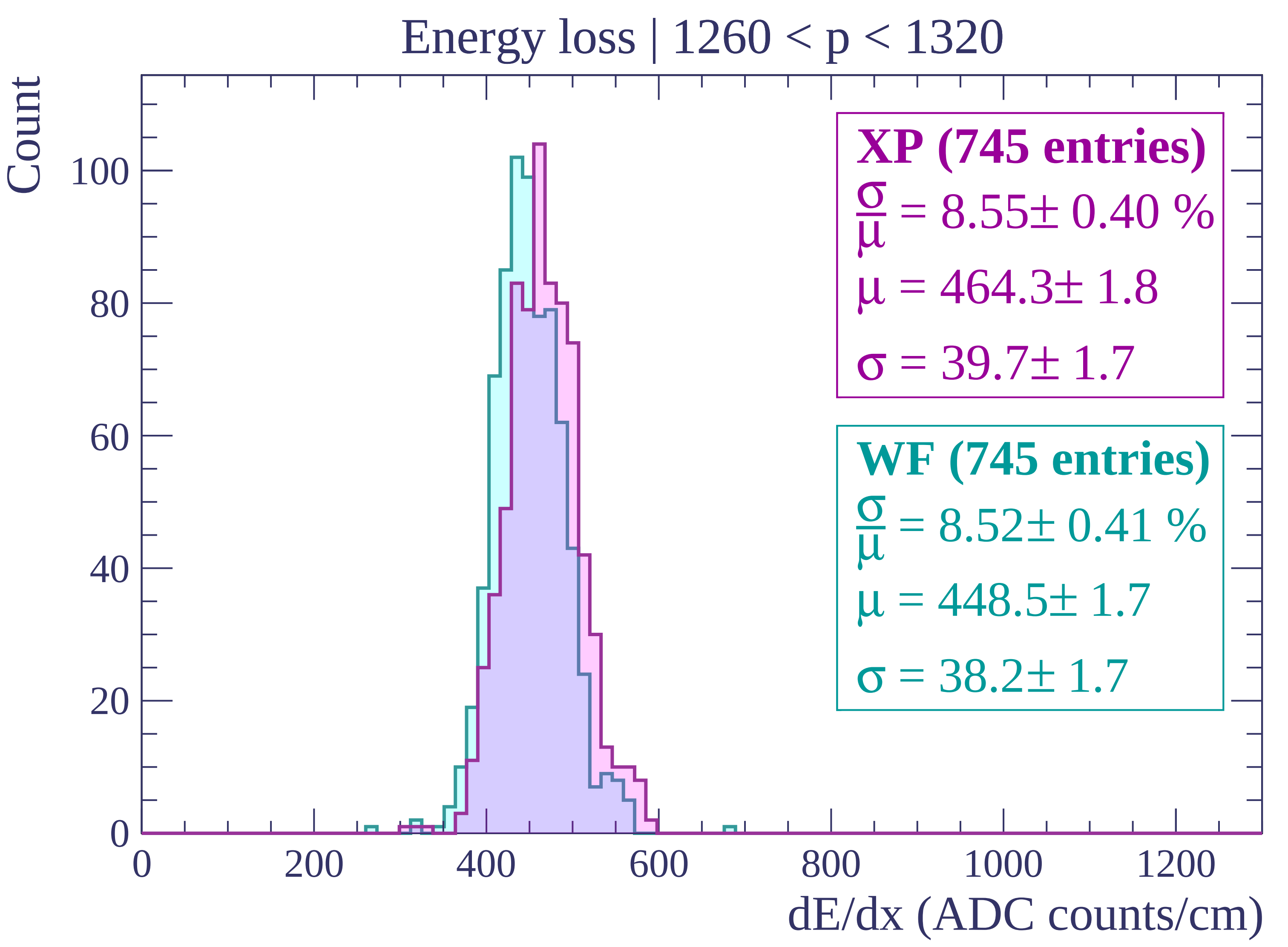

Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (737 entries)
$\frac{\sigma}{\mu}$ = 7.97± 0.43 %
μ = 454.4± 1.7
σ = 36.2± 1.8
WF (737 entries)
$\frac{\sigma}{\mu}$ = 8.11± 0.38 %
μ = 435.2± 1.5
σ = 35.3± 1.5

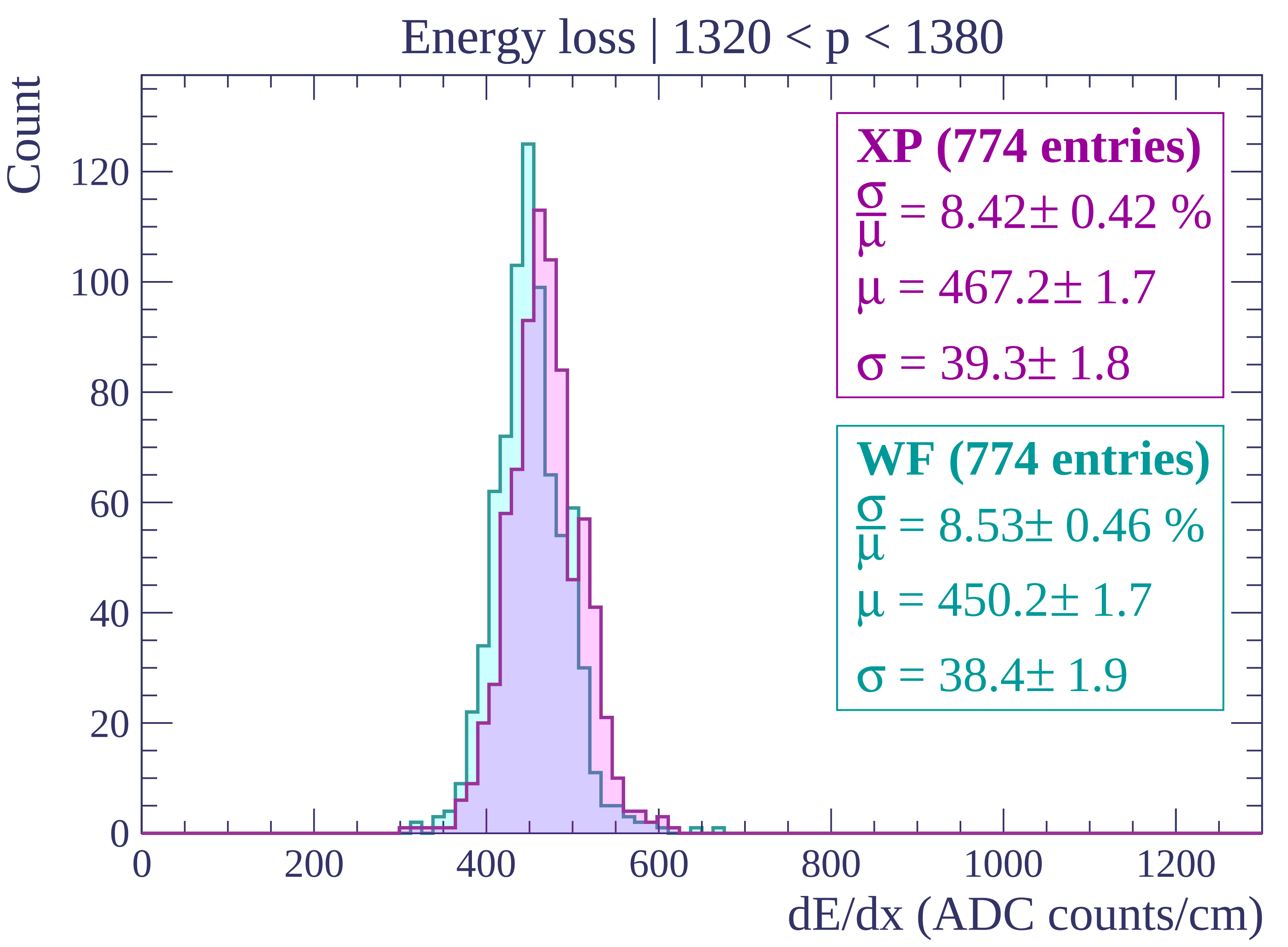

Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (746 entries)
σ/μ = 8.60± 0.37 %
μ = 453.4± 1.7
σ = 39.0± 1.5
WF (746 entries)
σ/μ = 8.55± 0.35 %
μ = 436.5± 1.6
σ = 37.3± 1.4

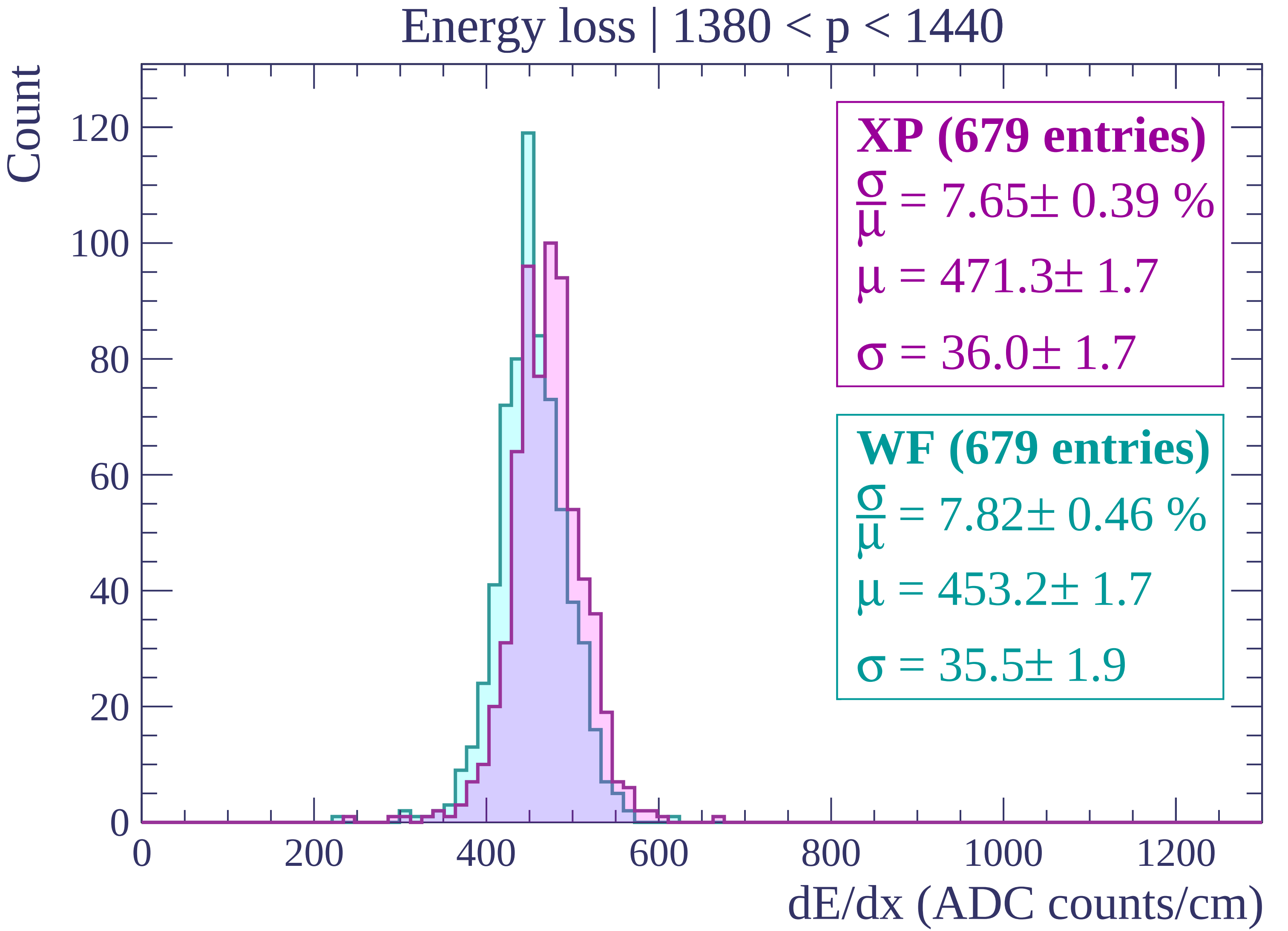

Energy loss | 1140 < p < 1200
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (708 entries)
σ/μ = 8.47± 0.38 %
μ = 457.9± 1.7
σ = 38.8± 1.6
WF (708 entries)
σ/μ = 8.88± 0.60 %
μ = 440.3± 2.0
σ = 39.1± 2.5



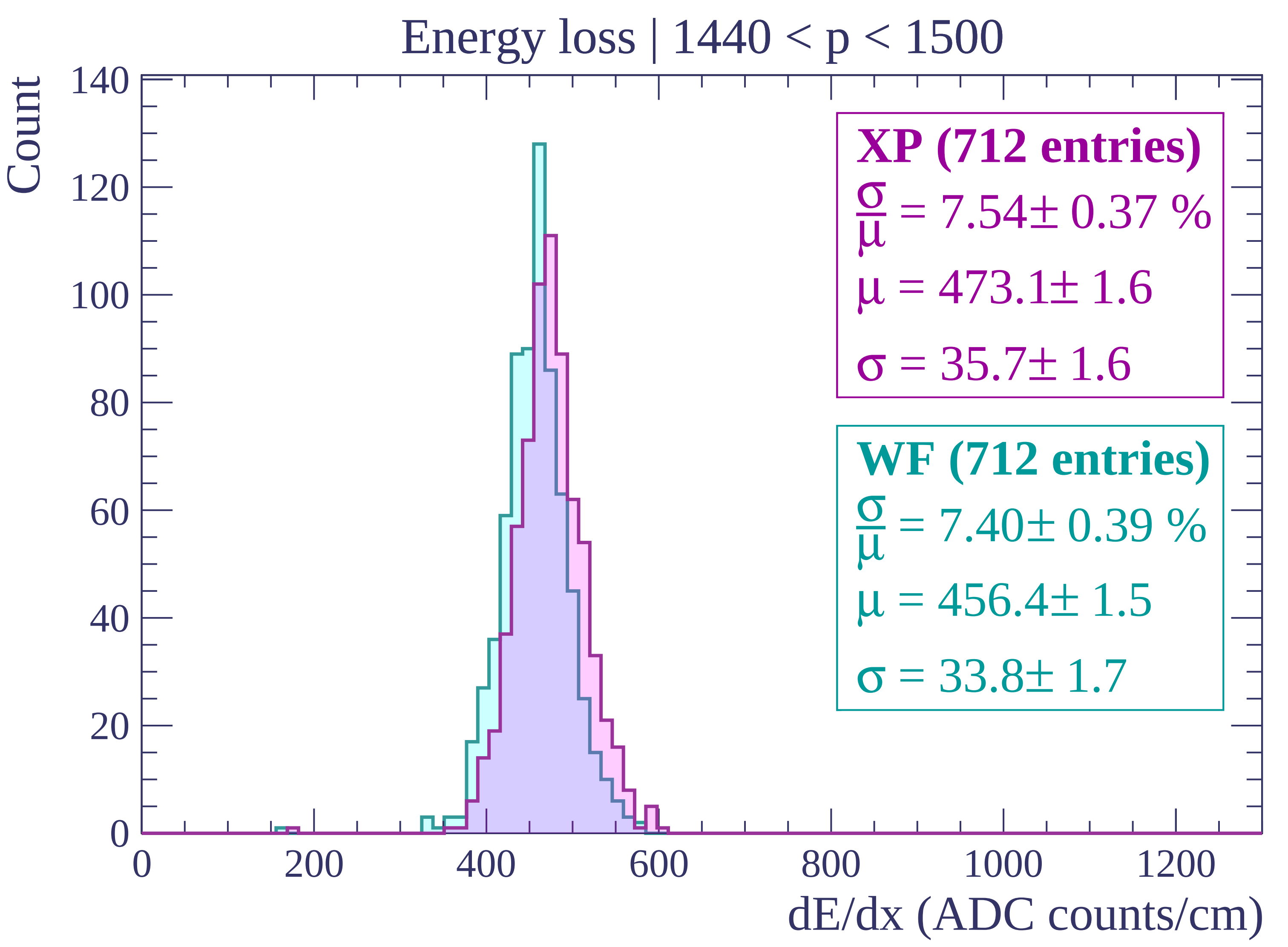
Energy loss | 1200 < p < 1260
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (738 entries)
$\frac{\sigma}{\mu}$ = 7.94± 0.40 %
μ = 460.5± 1.7
σ = 36.6± 1.7
WF (738 entries)
$\frac{\sigma}{\mu}$ = 8.25± 0.38 %
μ = 445.1± 1.6
σ = 36.7± 1.6

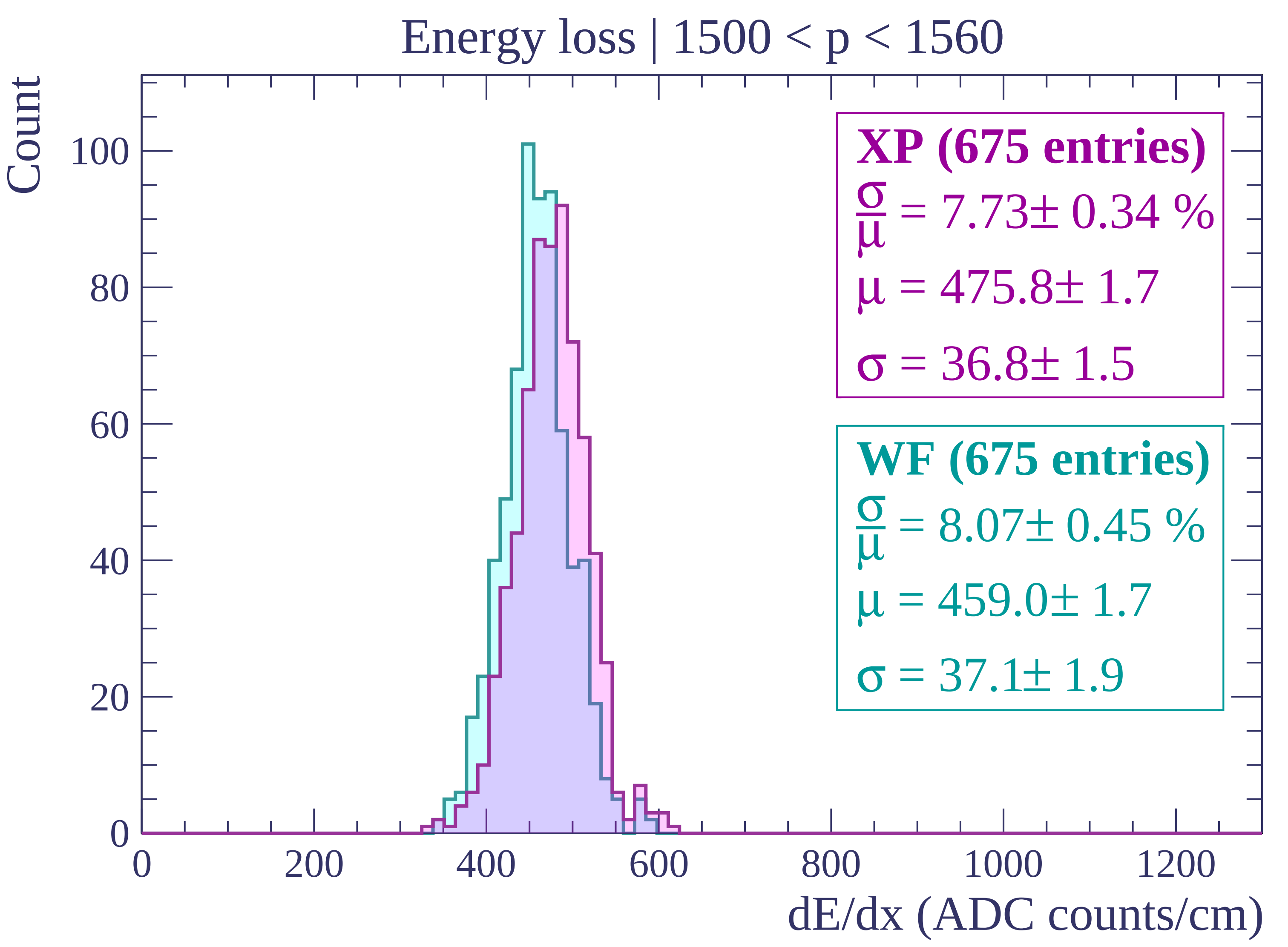

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (745 entries)
σ/μ = 8.55± 0.40 %
μ = 464.3± 1.8
σ = 39.7± 1.7
WF (745 entries)
σ/μ = 8.52± 0.41 %
μ = 448.5± 1.7
σ = 38.2± 1.7

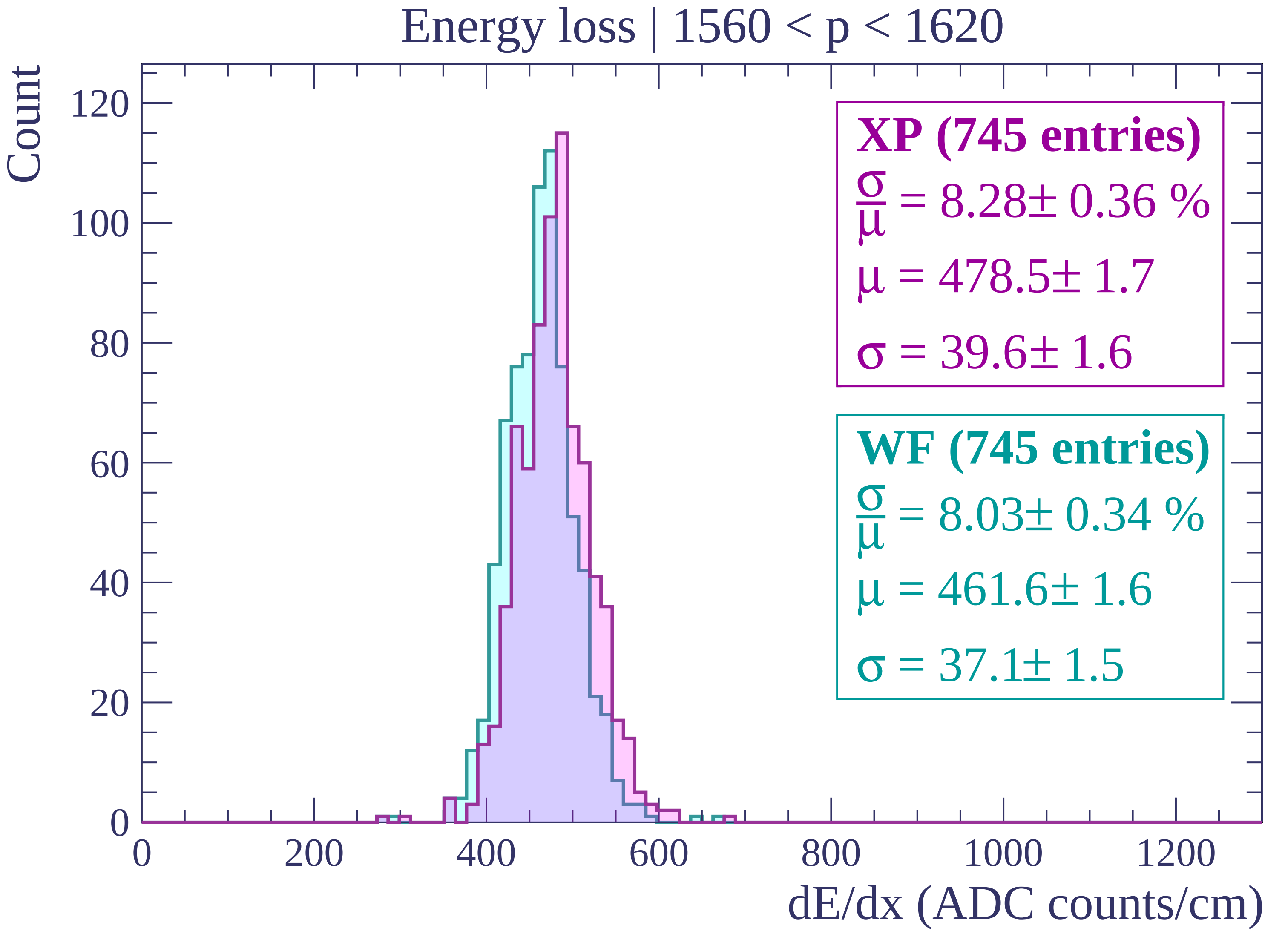

Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (774 entries)
σ/μ = 8.42± 0.42 %
μ = 467.2± 1.7
σ = 39.3± 1.8
WF (774 entries)
σ/μ = 8.53± 0.46 %
μ = 450.2± 1.7
σ = 38.4± 1.9

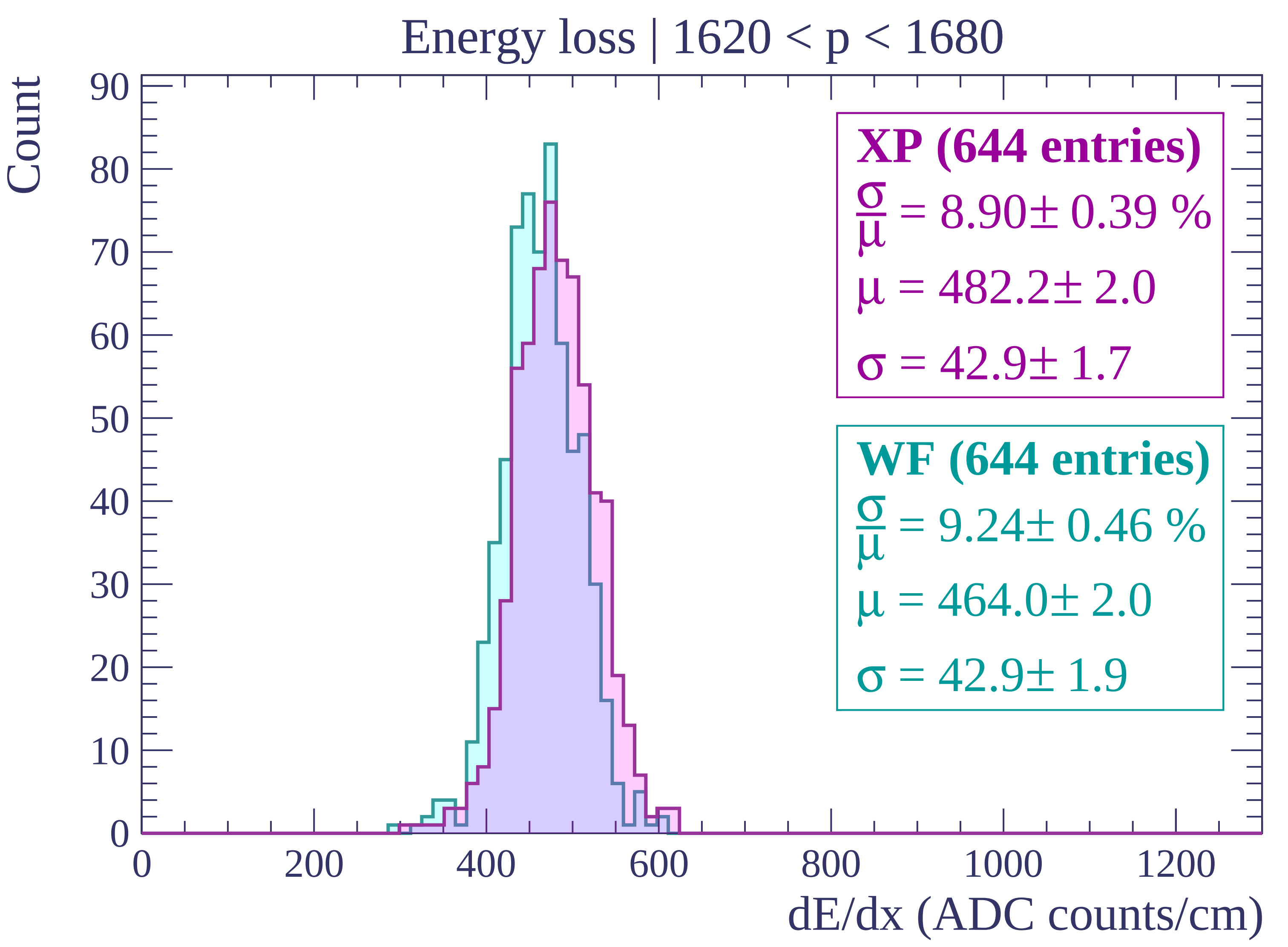


Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (679 entries)
$\frac{\sigma}{\mu}$ = 7.65± 0.39 %
μ = 471.3± 1.7
σ = 36.0± 1.7
WF (679 entries)
$\frac{\sigma}{\mu}$ = 7.82± 0.46 %
μ = 453.2± 1.7
σ = 35.5± 1.9

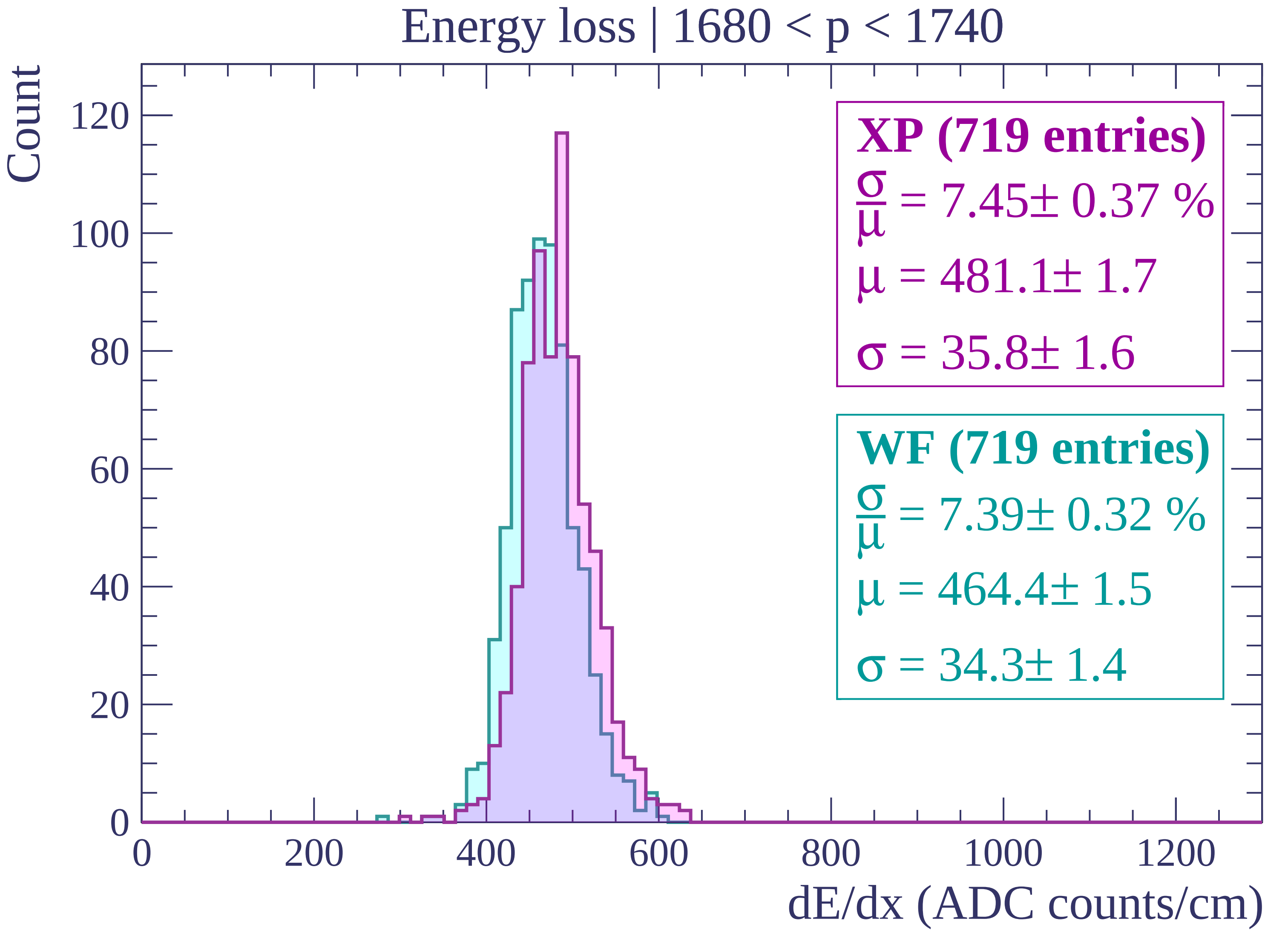

Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
XP (712 entries)
σ/μ = 7.54± 0.37 %
μ = 473.1± 1.6
σ = 35.7± 1.6
WF (712 entries)
σ/μ = 7.40± 0.39 %
μ = 456.4± 1.5
σ = 33.8± 1.7
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200


Energy loss | 1500 < p < 1560
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (675 entries)
σ/μ = 7.73± 0.34 %
μ = 475.8± 1.7
σ = 36.8± 1.5
WF (675 entries)
σ/μ = 8.07± 0.45 %
μ = 459.0± 1.7
σ = 37.1± 1.9

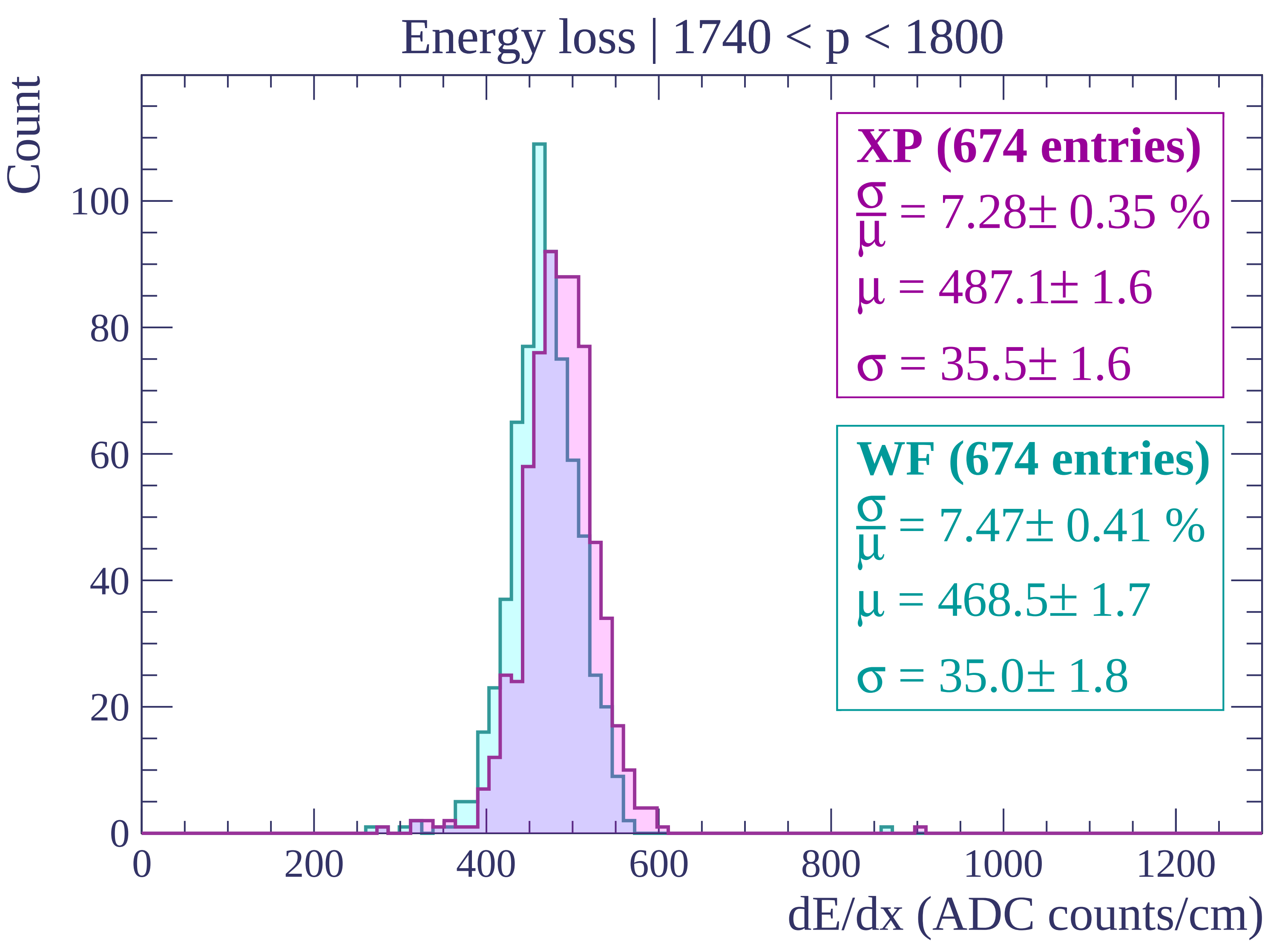

Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (745 entries)
$\frac{\sigma}{\mu}$ = 8.28± 0.36 %
μ = 478.5± 1.7
σ = 39.6± 1.6
WF (745 entries)
$\frac{\sigma}{\mu}$ = 8.03± 0.34 %
μ = 461.6± 1.6
σ = 37.1± 1.5



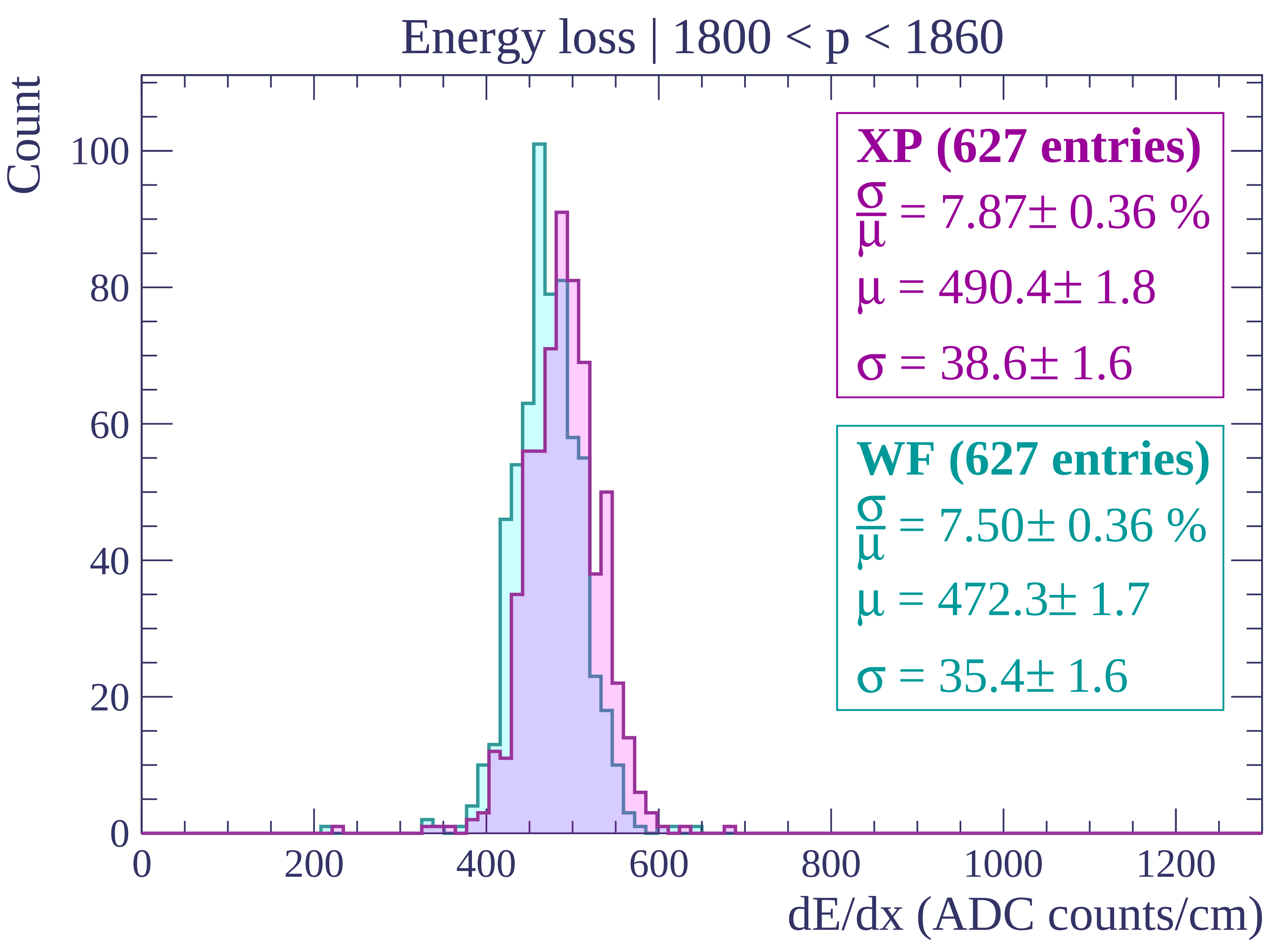
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (644 entries)
$\frac{\sigma}{\mu}$ = 8.90± 0.39 %
μ = 482.2± 2.0
σ = 42.9± 1.7
WF (644 entries)
$\frac{\sigma}{\mu}$ = 9.24± 0.46 %
μ = 464.0± 2.0
σ = 42.9± 1.9
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90



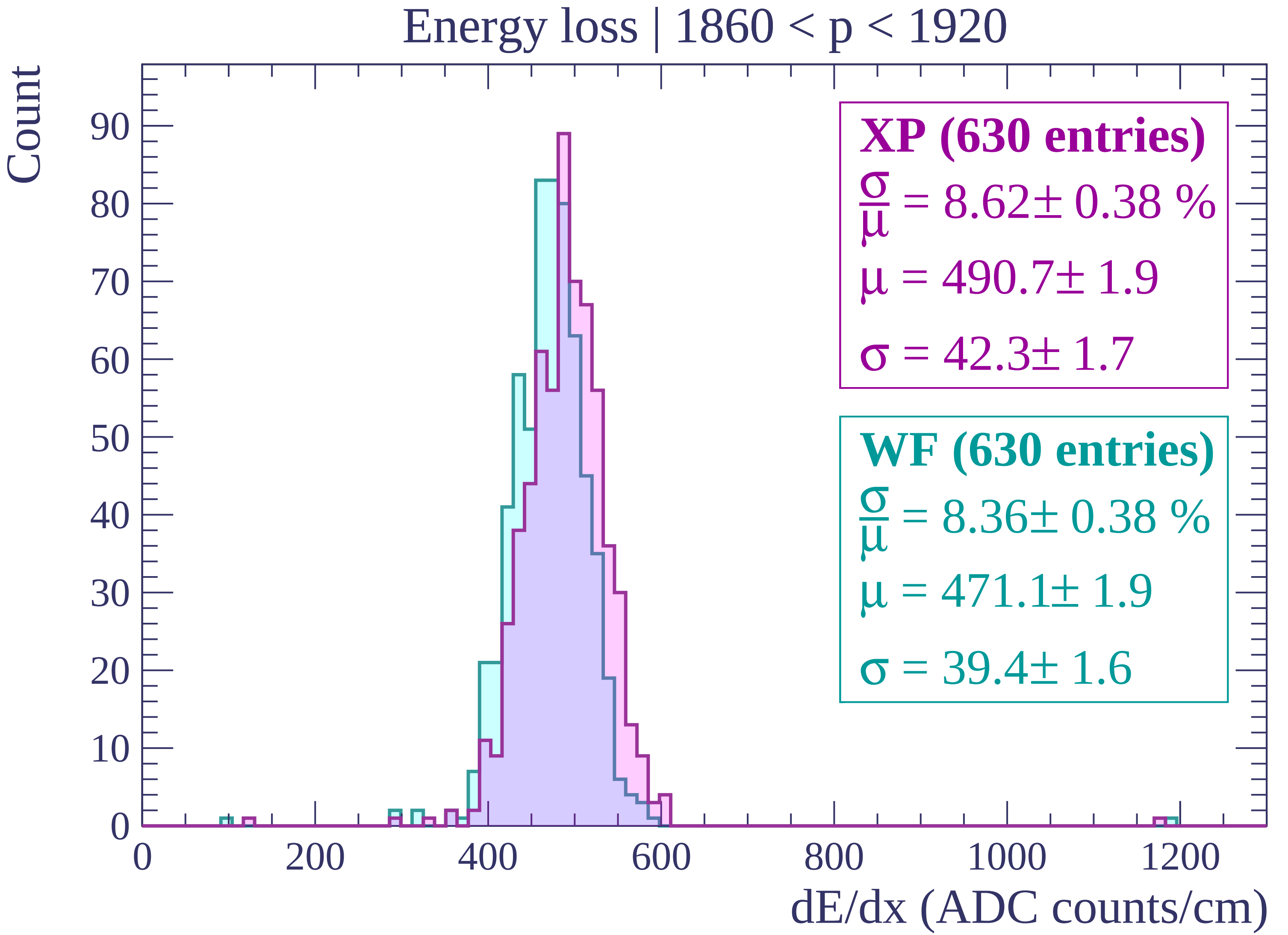
Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (719 entries)
σ/μ = 7.45± 0.37 %
μ = 481.1± 1.7
σ = 35.8± 1.6
WF (719 entries)
σ/μ = 7.39± 0.32 %
μ = 464.4± 1.5
σ = 34.3± 1.4

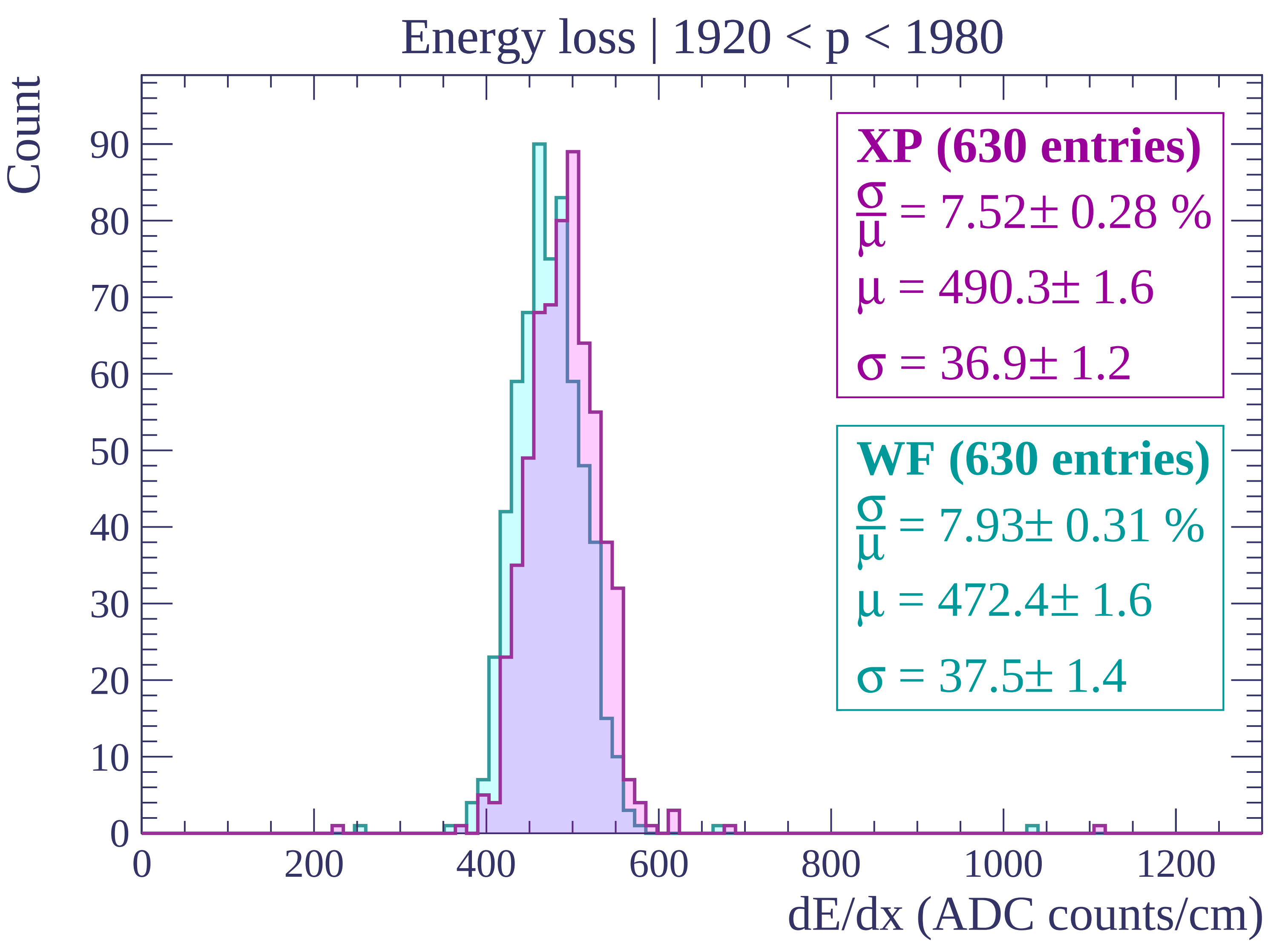


Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (674 entries)
$\frac{\sigma}{\mu}$ = 7.28± 0.35 %
μ = 487.1± 1.6
σ = 35.5± 1.6
WF (674 entries)
$\frac{\sigma}{\mu}$ = 7.47± 0.41 %
μ = 468.5± 1.7
σ = 35.0± 1.8

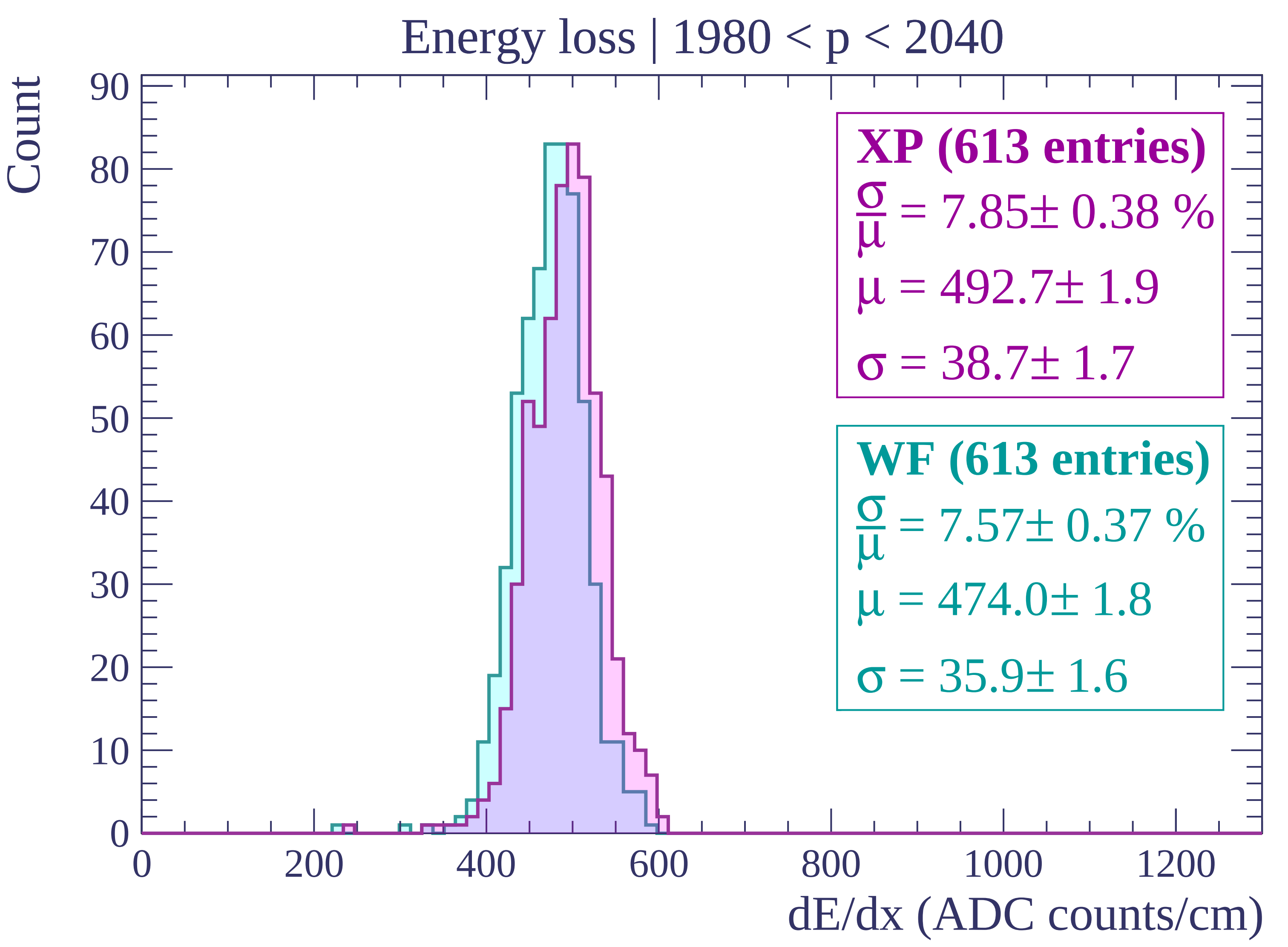


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (627 entries)
σ/μ = 7.87± 0.36 %
μ = 490.4± 1.8
σ = 38.6± 1.6
WF (627 entries)
σ/μ = 7.50± 0.36 %
μ = 472.3± 1.7
σ = 35.4± 1.6

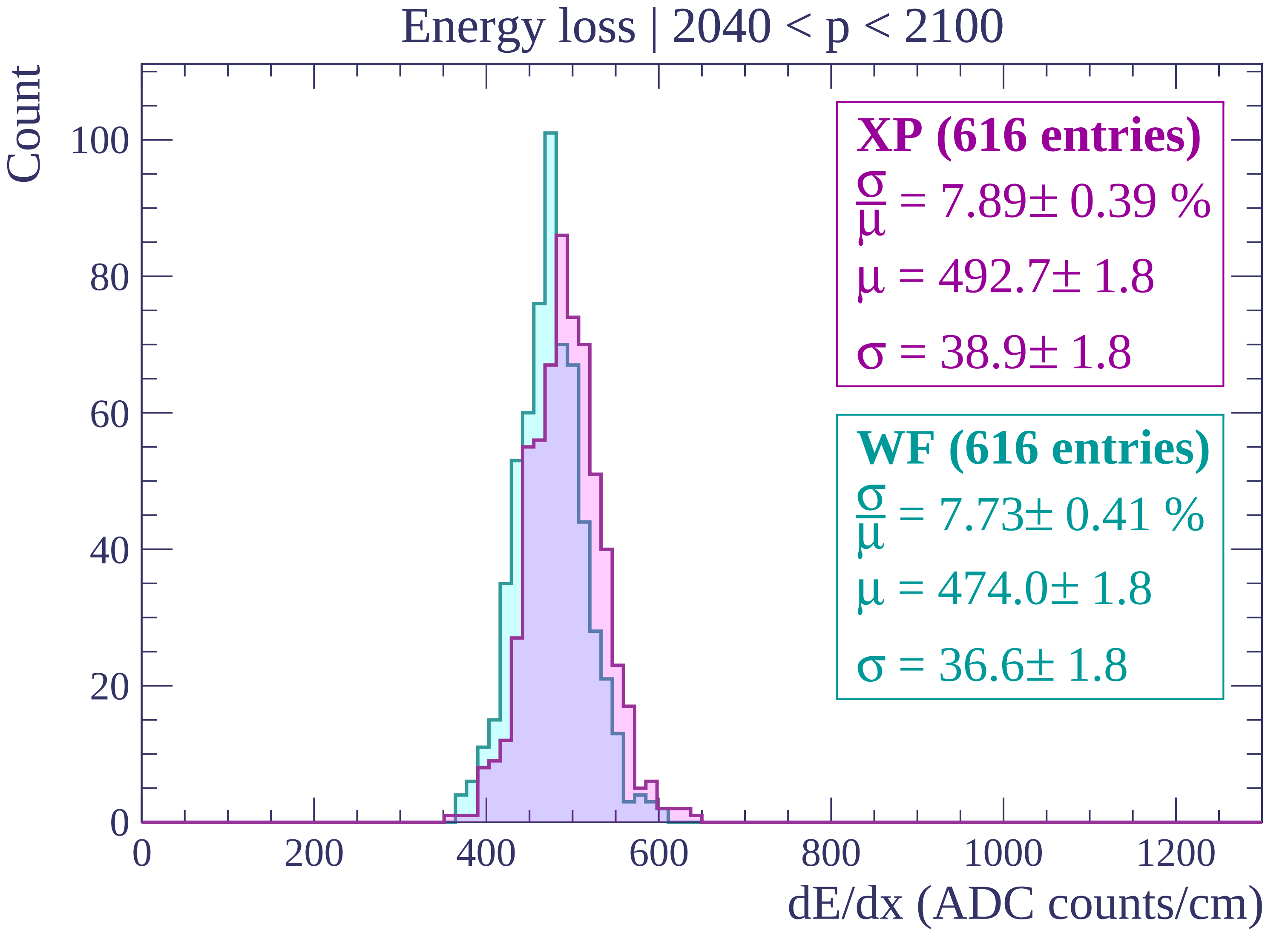

Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (630 entries)
σ/μ = 8.62± 0.38 %
μ = 490.7± 1.9
σ = 42.3± 1.7
WF (630 entries)
σ/μ = 8.36± 0.38 %
μ = 471.1± 1.9
σ = 39.4± 1.6

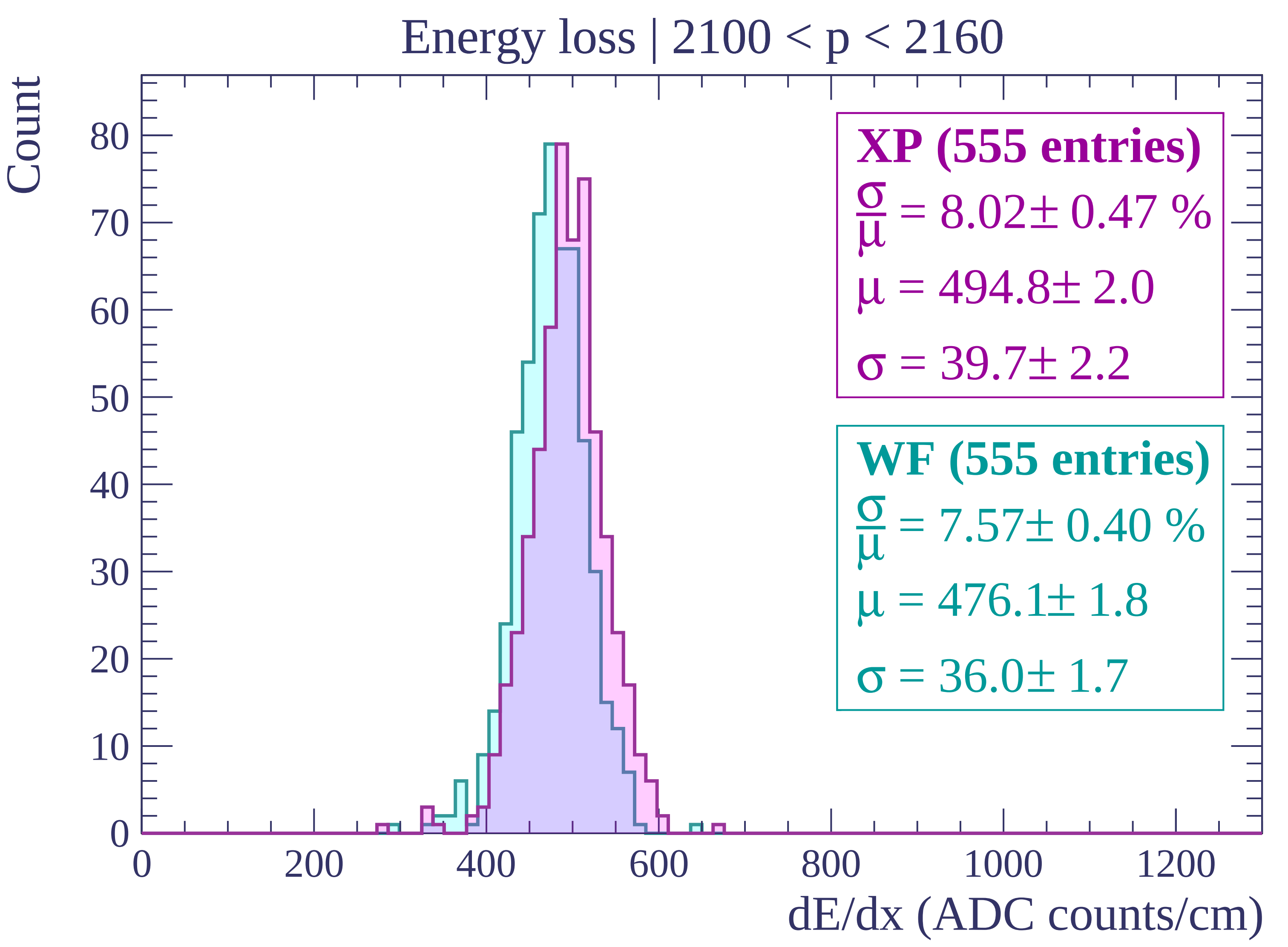


Energy loss | 1920 < p < 1980
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (630 entries)
σ/μ = 7.52± 0.28 %
μ = 490.3± 1.6
σ = 36.9± 1.2
WF (630 entries)
σ/μ = 7.93± 0.31 %
μ = 472.4± 1.6
σ = 37.5± 1.4

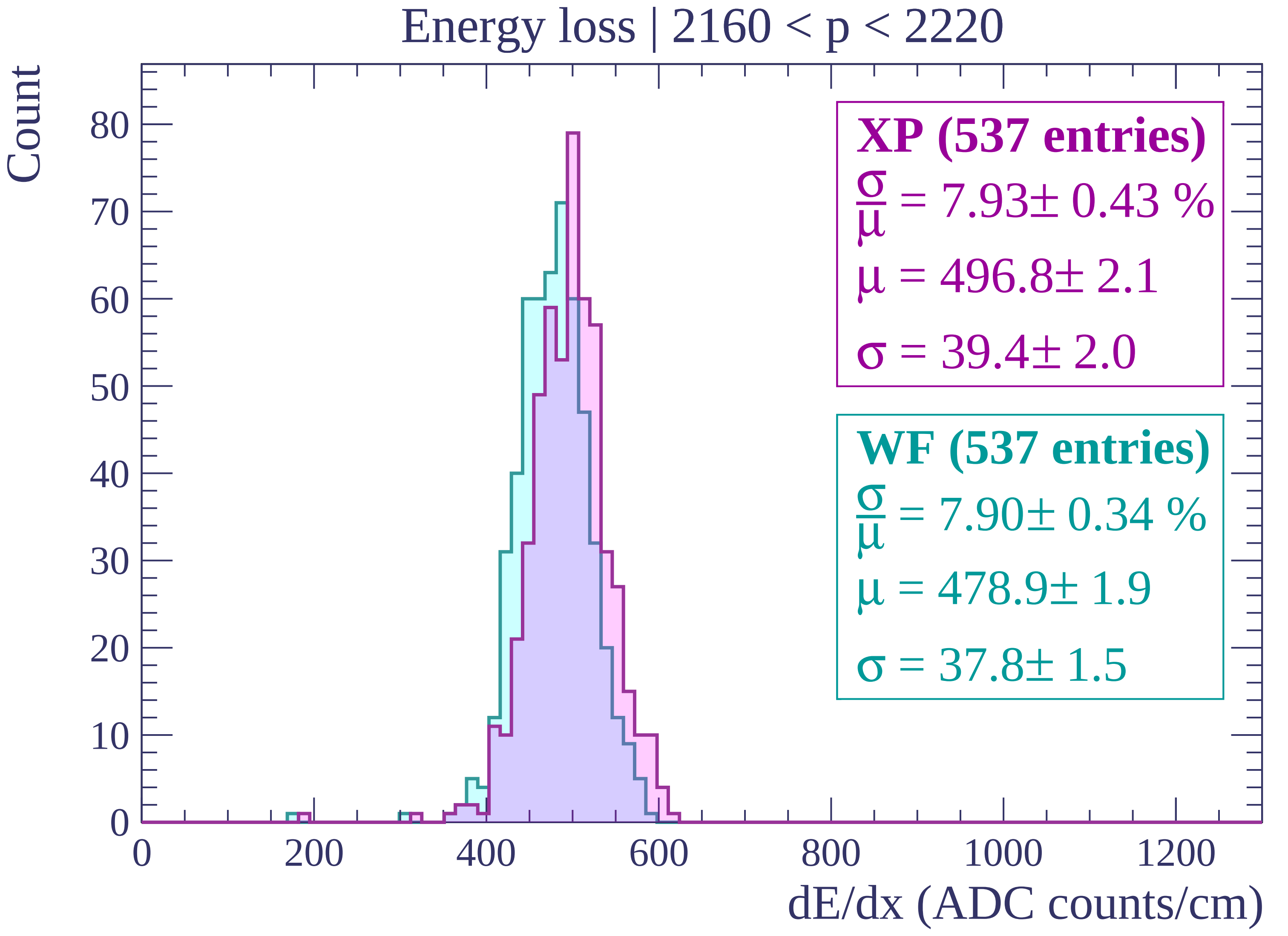


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (613 entries)
σ/μ = 7.85± 0.38 %
μ = 492.7± 1.9
σ = 38.7± 1.7
WF (613 entries)
σ/μ = 7.57± 0.37 %
μ = 474.0± 1.8
σ = 35.9± 1.6

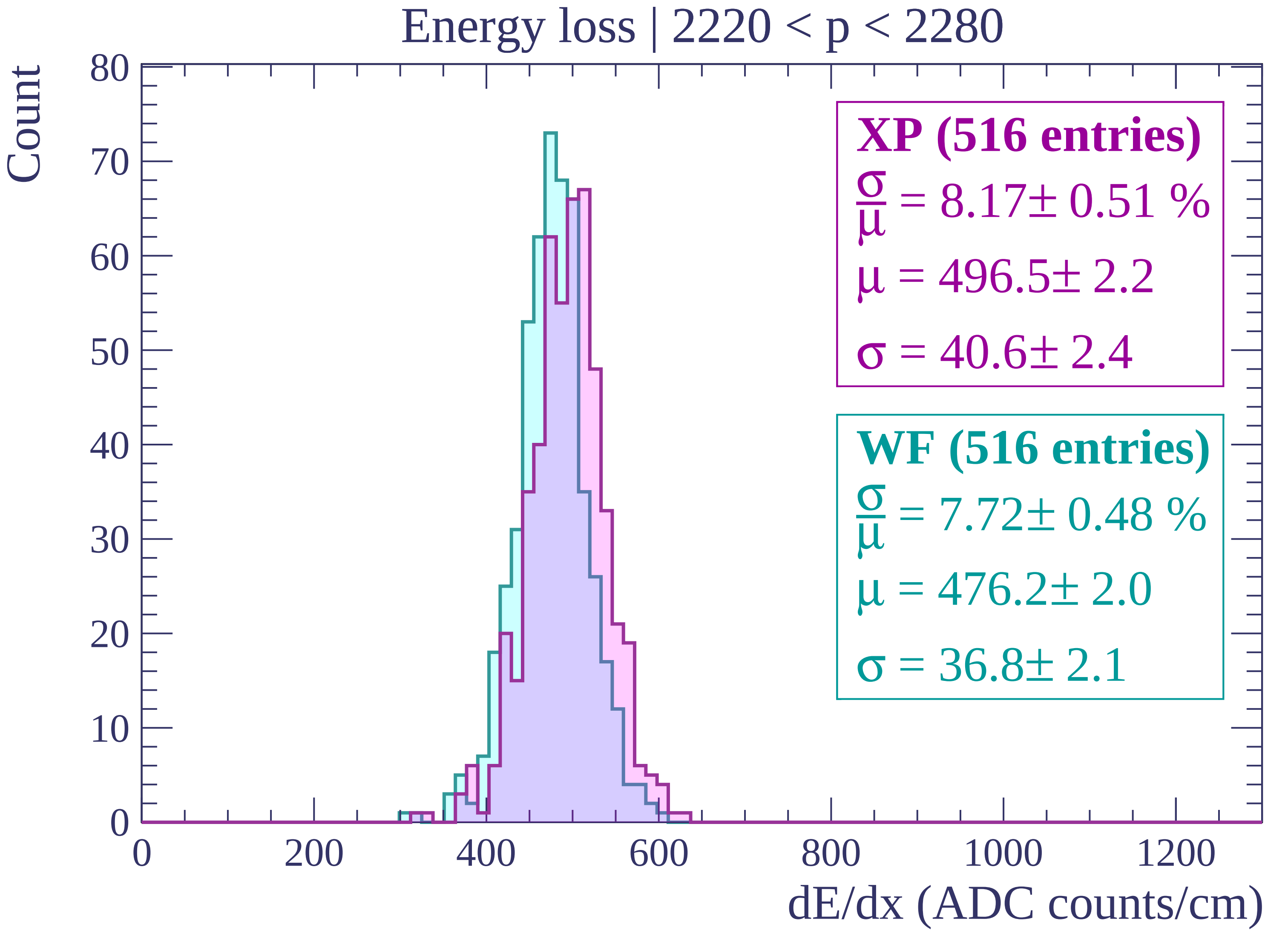

Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (616 entries)
$\frac{\sigma}{\mu}$ = 7.89± 0.39 %
μ = 492.7± 1.8
σ = 38.9± 1.8
WF (616 entries)
$\frac{\sigma}{\mu}$ = 7.73± 0.41 %
μ = 474.0± 1.8
σ = 36.6± 1.8
0
20
40
60
80
100
0
200
400
600
800
1000
1200

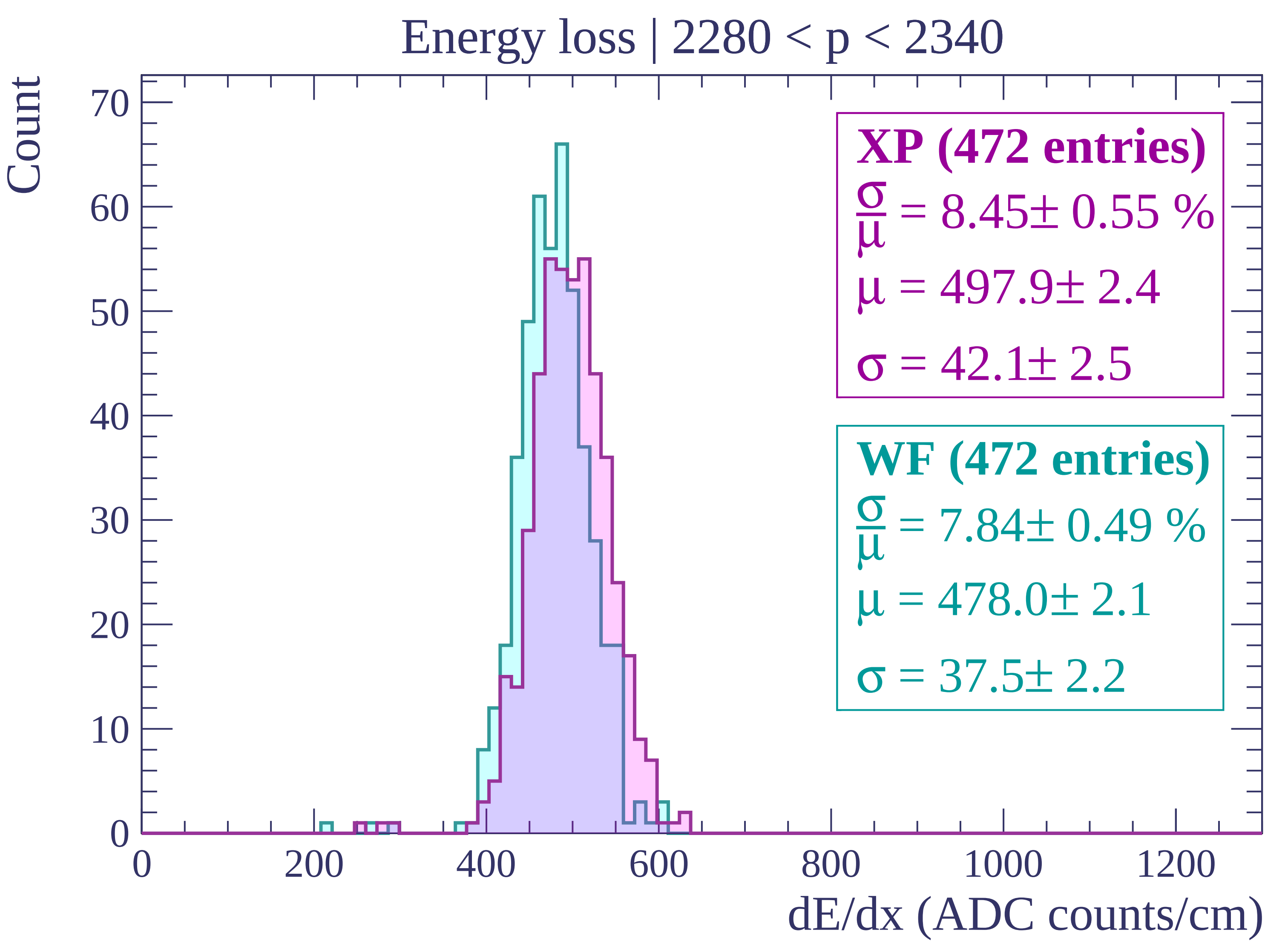


Energy loss | 2100 < p < 2160
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (555 entries)
σ/μ = 8.02± 0.47 %
μ = 494.8± 2.0
σ = 39.7± 2.2
WF (555 entries)
σ/μ = 7.57± 0.40 %
μ = 476.1± 1.8
σ = 36.0± 1.7

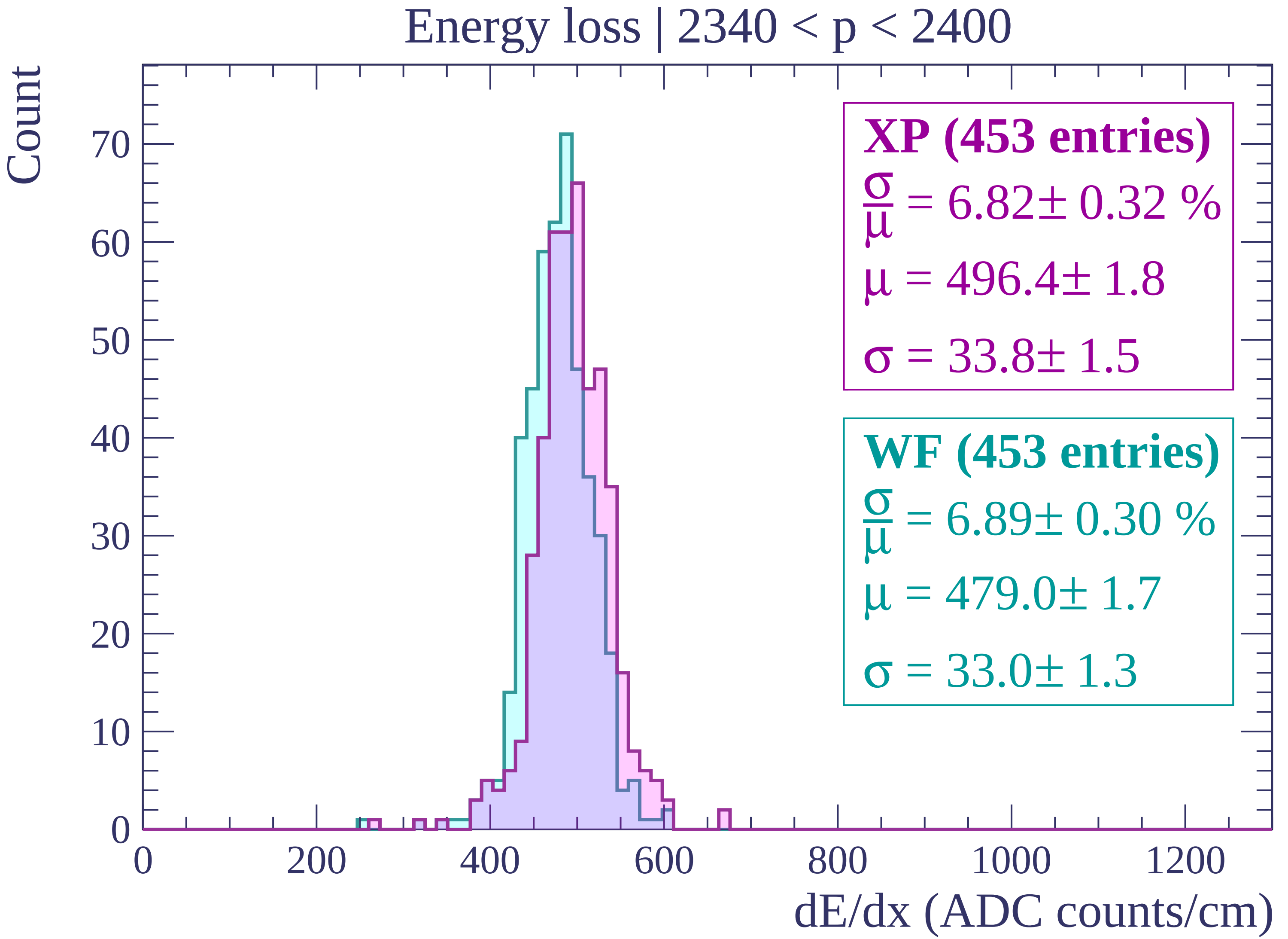


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (537 entries)
σ/μ = 7.93± 0.43 %
μ = 496.8± 2.1
σ = 39.4± 2.0
WF (537 entries)
σ/μ = 7.90± 0.34 %
μ = 478.9± 1.9
σ = 37.8± 1.5

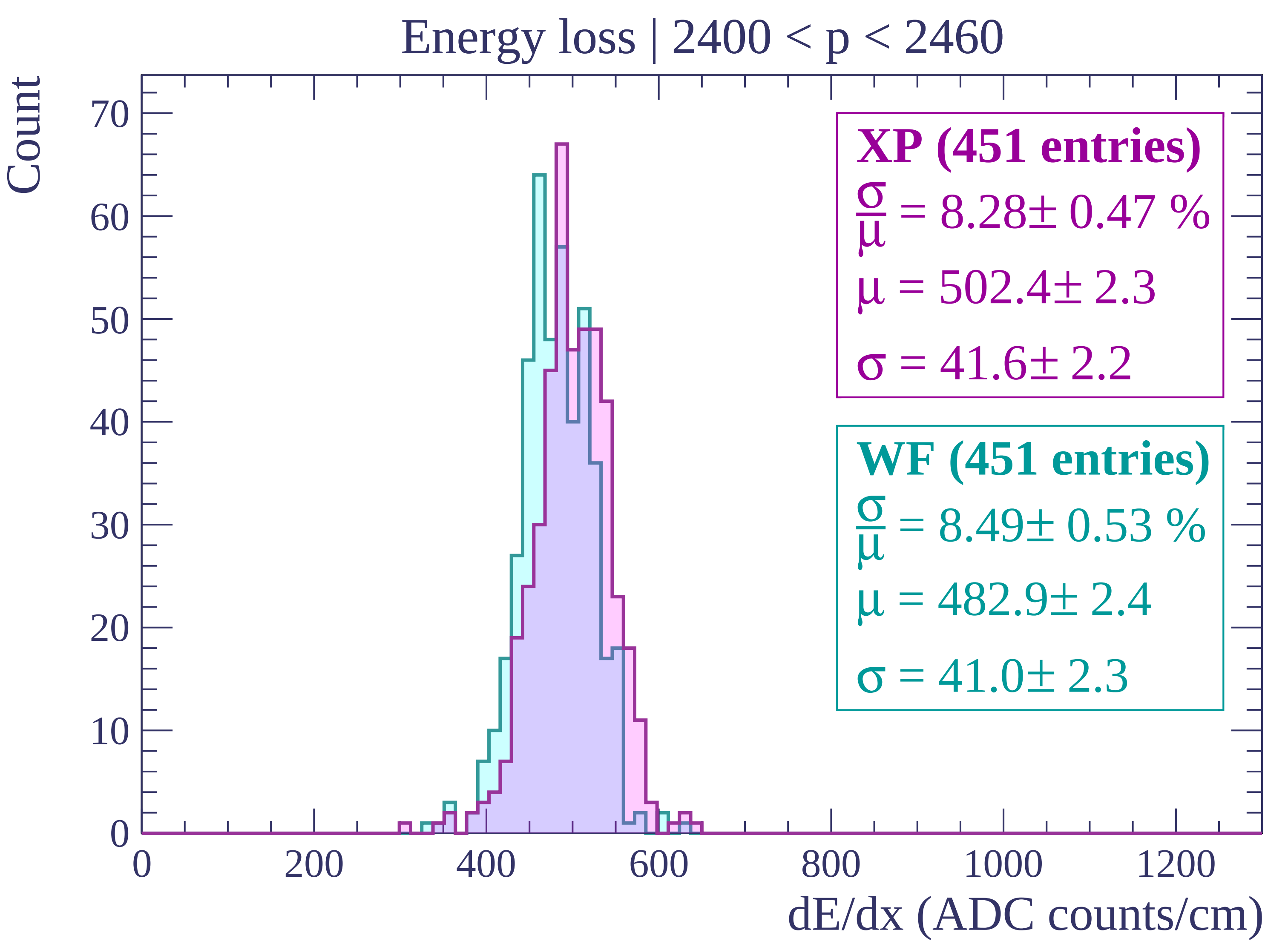

Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (516 entries)
σ/μ = 8.17± 0.51 %
μ = 496.5± 2.2
σ = 40.6± 2.4
WF (516 entries)
σ/μ = 7.72± 0.48 %
μ = 476.2± 2.0
σ = 36.8± 2.1



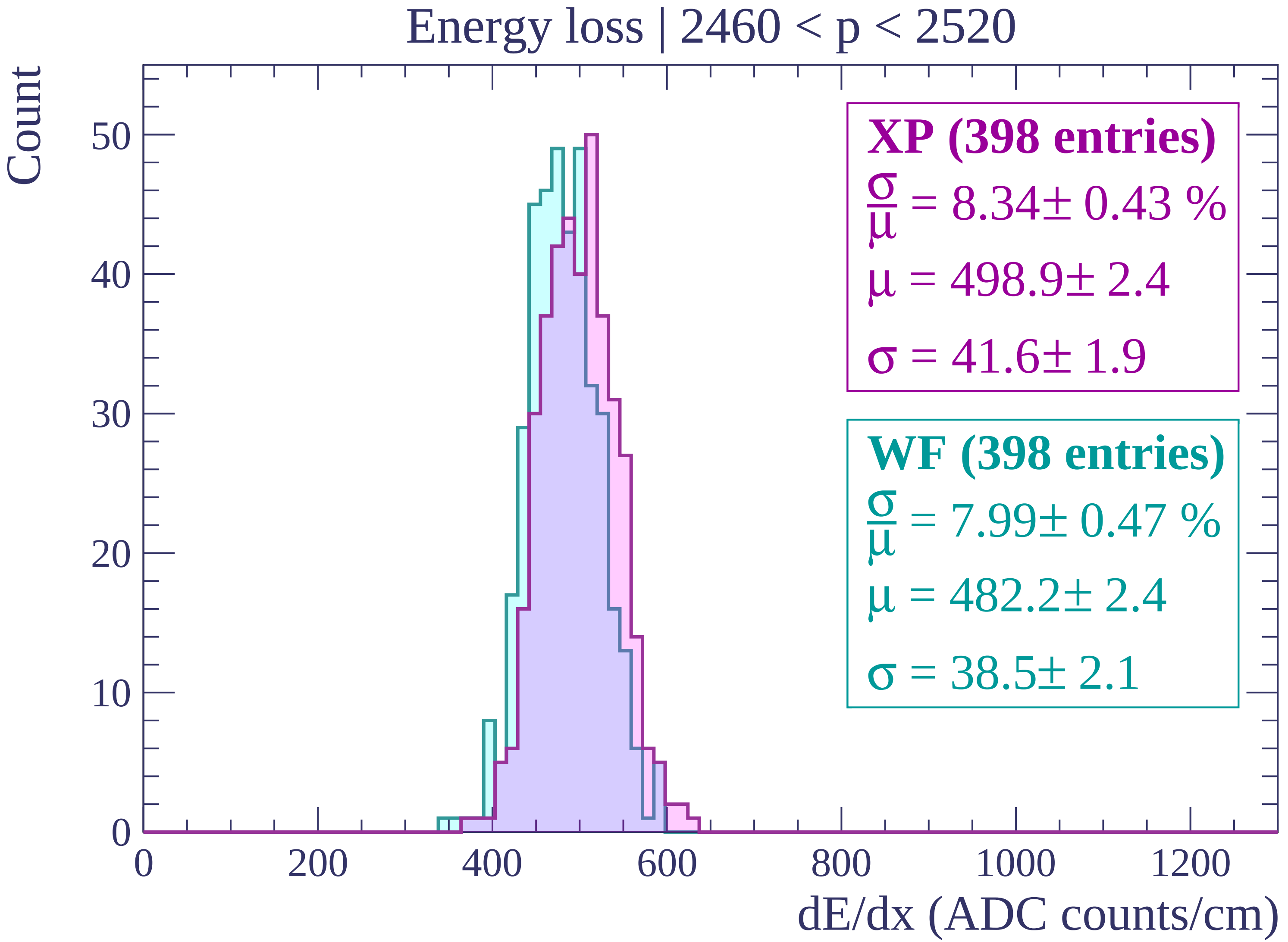
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (472 entries)
σ/μ = 8.45± 0.55 %
μ = 497.9± 2.4
σ = 42.1± 2.5
WF (472 entries)
σ/μ = 7.84± 0.49 %
μ = 478.0± 2.1
σ = 37.5± 2.2



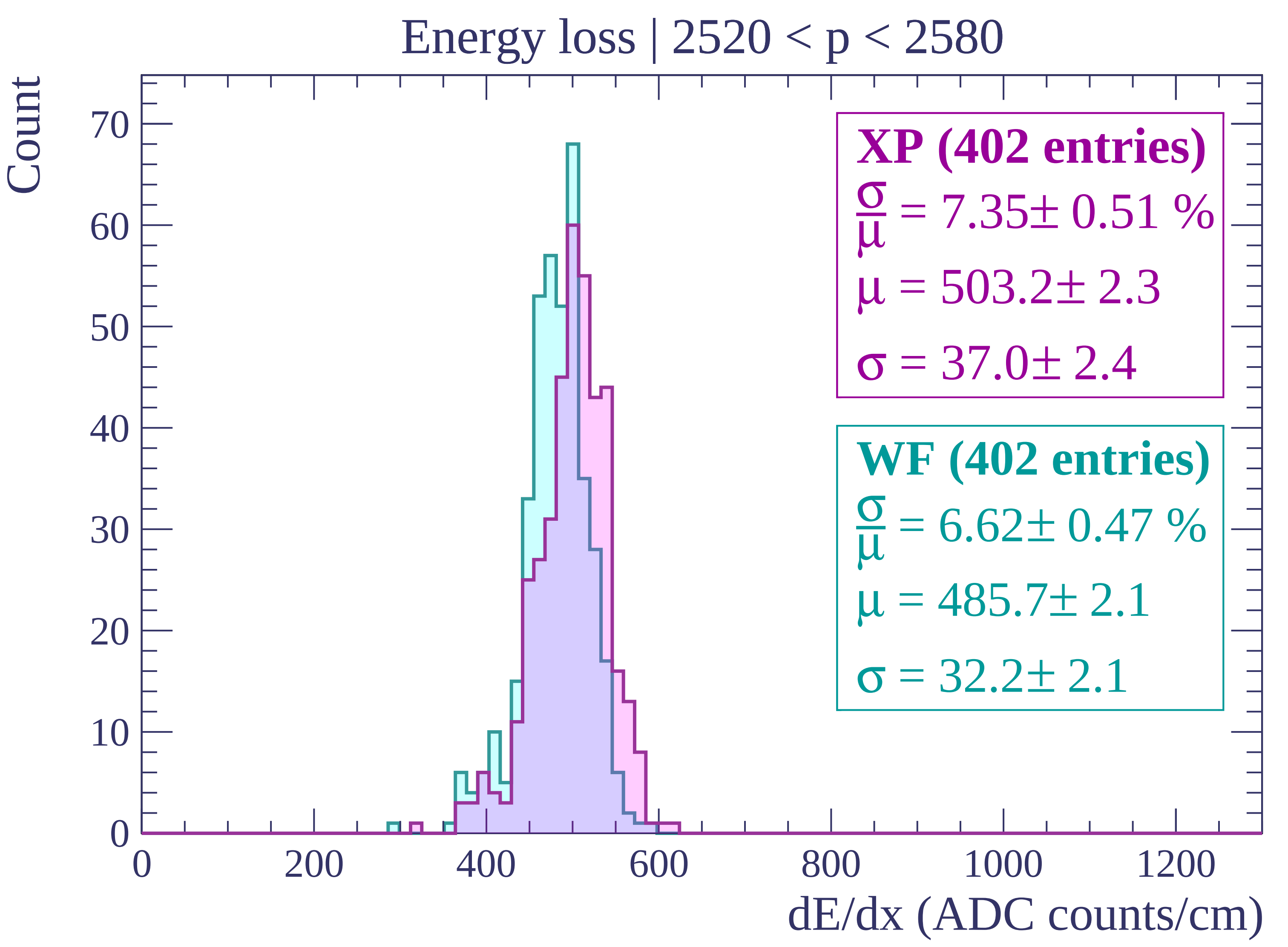
Energy loss | 2340 < p < 2400
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (453 entries)
σ/μ = 6.82± 0.32 %
μ = 496.4± 1.8
σ = 33.8± 1.5
WF (453 entries)
σ/μ = 6.89± 0.30 %
μ = 479.0± 1.7
σ = 33.0± 1.3



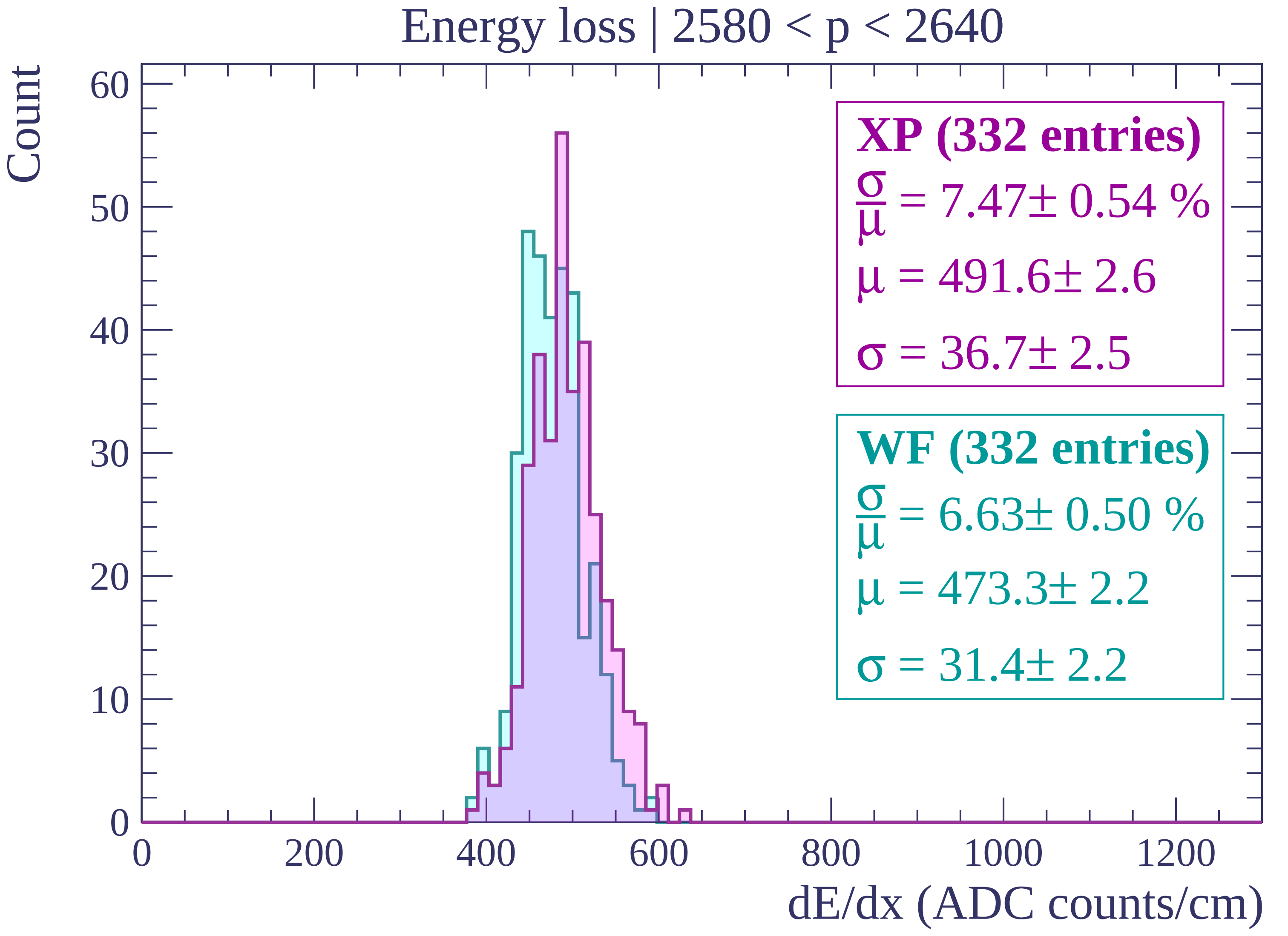
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (451 entries)
σ/μ = 8.28± 0.47 %
μ = 502.4± 2.3
σ = 41.6± 2.2
WF (451 entries)
σ/μ = 8.49± 0.53 %
μ = 482.9± 2.4
σ = 41.0± 2.3

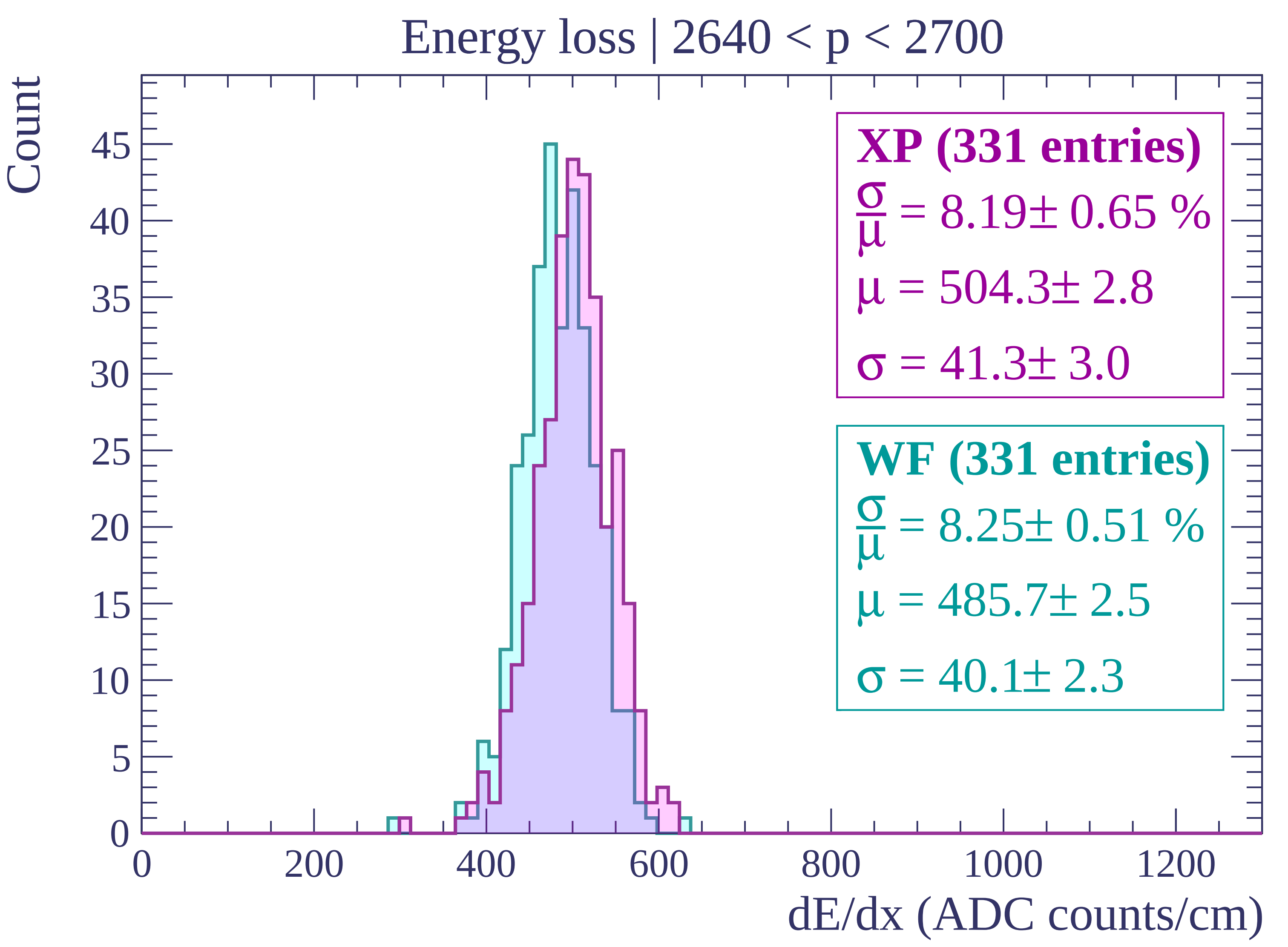


Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (398 entries)
σ/μ = 8.34± 0.43 %
μ = 498.9± 2.4
σ = 41.6± 1.9
WF (398 entries)
σ/μ = 7.99± 0.47 %
μ = 482.2± 2.4
σ = 38.5± 2.1

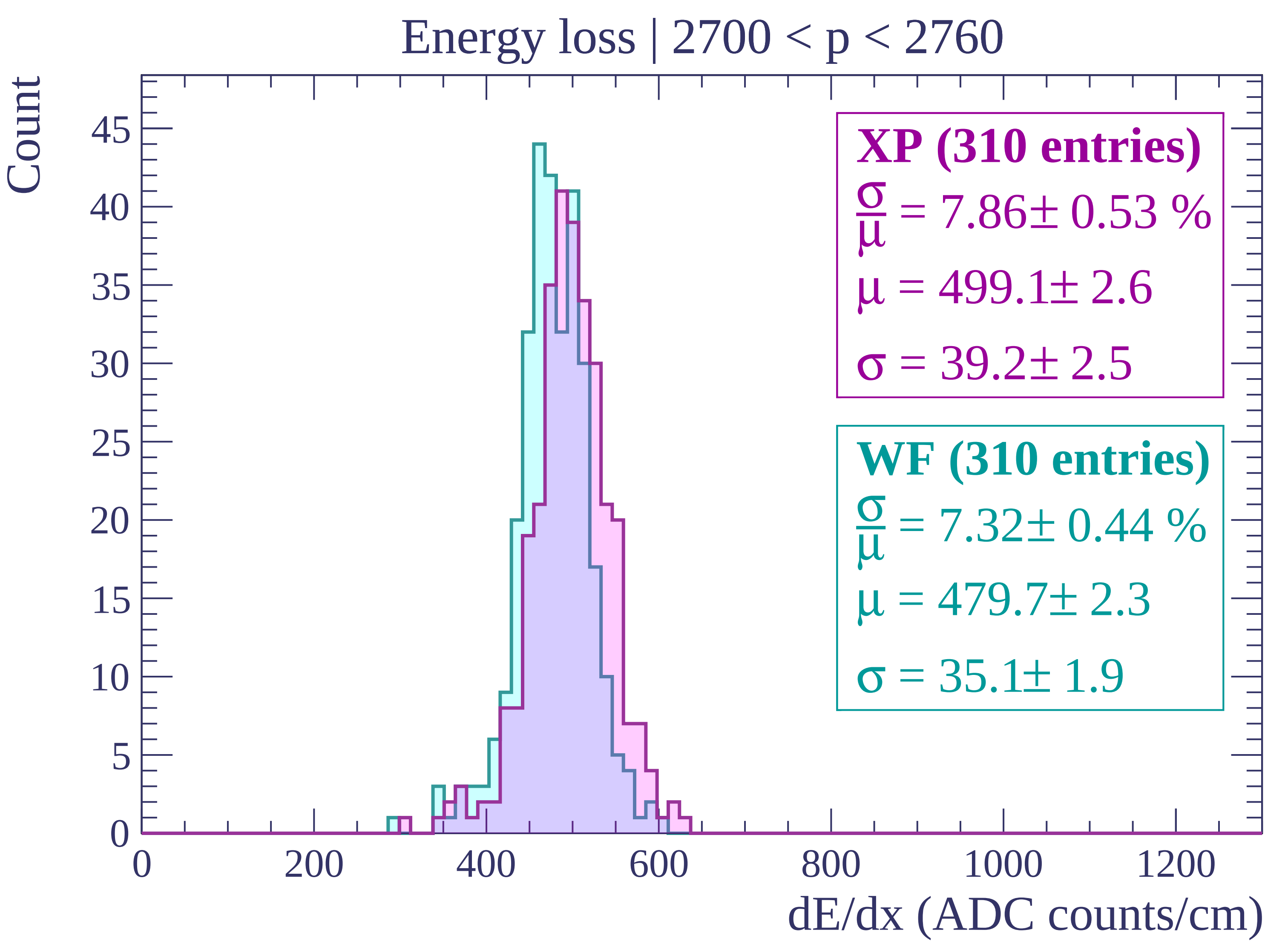


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
XP (402 entries)
σ/μ = 7.35± 0.51 %
μ = 503.2± 2.3
σ = 37.0± 2.4
WF (402 entries)
σ/μ = 6.62± 0.47 %
μ = 485.7± 2.1
σ = 32.2± 2.1
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

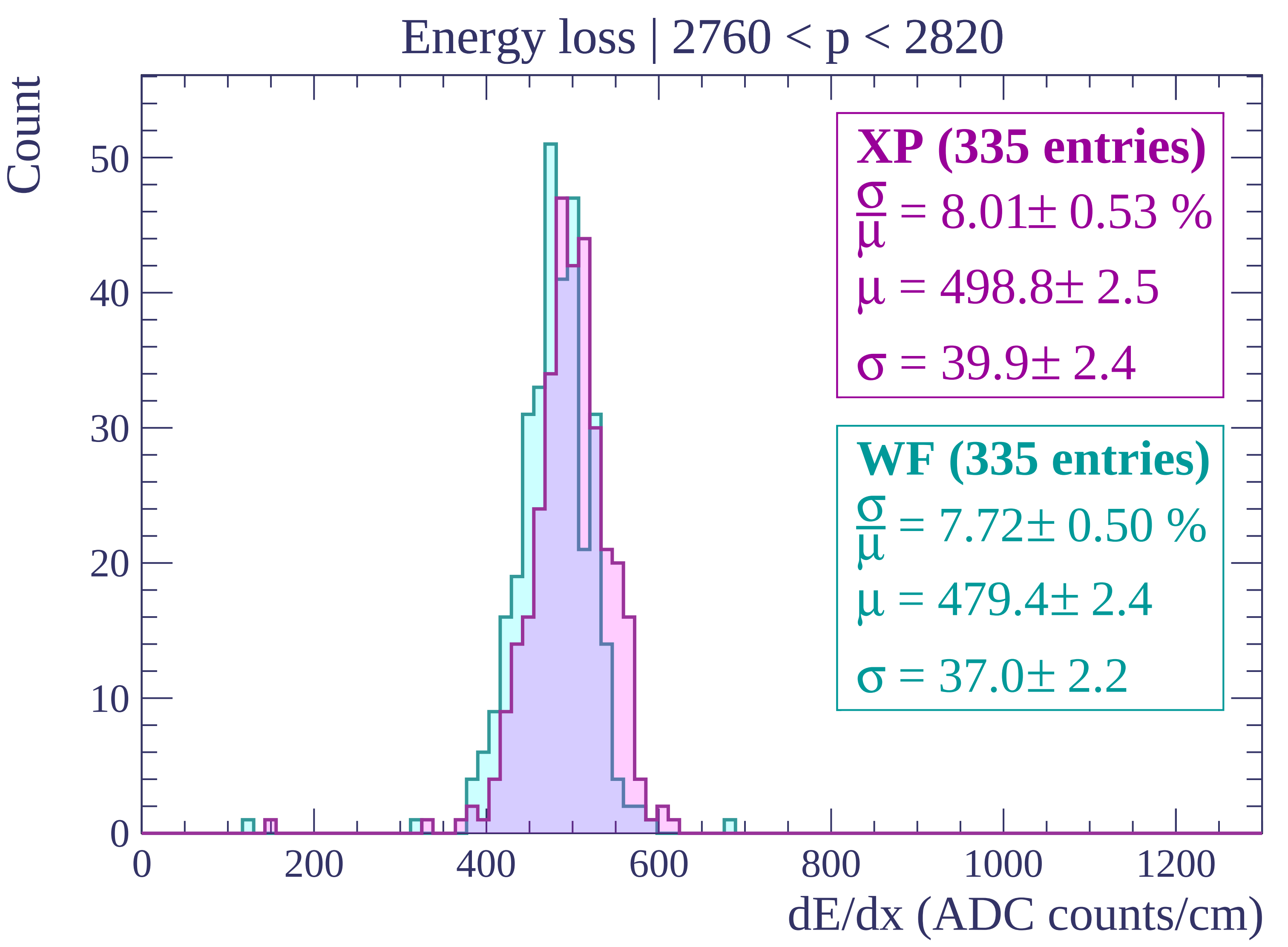

Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (332 entries)
σ/μ = 7.47± 0.54 %
μ = 491.6± 2.6
σ = 36.7± 2.5
WF (332 entries)
σ/μ = 6.63± 0.50 %
μ = 473.3± 2.2
σ = 31.4± 2.2

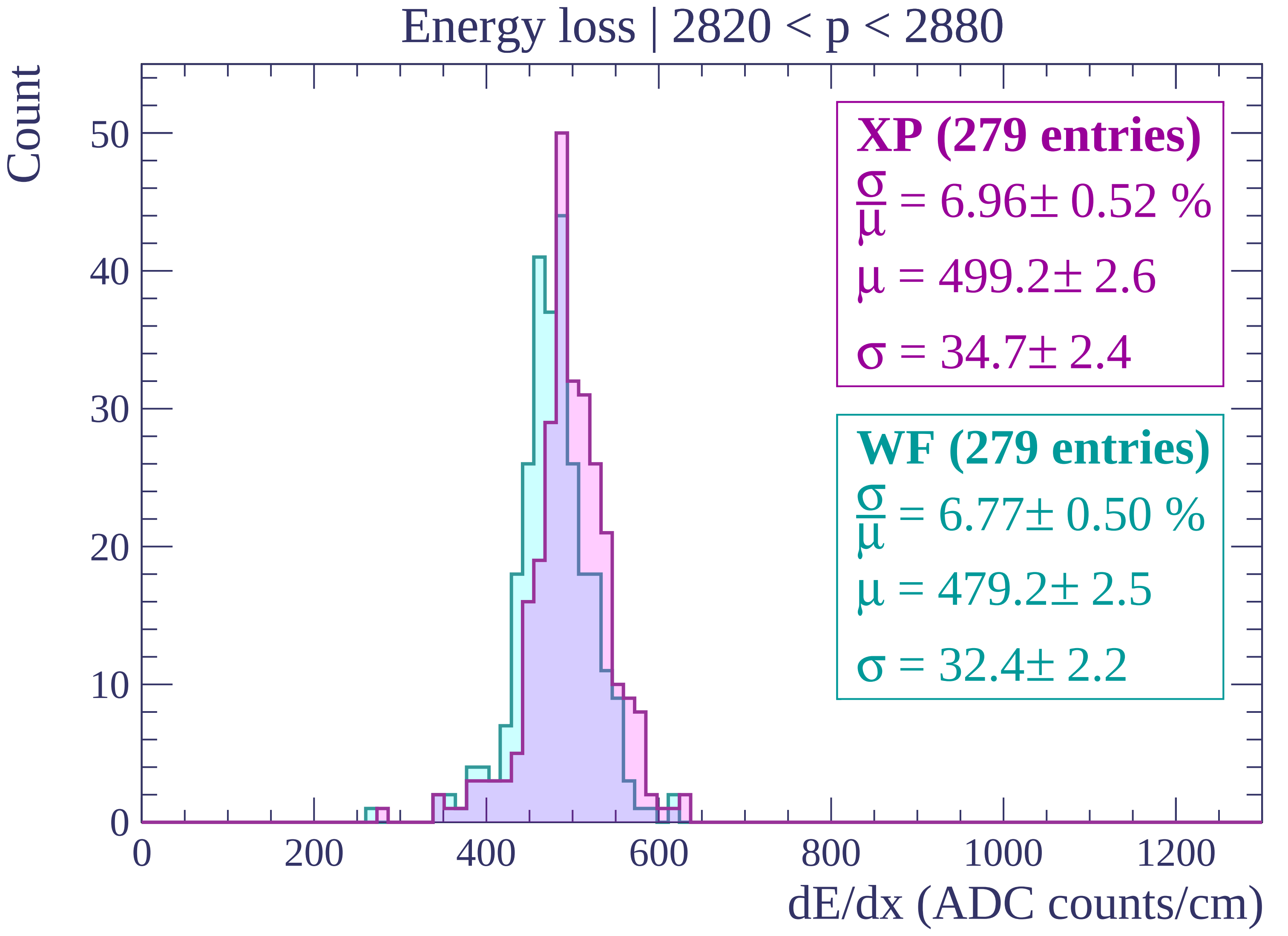


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP (331 entries)
$\frac{\sigma}{\mu}$ = 8.19± 0.65 %
μ = 504.3± 2.8
σ = 41.3± 3.0
WF (331 entries)
$\frac{\sigma}{\mu}$ = 8.25± 0.51 %
μ = 485.7± 2.5
σ = 40.1± 2.3

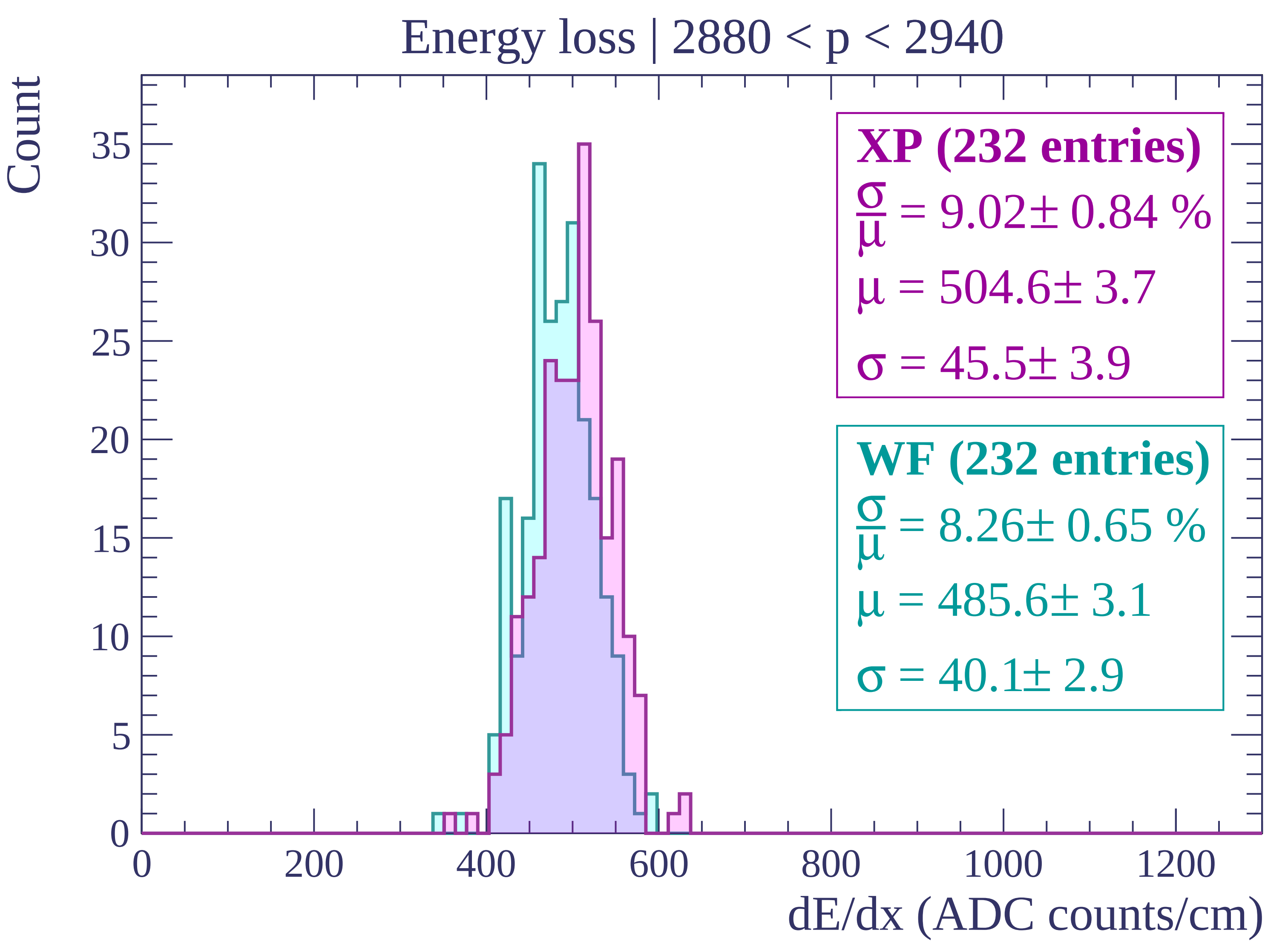


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP (310 entries)
σ/μ = 7.86± 0.53 %
μ = 499.1± 2.6
σ = 39.2± 2.5
WF (310 entries)
σ/μ = 7.32± 0.44 %
μ = 479.7± 2.3
σ = 35.1± 1.9

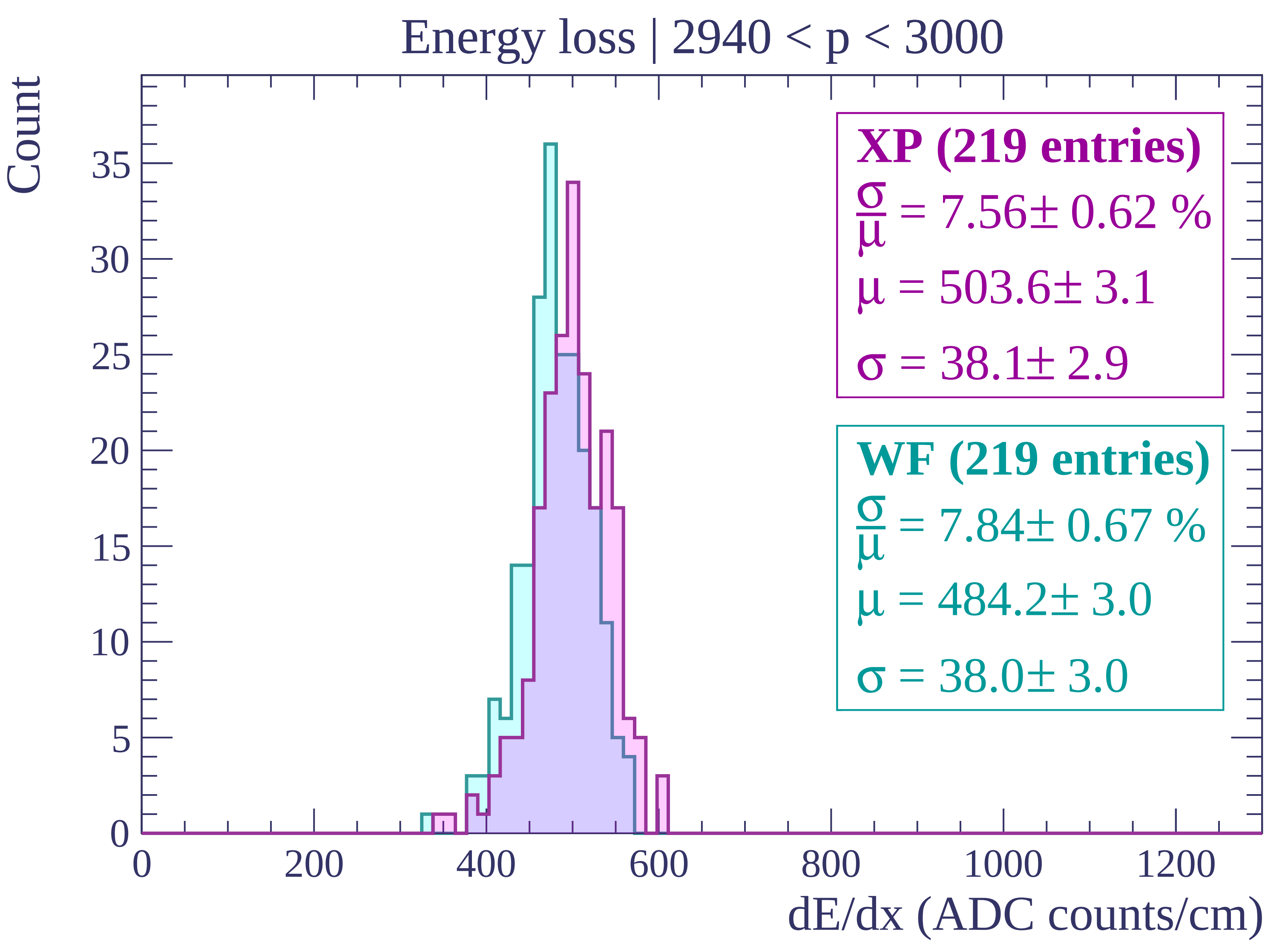

Energy loss | 2760 < p < 2820
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (335 entries)
σ/μ = 8.01± 0.53 %
μ = 498.8± 2.5
σ = 39.9± 2.4
WF (335 entries)
σ/μ = 7.72± 0.50 %
μ = 479.4± 2.4
σ = 37.0± 2.2



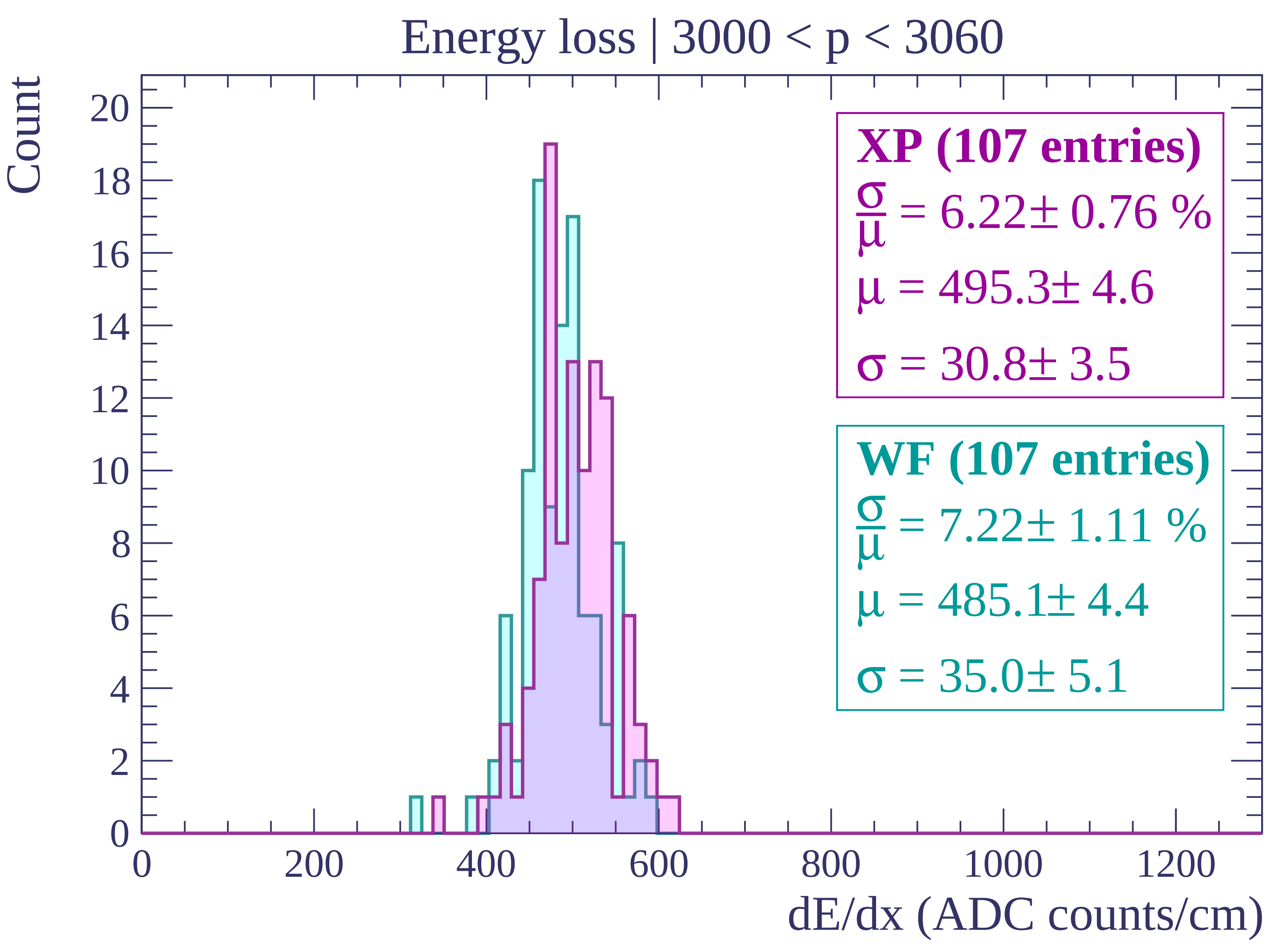
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (279 entries)
$\frac{\sigma}{\mu}$ = 6.96± 0.52 %
μ = 499.2± 2.6
σ = 34.7± 2.4
WF (279 entries)
$\frac{\sigma}{\mu}$ = 6.77± 0.50 %
μ = 479.2± 2.5
σ = 32.4± 2.2



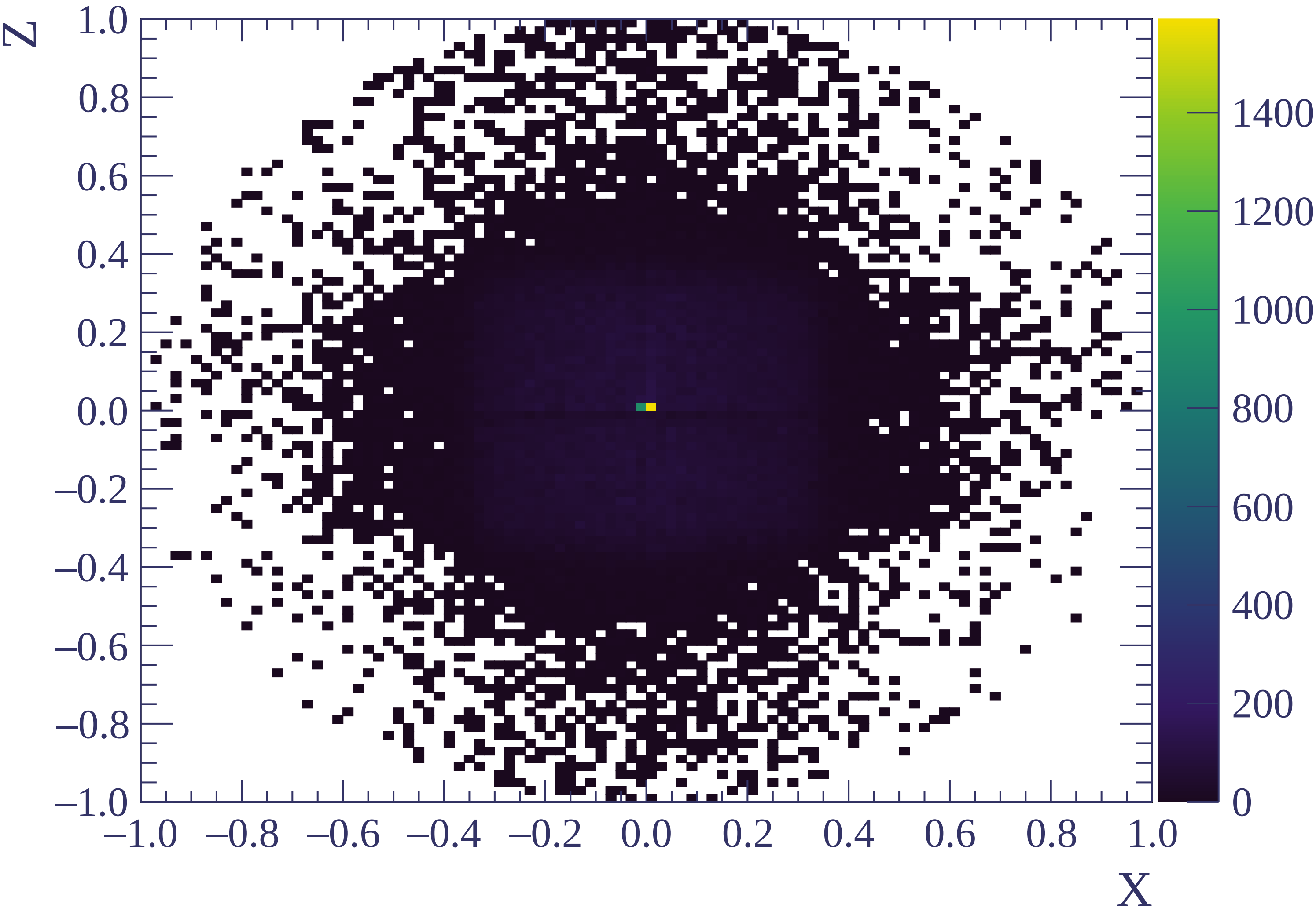
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (232 entries)
σ/μ = 9.02± 0.84 %
μ = 504.6± 3.7
σ = 45.5± 3.9
WF (232 entries)
σ/μ = 8.26± 0.65 %
μ = 485.6± 3.1
σ = 40.1± 2.9

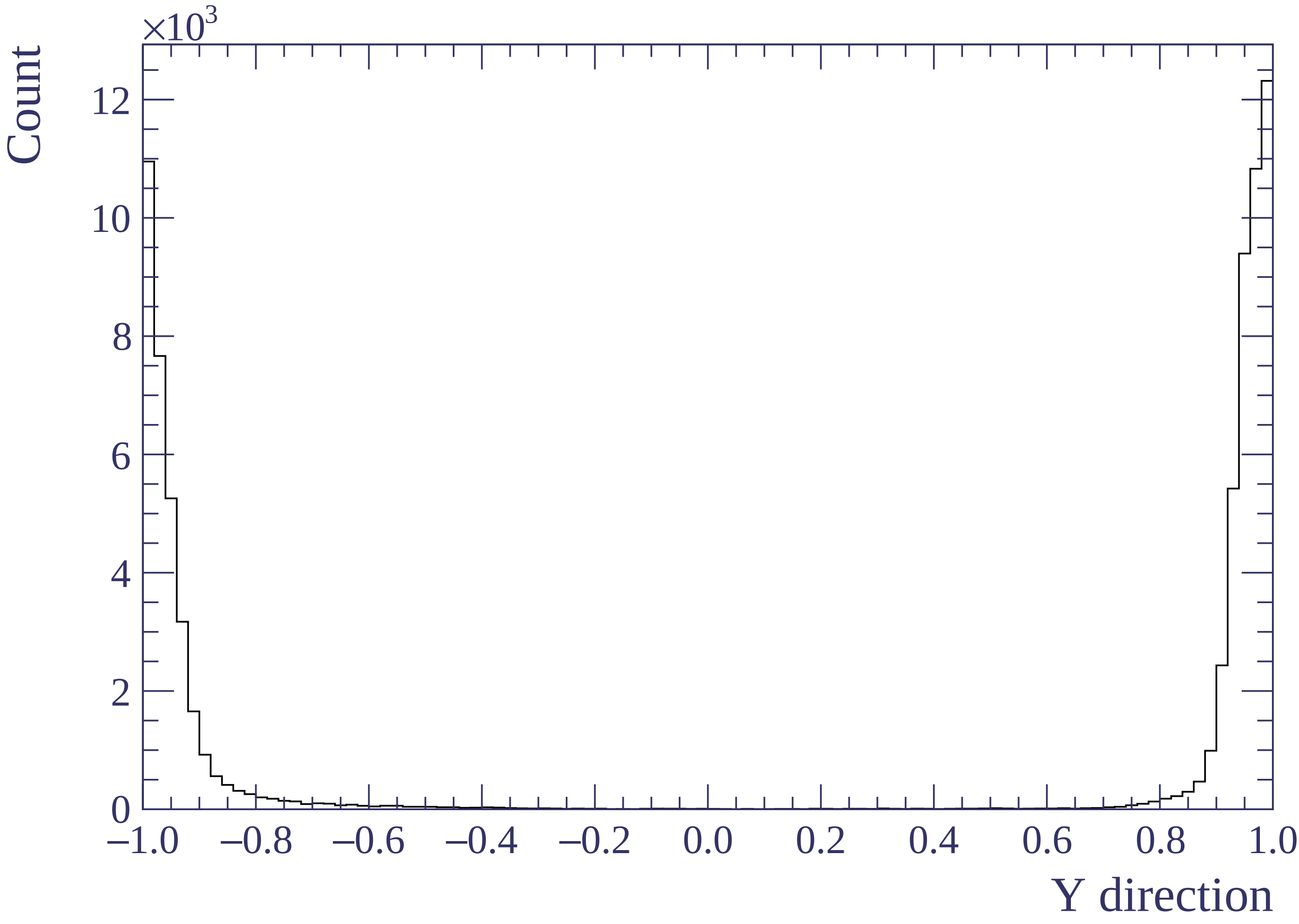


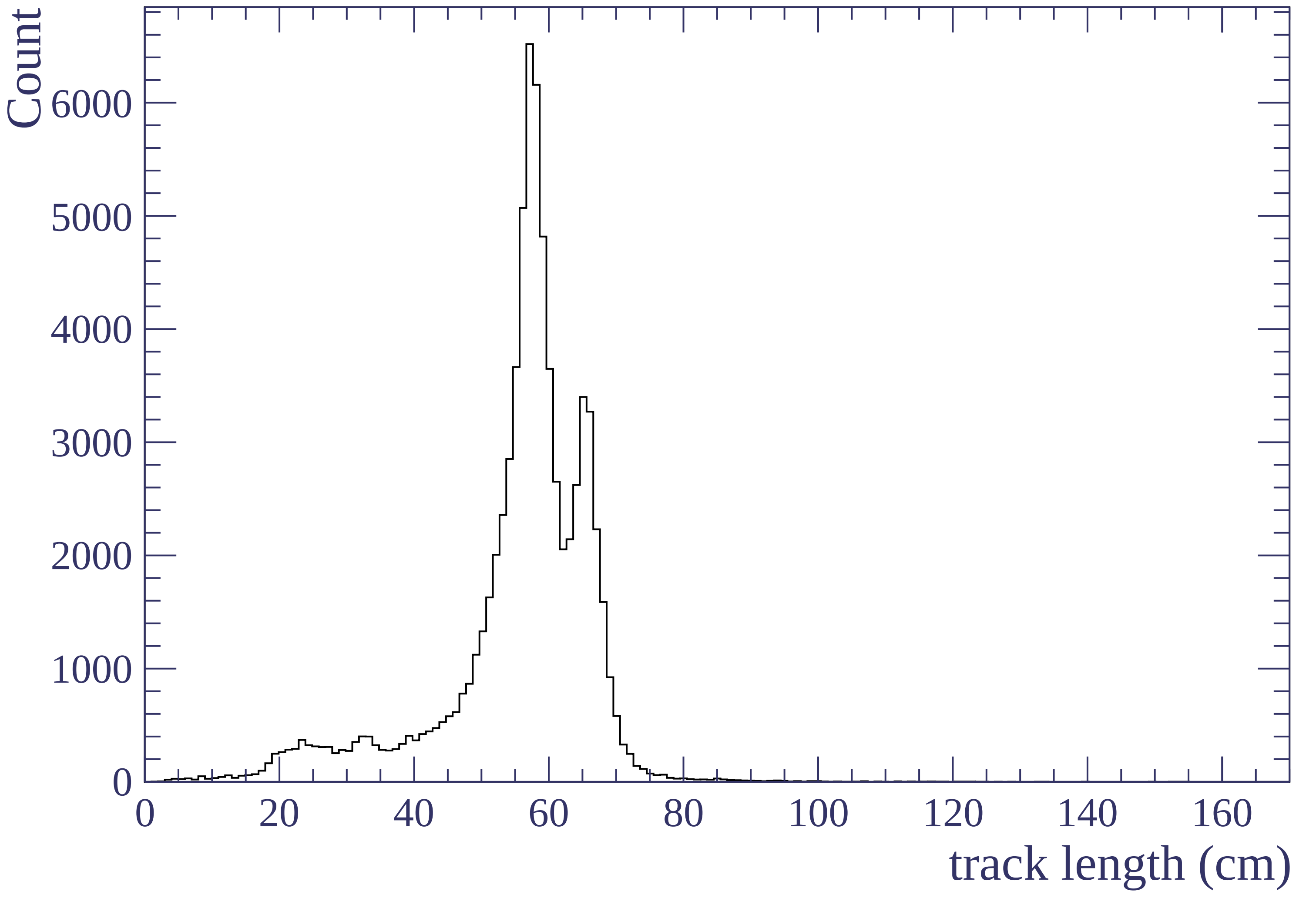
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (219 entries)
σ/μ = 7.56± 0.62 %
μ = 503.6± 3.1
σ = 38.1± 2.9
WF (219 entries)
σ/μ = 7.84± 0.67 %
μ = 484.2± 3.0
σ = 38.0± 3.0



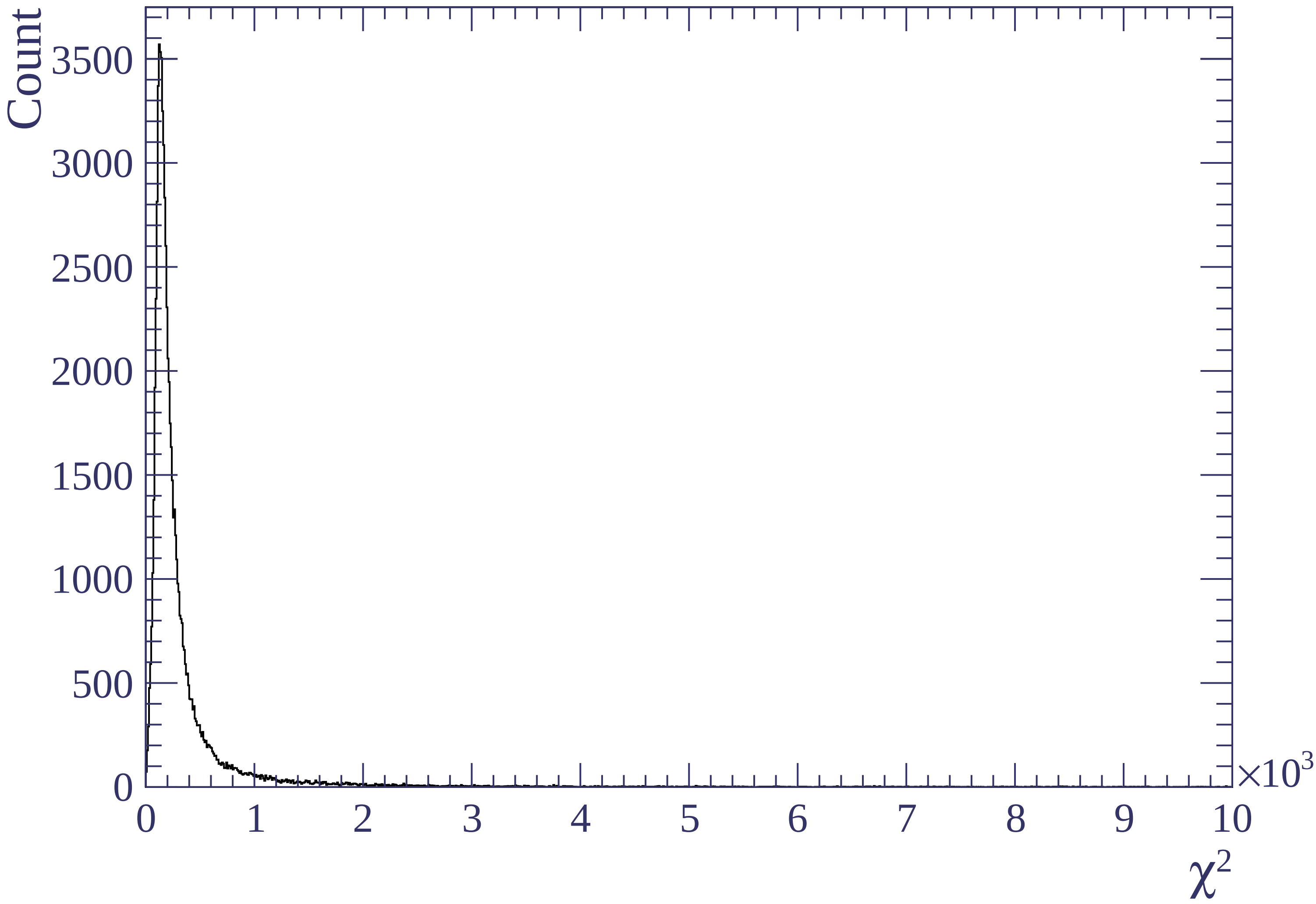
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP (107 entries)
σ/μ = 6.22± 0.76 %
μ = 495.3± 4.6
σ = 30.8± 3.5
WF (107 entries)
σ/μ = 7.22± 1.11 %
μ = 485.1± 4.4
σ = 35.0± 5.1

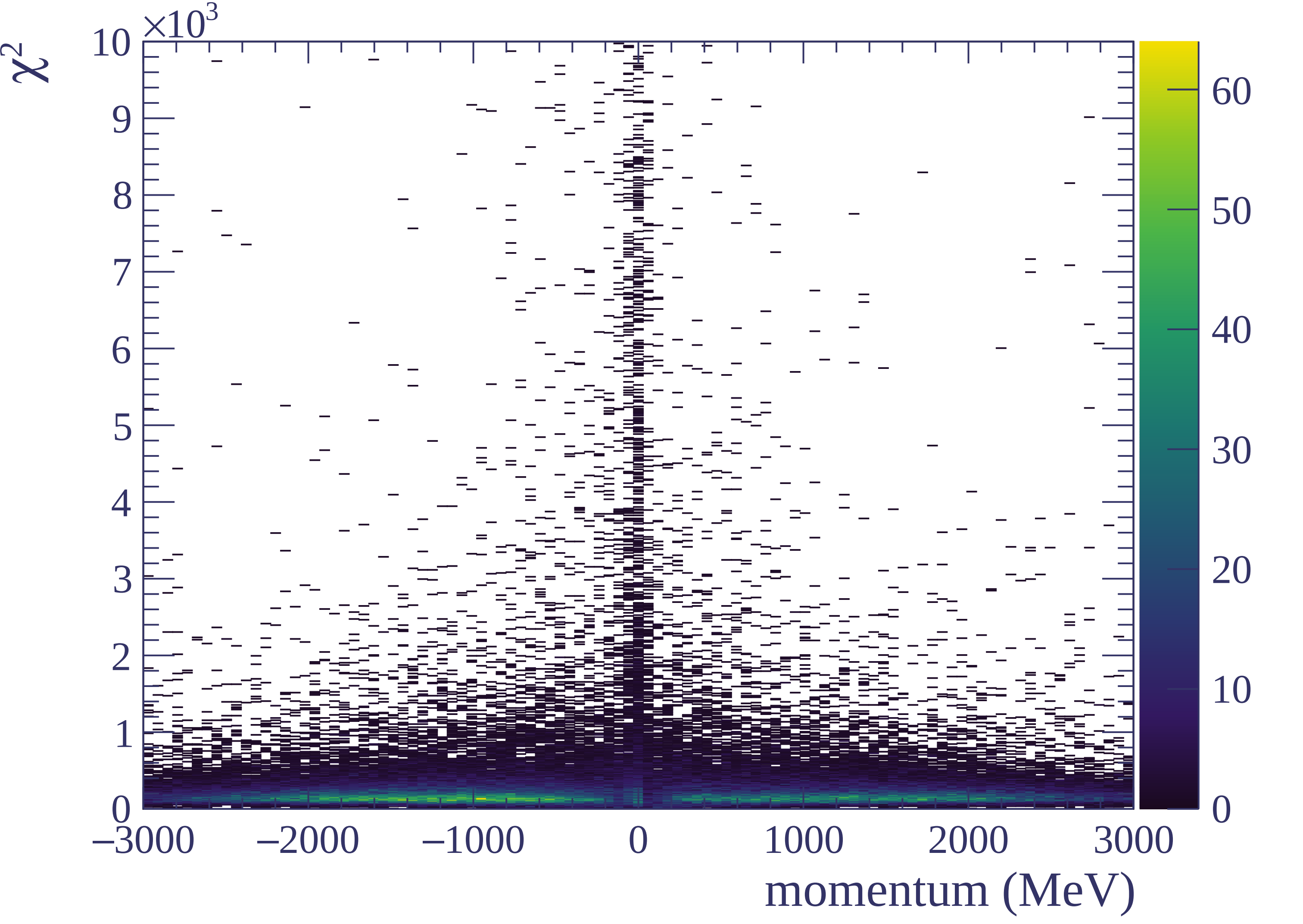

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
1400
1200
1000
800
600
400
200
0


×10^3
Count
0
2
4
6
8
10
12
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

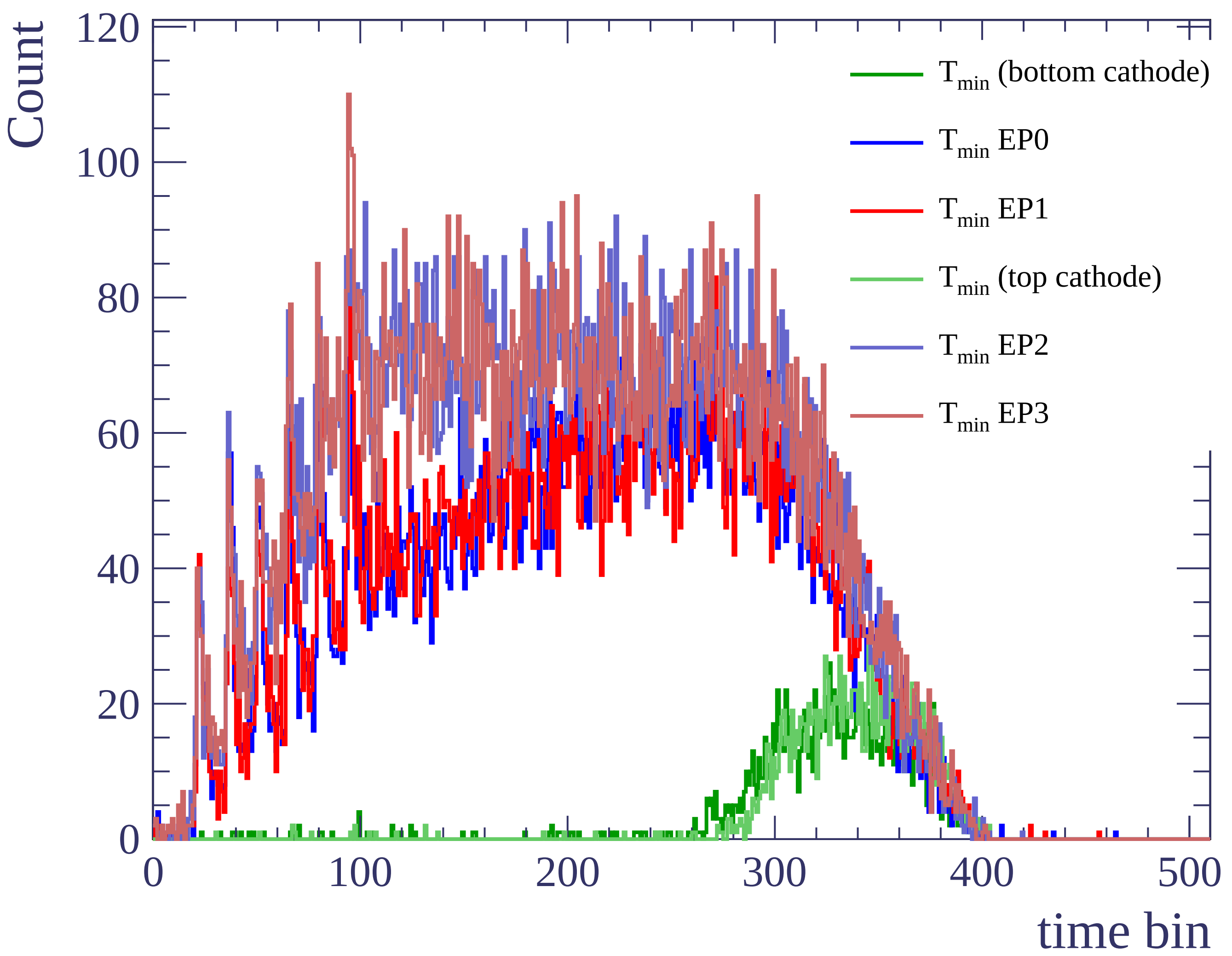


Count
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

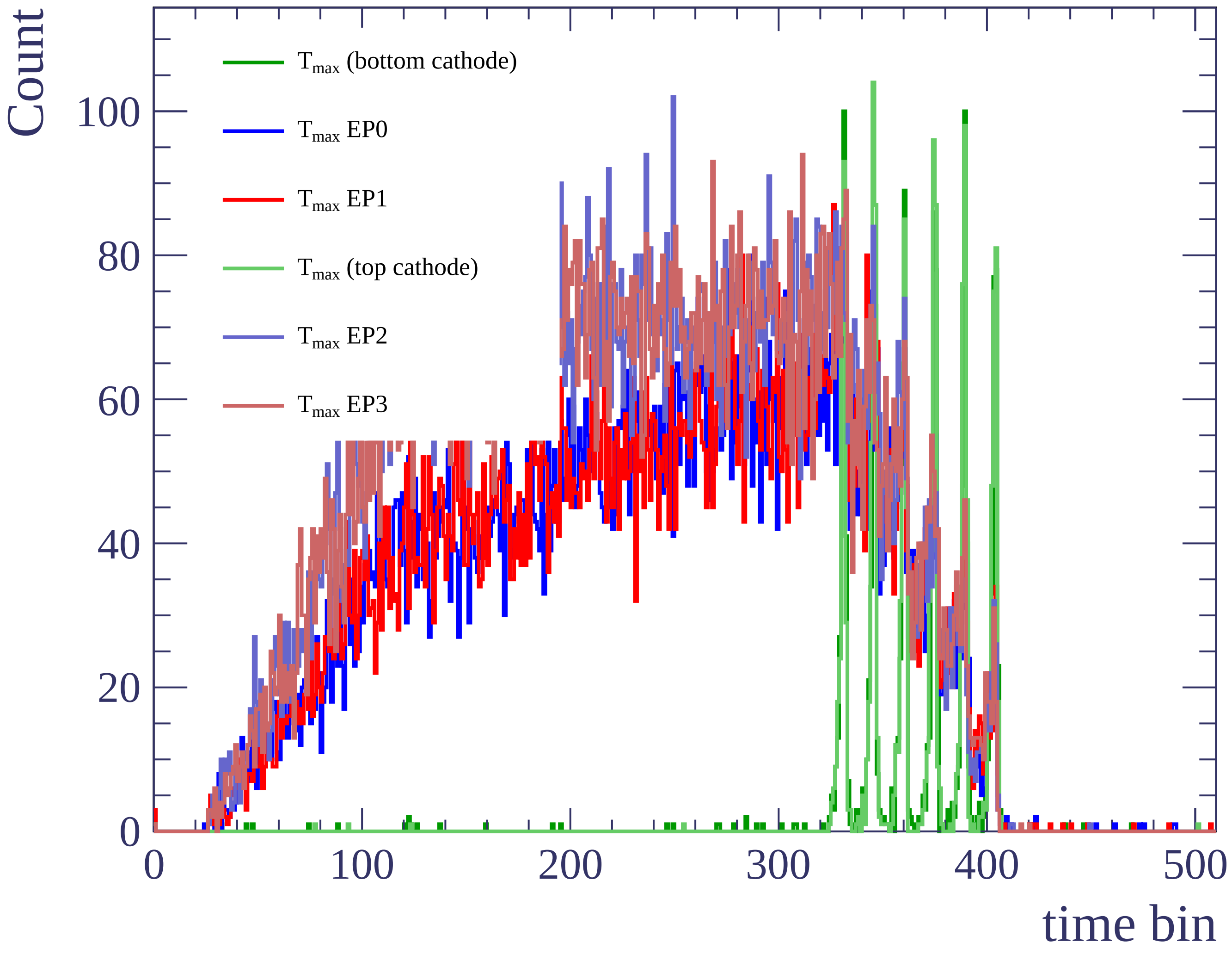

Count
3500
3000
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
60
50
40
30
20
10
0



Count
120
100
80
60
40
20
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
time bin
0
100
200
300
400
500
20
40
60
80
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3